

B.Ed. (2.5/4 Years)

TEACHING OF MATHEMATICS

Course Code 6450



Department of Science Education
ALLAMA IQBAL OPEN UNIVERSITY ISLAMABAD

TEACHING OF MATHEMATICS

For BEd (2.5 and 4 Years)

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**Department of Science Education
Faculty of Education
ALLAMA IQBAL OPEN UNIVERSITY, ISLAMABAD**

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FOREWORD

Learning is natural to the human beings, but in order to catalyze the process of learning the efforts of teachers contribute a lot towards educational attainments. It largely depends upon the skills of the teachers which they attained during the pre-service education. Mathematics most of the time considered difficult to understand it may be due to method and techniques of teaching that are less interactive. Therefore there is a need that prospective teachers may be equipped with the skills and techniques to engage the learners for the enhancement of his/her concepts of mathematics. This course aims to involve and provide exposure to the prospective teachers to different techniques that consequently help the teacher to make the mathematical concepts easier for the learners. There is no exaggeration to highlight that mathematics teaching is an integral and indispensable part of the teaching learning process. Therefore, for a teacher it is highly significant to understand the concepts of mathematics and nature of instructional techniques. Teachers should be able to plan and implement instructions in such a way that it instill positive attitude towards the subject of mathematics that consequently enhance the mathematics learning process. The development of this course intends towards the professional development of the prospective teachers in mathematics. The knowledge and skills gained during the course may help them while practicing in the classroom and also assist to develop a more positive attitude towards mathematics.

In the end, I am happy to extend my gratitude to the course team Course Development Coordinator, Dr. Muhammad Tanveer Afzal, Unit Writers and Reviewers for the development of this course book despite of the time constraint. I am also thankful to editing cell for the timely editing of the course. Any suggestions for the improvement of this course will be warmly welcomed.

August, 2019

Prof. Dr. Nasir Mahmood
Chairman/Dean

INTRODUCTION

For the development of problem solving skills, Mathematics plays pivotal role. Mathematics has very close and basic functions to play in most of other subjects such as Physics, Chemistry, and Astronomy. Mathematics is compulsory component of the curriculum being taught at every stage. Therefore teaching of mathematics is the key to develop the attitude of the learner towards this subject. The prospective teachers may be competent in their concepts about mathematics, but it's more important to consider that how these concepts should be taught in the actual classroom? And which instructional techniques should be utilized by considering the level of understanding of learners? This particular course is aimed to answer above two questions and equip the prospective teachers with the skills that may enhance the level of teaching of mathematics.

This course will equip the prospective teacher with knowledge and additional skills to teach mathematics in grades 1 to grade 10th. They will become familiar with the mathematics curriculum and expected student learning outcomes as much more problems, solved and un-solved examples have been incorporated. It is hoped that the prospective teacher will find the course as interactive for the development of skills of teaching. This will enable the prospective teacher to have a better concept of the subject and its techniques. This course will also increase the pedagogical education in mathematics as well as teacher guidance. The particular nine units course covers, nature scope and importance of mathematics and its teaching, brief history of mathematics and its relationship with other subjects has also been incorporated. Various Methods of Teaching Mathematics and application has been discussed with practical examples.

Laboratory learning and lesson planning will enable the teacher educators to actively involve the prospective teachers for development of attitude and skills for teaching. Different assessment techniques to be incorporated in lesson plans are also part of this course. Examples for development of concepts of sets, algebra, trigonometry and geometry are also placed to make course more comprehensive. The text includes relevant examples for the elaboration of the concepts and the activities are placed for the hands on works, which consequently, help to develop the attitude and the skills of the prospective teachers.

In the end, I am thankful to the Course Team, Editor and especially the unit writers for quick and timely response and effort.

Dr Muhammad Tanveer Afzal
Course development Coordinator

OBJECTIVES

Mathematics classrooms are the places where learners are engaged to appropriate tasks for the development of skills like, problem solving and analysis of the different situations. Teaching of mathematics to prospective teachers aimed to equip them to inculcate these skills among the learners. Therefore this particular course covers broad range of strategies for teaching of mathematics with following specific objectives.

After studying this course, you will be able to:

1. Recognize nature, scope and importance of mathematics and its teaching.
2. Utilize different methods and approaches for teaching of Mathematics.
3. Develop interactive lesson plans for teaching of mathematics.
4. Use instructional material and AV aids for teaching different concepts of mathematics.
5. Assess the learners' abilities in mathematics for adjusting instructions.
6. Provide practical examples for development of concepts of sets among learners.
7. Involve learners for solution of algebraic problems.
8. Create awareness and interest for solution of trigonometric and algebraic problems.

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Unit - 01

NATURE, SCOPE AND IMPORTANCE OF MATHEMATICS

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INTRODUCTION

Mathematics is considered as mother of sciences, means that it is difficult to understand other subjects without knowledge and skills of mathematics. Mathematics is logical in nature and has variety of skills that need to be mastered for studying basic sciences or even social sciences. Therefore, there is a need that children should be taught basic mathematics with understanding rather than rote memorization of concepts. This demands the alignment of mathematical concepts with the pedagogy adopted by the teachers. It is quite interesting that if teacher is competent in content of mathematics, he/she can create logical link with the pedagogies to be used in the classrooms. The overall aim of the pedagogy of mathematics teaching is to create mathematical potential among the learners; mathematical potential is sum of attitude, skill and logical thinking. It helps the learners to make sense of different concepts and procedures to solve the problems under consideration. It means that a good mathematics teacher is who help the learners to think, understand and apply variety of procedures to solve the problems. Here teachers have to understand that procedure and engagement of the learners to the meaningful activities are necessary components of learning. Research has shown different conceptions of the teachers about the nature of mathematics; some teachers consider it as a static discipline while the others consider it as ever changing discipline due to new discoveries. These concepts lead to different philosophy of teaching of mathematics. This demands that prospective teachers should be aware of how mathematics developed? How mathematical pedagogy influence learners knowledge and understanding? and how mathematics is related with other disciplines? In this unit we have tried to answer these questions, with the aim that having knowledge of discipline and pedagogy the prospective teachers will be in position to teach mathematics as per need and desire of the learners.

LEARNING OBJECTIVES

After the completion of this unit you will be able to:

1. Explain the importance, need and scope of the Mathematics.
2. Relate the Mathematics with other disciplines.
3. Explore the history of Mathematics.

1.1 NEED AND IMPORTANCE OF TEACHING MATHEMATICS

A teacher of Mathematics has to face many students with individual difference in the same classroom. Paul Ernest 2010 declared some reasons why mathematics is necessary to teach. Necessary Math is taught for the purpose of employment and economy. He included basic numeracy practical and theoretical knowledge and advanced specialist knowledge. In the second reason he included social and personal mathematics for the development of mathematical confidence and social confidence through mathematics.

1.1.1 Logic

The baseline of mathematics is logic. It has a main role in all Sciences. Both Inductive logic and deductive logic are the essentials of mathematics. Detective logic moves from specific to general while inductive logic moves from general to specific principles. The recognition of small set of fundamental ideas and rules from which all other attractive ideas and rules in that area that can be logically inferred is the main principle of theoretical mathematics. In various subjects the use of mathematics is different. So for the development of mathematical thinking and to pick up the mathematical skills, it is necessary to know the mathematical concepts.

Locke stated, “Mathematics is a way to settle in the mind a habit of reasoning”. Courant and Robin gave a more comprehensive definition of Mathematics “Mathematics is an expression of the human mind which reflects the active will, the contemplative reason and the desire for aesthetic perfection. Its basic elements are logic and intuition, analysis and construction, generality and individuality”

‘Number work’ and ‘Computation’ only is not mathematics, but it is more about making generalization, checking relationships, and enhancing logical thinking and reasoning. According to the National Policy on Education (NPE) (1986) “Mathematics should be visualized as the vehicle to train a child to think, reason, analyze and to articulate logically”. Mathematics should be revealed as an art or kind of beauty and as a mean of thinking or human achievement.

1.1.2 Creativity

Nature of mathematics gives an appropriate platform for enhancing creativity. Mathematicians and other scientists, tried to originate the unrelated parts of mathematics from one another or from the general theory. The excellence of mathematics lies in the simplest delineation of proofs. Greater relationship is created between its parts that were individually derived with the advancement of mathematical knowledge. For example, between the symbolic representation of algebra and special representation of geometry connection has been developed. We can achieve rightness in our calculations due to these cross connections and combinations.

Laycock (1970) stated that “mathematical creativity as an ability to examine a given

problem from different perspective, see patterns, differences and similarities, generate multiple ideas and choose a proper method to deal with unfamiliar mathematical situations” (cited in Idris & Nor, 2010).

1.1.3 Patterns and Relationships

Mathematics is considered as science of patterns and relationships. It finds out the possible relationship among abstractions due to theoretical subject. It does not matter whether these abstractions have relevant to the real world. Mathematics is thought as an applied science and to solve problems generated from the real world of experience is the focal point of Applied Mathematics. On the other side theoretical mathematicians are not restricted to the real world, but it provides better understanding about the world in the long term.

In all parts of mathematics patterns and relationships seems, not only in the area of formulas and functions but also in digits and chance, in geometry and figures. With the emancipating the concepts of functions and relations from the narrow confines of Algebra I, the Minnesota Frameworks foment teachers to give knowledge to students about the ligaments of mathematics, the connective tissue that retains the different parts combined.

Instructors give greater emphasize on authentic problems to associate mathematics with real world, which, encourage examination of realistic tasks for which mathematical relationships can provide significant awareness as compared to the conventionalized word jumbles of traditional algebra. Absolutely real data will be raw data, obscuring the perfect patterns that emerge from mathematical formulas. Both in applied and theoretical contexts, by examining patterns and relationships for the whole K-12 curriculum, students will see mathematics as an influential tool for examining the real problems not just as counterfeit game with signs. Patterns not only involved students in all areas of mathematics (arithmetic, algebra, geometry, probability, statistics, etc.), but also in all sorts of mathematical activity:

- Recognize: Determine mathematical possibility in different contexts;
- Visualize: Identify patterns in data and in non-mathematical contexts;
- Verbalize: Expression of the nature of patterns seen with the eye in words;
- Symbolize: The relationships discovered in patterns formalized in mathematical symbols;
- Analyze: predict new patterns by relating one pattern to another.

Patterns help learners not only to solve problems of mathematics but also develop ability of logic to authenticate mathematical statements. The potential of clarity allowed by algebraic reasoning glows like a light of rationality in a world darkened by a steady rain of conflicting claims, Patterns, relationships, and functions of algebra in schools: to assist pupils learn to think, and to produce them instruments to solve important problems.

1.1.4 Mathematical Inquiry

There are three levels of the expression of ideas or for the solving the problems.

- Abstractly representation of some components of things.
- To find new relationship manipulating the abstractions by rules of logic between them.
- Noticing whether the latest relationship is useful for the genuine things

The description of each mention here

1.1.5 Abstraction and Symbolic Representation

Mostly, mathematical reasoning begins with the procedure of abstraction by seeing common things between two or more facts or objects. Symbols like figures, words, signs, digits, some geometric formations, and letters are used for such common (whether hypothetical for concrete) things. For example from human faces a circle is an abstraction originated. For example 1, 2, and 3 the whole numbers are abstraction that presents the sequence of things within some realm or the extent of set of things. Likewise abstractions support mathematicians to think on some qualities of things and minimize the need to keep other attributes constantly in mind.

Since numbers cannot be observed by five senses because they are abstractions, with their names they are often puzzled. In using correct terms a teacher should be very careful, since this enables learners to think and learn well. For recognition the difference between indeed functioning with numbers and only manipulating signs showing those numbers, it is significant that a student comprehends the difference between a number and a numeral. In regard to correctness of language this is only one item. There are many else, like differentiating between the line and picture of a line, a point and the dot that is used to show a point, to list some.

1.1.6 Manipulating Mathematical Statements

The next phase after abstraction and symbolic representation of them is to connect and re connect those symbols in various methods accordingly to rightly described rules. By keeping predetermined goal in mind it is done sometimes. On the other side it is done for reasons of examination to observe what occurred when particular manipulations are being done. It is a continuing process. The intuitive meaning of constituent words and symbols can be identified easily from by an appropriate manipulation; and as opposed to this it is quite possible that by continuous trial and error useful series of manipulations have to be done. Normally, to express ideas or propositions, the string of symbols is joined into statements. For instance, for the area of any square the symbol A can be used and for the length of the side of square the symbol S may be used to design the proposition $A = S^2$. This equation describe in what way area is linked with the side and also shows that it relies on nothing more. This process of abstract relationship is not latest it is thousand years old and is still ongoing.

1.1.7 Application

“Knowledge is power” only when it is applicable. The requirement of the study of mathematics is the student can apply the skills to obtained new situations. The knowledge obtained by learners is vastly used for solving problems. By using and verifying the

mathematical ideas the learners may always ensure the authenticity of the mathematical rules and relationships. The knowledge and its use should be relevant to everyday situations wherever possible. When concepts and principles are connected to actual practical applications they become more operative and meaningful only. The learning of mathematics will make more purposeful and significant by practice.

The procedure of mathematics can direct to the advancement to any model. Any mathematical relationship developed by manipulating abstract statement that may be give something reliable about the thing being modeled. For instance if milk in three (3) glass is mixed into four (4) glass of milk and the Abstract mathematical operation $3+4=7$ is applied to sum the 3 and 4 glass, 7 glass of milk is the right answer. Although, if three (3) glass of milk is mixed into four (4) cups of sugar and the similar operation is applied, the sum of 3 and 4 is 7 that is not right answer. The answer is just 5 glass of much sweet tea. For first situation this simple addition of volume is correct but not for the second one. To understand and use mathematics well, it is essential to know the mathematical validity of abstract operation and to also consider how they will correspond to the features of the things represented. Now a day Mathematics is more beneficial for society than past. So it is known as applied science. Mathematician sometimes pays heed to such problems which are created from day to day life. The connection of scientific clients with mathematics has expanded their relationship. Now here is slightly space between an idea and its application.

Activity-1

Write down your views about nature of mathematics.

1.2 NATURE OF MATHEMATICS

The word Mathematics comes from the Greek word *máthēma* it means "related to learning" or "studious". So the study of properties of numbers and sets, measurement, relationships, using numbers and symbols is called mathematics. The branches of mathematics are arithmetic, geometry, algebra, and calculus. Mathematics shows secret patterns that assist us to understand the world around us. Mathematics is not only study of arithmetic and geometry but considered a diverse discipline that deals with measurements, data, and observations from science; with assumption, deduction, conclusion and proof; and also with mathematical models of natural phenomena, of human behavior, and of social systems.

From a practical point, mathematics considered as science of pattern and order. Molecules or cells are not its domain, but algorithms, numbers, formulas, and change. It is science of abstract objects. Mathematics based on logic not on observation. As its paradigm of truth, for discovering truth it uses observation, simulation, and even experimentation.

In education the particular role of mathematics is the consequence of its universal applicability. The conclusions of mathematics theorems and theories are important and

useful; the right results are also splendid and elegant. Mathematics is considered as science that provides a base of truth as well as a standard of certainty by theorems.

Additionally to theories and theorems, mathematics comprising on modeling, abstraction, optimization, logical analysis, conclusion from data, and usage of symbols, gives unique methods of thinking which are flexible and influential. In this technological age mathematical methods of thinking builds mathematical power a competency of mind of growing value that allows one to read critically, to identify errors, to identify bias, to measure risk, and to suggest alternative solutions. The information- loaded world in which we survive mathematics enables us to comprehend better this world.

During the first half of the twentieth century, by the power of abstraction and deduction mathematical development was aroused primarily, Isaac Newton developed mathematical principles of physical science to extract full benefit by effort of climaxing more than two centuries. Now the historic relationships of mathematics with science are increasing speedily as the century end.

The development of latest and influential methods of applied mathematics started after Second World War. The growth of mathematics increases in both education and research by postwar government investment in mathematics and fueled by Sputnik. Mathematicians have an algorithmic perspective and a powerful tool for examining patterns and testing assumptions became by the development of electronic computing.

Mathematics is inbuilt within the logical power of human beings it is practiced for intrinsic interest as well as for practical purposes. For others the importance of the mathematics depends on its applicability.. Mathematics is involved everywhere either in the simple task or to complicated one. We execute many functions which need decisions including problem solving. The value of mathematics increases as compare to past. The interaction between scientific client and mathematics has enhanced by increasing continuously scientific society. Now a day as compare to previous centuries scientists are more concerned about Mathematics.

Remember

- Mathematics is inbuilt within the logical power of human beings
- There is need to comprehend the nature of mathematics for scientific literacy due to crucial role of mathematics is modern culture.
- The baseline of mathematics is logic.
- In detective logic, moves from specific to general examples
- In inductive logic, moves from general to specific principles.
- Creativity is always use in solving mathematics
- Mathematics is considered as science of patterns and relationships
- There are three levels of the expression of ideas or for the solving the problems.
 - Abstractly representation of some components of things.

- To find new relationship manipulating the abstractions by rules of logic between them.
- Noticing whether the latest relationship is useful for the genuine things

1.3 SCOPE OF MATHEMATICS

Mathematics is a very exciting subject. We can use it in daily life as we deal with seller by using mathematics when we go to purchase anything. By applying mathematics we can prove anything. In future, if anyone has interested in mathematics and wants to go with mathematics then researcher, mathematics professor or mathematician profession can be chosen. Hence we cannot ignore the scope of studying mathematics.

Man has created Mathematics as other inventions exist to fulfill certain human needs and desires. It is very hard to say that in the history of mankind where and when mathematics had its birth. The fact that it has been constantly pursued till many centuries, that it has increased attentions and that greatest intellectual interest of mankind indicates that its application very influential for mankind. This concluded by everything from which we familiar about the starting of mathematics. Before the starting of the Christian era, both the Egyptians and the Babylonians were in possession of organized approaches of measuring space and time. They had well aware about the rudimentary geometry and rudimentary astronomy. To meet the functional needs of an agricultural population the rudimentary mathematics was developed. In the southern part of Central Asia with the Indus and Ganges rivers same trying seems in early times and there is needed mathematics in Eastern Asia Projects of engineering, financing, flood prevention, irrigation and navigation. To serve agricultural needs an operable calendar had to be designed. This is possible to led the positional notations for whole numbers and after to the same notation for fractions by defining Zero. It proved as present of this time that place value system finally devised. These attainments and many other of a same nature are the victory of the human mind. It became more intricate as they answered to the needs of human society. Ancient men can hardly be said to have created or developed their arithmetic, in which they in fact lived. The men who formed the stones in lifting the Temple of Mathematics were extensively spread in Egypt, India, Babylon and China. These workmen faced nature and worked according to it. Their products therefore, participated in the unity of nature and spread in time and space. To meet the cultural needs of time the man has himself created mathematics. Practically to represent numbers every ancient group have created words. But it was only when trade, architecture, taxation and other civilized contracts that the number systems were developed by ancient civilizations such as the Sumerian, Babylonian, the Chinese and the Mayan. As a result, one of the most significant cultural elements of our society mathematics has expanded. Without mathematics it hardly be possible our advanced life style. Imagine without using a number in some manner or the other trying to get through the day. He is a man as disabled, if a person does not have the ability to compute. For example, it is necessary to know the time and tell the exact. It is complicated to tell the time yet nearly everyone learns to see and tell it. Before long, we shall alter the old fashioned clock with digital

readings to read time. In money, in 'weights and measures a degree of estimation is very important. We manage many mechanized appliances which need geometrical or special skills in our daily routine works including classifying, ordering and organizing processes. Interpreting scales and reading of maps, diagrams have become a crucial part of our intellectual equipment. By using mathematics we can understand and explain matters like income tax and read information provided to us by the mass media in numbers or in the shape of graphs and we can also comprehend the use of phrases such as rising prices, per capita index, stock market index, income and inflation, so the knowledge of mathematics is useful for us. In common daily speaking language, It is not essential to provide a complete list to demonstrate the situation of advantages of "mathematics for survival" or "useful mathematics".

1.4 RELATION OF MATHEMATICS WITH OTHER SUBJECTS

Mathematics is considered as a language of science. According to Italian physicist and astronomer Galileo Galilei “Mathematics is the language in which God has written the universe.” Unless we have well-known the language and vocabulary of it, the universe cannot be read and become. Mathematics as an international language, it does not matter from where you belong and what your speaking language, the structure of mathematics is the similar. For representation of the parameters and variables, we make use of alphabet. commonly integers are showed by i, j, k, l, m, n . and Real numbers are indicated by $a, b, c, \alpha, \beta, \gamma$. Complex numbers are pointed out by w and z . Unknowns are showed by x, y, z . functions' names are usually specified by f, g, h .

1.4.1 Relationship with Science subjects

Here, the Role of Mathematics is broadly admitted in some main subjects.

1.4.1.1 Mathematics in Physics

In Physics, the mathematical form has used for every rule and principle. For given the final shape to the rules of physics mathematics have used. It shows them in a practical form. At every step in physics, mathematical calculations uses. For substances in physics as in mathematics the units of measurement are used to. For expansion of gases the Chare's law of is established upon mathematical calculations. Graduation of the stem of thermometer and then the transformation of scales is possible through mathematical.

In mathematical physics, through observations and physical experiments about energy, temperature, mass, momentum, temperature, force, heat some primary Maxim are abstracted, and then logical inference, the tricks of abstraction and generalization are employed. The branch of mathematical analysis stresses on devices and skills of special use to physicists and engineers. It emphasis on integral equations, vector spaces, differential equations, matrix algebra, integral transforms, complex variables and infinite series. In electromagnetism, quantum mechanics and classical mechanics, its technique can be adapted to applications.

In mathematics, a particle is a point-like, perfectly stiff, solid object. Particle mechanics deals with the inferences of subjecting particles to forces. It comprises celestial mechanics that is study of the motion of celestial objects. In mechanics the role of Space Dynamics is very significant. Here we mean the trajectories that considered time optimal i.e. for going from one point to another it take the minimum time and with zero velocity the object begins journey and reaches the target point.

1.4.1.2 Mathematics in Chemistry

In physical chemistry Math is also play an important role particularly advanced topics like quantum or statistical mechanics. Quantum depends highly on group theory. Physical topics such as Hilbert spaces and Hamiltonian operators require the linear algebra and knowledge of mathematical. Statistical mechanics depends highly on probability theory. A significant amount of mathematics also applies in other fields of chemistry. For instance, to obtain spectra, most modern IR and NMR spectroscopy machines apply the Fourier transform. Even biochemistry has significant topics such as binding theory and kinetics which depends highly on mathematics.

Certain mathematical laws are used to govern the all chemical combinations and their equations. Mathematical calculations are used to govern the creation of chemical compounds. For example, water is a compound and the formation of water is feasible when exactly two atoms of hydrogen merge with one atom of oxygen. The formulation is impossible without this rigorous observance of the mathematical fact.

1.4.1.3 Mathematics in Biological Sciences

Mathematics is applied in all major field of science including biology. It is a field of research that examines mathematical representation of biological system. Biomathematics deals with challenging and solve problems in the field of mathematical genetics, mathematical ecology, mathematical neuron- physiology, formation of computer software for particular biological and medical problems, mathematical theory of epidemics, application of mathematical programming and reliability theory in biosciences and mathematical problems in biomechanics, bioengineering and bioelectronics.

1.4.1.4 Mathematics in Engineering and Technology

The application of mathematics is very familiar in engineering. It is believed as the foundation of engineering. It is very important in surveying, grading, estimating, designing, and construction etc. The durability of constructed things can be expanded with the application of geometric principles that used for design and constructions. Inferences can often be verified in engineering by using it.

Mathematics has many contributions to mechanics of rigid bodies, aero-dynamics, heat transfer, hydro-dynamics, turbulence, aero-dynamics, lubrication, elasticity and in the formation of civil aeronautical, mechanical and chemical engineering. Through its usage to information theory, automatic control systems, cybernetics, analysis and synthesis of networks, construction of digital computers, it has become of high attention to electrical

engineers. For manufacturing magneto-hydrodynamic generates, flow meters, and for experiments in controlled nuclear fusion, the latest mathematical sciences of magneto-hydrodynamics and plasma dynamics are applied.

1.4.2 Relationship with Arts Subjects

"Mathematics and art are just two different languages that can be used to express the same ideas." It is believed that in the mathematics language the universe is written. Triangles, circles, and other geometric figures are its characters. The ancient Goethic Architecture is developed on geometry and the Egyptian Pyramids, the greatest feat of human architecture and engineering, were developed on mathematics. Artists, who want to study the nature, should understand the mathematics fully. On the other side, by using the lens of geometry and rationality, mathematicians have tried to understand and explain art. This branch of mathematics deals with the nature of geometric objects by enabling them to misrepresent and alteration.

Mathematical literacy is necessary for continuously increasing the individual activities as well as social. For obvious thinking mastery of the essential processes is required. The social sciences are also starting to trust on mathematics.

For defining economic phenomena mathematical language and approaches are applied commonly. Marshall stated "The direct application of mathematical reasoning to the discovery of economic truths has recently rendered great services in the hand of master mathematicians." Statistical techniques are employed to predict different problems that can occurred and described statistically like, Trend of exports and imports, Volume trade, population trends, Trade Cycles industrial trends and Expenses of public money etc.

In the field of economic theory and econometrics, a lot of mathematical work is being done around the world. Tools of matrices, probability and statistics are employed in econometrics. A large number of mathematical thinking utilized for national economic planning and a lot of mathematical models used for planning have been developed.

1.4.3 Relationship with Business Education

The relationship between Mathematics and Business is powerful that without the previous existence, the latter cannot flourish. It makes sense to think that in the absence of knowledge of simple addition and subtraction one cannot administer any business. How much we assumed to calculate profit and loss? We need to calculate profit and loss in percentage in next growing stage it means we should have knowledge of percentages too. In business, these are the initial stage of employing mathematics. An accountant of a business firm has much mathematics knowledge and ability to explain in better way. If ask an accountant about the connection, he can spend couple of hours in explaining it. Different organizations employ mathematics in inventory management, sales forecasting, marketing, accounting, and financial analysis. The application of mathematics is an essential not just an accomplice in many stages and conditions.

At the very first stage while initiating the business mathematics uses to estimate the production cost. For attractive income, all the main business firms estimated in advance

the production cost of all steps and continue reviewing the cost according to the market conditions and economy.

The next step after initiating the business is using different accounting tools to analyze the finances. When change the number to show an increase or decrease in future sales (revenue) then it's very significant to comprehend the impact of accounting records. Also when a competitor initiates marketing an inexpensive product, you may require estimating the amount by which your volume must increase if you minimize prices. To improve sales you may require knowledge of mathematics if you can able to enlarge your business. For prediction of future performance it is essential part of the financial planning of a business is sales forecasting by using old and latest sales statistics. Job cost sheets are prepared regularly to monitor the expenses on a job. All this is executed with the help of different mathematical methods (not just including basic addition, subtraction, multiplication, division, percentages etc.)

Now, here are the descriptions of the importance of diverse mathematical fields in the business.

1. **Statistics:** it is the most well-known branch of math. In business, with the help of knowledge of statistics charts, graphs, histograms are created.
2. **Linear Programming:** In a mathematical model it is use for achieving the best outcomes (maximum profit or lowest cost) and by linear relationships demands are represented.
3. **Probability:** It is the science of incertitude or chance. A probability value ranges between 0 and 1. It presents the chance that a specific event will occur. A probability value of 0 shows there is no chance that an event will occur and a value of 1 show there is 100 percent chance that the event will occur. For a business decision it employed to know the degree of risk.
4. **Calculus:** In business, we discover several such variables where one variable is an operation of the other. For instance, the demanded quantity called function of price. Some other demanded variables are supply, price, cost and quantity. Calculus assists in searching the rate at which one such quantity changes from other. Marginal analysis is the most direct utilization of differential calculus in economics and commerce. In this context, in finding maximum profit or minimum cost differential calculus is helpful. To find the cost function integral calculus is used.
5. **Algebra:** we cannot use statistics if we have no knowledge about algebra. Now it is clear how much strong relationship between mathematics and business. There are many other fields which are compulsory in a business in addition to the above mentioned fields. We cannot ignore the importance of mathematics in business field.

Activity: 2

Collect some examples of relationships of mathematics with other subjects.

1.5 HISTORY OF MATHEMATICS

The history of mathematics is very old. It is very difficult to conclude that when did mathematics start. From the history we came to know that before Greeks there were many cultures which have used in mathematical operations including simple counting and measuring onward and they solved problems of differing degrees of difficulty. But how mathematics started properly? The ancient history of Greek mathematics is difficult to reconstruct with certainty. In contrast, the history of the much more ancient civilization of Iraq (Sumer, Akkad, Babylon) in the years from 2500 to 1500 BCE provides a quite detailed, if still patchy record of different stages along a route which leads to mathematics of a kind. In the civilization of human beings, the role of mathematics is very important. The history of mathematics educates the learner about the development of this subject over a period of time. Therefore it is the source of knowledge and enlightens the learner.

The importance of the history of mathematics is as follows.

- It enables the learner to recognize the development of human beings over a period of time. The learner would know the discoveries of mathematical facts.
- The history of mathematics reveals that it has connection with other branches of science.
- The teaching of the historical development can bring a change in the classroom environment and get create interest among students.
- It enables the learner to know that all the branches of mathematics are interlinked with one another.
- The historical background of mathematics will provide reference to understand the terms, conventions and concepts.
- Historical review of mathematics is revealed that the significant development of mathematics was conditioned by human needs.
- Some mathematical concepts can be introduced in a better way by discussing the history of those concepts.

1.5.1 Babylonians

One of the most ancient mathematical text is available of Babylonian mathematics. It contains some far more advanced mathematics. The Babylonian mathematics refers to any mathematics of the people of Mesopotamia (modern Iraq) from the days of the early Sumerians through the Hellenistic period almost to the dawn of Christianity. Its name "Babylonian mathematics" is due to its place of study. Afterwards Mesopotamia especially Baghdad under the Arab Empire once again became a vital center of study for Islamic mathematics. It contained quite Complex algebraic problems featuring on cuneiform tablets in Cuneiform script. The calculation depends largely on tables. from around 2500 BC onwards, Sumerians wrote multiplication tables on clay tablets and dealt with geometrical exercises and division problems the earliest traces of the Babylonian numerical also date back to this period. The majority of recovered clay tablets dates from 1800 to 1600 BC and cover topic which include fractions, algebra, quadratic and cubic

equations, and the calculations of regular reciprocal pairs. The tablets also include multiplication tables and methods for solving linear and quadratic equations. The Babylonian tablet YBC 7289 gives an approximation of $\sqrt{2}$ accurate to 5 decimal places. In Babylonian Mathematics typical question was expressed in geometrical terms, but the nature of its solution is essentially algebraic. The numerical system with a base of 60 is used in Babylon mathematics. The modern day usage of 60 second in a minute, 60 minute in an hour and 360 (60×6) degrees in a circle are derived from this base system. Babylonian advances in mathematics were facilitated by the fact that 60 has many divisors. Also, unlike the Egyptians, Greeks and Romans the Babylonian had a true place value system, where digits written in the left column represented large values, much as in the decimal system. They lacked, however, an equivalent of the decimal point, and so the place value of the symbol often had to be inferred from the context.

1.5.2 The Sophist School

Greeks introduced democracy in the country in 480 B.C. due to that every citizen tried to be a politician. Therefore education had to be provided to every common man. Therefore there was demand for teachers. The supply came principally from Sicily, where Pythagorean doctrine had spread. The teachers were called "Sophists or Wisemen".

The following subjects were taught.

- Geometry
- Language
- Philosophy astronomy

Sophists took up the "geometry of circles", which was neglected by Pythagoreans. The following problems were tried by them to be solved.

- To square the circle
- To double the cube
- To trisect an arc or an angle

Democritus, Zeno, Bryson and Antiphon were the famous mathematicians of that time.

1.5.3 The Platonic School

In 429 B.C. at Athens Plato was born and he died in 348 B.C. Plato travelled to other countries and met with great philosophers and mathematicians of those countries. He started his school in 389 B.C. Plato sought in arithmetic and geometry the key to universe. He always emphasized on the relationship of philosophy and mathematics continuously. Plato called a maker of mathematicians because he produced many mathematicians in his school. Manaechnus, Dinostratus, Leodames and Eudoxus were his students and they conducted many researches in this field.

1.5.4 The First Alexandrian School

The first Alexandrian school was a great centre of learning. In 338 B.C. Ptolemy founded

the University of Alexandria. He invited the Euclid to open mathematical school there. The elements on the history of geometry written by Euclid before two thousand years ago, still regarded as the introduction of mathematical science. Archimedes was also a famous mathematician of that era. He wrote many books like the measurement of circle, on spirals, the sand counter, center of plane gravities and conoids and spheroids. Aristotle found the property of lever, but he could not establish its true theory. Archimedes's proof of the property of the lever holds its place in many text books of this day.

After forty years of Archimedes, Apollonius emerged as a mathematician. He wrote eight books on "conic section". Euclid, Archimedes and Apollonius made the Geometry to high state of perfection. Hipparchus was known as a greatest astronomer of antiquity during 161 B.C. and 127 B.C. He made arithmetical and graphical devices of solving geometrical problems in a plane and on a sphere.

1.5.5 The Second Alexandrian School

The Second Alexandrian School was started in Christian era. Claudius, Ptolemy, Theon of Smyrna, Siophantus and Pappus made it famous. Claudius Ptolemy was a popular astronomer and he belonged to Egypt. He was famous because of his work "Almagest" in thirteen books. His central theme was that the earth is the centre of univers and that the sun and other planets revolve around it. Pappus was born around 340 AD in Alexandria. He was the last mathematician of this school. He wrote many commentaries like Commentary on the Almagest, A Commentary on the Aralemma of Diodorus and Commentary on the Euclid's Elements. "Mathematical Collection" was the famous work of Pappus. In that same age Diophantus was also a famous and knowledgeable last mathematician. Firstly he perform the functions $(X - 1) \times (X - 2)$ without references of geometry. He Also dealt with the solution of simultaneous equations.

1.5.6 Egyptian Mathematics

The Egyptian famous mathematical works are Rhind papyrus (c. 1650 BC), Moscow papyrus (c. 1890 BC) and Berline papyrus (c. 1300 BC). Rhind papyrus considered the most famous Egyptian mathematical text. It was an instructional manual in arithmetic and geometry for learners. It contained the formulas and methods of multiplications, fraction and division; also include the composite and prime numbers; arithmetic, geometric and harmonic means; and simplistic understanding of both the Sieve of Eratosthenes and perfect number of theory. Moscow papyrus consisted of such problems which were apparently intended as entertainment and today known as "word problem" or story problem. Berline papyrus provides evidence that the ancient Egyptians could solve a second-order algebraic equation.

1.5.7 Greek and Hellenistic Mathematics

Greek mathematics is considered to begun Thales of Miletus (c. 624- c. 546 BC) and Pythagoras of Samos (c. 582 – c. 507 BC). It was written in Greek language. Greek mathematics of the period following Alexander the Great is sometimes called Hellenistic mathematics. This mathematics was comparatively considered more sophisticated than the mathematics of earlier cultures. The Greeks used logic to derive conclusion from

definitions and proverb, and used mathematical thoroughness to prove them. Thales used geometry to solve problems such as calculating the height of pyramids and the distance of ships from the shore. He is credited with the first use of deductive reasoning applied to geometry, by deriving four corollaries to Thales' Theorem. According to myth, Pythagoras traveled to Egypt to learn mathematics, geometry and astronomy from Egyptian priests. It was Pythagoreans who created the term "mathematics" and with whom the study of mathematics for its own sake begins. The Pythagoreans are credited with the first proof of the Pythagorean Theorem. In 384 – c.322 BC Aristotle wrote down the laws of logic. In 300 BC Euclid the book of Euclid "Elements" was known to all educated people in the West until the middle of the 20th century. Sieve of Eratosthenes discovered prime numbers. Archimedes considered the greatest mathematics of ancient times. He used methods of exhaustion to calculate the area under the arc of a parabola with the summation of an infinite series, and gave remarkably accurate approximations of Pi (π).

1.5.8 Chinese Mathematics

Chinese mathematics is different from that of rest of the world. It is an independent development. One of the different features of this is the use of decimal positional notation system, the so called "rod numerals" in which distinct ciphers were used for numbers between 1 and 10, and additional ciphers for power of ten. Thus the number 123 would be written using the symbol for "1", followed by the symbol for "100", then the symbol for "2" followed by the symbol for "10", followed by the symbol for "3". This was the most advanced number system in the world at the time, apparently in use several centuries before the Common Era and well before the development of Indian numeral system. The oldest existence work on compiled by the followers of Mozi (470 – 390 BC). The Mo Jing described various aspects of many fields associated with physical science, and provided a small number of geometrical theorems as well. In 202 BC- 220 AD) Han Dynasty produced works of mathematics with presumably expanded on works that are now lost. The most important of these is The Nine Chapters on the Mathematical Art, the full title of which appeared by AD 179, but existed in part under other titles beforehand. Liu Hui commented on the work by the 3rd century AD, and gave a value of π accurate to 5 decimal places. The high water mark of Chinese mathematics occurs in the 13th century, with the development of Chinese Algebra. The most important text from that period is the Precious Mirror of the four Elements, during with the solution of simultaneous higher order algebraic equations using a method similar to Horner's methods.

1.5.9 Indian Mathematics

The oldest extant mathematical records from India are the Shatapatha Brahmana (c. 9th century BC but estimates of the date vary widely). The Sulba Sutras (c. 800 BC-200 AD), appendices to religious texts which give simple rules for constructing altars of various shapes, such as squares, rectangles and parallelograms. The Sulba Sutras give methods for constructing a circle with approximately the same area as a given square, which imply several different approximations of the value of π , in addition, they compute the square root of 2 to several decimal places, list Pythagorean triples, and give a statement of the Pythagorean theorem. Panini (c. 5th century BC) formulated the rules for Sanskrit

grammar with notation similar to modern mathematical notation, and used metarules, transformation and recursion. Pingala (3rd- 1st centuries BC) Pingala used a device corresponding to a binary numeral system. His discussion of the combination of meters corresponded to an elementary version of the binomial theorem. Pingala's work also contains the basic ideas of Fibonacci numbers. Surya Siddhanta (c. 400) introduced the trigonometric functions of sine, cosine and inverse sine, and laid down rules to determine the true motions of the luminaries, which conforms to their actual position in the sky. Aryabhata (5th century AD) wrote the *Aryabhatiya*, a slim volume, written in verse, intended to supplement the rules of calculation used in astronomy and mathematical mensuration, though with no feeling for logic or deductive methodology.

1.5.10 Islamic Mathematics

The Islamic Empire established across Persia, the Middle East, Central Asia, North Africa, Iberia and parts of India from the 8th Century onwards made significant contributions towards mathematics. One consequence of the Islamic prohibition on depicting the human form was the extensive use of complex geometric patterns to decorate their buildings, raising mathematics to the form of an art. Muhammad Al-Khwarizmi was one of the greatest early Muslim Persian mathematicians, in the 9th Century an early Director of the House of Wisdom. His strong advocacy of the Hindu numerical system (1 - 9 and 0) is the most important contribution in mathematics was, which he recognized as having the power and efficiency needed to revolutionize Islamic (and, later, Western) mathematics, and which was soon adopted by the entire Islamic world and Europe as well. Algebra was another contribution of Al-Khwarizmi.

During the Islamic golden age Abu Al-Hasan, also known as Alhazen, lived from 965 till 1040. Al-Hasan lived in the 10th and 11th centuries, and had success in many other fields too, including Astronomy, Philosophy and Engineering. He did a lot of work on the number theory, and came across theories based upon perfect numbers. Still, these theories were not credited at the time, as most thought that he did not have enough evidence to prove it was true. He created theories linking algebra to geometry. His theory for adding the first 100 numbers was proved with the help of geometry, as it was his focal point.

Omar Khayyam lived from 1048 till 1131, and was a key figure in the Mathematical area during the Islamic Golden Age. The most famous treatises of Algebra was written by Omar Khayyam. Omar Khayyam created the first theories known about parallels in quadrilaterals. He also found out how to solve roots of any degree, laying down some of the principles of Algebra about a century and a half after Al Khwarizmi produced the cores of Algebra. Khayyam had notable works in geometry, particularly on the theory of proportions (Famous Scientists. org, 2013).

Sharaf al-Dīn lived from 1135-1213. He was a Persian mathematician and astronomer who lived during the Islamic Golden Age. Sharaf Al-din dedicated most of his work to algebra and astronomy. He further worked upon the idea of algebra from mathematicians like Al-Khwarizmi and Omar Khayyam. As an astrologist, he created some algorithms

and observation reports for the exact bearings of the stars at different times of the year. He solved extremely bulky arithmetic problems like $2 \times 2 \times 2 \times 2 \times 2 \times 2$ and so on, using exponents 2^7 . He founded algebraic identities such as $a^2 + b^2 = (a+b)^2$ and many other concepts. He was the first to define the different degrees of equations and solve them. His contributions have been very useful to the development of Mathematics.

1.5.11 Roman and medieval European Mathematics

From the 4th to 12th Centuries, European knowledge and study of arithmetic, geometry, astronomy and music was limited mainly to Boethius' translations of some of the works of ancient Greek masters such as Nicomachus and Euclid. All trade and calculation was made using the clumsy and inefficient Roman numeral system, and with an abacus based on Greek and Roman models.

By the 12th Century, though, Europe, and particularly Italy, was beginning to trade with the East, and Eastern knowledge gradually began to spread to the West. Robert of Chester translated Al-Khwarizmi's important book on algebra into Latin in the 12th Century, and the complete text of Euclid's "Elements" was translated in various versions by Adelard of Bath, Herman of Carinthia and Gerard of Cremona. The great expansion of trade and commerce in general created a growing practical need for mathematics, and arithmetic entered much more into the lives of common people and was no longer limited to the academic realm.

Europe's first great medieval mathematician was the Italian Leonardo of Pisa, better known by his nickname Fibonacci. Although best known for the so-called Fibonacci Sequence of numbers, perhaps his most important contribution to European mathematics was his role in spreading the use of the Hindu-Arabic numeral system throughout Europe early in the 13th Century, which soon made the Roman numeral system obsolete, and opened the way for great advances in European mathematics.

The German scholar Regiomontanus was perhaps the most capable mathematician of the 15th Century, his main contribution to mathematics being in the area of trigonometry. He helped separate trigonometry from astronomy, and it was largely through his efforts that trigonometry came to be considered an independent branch of mathematics. His book "De Triangulis", in which he described much of the basic trigonometric knowledge which is now taught in high school and college, was the first great book on trigonometry to appear in print.

During the 15th Century German philosopher, mathematician and astronomer, whose prescient ideas on the infinite and the infinitesimal directly influenced later mathematicians like Gottfried Leibniz and Georg Cantor. He also held some distinctly non-standard intuitive ideas about the universe and the Earth's position in it, and about the elliptical orbits of the planets and relative motion, which foreshadowed the later discoveries of Copernicus and Kepler.

1.5.12 Modern Mathematics (19th century)

All over the 19th century mathematics became increasingly abstract. Joseph Fourier's study, at the beginning of the 19th Century, of infinite sums in which the terms are trigonometric functions were another important advance in mathematical analysis. Periodic functions that can be expressed as the sum of an infinite series of sine and cosines are known today as Fourier Series, and they are still powerful tools in pure and applied mathematics. Fourier (following Leibniz, Euler, Lagrange and others) also contributed towards defining exactly what is meant by a function, although the definition that is found in texts today - defining it in terms of a correspondence between elements of the domain and the range - is usually attributed to the 19th Century German mathematician Peter Dirichlet.

The German Bernhard Riemann worked on a different kind of non-Euclidean geometry called elliptic geometry, as well as on a generalized theory of all the different types of geometry. Riemann, however, soon took this even further, breaking away completely from all the limitations of 2 and 3 dimensional geometry, whether flat or curved, and began to think in higher dimensions. His exploration of the zeta function in multi-dimensional complex numbers revealed an unexpected link with the distribution of prime numbers, and his famous Riemann Hypothesis, still unproven after 150 years, remains one of the world's great unsolved mathematical mysteries and the testing ground for new generations of mathematicians.

British mathematics also saw something of resurgence in the early and mid-19th century. Although the roots of the computer go back to the geared calculators of Pascal and Leibniz in the 17th Century, it was Charles Babbage in 19th Century England who designed a machine that could automatically perform computations based on a program of instructions stored on cards or tape. His large "difference engine" of 1823 was able to calculate logarithms and trigonometric functions, and was the true forerunner of the modern electronic computer. Although never actually built in his lifetime, a machine was built almost 200 years later to his specifications and worked perfectly. He also designed a much more sophisticated machine he called the "analytic engine", complete with punched cards, printer and computational abilities commensurate with modern computers.

Another 19th Century Englishman, George Peacock, is usually credited with the invention of symbolic algebra, and the extension of the scope of algebra beyond the ordinary systems of numbers. This recognition of the possible existence of non-arithmetical algebras was an important stepping stone toward future developments in abstract algebra.

20th Century

The 20th Century continued the trend of the 19th towards increasing generalization and abstraction in mathematics, in which the notion of axioms as "self-evident truths" was largely discarded in favor of an emphasis on such logical concepts as consistency and completeness. It also saw mathematics become a major profession, involving thousands

of new Ph.D.s each year and jobs in both teaching and industry, and the development of hundreds of specialized areas and fields of study, such as group theory, knot theory, sheaf theory, topology, graph theory, functional analysis, singularity theory, catastrophe theory, chaos theory, model theory, category theory, game theory, complexity theory and many more.

The eccentric British mathematician G.H. Hardy and his young Indian protégé Srinivasa Ramanujan, were just two of the great mathematicians of the early 20th Century who applied themselves in earnest to solving problems of the previous century, such as the Riemann hypothesis. Although they came close, they too were defeated by that most intractable of problems, but Hardy is credited with reforming British mathematics, which had sunk to something of a low ebb at that time, and Ramanujan proved himself to be one of the most brilliant (if somewhat undisciplined and unstable) minds of the century.

The early 20th Century also saw the beginnings of the rise of the field of mathematical logic, building on the earlier advances of Gottlob Frege, which came to fruition in the hands of Giuseppe Peano, L.E.J. Brouwer, David Hilbert and, particularly, Bertrand Russell and A.N. Whitehead, whose monumental joint work the “Principia Mathematica” was so influential in mathematical and philosophical logicism.

Activity: 3

Prepare a list of all Mathematicians and write the concepts of mathematics and they presented.

SUMMARY

- Teaching of mathematics should be in such a manner that it can relate mathematics with other subjects.
- Science provides mathematics interesting problem to investigate with; on the other hand mathematics provide science with powerful tools to use in data analysis.
- There are so many commonalities in mathematics and Science.
- The advancement in technology is due to the utilization of mathematical concepts.
- Scientific ideas can be expressed in the symbolic language of mathematics.
- The solution of Physics problems requires a good understanding of mathematical concepts.
- For the analysis of various experiments, biologists need mathematics.
- Mathematical laws are helpful in the composition of all Chemicals.
- Mathematics is pre-requisite for all engineering classes.
- Mathematics helps to know how to determine prescriptions, calculating how much of a medication to give a patient at a time and reading X rays all required the usage of math.
- For the measurements of lands, mathematics is rule for free size calculation.
- In Social Sciences, graphs, charts and other tables are required to describe and explain different phenomenon.

- To explain psychological data from experimentation there is need of statistics and interpret data.
- To solve economic problems, theory of probability of statistics is very helpful. Logical expression can be proved with the help of mathematics without any personal basis.
- Symmetry and harmony is common in mathematics and fine arts.
- In Geography, to describe physical conditions of any society, mathematics is a valid tool.

EXERCISE

1. Explain the need of mathematics in school curriculum.
2. Highlight the significance and importance of mathematics in daily life.
3. Elaborate the scope of mathematics and its relationship with basic science subjects.
4. Elaborate the scope of mathematics and its relationship with social science subjects.
5. Highlight the importance of teaching of mathematics in the light of learners' understanding of mathematics.
6. Elicit the recent development in mathematics pedagogy by providing examples.
7. Discuss the discoveries of Muslim scientists in the field of mathematics.
8. Elaborate the discoveries of Greek scientists in mathematics field.
9. Write brief note on development of mathematics and its impact on common person.
10. Discuss the impact of mathematics on development of other science disciplines.

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Unit - 02

METHODS OF TEACHING MATHEMATICS

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INTRODUCTION

An information- and technology-based society requires individuals, who are able to think critically about complex issues, analyze and adapt to new situations, solve problems of various kinds, and communicate their thinking effectively. The study of mathematics equips students with knowledge, skills, and habits of mind that are essential for successful and rewarding participation in such a society. To learn mathematics in a way that will serve them well throughout their lives, students need classroom experiences that help them develop mathematical understanding; learn important facts, skills, and procedures; develop the ability to apply the processes of mathematics; and acquire a positive attitude towards mathematics. Although many students dread math class, or find it boring, it is important for teachers to do whatever they can to make math class fun. Students who enjoy math class perform better. It is therefore important for teachers to do all they can to help students have fun while learning math. If a teacher does not enjoy a subject, the students will not enjoy learning in that class either. The more positive energy a teacher has about a subject, the more enjoyable the class will be. A teacher who dislikes math may have a lower energy level than the teacher who enjoys math, and assign more seat-work. The more energy the teacher puts into planning and teaching, the more enjoyable the lessons will become, as students are getting out of their seats more.

OBJECTIVES

After study of this unit, you will be able to apply following Methods of Teaching of Mathematics in their classrooms;

1. Inquiry Method.
2. Inductive Method
3. Deductive Method.
4. Lecture Method effectively
5. Co-Operative Learning
6. Collaborative Method
7. Problem Solving Method
8. Precision Method
9. Drill Method

2.1 VARIOUS METHODS OF TEACHING MATHEMATICS

The students of present age need to learn mathematics today is not the same that their parents and grandparents needed to learn. Students of present technological era will face new demands for mathematical proficiency when they become adults so mathematics is a realm no longer restricted to a select few but our teaching ones must learn to think mathematically, and they must think mathematically to learn.

Teaching mathematics is not a simple art or just knowing how to teach alone. Teaching mathematics is an art of knowing how to teach effectively and efficiently to ensure meaningful learning and the desired results.

The teaching of mathematics also involves different methods. The mathematics is taught at different levels of school education. No one single method is fully suitable or appropriate for each level. Thus, different methods are used for the students of different level. There are different methods of teaching mathematics for primary, secondary and higher level education as the students of each level differ in age, maturity, mental abilities, mental development, mathematical understanding etc. therefore, one method cannot be applied to teach mathematics to all the classes. Besides this, with the use of one method, all the students of a class cannot be equipped with equal amount of knowledge because individual difference lies among them. Individual difference is an important psychological phenomenon which affects teaching and its outcome to a great extent. According to Thorndike, “There is a much difference between lower and higher categories of students of the same class. Higher category students learn six times more than the lower category students or lower category students learn only one sixth in comparison of higher category students.” Therefore, a teacher has to apply different methods to teach same topic to same class so that every student could understand and learn the given content.

2.1.1 Inquiry Method

Inquiry-based teaching/learning is a teaching approach that relies on student-centered activities and questioning, rather than the traditional teacher-centered approach, relying on textbooks and lecturing. The teacher's role is more as a mentor than authority; teacher uses well-crafted problems and the minimal amount of information the students will need, leading them to discover the answers and come to their own understanding of the ideas.

This teaching method is rooted in the scientific method of investigating phenomenon in a structured and methodical manner.



Scientific Method

Related to teaching and learning, it is an information-processing model that allows pupils to discover meaning and relevance to information through a series of steps that lead to a conclusion or reflection on the newly attained knowledge. In most cases, teachers use a "guided inquiry" method to facilitate the learning experience and structure the inquiry around specific goals of instruction. Teacher develops a set of questions or encourages the students to come up with some questions that are on-point for the concept teacher wants them to understand. Then teacher has the students gather information from either resources. Teacher provides or ones they find on their own. When the students have enough information, they apply it to the problem by breaking it out into categories or making an outline that organizes the information in order of importance to the subject. At that point the teacher can lead a class discussion, which gives him a chance to emphasize how the information relates to the topic and helps the students see how the data they've collected goes toward answering the questions. Ultimately the class will reach a conclusion that uses their research to answer the original questions, much as a scientist uses experimental results to confirm or rule out a hypothesis.

Teaching math involves both grasping the concepts and applying the procedures to theoretical and practical problems. Inquiry-based learning focuses primarily on the big ideas first. The math teacher encourages students to look for patterns and relationships and try out different approaches to solving the problems he presents to them. Teacher encourages the students to be able to explain how they solved the problem, rather than just get the right answers. Since even teaching children have some grasp of the idea that things cost money, the teacher can use money to talk about math concepts and skills, from counting to addition and subtraction.

Is it correct? $24 + 21 = 45$ and $42 + 12 = 54$. The digits in the answers are reversed.

Is it something to do with doubling numbers 1, 2, 4?

$24 \times 21 = 42 \times 12$

How do you multiply two 2 digit numbers?

One number on the right is half one on the left and the other is double.

The digits on the left have been reversed to make the numbers on the right.

Do these rules always work?

Inquiry-based math activities involve student-centered learning.

Inquiry-based teaching involves student-centered learning, in which the student learns through questioning and gaining information. This differs from traditional learning that takes place in schools, where teachers instruct or give information to students. Inquiry-based learning activities usually involve students working in groups, which can lead to loud noise and excessive movement around the room. Using inquiry-based learning in a mathematics class develops critical thinking skills, improves student attitudes and facilitates mathematical discovery. Teachers are often concerned about having enough class time to complete inquiry-based learning activities. Limited space is another concern which can be fixed by rearranging desks. Arranging desks in a horseshoe shape will allow the teacher to easily move around the classroom and monitor all students. Teachers should be sure all equipment is prepared and should use students to help distribute supplies.



Grouping students for inquiry-based learning activities can be difficult because of students' diverse learning levels. Students can be grouped by ability levels and given a specific activity based on their ability, or groups can be designed with mixed abilities so higher level students can tutor their peers. Peer tutoring can also be used for helping to catch up students who were absent. Many teachers are also concerned about students' difficulty in working collaboratively. In this case, teachers may have students' complete team building activities prior to the lesson to encourage cooperation.

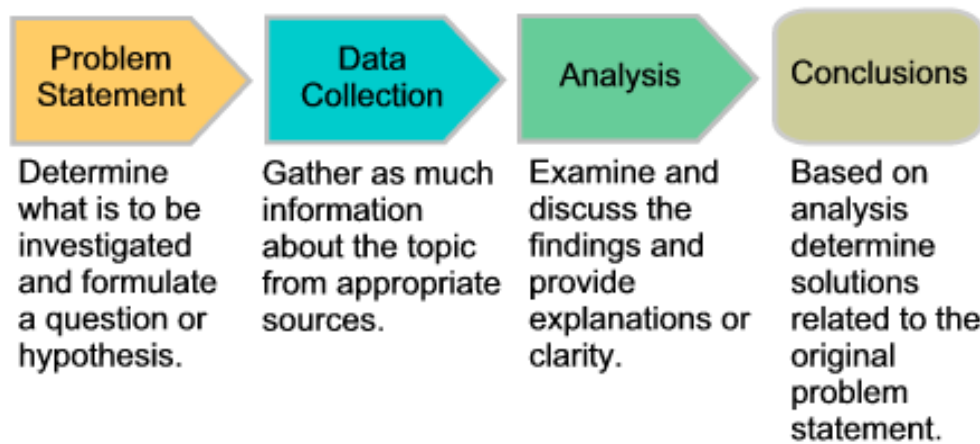
Inquiry-based activities are easier to complete with the help of outside resources. Teachers should talk with administrators and gain their support in gathering materials. Parents can also be used to help with set-up, clean-up or monitoring during the lesson. Parents or older students can help to organize supplies or rearrange the classroom if necessary. Parent teacher organizations or other community members are often willing to help donate supplies for activities.

Principles of Inquiry-based Learning

The main components of inquiry-based learning include:

- a question(s) related to the topic of inquiry to be explored (problem statement), followed by an investigation and gathering of information related to the question (data collection),
- continuing with a discussion of findings (analysis),
- commencing with a reflection on what was learned (implications/conclusion).

Inquiry-based Teaching Strategy



Procedures of Inquiry based Teaching

1. The first step in any inquiry is the formulation of a question or set of questions related to the topic of inquiry. The question can be posed by the teacher or by the pupil(s). Sometimes the question is referred to as a hypothesis or a problem statement.

2. Once a question is posed, pupils are encouraged to investigate the topic by gathering information from sources that either the teacher provides or within learning resources or tools that are readily available to the pupils.
3. When enough information related to the topic of inquiry is gathered, it is organized in categories or outlined by highlighting the important information relative to the topic. This helps the pupil make connections with new learning and prior learning.
4. The information is discussed and analyzed for further understanding. The teacher can direct the discussion and highlight the implications that arise from the investigation and show how it relates to the solution of the problem.

Conclusions are made and related back to the original question. Student reflections are encouraged and serve as a way to relate back to the inquiry and retrace the steps that led to the conclusion. This also serves to reinforce the model so that pupils can repeat the process in any problem-solving situation.

Characteristics of Inquiry-Based Teaching

1. **Process focus:** When students solve problems themselves, they internalize conceptual processes. Inquiry-based teaching prioritizes process over product.
2. **Investigation:** The teacher may pose a problem derived from the class content or students' questions. The students then investigate the issue to find an answer.
3. **Group learning:** Students may work in pairs or in small groups when exploring a problem. Students assist one another throughout the learning process, which enables them to share and build upon ideas as well as articulate how they arrived at a solution.
4. **Discussion monitoring:** As the students work together, the teacher can move from group to group, listening to their discussions. Teachers may ask questions to gauge students' understanding and correct any misconceptions.
5. **Real-life application:** Students solve math problems that have a meaningful life application. For example, a teacher may present a multiplication problem as an interesting story: "Brittany has 2 bags of candy. Each bag has 4 candies inside. How many candies does Brittany have altogether?"

Activity No. 1

The following number game is an excellent opening exercise for a whole class to tackle. It is a discovery exercise in which all students can participate. As the grid is filled in, some patterns will emerge which can be seen even by students who consider themselves 'weak' in math. Thus, everyone in the class gets a chance to find a correct answer. But the actual rule for filling in the grid also yields some results which may prove challenging to even the math 'whizzes' in the class.

Procedure

Present students with the following grid written on the chalkboard. Their job is to fill in the grid.

12												
11												
10								80				
9												
8												
7												
6												
5				20								
4												
3							21					
2												
1												
	1	2	3	4	5	6	7	8	9	10	11	12

Students make suggestions for numbers they would like to place in the grid. The grid is set up like a set of coordinate axes, so students name the box they would like to fill by saying the 'x' number first, followed by the 'y' number. For example, box (4,5) has already been filled with the number 20. This gives students practice for naming coordinates in a standard x,y graph. If a student guesses a number correctly, that number is placed in the grid. We make the rule that only students who raise their hands get to put numbers in the grid. Students who just call out numbers are ignored. In this way, the teacher can 'reward' specific students by allowing them to fill in several numbers in a row once they find a specific pattern. It also makes it possible to get more students involved, rather than letting a few aggressive students dominate the action.

Merits of Inquiry Method

- Teacher can keep a vigil check on the activities of the students as it is the teacher who evokes the responses in the students through the puzzling event.
- Through this method, students get the opportunity to learn various kinds of information on their own. They do not rely on the readymade information provided

by the teacher in any way. Thus, this method helps in making the students creative in their own way.

- c. As students do not accept hypothesis designed by them on the basis of information provided by teacher, thus they learn to verify the hypothesis after reading and experimentation processes.

Demerits of Inquiry Method

- a. It is not possible to use this method under the structured school curriculum as it is slow in nature and requires a lot of time.
- b. This method can only be used properly if the teacher who is making use of it is creative. Not only this, if the teacher does not know how to arrange practical experiments work, then also he cannot make use of this method properly.
- c. As students of different mental capabilities attend the same class in the school, thus it is not possible for all of them to learn various information's through this method effectively.
- d. If all the students do not take participate in question asking function, then the class room will become dominated by few students, as a result of which other less able students will feel a sense of neglect.

When to Use:

- Teacher has ample time to cover the syllabus
- Relevant resources are available
- This method should be used by the teacher when he intends to develop spirit of enquiry in the students, as it will motivate them to find out for themselves the answers of various questions arising in their mind by making various kinds of enquiries instead of getting or accepting readymade information from the teacher.

Activity No. 2

Ask some questions starting with “why/how” at the end of lesson.
e.g. How you use the formula learnt today in the daily life?

2.1.2 Inductive Method

Inductive Method is based on process/principle of induction. Induction means to offer/establish a general/universal truth by showing that if it is true for a particular case and is further true for a reasonably adequate number of cases then it is true for all such cases. In induction, first take a few examples and greater than generalize. Thus, in this method, at first stage a problem is solved on the basis of previous knowledge, thinking, reasoning and insight of the learner. Inductive method proceeds from particular examples to general rules of formulae, concrete illustration to abstract rules, known to unknown and simple to complex. Inductive approach is psychological in nature.

In inductive method of teaching, the children follow the subject matter with great interest and understanding. This method is more useful in arithmetic teaching and learning.

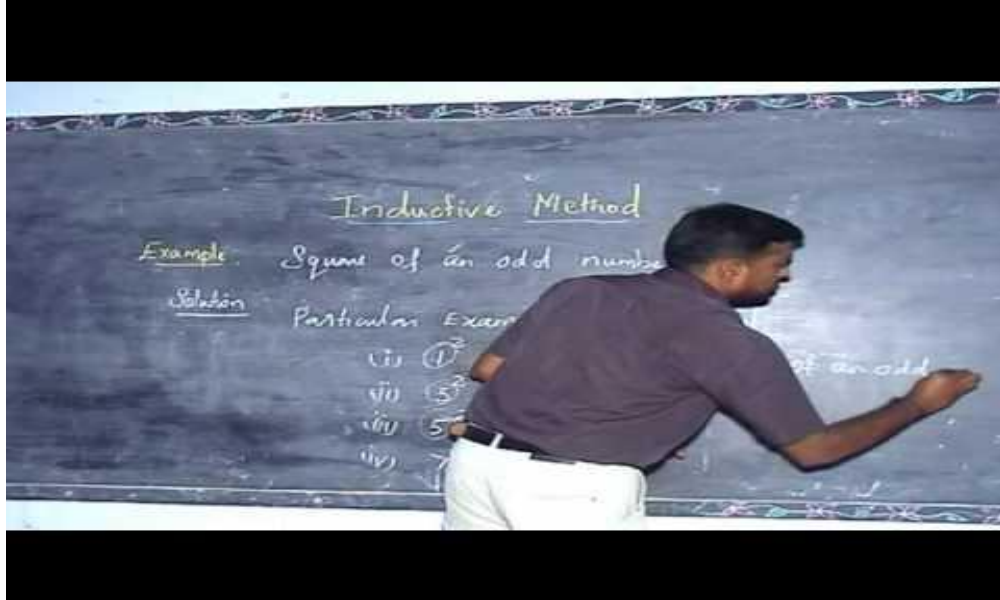
Inductive approach proceeds from

- Particular cases to general rules of formulae

- Concrete instance to abstract rules
- Known to unknown
- Simple to complex

Following steps are used while teaching by this method:-

- Presentation of Examples:** In this step teacher presents many examples of same type and solutions of those specific examples are obtained with the help of the student.



- Observation:** After getting the solution, the students observe these and try to reach to some conclusion.



- c. **Generalization:** After observation the examples presented, the teacher and children decide some common formulae, principle or law by logical mutual discussion.

Sum of first one natural number
 $1 = 1 = \frac{1(\text{Next natural number})}{2} = \frac{1(2)}{2} = 1$

Sum of first two natural numbers
 $1 + 2 = \frac{2(\text{Next natural number})}{2} = \frac{2(3)}{2} = 3$

Sum of first three natural numbers
 $1 + 2 + 3 = \frac{3(\text{Next natural number})}{2} = \frac{3(4)}{2} = 6$

Hence
Sum of first fifty natural numbers = $\frac{50(51)}{2} = 1275$

- d. **Testing and verification:** After deciding some common formula, principle or law, children test and verify the law with the help of other examples. In this way children logically attain the knowledge of inductive method by following above given steps.



Example 1:

Square of an odd number is odd and square of an even number is even.

Solution:

Particular concept:

$$\begin{array}{lll} 1^2 = 1 & 3^2 = 9 & 5^2 = 25 \quad 7^2 = 49 \text{ -----> Equation 1} \\ 2^2 = 4 & 4^2 = 16 & 6^2 = 36 \quad 8^2 = 64 \text{ -----> Equation 2} \end{array}$$

General concept:

From equation 1 and 2, we get (i) Square of an odd number is odd (ii) Square of an even number is even.

Example 2:

Sum of two odd numbers is even

Solution:

Particular concept:

$$\begin{array}{l} 1+1=2 \\ 1+3=4 \\ 1+5=6 \\ 3+5=8 \end{array}$$

General concept:

In the above we conclude that sum of two odd numbers is even

Example 3:

Law of indices $a^m \times a^n = a^{m+n}$

Solution:

$$\begin{array}{lll} \text{We have to start with } a^2 \times a^3 & = & (a \times a) \times (a \times a \times a) \\ & = & a^5 \\ & = & a^{2+3} \\ a^3 \times a^4 & = & (a \times a \times a) \times (a \times a \times a \times a) \\ & = & a^7 \\ & = & a^{3+4} \\ \text{Therefore } a^m \times a^n & = & (\text{axax...m times}) \times (\text{axa ...n times}) \\ a^m \times a^n & = & a^{m+n} \end{array}$$

MERITS

- It enhances self-confident
- It is a psychological method.
- It is a meaningful learning
- It is a scientific method
- It develops scientific attitude.
- It develops the habit of intelligent hard work.

- It helps in understanding because the student knows how a particular formula has been framed.
- Since it is a logical method so it suits teaching of mathematics.
- It is a natural method of making discoveries, majority of discoveries have been made inductively.
- It does not burden the mind. Formula becomes easy to remember.
- This method is found to be suitable in the beginning stages. All teaching in mathematics is conducive in the beginning.

DEMERITS

- Certain complex and complicated formula cannot be generated so this method is limited in range and not suitable for all topics.
- It is time consuming and laborious method
- It is lengthy.
- Its application is limited to very few topics
- It is not suitable for higher class
- Inductive reasoning is not absolutely conclusive because the generalization made with the help of a few specific examples may not hold well in all cases.

Applicability of inductive method

Inductive approach is most suitable where

- Rules are to be formulated
- Definitions are to be formulated
- Formulae are to be derived
- Generalizations or law are to be arrived at.

Activity No.3

Give some numeric expressions, as given in the examples, and ask students to induct a formula.

2.1.3 Deductive Method

Deductive method is based on deduction. Deductive method proceeds from general rules of formulae, concrete illustration, and abstract rules to particular examples, unknown to known and complex to simple. At first the rules are given and then students are asked to apply these rules to solve more problems. This approach is mainly used in Algebra, Geometry and Trigonometry because different relations, laws and formulae are used in these sub branches of mathematics. In this approach, help is taken from assumptions, postulates and axioms of mathematics. It is used for teaching mathematics in higher classes.

Deductive approach proceeds from

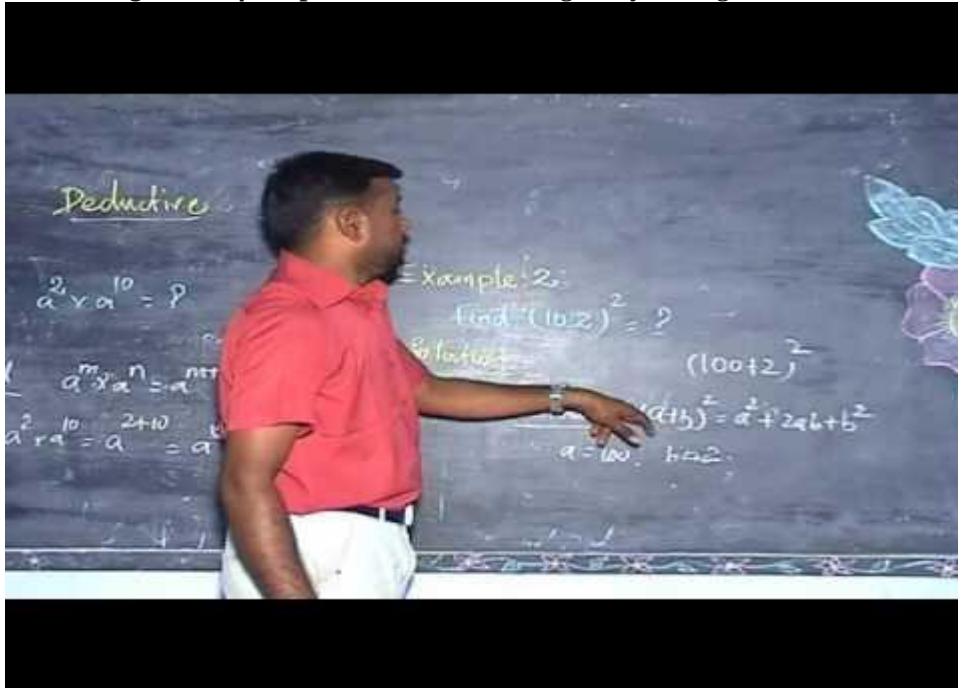
- General rule to specific instances
- Unknown to known
- Abstract rule to concrete instance
- Complex to simple

Steps in deductive approach

Deductive approach of teaching follows the steps given below for effective teaching

- Clear recognition of the problem
- Search for a tentative hypothesis
- Formulating of a tentative hypothesis
- Verification

Clear recognition of the problem: Teachers begins by stating the rule



Search for a tentative hypothesis: Teacher give details of possible answers



Formulating of a tentative hypothesis: At this stage, teachers suppose a/many correct answer (s)



Postulate (v.) to hypothesize; propose

Verification: At the final stage, hypothesis is tested and verified each supposed correct answer and arrived at the solution.

Decision is:	The Null Hypothesis is	
	True	False
Accept H_0	$(1-\alpha)$ Confidence Level	β
Reject H_0	α	$(1-\beta)$ Power of the test

Example 1:

Find $a^5 \times a^3 = ?$

Solution:

General: $a^p \times a^q = a^{p+q}$

Particular: $a^5 \times a^3 = a^{5+3} = a^8$

Example 2:

Find $(101)^2 = ?$

Solution:

General: $(a+b)^2 = a^2 + b^2 + 2ab$

Particular: $(100+1)^2 = 100^2 + 1^2 + (2 \times 100 \times 1)$
 $= 10000 + 1 + 200 = 10201$

MERITS

- It is time saving and short method.
- It is applicable for entire course.
- It is useful for drill work and revision
- It uses learner's memory
- It is suitable for all learners
- It helps all types of learners
- It provides sufficient practice in the application of various mathematical formulae and rules.

DEMERITS

- It is a non-psychological method.
- It is difficult to understand
- It taxes the pupil's mind.
- It is not scientific method
- It is suitable for experienced learners.
- It promotes cramming.

Applicability of Deductive Approach

Deductive approach is suitable for giving practice to the student in applying the formula or principles or generalization which has been already arrived at. This method is very useful for fixation and retention of facts and rules as it provides adequate drill and practice.

Activity No.4

Find solution of $(1007)^2$ and $(93)^2$ using formulae $(a \pm b)^2$

2.2 EFFECTIVE USE OF LECTURE METHOD

Lecture method is an oldest method of teaching based on the philosophy of idealism. It is the simplest method for teachers and does not require any arrangements. It brings a personal contact and touch to impress or influence the pupils. This method refers to the explanation of the concept/topic to the students. The emphasis is on the presentation on the topic. The teacher clarifies the topic/concept to the students by using gestures, simple devices, by changing voice, change in position and facial expressions. Teachers are more active and students are passive but the teacher also asks questions to keep the students attentive.

This method is economical and can be used among large number of students. It saves time and also covers syllabus. It gives the students training in listening. It provides an opportunity for better clarification of the topics and laying stress on significant ideas.

This method is useful in large classes, but it also has some limitations. There is very little scope for pupil activity. This method makes the students passive listeners. It is against the principle of learning by doing. It does not take into consideration individual differences. It does not develop power of reasoning of the students. It becomes monotonous to the students.

Though this method has much limitation but still it is the most used method. The lecture method can be made effective by using following points matter should be arranged properly. Teacher should have process in between the lesson. There should be abundant repetition of the topic. The teacher should encourage the students to ask questions. Teaching aids should also be used to make the lecture effective. The content of the lecture should be logical and according to the standard of the students.

Following steps will make the lecture effective.

A. PREPARE IN ADVANCE

- **Visit teacher classroom in advance.** Teacher should familiarize classroom setting. Decide how he/she will move from one place to another and where he/she will stand. Check availability of audio-visual equipment in the classroom. Teacher may request, if not available, it from administration.

Presence of audio-visual equipment and its operator is necessary or teacher himself/herself know how to use the audio-visual equipment.

- ***Have a back-up plan.*** If teacher are using technology, have a back-up plan ready in case electricity failure or teacher run into technical difficulties. Technology problems can negatively affect teacher credibility, even if they are beyond teacher control.
- ***Plan teacher lecture and visual aids beforehand.*** Outline how teacher will introduce, explain, and summarize the main ideas. Select examples and prepare how teacher will show students the relationships between the main ideas.
- ***Prepare speaking notes.*** Prepare notes that work for teacher (e.g., a detailed outline, a list of major points, key definitions, proofs, solved problems, examples, etc.). To better engage students, avoid reading from a script, a computer screen, or overhead projector.
- ***Include delivery reminders in teacher notes.*** Include cues to remind teacher self to smile, look at the whole class, and pause after posing a question, etc.
- ***Practice teacher lecture.*** Practice to ensure that teacher have an appropriate amount of material and activities for the time available. Resist the common error of including too much material in a lecture. Students' questions and learning activities can take up to 50% more time than teacher may first think.
- ***Bring a bottle of water.*** The water will soothe a sore or dry throat. Taking a sip is also a good way to buy thinking time before responding to a student question.

B. STRUCTURE THE LECTURE CLEARLY

- ***Be transparent. Show teacher students "the big picture."*** Don't assume that trainee teachers know the teacher lecture's pedagogical purpose. Explain them lecture is related to previously-learned topics, themes and goals in general. Begin the class with recap of the main points of previous lecture.
- ***Make explicit transitions between topics with mini-summaries.*** Relate present material to previously learned contents and link with future lectures. By linking new and previously learned content material, teacher helps trainee students to understand and organize new information in their minds.
- ***Cover only a few main points in each lecture.*** Plan to cover only three or four points in a fifty-minute lecture and four or five points in a seventy-five-minute class. Select key points that introduce, complement, and/or clarify the course readings, assignments, and goals. Focus on presenting central points or general themes that tie together as many topics as possible.
- ***Avoid merely repeating the course readings.*** Elaborate on readings using new examples and sample exercises or problems.
- ***Be flexible when following teacher notes.*** Watch students' level of interest and confusion and be ready to adapt teacher lecture accordingly. Teacher notes are there if needed, but the lecture should arise out of teacher interaction with the students, not the notes themselves.

C. STRIVE TO ENGAGE TEACHER WHOLE CLASS

- ***Be aware of shifting engagement levels.*** During lecture, students engage in mind wandering approximately 33% of the time; however, this amount varies according to several factors, including engagement.
- ***Ask first, then tell.*** Prompt students to engage by asking questions rather than simply telling them information. For example, rather than telling students the findings from a study, ask them to predict what the study found based on what they know so far.
- ***Allow breaks during long classes.*** Encourage students to move around, talk with one another, or just to relax quietly. Creating breaks also allows students to catch up on and digest what has been discussed.
- ***Use questions to prompt students to think about how the material relates to their life experience.*** Relate the content to students' interests, knowledge, experiences, and their future occupation in the discipline. Making the material relevant helps students retain the information.
- ***Invite student questions and use them in class.*** Invite students to submit questions in person and/or online via the learning management system. Incorporate the answers to students' questions into teacher lecture, or introduce an activity that allows students to discover the answers for themselves.
- ***Ask students for feedback.*** Provide opportunities for students to share feedback, in written form, and anonymously. An effective low-tech method is to circulate and then collect note cards, which students can use to record questions or comments. More high-tech methods of encouraging feedback include using online polling tools.
- ***Consider posting teacher partial notes or slides online before or immediately after class.*** Teacher might also consider making the audio or video of teachers lectures available online. Videos can be captioned using free online tools like Teacher Tube's automatic caption creator. Captions increase accessibility and can help students whose first language is not English.
- ***Encourage students to take notes.*** To help students make good notes, provide a clear structure for the lecture and use a pace that allows them to keep up. Rather than writing extensive notes that students must copy word for word, write key terms on the board or slides to facilitate students' own processing of the information, or provide skeletal course notes for the students to annotate. Pause regularly so that students can ask for clarification.
- ***Use inclusive practices.*** Be mindful of potential biases and stereotypes conveyed in the images, phrases, pronouns, examples, images, etc. that teacher use in class.
- ***Prepare accessible teaching materials.*** The font size of presentation should be at least 24 point.

D. USE EFFECTIVE PRESENTATION STRATEGIES

- ***Maintain regular eye contact with the entire class.*** By doing so, teacher create connections with them, are able to gauge their note-taking, and discourage distracting class noise.
- ***Teacher faces students while speak.*** It helps many students to be able to see teacher face and mouth while teacher speak
- ***Use a sound system for large classes.*** Amplifying teacher voice will help all students — not just students with hearing impairments — and will also put less stress on teacher vocal cords.
- ***Speak politely and clearly with conversational tone.*** Think of the lecture as an opportunity to speak with the students, not at them.

E. USE EFFECTIVE VISUAL AIDS

- ***Use visual aids to stimulate and focus students' attention.*** Multimedia aids using sound, colour, and/or animations can help to attract and maintain students' attention, particularly in large classes where the impersonal situation makes students feel less involved. Visual aids should be a support for, not the focus of, teacher lecture. They also should not replace teacher personal interaction with the students.
- ***Avoid writing everything that teacher say on teacher slides.*** Consider providing partial or skeleton slides that leave space for students to write down examples and other notes.
- ***Follow the guidelines on good slide design.*** If teacher are using overheads or PowerPoint, aim for twelve to twenty slides for a fifty-minute lecture. Be conscious of speeding through the slides and/or overloading students with content—common problems with these types of media. .
- ***Reveal visual information gradually rather than all at once.*** This keeps students focusing on teacher oral development of each point, instead of rushing to copy down the material.
- ***Consider creating visual aids during the lecture.*** Solving problems, showing processes, or building models in real time is often clearer for students than seeing completed work. Teacher can also create visuals to reflect the outcomes of interactive exercises, thereby validating the students' input. The act of writing also helps teacher to pace the lecture appropriately.
- ***Write down key words and names.*** Many students try to write down everything they see. If information does not need to be copied down, mention that to the students, or consider whether it is important enough to include in the first place. Consider providing handouts that give an outline of the lecture material for students to annotate.
- ***If teacher show a video in class, ensure that captions are turned on.*** Doing so helps all students.
- ***When using a projector, dim the lights appropriately.*** If the lights are not sufficiently dim, the projected image will not be visible. But if teacher are going to be verbally commenting on the projected images, ensure that students with hearing issues will still be able to see teacher face and lips.

Thus, we conclude that the lecture method can be very effective, if it is made more interactive and interesting for students.

2.3 CO-OPERATIVE LEARNING

Cooperative learning (sometimes called small-group learning) is a learning and teaching style/teaching strategy/educational approach/instructional strategy that contrasts greatly with traditional direct instruction. In cooperative learning, the students start the problem and work it out together as group work with commitment. The teacher then provides closure, after students have presented their ideas and shown how they have connected the ideas and added academic vocabulary. There is much more to cooperative learning than merely arranging students into groups, and it has been described as "structuring positive interdependence (Students feel responsible for their own and the group's effort)". It serves to improve students' problem solving and creative thinking skills as well as interpersonal and communication skills all of which help the learner to become a team player. Cooperative learning provides opportunities for productive struggle, in which students learn from their mistakes through explanations from their peers and teacher. The classroom environment must be such that students feel safe to make mistakes.

How to use it

1. Determine if teacher have the means to conduct a cooperative lesson.
2. Determine how many students will comprise a team.
3. Pre- plan resources and enough time.
4. Think about how teacher will arrange the classroom furniture.
5. Think about what teacher role as a teacher will be.
6. Group learners in a heterogeneous manner. Of about 5-7 and encourage discussion and solving problems.
7. Provide enough time and materials to aid the discussion.
8. Move from group to group giving assistance and encouragement ask provoking questions.
9. Allow learners time to report to entire class for further discussion





WHAT ELSE DOES IT DO?

Schools are using similar strategies with both students and teachers to do the following:

- Develop and use critical thinking skills and teamwork;
- Promote positive relations among different ethnic groups;
- Implement peer coaching;
- Establish environments where academic accomplishments are valued; and even
- Cooperatively manage schools.

2.4 COLLABORATIVE METHOD

Collaborative learning is a method of teaching and learning in which students' team together to explore a significant question or create a meaningful project. Cooperative learning is a kind of collaborative method. Collaborative learning can take place any time students' work together -- for example, when they help each other with homework. In collaborative method, a group of students discussing a lecture or students from different schools working together over the Internet on a shared assignment are both examples of collaborative learning.

Benefits of Collaborative Method

Celebration of diversity. Students learn to work with all types of people. During small-group interactions, they find many opportunities to reflect upon and reply to the diverse

responses fellow learners bring to the questions raised. Small groups also allow students to add their perspectives to an issue based on their cultural differences. This exchange inevitably helps students to better understand other cultures and points of view.

Acknowledgment of individual differences. When questions are raised, different students will have a variety of responses. Each of these can help the group create a product that reflects a wide range of perspectives and is thus more complete and comprehensive.

Interpersonal development. Students learn to relate to their peers and other learners as they work together in group enterprises. This can be especially helpful for students who have difficulty with social skills. They can benefit from structured interactions with others.

Actively involving students in learning. Each member has opportunities to contribute in small groups. Students are apt to take more ownership of their material and to think critically about related issues when they work as a team.

More opportunities for personal feedback. Because there are more exchanges among students in small groups, teacher students receive more personal feedback about their ideas and responses. This feedback is often not possible in large-group instruction, in which one or two students exchange ideas and the rest of the class listens.

20 COLLABORATIVE LEARNING TIPS AND STRATEGIES FOR TEACHERS

teachthought



(<https://www.teachthought.com/pedagogy/20-collaborative-learning-tips-and-strategies/>)

1. **Establish clear group goals:** Effective collaborative learning involves the establishment of group goals, as well as individual accountability. This keeps the group on task and establishes an unambiguous purpose. Before beginning an assignment, it is best to define goals and objectives to save time.
2. **Keep groups mid-sized:** Small groups of 3 or less lack enough diversity and may not allow divergent thinking to occur. Groups that are too large create 'freeloading' where not all members participate. A moderate size group of 4-5 is ideal.
3. **Establish flexible group norms:** Research suggests that collaborative learning is influenced by the quality of interactions. Interactivity and negotiation are important in group learning. In the 1960's studies by Jacobs and Campbell suggested that norms are pervasive, even deviant norms were handed down and not questioned.
If teacher notice a deviant norm, teacher can do two things: rotate group members or assist in using outside information to develop a new norm. Teacher may want to establish rules for group interactions for teacher students. Older students might create their own norms. But remember, given their durable nature, it is best to have flexible norms. Norms should change with situations so that groups do not become rigid and intolerant or develop sub-groups.
4. **Build trust and promote open communication:** Successful interpersonal communication must exist in teams. Building trust is essential. Deal with emotional issues that arise immediately and any interpersonal problems before moving on. Assignments should encourage team members to explain concepts thoroughly to each other. Studies found that students who provide and receive intricate explanations gain most from collaborative learning. Open communication is key.
5. **For larger tasks, create group roles:** Decomposing a difficult task into parts to saves time. Teacher can then assign different roles. A great example in my own classroom was in science lab, fifth grade student assumed different roles of group leader, recorder, reporter, and fact checker. The students might have turns to choose their own role and alternate roles by sections of the assignment or classes.
6. **Create a pre-test and post-test:** A good way to ensure the group learns together would be to engage in a **pre and post-test**. In fact, many researchers use this method to see if groups are learning. An assessment gives the team a goal to work towards and ensures learning is a priority. It also allows instructors to gauge the effectiveness of the group. Changes can be made if differences are seen in the assessments over time. Plus, teacher can **use Bloom's taxonomy** to further hone in on specific skills.
7. **Consider the learning process itself as part of assessment:** have considered how cooperative learning helps children develop social and interpersonal skills. Experts have argued that the social and psychological effect on self-esteem and personal development are just as important as the learning itself. Assessing the process itself provides motivation for students to learn how to behave in groups. It shows students that teacher value meaningful group interactions and adhering to norms.
8. **Consider using different strategies, like the Jigsaw technique.** The jigsaw strategy is said to improve social interactions in learning and support diversity. The workplace is often like a jigsaw. It involves separating an assignment into subtasks, where individuals research their assigned area. Students with the same topic from different groups might meet together to discuss ideas between groups.

9. ***Allow groups to reduce anxiety:*** When tackling difficult concepts, group learning may provide a source of support. Groups often use humor and create a more relaxed learning atmosphere that allow for **positive learning experiences**. Allow groups to use some stress-reducing strategies as long as they stay on task.
10. ***Establish group interactions:*** The quality of discussions is a predictor of the achievement of the group. Instructors should provide a model of how a successful group functions. Shared leadership is best. Students should work together on the *task and maintenance functions* of a group. Roles are important in group development. Task functions include:
 - Initiating Discussions
 - Clarifying points
 - Summarizing
 - Challenging assumptions/devil's advocate
 - Providing or researching information
 - Reaching a consensus

Maintenance involves the harmony and emotional well-being of a group. Maintenance includes roles such as sensing group feelings, harmonizing, compromising and encouraging, time-keeping, relieving tension, bringing people into the discussion, and more.
11. ***Use real-world problems:*** Experts suggest that project-based learning using open-ended questions can be very engaging. Rather than spending a lot of time designing an artificial scenario, use inspiration from everyday problems. Real world problems can be used to facilitate project-based learning and often have the right scope for collaborative learning.
12. ***Focus on enhancing problem-solving and*** critical thinking skills: Design assignments that allow room for varied interpretations. Different types of problems might focus on categorizing, planning, taking multiple perspectives, or forming solutions. Try to use a step-by step procedure for problem-solving. Educationists explain one generally accepted problem-solving procedure:
 1. Identify the objective
 2. Set criteria or goals
 3. Gather data
 4. Generate options or courses of action
 5. Evaluate the options using data and objectives
 6. Reach a decision
 7. Implement the decision
13. ***Keep in mind the diversity of groups***
 Mixed groups that include a range of talents, backgrounds, **learning styles**, ideas, and experiences are best. Studies have found that mixed aptitude groups tend to learn more from each other and increase achievement of low performers. Rotate groups so students have a chance to learn from others.
14. ***Consider demographics:*** Equally, balanced gender groups were found to be most effective. Some research suggests that boys were more likely to receive and give elaborate explanations and their stances were more easily accepted by the group. In majority male groups girls were ignored. In majority girl groups, girls tended to direct questions to the boy who often ignored them. Teacher may also want to specifically discuss or establish gender equality as a norm. This may seem

obvious, but it is often missed. It may be an issue teacher may want to discuss with older students.

15. **Use scaffolding:** At the beginning of a project, teacher may want to give more direction than the end. Serve as a facilitator, such as by gauging group interactions or at first, providing a list of questions to consider. Allow groups to grow in responsibility as times goes on. In teacher classroom, this may mean allowing teams to develop their own topics or products as time goes on. After all, increased responsibility over learning is a goal in collaborative learning.
16. **Include different types of learning scenarios:** Studies suggests that collaborative learning that focuses on rich contexts and challenging questions produces **higher-order reasoning**. Assignments can include laboratory work, study teams, debates, writing projects, problem-solving, and collaborative writing.
17. **Technology makes collaborative learning easier:** Collaboration had the same results via technology as in person, increased learning opportunities. Be aware that some research suggests that more exchanges related to planning rather than challenging viewpoints occurred more frequently through online interactions.
18. **Avoid 'bad group work':** As with any learning strategy, it's important to have a balanced approach. A recent research article, cites some criticism of collaboration for not allowing enough time for individual, creative thinking. Teacher may allow some individual time to write notes before the groups begin. This may be a great way to assess an individual grade.
19. **Be wary of "group think":** While collaborative learning is a great tool, it is always important to consider a balanced approach. At times, group harmony can override the necessity for more critical perspectives. Some new research suggests that groups favored the more confident members. Changing up groups can help counter this problem.
20. **Value diversity:** Collaborative learning relies on some buy-in. Students need to respect and appreciate each other's viewpoints for it to work. For instance, class discussions can emphasize the need for different perspectives. Create a classroom environment that encourages independent thinking. Teach students the value of multiplicity in thought. Teacher may want to give historical or social examples where people working together were able to reach complex solutions.

2.5 PROBLEM SOLVING METHOD

Is the ability to apply mathematics to a variety of situations in which we give the name problem solving. it helps learners to develop the ability to think mathematically. As you teach mathematics to learners you should let them solve problems drawn from their own experience. We need to see learners doing mathematics not to listen and watch mathematics being done by the teacher.

You should adopt a teaching method after considering the nature of children, their interests, maturity and resources available. The method adopted must

1. Ensure maximum participation
2. Proceed from concrete to abstract
3. Provide knowledge at the understanding level of learners

Note for effective teaching and learning

- All learners must be involved.
- Teaching must be made practical
- Should not bore the learners
- Present the topic consistently
- Derive examples from the learners' experience and explanations must be clear.

Problem Based learning and Mathematics

Mathematics is the science of things that have a pattern of regularity and logical order and finding and exploring the regularity. Mathematics is the foundation of science and technology and the functional role of mathematics to science and technology is multifarious, that no area of science, technology and business enterprise escapes its application. Besides its importance it is observed that mathematics is one of the most poorly taught, widely hated and abysmally understood subject in schools. Students particularly girls run away from the subject. Researchers stated attributed students' poor performance to factors such as the society view that mathematics is difficult and shortage of qualified teachers. Problem based learning is a model which centered on students, develops active and motivated learning, problem solving skills and broad field knowledge, and based on the deep understanding and problem solving.

Problem based learning method/model

In the problem based learning model the students' turn from passive listeners of information receivers to active, free self-learner and problem solvers. It also shifts the emphasis of educational programs from teaching to learning. It enables the students to learn new knowledge by facing the problems to be solved instead of feeling boredom. Problem based learning affect positively certain other attributes such as problem solving, information acquisition, and information sharing with others, group works, and communication etc. Again problems solving is a deliberate and serious act, involves the use of some novel method, higher thinking and systematic planned steps for the acquisition set goals. The basic and foremost aim of this learning model is acquisition of such information which based on facts (Yuzhi, 2003 & Mangle, 2008).

According to Gallagher et. al,(1999) in problem based learning environment, students act as professionals and are confronted with problems that require clearly defining and well-structured problems, developing hypothesis, assessing, analyzing, utilizing data from different sources, revising initial hypothesis as the data collected developing and justifying solutions based on evidence and reasoning. The practice of problem based learning is richly diverse as educators around the world and in a wide range of disciplined have discovered it as a route to innovating education, The educators used problem solving method as an educational tool to enhance learning as a relevant and practical experience, to have students' problem solving skills and to promote students' independent learning skill. Eng (2001) opined problem based learning as a philosophy aims to design and deliver a total learning environment that is holistic to student-centered and student empowerment.

Teachers' Role in Problem Based Learning

The most important achievement of a teacher is to help his/her students along the road to independent learning. In problem based learning, teacher acts just as facilitator, rather than a primary source of information or dispenser of knowledge. Roh, (2003) argued that within problem based learning environments, teachers' instructional abilities are more critical than in the traditional teacher-centered classrooms. Beyond presenting mathematical knowledge to the students, teachers in problem based learning environments must engage students in marshalling information and using their knowledge in applied and real settings. Evidence of poor performance in mathematics by elementary school students highlight the facts that the most desired technological, scientific and business application for mathematics cannot be sustained. This makes it paramount to seek for a strategy for teaching mathematics that aims at improving its understanding and performance by students practically (Okigbo & Osuafor, 2008). Problem solving as a method of teaching may be used to accomplish the instructional roles of learning basic facts, concepts, and procedure, as well as goals for problem solving. Problem solving is a major part of Mathematics has many applications and often those applications represent important problems in, mathematics. We include problem solving in school mathematics because it can stimulate the interest and enthusiasm of the students (Wilson, 1993).

2.6 PRECISION METHOD

Mathematics is a core subject of curriculum around the globe. Innovative approaches and strategies have been advocated to achieve high standards of mathematics stated in curriculum document. One approach that has been advocated is Precision Teaching, supported by its successful application in many schools in different countries.

The **precision teaching**PT refers to the changes made to instructional strategies based on continuous and frequent monitoring, and analysis of student performance (Binder, 1996, White, 1984). Therefore, PT is not so much a method of instruction but it is a method of evaluating the effects of instructional efforts (Lindsley, 1991, West, 1995). Precision teaching has been investigated in several formats e.g. computer mediated programs. It is also beneficial with students with mild handicapping conditions. PT provides regular and instant feedback to teacher. This kind of immediate feedback for teachers is important for monitoring student performance and maintaining the highest level of effective teaching.

Bases of Precision Teaching

- Student academic performance is utilized as a guide for teachers to develop and refine instructional strategies.
- Direct and continuous measurement of behavior. The focus on directly observable behavior includes daily assessment of student performance recorded on teacher-made probes.
- The metric of measurement is rate of response. Precision teachers use rate of response to determine the progress of a student through an instructional program.
- Outcome measures of student performances are represented graphically. Evaluation of student progress is enhanced (i.e., decisions regarding student performance are

- made in an efficient and accurate fashion) through the use of a graph or chart.
- An emphasis on a behavior analytic view of student behavior. Student strengths and weaknesses on any task can be lawfully determined based on a functional understanding of the student's response.
- Continuous monitoring of the impact of the environment (e.g., instructional activities) on student performance. Teachers can only modify a program of instruction if they have data to support hypotheses of progress or non-progress.
- Develop functional methods to build appropriate behavior rather than primarily focusing on which student behavior to eliminate. The student is provided with instructional methods that foster generative effects rather than methods that only respond to the students' skill deficit (Johnson & Layng, 1992).

2.7 DRILL METHOD

Drill Method (DM) or Practice Method (PM) as an instructional strategy is familiar to all educators. It DM in teaching involved simple pair-associate learning to overcome problem. Drilling and practicing method emphasizes repeat activity of the facts or the efficiency gained. The main objective of drill and practice method is to achieve the degree of mastering the skills of the students while ensuring their immortality. Doing drill as frequent as possible helps the student to use a technique of repetition and to remember what they have studied. Drill-and-practice, like memorization, involves repetition of specific skills, such as addition and subtraction, or spelling. To be meaningful to learners, the skills built through drill-and-practice should become the building blocks for more meaningful learning. Researches explained that there are many strategies can be used when the teacher wants to teach mathematics. Student who begin secondary school have already have knowledge and ideas about many aspects of math procedures from their experiences both in primary classes and outside school environment. Students are likely to use these ideas to help themselves to make sense of their experiences in mathematics lessons. Dooley et al. (2014) in their research mentioned that teachers who uphold constructivism beliefs and apply drill and practice method are more likely to detect student alternative conceptions and potentially to do effective teaching strategies.

This explains that to solve the problem of students less interested in mathematics subjects as they cannot relate math knowledge to their lives, should actively participate in learning math subject by using drill and practicing. Improving weak mathematics concepts using drill and practicing is suitable because after learning a concept or as practicing, weak students could understand the fact clearly. Therefore, good drills and practicing need to meet the requirement of learning objectives. Teachers can carry out convalescence activities for oral problems such as references to their pupils in the classroom through drills. Drill and practicing is carried out repeatedly in the aspects of oral proficiency to be taught.

How can do it?

Drill and practice software packages offer structured reinforcement of previously learned concepts. They are based on question and answer interactions and should give the student

appropriate feedback. Drill and practice packages may use games to increase motivation. Teachers who use computers to provide drill and practice in basic skills promote learning because drill and practice increases student acquisition of basic skills. In a typical software package of this type, the student is able to select an appropriate level of difficulty at which questions about specific content materials are set. In most cases the student is motivated to answer these questions quickly and accurately by the inclusion of a gaming scenario, as well as colourful and animated graphics. Good drill and practice software provides feedback to students, explains how to get the correct answer, and contains a management system to keep track of student progress.

General Techniques of Drill

- Make all practice periods short
- Make use of incidental drill situations
- Use the new and older skills already established in its operation
- Give practice in “mental arithmetic”
- Use prepared workbooks and worksheets based on the skills to be acquired
- Give additional practice through life-situation units
- Give a variety of paper- and- pencil exercises
- Give praise for improvement in skill performance
- Insist on accuracy at first, then speed
- Homework should be given to provide drill
- Never expect perfect outcome from all

Advantages Drill Method

- It ensure perfection of skills
- It develops confidence and sense of achievement
- It makes learning more permanent as it is made habit at heart and practical in limbs by doing
- It is good for beginners
- In most times, it’s interesting
- It allows learners to learn by themselves.
- It holds what has already been grasped
- Reinforcement is present in drill.

Disadvantages Drill Method

- It promotes rote memorization more than meaningful understanding and to some extents, dulls creativity.
- It is not good for all topics
- If done in excess, it is likely to destroy initiative.
- It is not effective if students are unmotivated
- Frequency of repetition in drill sometimes makes it boring.
- It is to some degree dependent on attributes of other processes of education.
- There is limitation and specification of what is to be taught through drill.
- Even having to spend long time on repetitive tasks is a sign that learning is not taking place

SELF-ASSESSMENT

Write five merits and three demerits of each of following teaching methods

1. Inquiry Method.
2. Inductive Method
3. Deductive Method.
4. Lecture Method effectively
5. Co-Operative Learning
6. Collaborative Method
7. Problem Solving Method
8. Precision Method
9. Drill Method

SUMMARY

To learn mathematics in a way that will serve them well throughout their lives, students need classroom experiences that help them develop mathematical understanding; learn important facts, skills, and procedures; develop the ability to apply the processes of mathematics; and acquire a positive attitude towards mathematics. Although many students dread math class, or find it boring, it is important for teachers to do whatever they can to make math class fun. Students who enjoy math class perform better. It is therefore important for teachers to do all they can to help students have fun while learning math. Teachers have to select a teaching method(s) that best fit for topic under consideration and cognition level of the students. Inductive and deductive methods are most important for entire mathematics. Problem solving method is best fit for real life problems/question. Drill method is also important for practice.

FURTHER READINGS

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Unit - 03

LABORATORY LEARNING AND LESSON PLANNING

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INTRODUCTION

This unit is expected to provide the students and teachers with a skill of knowledge to teach mathematics, so that they become familiar with the 21st Century skills of teaching. Pupil teachers will acquire an understanding to use a wide range of instructional methods in order to encourage active learning mathematics which includes creation and usage of teaching and learning materials. Therefore, they are expected to plan the lessons and activities in order to engage the students in practice teaching of mathematics.

OBJECTIVES

After studying this unit, you are expected to:

- Understand the importance of mathematical laboratory in schools;
- Identify the activities for making mathematics creative;
- Develop individual and group learning environment in mathematical laboratory;
- Design activities of making, doing and discovering mathematics for school students.
- Draw a Mathematical Laboratory manual for each grade;
- Design model lesson plans of teaching mathematics for different grade levels.

3.1 MATHEMATICAL LABORATORY

Teaching of mathematics does not only aim to transmit the knowledge from the mind of the teachers to the notebook of the students using the tip of the pens. The current scenario of Mathematics teaching in schools is textbook dominated, which mostly concerns the manipulation of symbols, infinite recurrences, worthless memorization leading to an overall lack of concentration are some factors which contributes towards the negative attitudes in mathematics. However, the teachers, experts as well as practitioners of mathematics education are providing thoughts to the current method of teaching mathematics and to reform it to the Learner Centered. Mathematical laboratory has now become an unavoidable constituent to be engrossed as one of the conventional methods of teaching of mathematics which is anticipated to make the learning of mathematics as Learner Centered.

Since, teaching of mathematics deals with the process therefore a phenomenal change is required in the form of mathematical laboratory. It can provide association between teachers and students to provide a chance to recognize and unleash the beauty, significance and relevance of mathematics as a subject in a real-world scenario.

Mathematics laboratory being a fascinating concept in the arena of mathematics education was familiarized in 1994. In many developed countries the mathematics laboratory plays a significant role in regular curriculum. It has been brought up from a long way in elevating mathematics at primary, secondary & at higher secondary levels. Therefore, it can be anticipated to enhance the teaching and learning of mathematics at school stages an activity based and more experimentation oriented.

Mathematics laboratory is such a place where some mathematical activities are performed on similar basis to any other science laboratory. In a science laboratory the students get proactive practicing experiments. In the same way, this should help students to get a hands-on exposure and find the scope for new novelties. A place developed with manipulated activities and resources to which pupils have complete access to handle along with performing mathematical experimentations while solving the mathematical puzzles as well as playing mathematical games and becoming involved in other creative tasks. This laboratory approach allows the pupils to set up a mathematical experiment for the persistence of discovering some mathematical principles, patterns or processes. These activities or tasks may be carried out by the teacher or the students itself in order to enhance the interest and develop more satisfactory attitudes towards mathematics as a subject. The more participation in the laboratory work the more will be the creative and constructive competencies in the students. Also, a well-resourced laboratory is very essential for its effective functioning in order to secure favourable results. For this it should contain the following materials:

Materials and Equipments

Concrete materials: Mathematics laboratory should comprise of some basic essential materials for instance: balance, seeds, beads, sticks, weighs, pebbles, ball frames,

measuring tapes, pins, scissors, abacus, cardboards, chart paper, board pins, graph paper as well as other items which are useful for signifying the elementary mathematics perceptions.

Pictures and charts: The charts which shows the influences of mathematics to distinct aspects of life, histories & Pictures as well as the profiles of mathematicians are very supportive in exciting the students and can be placed in the mathematics laboratory.

Models: A mathematical laboratory must cover several mathematical models that offers opportunities for students to discover and recognize different concepts and fundamentals.

Black & Bulletin Board: There should on the minimum 1 black board inside for lettering the instructions, sketching the geometrical figures and illustrations which are essential for accomplishing the laboratory and bulletin board in the laboratory to exhibit several illustrations which are predominantly concerned with mathematics. Knowledge involving mathematics and its applied forms which can be collected from distinct sources such as: journals, magazines and newspapers which can be exhibited for the advantage of all pupils.

Drawing, Weighing and Measuring instruments - Mathematics laboratory necessitates students to sketch and draw the geometrical figures and graphs in order to weigh and measure different items. Therefore, there must be an established set of drawing instruments for instance: protractors, rulers, compasses and etc. for the teacher in order to validate the drawing on the blackboard. The boxes of instruments for use of students. Similarly measuring tapes, different types of balances, graduated cylinder and measuring jar must be available inside the mathematics laboratory.

Surveying instruments- Surveying is a significant factor in mathematical laboratory work. Placing out right angles or determining the angular distance, approximating the altitude of a building, guessing the distance of an object, evaluating the angle of elevation and depression are handy activities which makes the usage of surveying instruments. The following instruments are very beneficial in this aspect and must have a specific place inside the laboratory.

- Angle mirror glass for placing out right angles in the field.
- Clinometers aimed at approximating the:
 - a. Angle of elevation
 - b. Height of an object.
- Transit for measuring the angular distance of two objects
- Level for finding differences in elevation.

Proportional dividers: Proportional dividers progresses on the rule of proportionally for similar triangles. It can be often used for broadening or reducing the pictures, graphs, figures, maps and etc.

Computers and other Computing devices: A computer is usually mandatory for a multimedia presentation of a lesson will be more effectual and exciting than any conventional other way of organizing a lesson. Devices like Slide rulers and calculators are useful devices to have in a mathematical laboratory. A slide ruler compromises of 2 or more than 2 logarithmic scales which slides on each other. Thus, it delivers quick means for different arithmetic operations and is rather useful in those schools which don't have the funds for to purchase calculators. Calculators are also very handy when it comes to doing wide-ranging computations with a more greater accuracy and efficiency.

Furnishing mathematics laboratory

Adequate furniture must be provided inside the laboratory to do different experiments and at that very moment it must display working models and some other resources for undertaking observations.

These days, the character of a teacher is abridged to as a facilitator. The methodology employed for teaching mathematics has its own laboratory environment which will convert the status of a teacher from being an active speaker to a facilitator. Mathematics laboratory is the latest design to make mathematics real. It is new in teaching and learning of mathematics which provides opportunity to the students to conceptualizing mathematical ideas through their own experiences. In mathematical laboratory teacher can stimulates and promotes the growth of analytical thought in students, which is important for succeeding in the modern world. Through mathematical laboratory, it is possible to build conceptual understanding to support the conversion from real concepts and early elementary operational levels to more advance forms of thought. It also better enable students to handle logical and complex information throughout their lives. Therefore it is the dire of today to provide math education according to the 21st-century and to develop mathematical skills in students through laboratory learning.

3.2 SCOPE OF LABORATORY LEARNING

Mathematics laboratory is such an activity which focusses its place where a student is cracking a problem situation through self-exploration and discovery. It delivers a solution based on different experience, interests and needs. Some of the ways in which a mathematics laboratory can contribute towards the learning of the subject are as follows:

- A good opportunity to comprehend and adopt the basic mathematical concepts via concrete objects and situations.
- Allowing the students to confirm or unleash various geometrical properties, concepts and facts employing different models or by cutting the paper and folding techniques.
- Serving the students to shape interest and confidence in learning the subject.
- An opportunity to validate the affiliation of mathematical concepts with everyday life events.
- Wider scope for individual contribution in the process of learning and becoming self-sufficient learners.
- Good Scope for larger participation for the cognitivists and hand which simplifies the cognitive abilities.
- The laboratory permits and inspires students to look beyond the box while discussing and brainstorming with each other and the teacher and integrates the concepts in a more effective manner.
- Envisioning the students to identify the source of mathematical ideas and participate in "Mathematics in the making".
- Enabling the teacher to establish, explain and strengthen the abstract mathematical ideas and concepts by using solid objects, charts, models, pictures, graphs, posters etc.

In a nutshell, the laboratory of mathematics serves as a motivation for teacher as well as for student, for focusing and reflecting on specific concepts. Therefore, mathematics is not considered as heavy subject but a supportive discipline. Finally, The 21st- century mathematics teachers should be readily acquainted with the modern day techniques of teaching. Therefore all attentions paid for teaching of mathematics that has practical relevance of mathematical laboratory within the school should relegate the significance of research for further education and give emphasis on it as the 1 of the centrals in 21st century for dealing mathematics in everyday life.

3.3 IMPORTANCE OF MATHEMATICS LABORATORY

Mathematical laboratory should have all the important items (such as: a set-square along with compass & models and etc) relating with the learning tasks in mathematics. The central purpose for teaching mathematics lies in the expansion of power of conceptual thinking and reasoning. If mathematical facts or proofs are proved physically then they can be more easily understood and can be effortlessly applied in novel circumstances and hence a more practical work in mathematics formulates the subject to be more fascinating for students. The primary importance of a mathematics laboratory is:

- To provide readily available rich manipulative materials to emphasis on “learning by doing”.
- To develop an attitude of enquiry.
- Remove the weakness of present day mathematics education.
- To develop much needed confidence in students.
- To generate interest in the subject.
- To make the students divergent thinkers.
- To make the children to look for pattern and ask questions.
- To make mathematics teaching and learning, interesting and purposeful for students.
- To compose mathematics teaching and learning, interesting and purposeful for students.
- To provide activities those arouses the curiosity of the students and maintain their interest in learning.
- To enable students to develop proper skills in handling equipments and gadgets.
- To make students appreciate the practical applicability of mathematical principle and laws.
- To help the students develop powers of observation analysis and drawing inferences.

The learning being the fundamental aim of teaching envisions a destination from where the efforts and opinions are centred to develop the understanding of text in students. Mathematics isn't static rather there is a history of continuous changes like other subjects of sciences. It is pertinent to valorising the knowledge and experience of students therefore, it should be completed via creativity and discovery. They are equally substantial in the role of development of procedural skills (such as: written, graphic,

mental, via computational tools), the memorization (for instance: the definitions and characteristics of mathematical objects) of mathematics aimed at its globalisation. In other words, the stress is laid down on laboratory learning to achieve all the objectives of teaching of mathematics. In it, there are 3 distinct types of learning environment should be provided to the students; Creative learning; Individual learning and Learning by discovery.

3.3.1 Creative Learning:

Mathematical creativity is such a skills to examine a given problematic situation from distinct viewpoints. Such as: seeing patterns, then finding differences and similarities, manipulating multiple ideas, concepts and choosing an appropriate method to coup up with the situation under consideration. It is based on the principle of learning "Mathematics By Making". Therefore teacher of mathematics needs to explore creative learning environment, in which experience of teachers themselves plays a main role in introducing creative mathematical activities. The mathematical laboratory is an ideal place for mathematical creativity in which teachers interact with students on a specific set of instructions. It is such a place where the students can understand and discover new mathematical concepts through a wide range of activities by means of laboratory materials.

The following are the ways in which a mathematics laboratory can influence to the creative learning:

- It stimulates the imagination of students to create activities for the particular topic of mathematics i.e. algebra or geometry.
- It sustains the interest of students and improves their performance in the subject.
- It enables students to utilize relevant concrete materials for constructing models of mathematical concepts.
- It provides an opportunity for the interaction of students with teachers to produce new ideas for arriving at accurate results without any hesitation.
- Activities give opportunity to display the understanding of mathematical concepts in everyday life scenarios.
- Students can made Low-cost mathematical kit from their activities in the laboratory.

List of Material used in Mathematical Laboratory for Creative Learning;

- Paper folding
- Paper cutting and pasting (Collage)
- Unit Cubes (wooden or any other Material)
- Geo-Board , rubber bands
- Geometry box
- Transparency sheets
- Graph paper
- Pins and threads
- Broom sticks

- Chart papers, glazed papers, sketch pens
- Stationary
- Geometry box

Creative Activities in Mathematical Laboratory;

To involve students in activities, teacher illustrate the topic from the text book of different grade levels give instructions about the mathematical concepts and allow the students to create charts, graphs, pictures, posters and models by using these concrete objects. Many activities can be carried out; in geometry following are the examples of some paper folding activities;

- The midpoint of a line segment,
- The perpendicular bisector of a line segment,
- The bisector of an angle,
- The median of a triangle
- Obtain a parallelogram by paper folding,
- Verify the midpoint theorem for a triangle, using paper cutting and pasting,
- To divide a given length of paper into a specified number of equal parts using a ruled graph paper,
- To illustrate the students that medians of a triangle meet at a point called the centroid, which always lies inside the triangle.

3.3.2 Individual Learning:

It is well known that schools need to prepare students right from the beginning to shape their future. The role of education in 21st- century is to be different and education should foster skills of problem-solving, reasoning and proof, communication, representation and connection etc. The demand of new century requires that all students should acquire proficiency with these skills and a positive attitude in mathematics. A teacher can design and implement activity-based mathematics instructions with a view to develop 21st-century skills in mathematical laboratory. Individual learning in mathematics laboratory can be developed in the students. It is based on the principle of learning "Mathematics by Doing". In it students can physically verify the mathematical facts. It is only the extension of inductive method. The potential of the student should be promoted to understand the problem of the text by individually using computing devices and computer. Computing device consists of slide rules and calculator. Since working with computers devices students take interest for doing arithmetical calculations of mathematics. The arithmetic part of school mathematics is mostly concerned with the calculations of numbers, e.g. finding the square root of five digit numbers with or without calculator, exponential and logarithmic values of numbers.

There are some ways in which mathematics laboratory can promote individual learning:

- It gives more scope for individual participation of students to become autonomous learners and allows learning his / her on space.
- It provides scope for greater involvement of both hand and the which facilitates cognition.

- It widens the experiential base, and prepares the ground for later learning of new areas in mathematics and of making appropriate connections.
- It generates self –reliance in students.
- It builds up confidence in doing mathematics.
- It produces quality of doing hard work in subject.
- It develops calculus skills that offer the possibility of solving many exercises correctly and obtain better result.

Material used in Mathematical Laboratory for Individual Learning;

Material used in math lab for doing individual practical is almost same which is used in creative activities,

- Geo-Board , rubber bands
- Cardboard
- Paper cutting and pasting
- Transparency sheets
- Graph papers
- Pins and threads
- Chart papers, glazed papers, sketch pens
- Computer devices

Individual Activity in Mathematical Laboratory;

Teacher will illustrate the mathematical concept of least common multiples to the students of elementary level and demonstrate the procedure of doing activity individually to whole class.

Procedure;

- i. Cut a15 units x 12 units cardboard. This will be the baseboard
- ii. Cut 12 units x 12 units cardboard and cut 5 more boards of this measure.
- iii. All the boards will have a margin of 1 unit width on all four sides.
- iv. In the baseboard, make a10 units x 10 units squares and make equal squares on it.(Fig. a).
- v. Paste the heading on top of the baseboard and write the numbers 1 to 100 on these squares
- vi. On the other 5 boards, make 100 equal squares leaving the margin (Fig.b).
- vii. On each board, represent the multiples of different numbers such as, 2, 3, 4, 5, 6 etc.
- viii. Drill the squares which represent the multiples of the number specified.
- ix. Paste to the respective headings written on the chart paper on these boards as follows:
 Multiples of 2
 Multiples of 3
 Multiples of 4
 Multiples of 5
 Multiples of 6

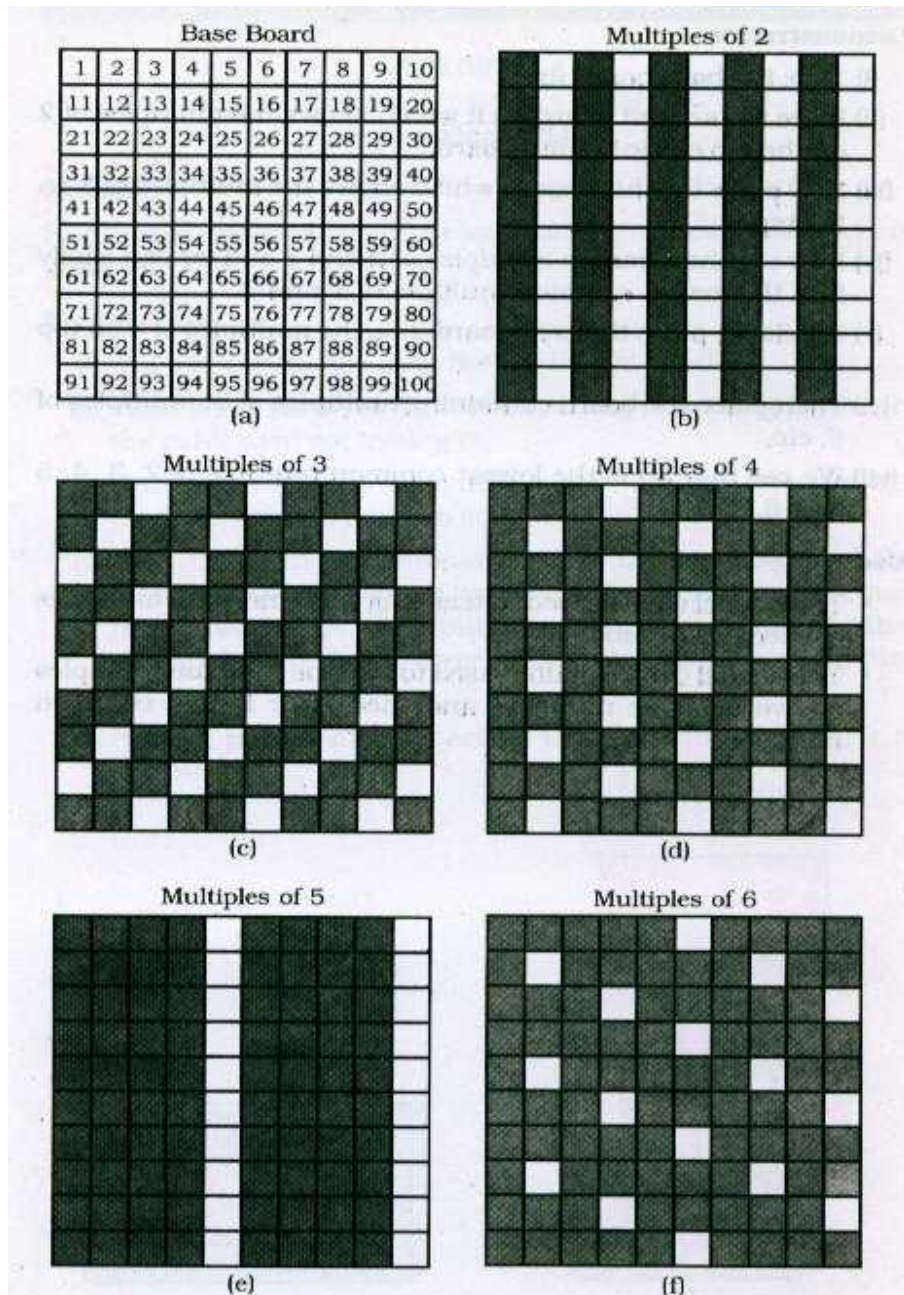


Fig. no. 3.3.2

Demonstration;

- i. Take the baseboard first.
- ii. Put the second board on it which shows the multiples of 2 in the top the second board.
- iii. Now place the third board on it which shows the multiples of 3 on the top.

- iv. Now see the common multiples of 2 and 3, so the students can find the lowest common multiples of 2 and 3.
- v. Similarly, place the next board, i.e. multiples of 4 on the top.
- vi. Then place the boards containing multiples of 5 and 6.

Use;

- i. This can be used in teaching the concept of multiples of specified numbers.
- ii. This is also used in teaching the concept of understanding the common multiples, and not common multiples of the given numbers.
- iii. This can be mainly used to find the least common multiples of different numbers.
- iv. This individual learning activity can be used in teaching mathematics at elementary level.
- v. After doing every activity, teacher should ask the students to make a practical notebook.

3.3.3 Learning by Discovery

Mathematics involves thinking logically and reasonably which leads people into discovering things. Practically, new discoveries cannot be made unless it is effectively taught through application of sufficient and efficient human and physical facilities. Education is imparted for achieving these things. The main goal of mathematics education in school is the mathematization of student's thought processes. School mathematics has two aims (i) narrow and (ii) broader or higher aim. The first aim is used to develop 'useful' capabilities, particularly related to numeracy-numbers, number operations, measurement, decimal and percentage. The higher or broader aim is to develop the student's resources to think and reason mathematically. It could be achieved through the use of innovative mathematics teaching strategies. Therefore the teacher of mathematics should design or plan of teaching a lesson enriched with activities for "Discovering Mathematics" to develop logical thinking and reasoning in the students. The place of discovering mathematics is mathematical laboratory in which teacher can plan tasks for the groups of students and gives them instructions for designing a project on any topic of mathematics. A plan to complete series of tasks is simply a project and the learning by completing the series of tasks is project base- learning, e.g. puzzles and games related to topics of mathematics as projects, a group activity of students finding the percentage of heights of students of grade 6, etc . The example of discovering mathematics is project- base learning. Teacher of mathematics should supervise the projects assigned to the group of students in mathematical laboratory. Laboratory can supports the students in the following ways:

- A new method of learning math through projects.
- It allows a variety in school mathematics learning.
- It engages students in a range of tasks, problems and investigation.
- Simple projects will lead to the development of different skills like thinking analytically and numerically.
- It develops discovery attitude in students.
- It provides self-motivation to execute the project of group.

Group Activities;

- To help the students establish interesting mathematical relationships by measuring the height and weight of the students of grade10 and find the average height and weight of students of grade10
- To collect the data of the students of grade 8 who write faster with left hand or right hand

Suggested Projects in school;

1. **Design a crossword puzzle with mathematical terms.**
To review mathematics vocabulary, to give the opportunity for creative expressions in designing puzzles, to act a means of monitoring the study of a given unit and to give recreation.
2. **Mathematics line designs.**
Using strings obtain interesting designs and patterns. Use threads and shapes made by cardboard; try to make designs on it by making slits on the cardboard. Observe different patterns on it.
3. **Computer project**
Using spreadsheet which is programed on a PC obtain the graph of the equation $ax^2 + bx + c = 0$ for different values of a ,b and c and note the interesting features and patterns. Interested students can also try for quadratic equations.

3.4 EXEMPLARY ACTIVITIES

Example 1

Topic: Area of a rectangle.

Definition: The area of a rectangle is the number of unit squares enclosed in it.

Objective: Deriving a formula for the area of a rectangle.

Previous knowledge:

1. Students are familiar with rectangles
2. Know about length and width of the sides of rectangle.
3. They can measure, cut and paste paper.

Material required: Cardboard paper, scissors and mathematical sets.

Procedure:

1. Draw a rectangle of any size on the cardboard paper (say 9 cm by 6 cm).
2. Mark each side at 1 cm interval and join a perpendicular from each mark to the opposite side.

1	2	3	4	5	6	7	8	9
2								
3								
4								
5								
6								

3. A network of 1 cm by 1 cm unit squares is formed.
4. Count the number of unit squares and take note. This is the area of the rectangle.
5. Now the length and the width used as a scale for drawing and take a note of the result (say $9\text{ cm} \times 6\text{ cm} = 54\text{ cm}^2$).
6. Compare the results of (4) and (5) above, are they similar or different?
7. If the results are similar, then the formula for the area of a rectangle is given by the multiple of the length and the width ($L \times W$) in square units which is equal to the number of unit squares contained in it as defined.

Observation:

The formula for the area of a rectangle as derived by the students themselves is internalized and appreciated.

Example 2

Topic: Circumference of a circle.

Definition: Distance covered around the circle.

Objective: Finding a value of Pi (π), hence deriving a formula for the circumference of a circle.

Previous Knowledge:

1. Students are familiar with circles.
2. Students can take simple measurements.

Requirements:

1. Students were asked to find circular based objects, draw out the circular base on a cardboard paper cut them out and bring the cuttings to the class for the practical lesson
2. Thread, exercise book, pen or pencil, mathematical set and a calculator

Procedure:

1. Chose one of your cut out circles and mark a point on the edge.
2. Place the thread on the mark and measure round the edge of the circle until you return back to the mark.
3. Measured out the length of the thread on a meter rule and take 2 or 3 readings. This is the circumference 'C' of the circle.
4. Mark a point 'O' at the center of the circle. Place a meter rule to measure from one edge of the circle through the center 'O' to the opposite side of edge in a straight line and take readings. This is the diameter 'D' of the circle.
5. Find the ratio of the circumference 'C' to the diameter 'D' (i.e. C / D) and take note.
6. Compare the result in (5) above of all the students with circles of different sizes. All their results tend to a constant $22/7$ or 3.142 . This constant is called Pi denoted by π .
7. It has been noted in (6) above that $C/D = \pi$. In other words students can internalize the definition of π is the ratio of circumference to the diameter of the circle

$$\text{i.e } \pi = C / D \rightarrow C = D \pi$$

But the distance from the center 'O' of the circle to the edge is equal to a radius 'R'.

Hence, a diameter $D = 2R$.

Thus we can write it as ; $C = (2R)\pi \rightarrow 2\pi R$

Therefore the circumference C of a circle is derived as; $C = 2\pi R$ or $D\pi$

8. Multiply the value of π with the result of diameter in (4) above and compare it with the result of (3) above.

Observation:

The results are the same for each particular circle in step (3) and (8). Students can easily use the formula for circumference of a circle to compute and will also be familiar with the derivation of the value of the constant π .

Example 3

Topic: Pythagoras Theorem

Theorem: The area of the square on the hypotenuse side of a right-angled triangle is equal to the sum of the areas of the squares on the other two sides (i.e. $c^2 = a^2 + b^2$)

Objective:

To Verify Pythagoras theorem by the method of paper folding, cutting and pasting.

Previous Knowledge:

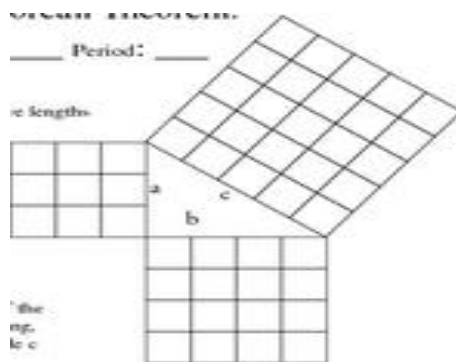
1. Students are familiar with sides of right-angled triangle.
2. By taking simple measurements, cuts and paste students can construct right – angled triangle.

Material required:

Cardboard paper, pencils, mathematical sets, gum and pair of scissors.

Procedure:

1. Take a normal size cardboard.
2. Construct a right –angled triangle ABC on cardboard with right angle at C and sides a, b, c, also c as hypotenuse.
3. Construct a square on each side such that the length of the sides of the square is equal to the length of the particular side of the triangle.
4. Cut out the squares on sides 'a' and 'b' .
5. Divide each square into unit squares and cut out.
6. Gather all the unit squares and carefully place them orderly inside the square on the side 'c' of the right-angled triangle.



Observation:

The unit squares cut from the squares on the sides 'a' and 'b' of the right-angled triangle completely fill the square on the side 'c' of the same triangle which verify the Pythagoras theorem and establishing the students' cognition.

Similar examples should be created by the teachers for the students of different grade levels on topics of algebra and geometry.

3.5 INTRODUCTION TO LESSON PLANNING

Planning is essential not only in teaching but in all spheres of life. In the teaching profession planning is often connected with lessons. Planning of a lesson is an important equipment of a teacher in a school or in a college. A lesson plan is strictly individual; it is indeed the creation of the teacher who plans out the lesson plan. A plan is a work of art involving much imagination and study. The plan is an unfolding of the teacher's soul, it contains the life-blood of the teacher. Lesson plan is a kind of discipline, which has to be learnt in the training college.

A lesson is not giving of instructions, or mere doing out of facts. It becomes an occasion for learning, thinking and understanding as well as judging. Lesson planning constitutes essential learning experiences for all teachers in training because pre-planning is essential for quality teaching. But it raises an important issue: how lesson planning is to be done?

The question has two dimensions, viz:

1. What is going to be the broad pattern of lesson plan?
2. How much and what kind of details in their lesson notes are to be required by the pupil-teacher.

However, the lesson plan is a good servant but a bad master. It is a means to an end by itself. The teacher should be able to discard his plan if a sudden situation demands it.

R.L. Stevenson states the importance of lesson plan as, "To every teacher I would say, always plan out your lesson beforehand but do not be slave to it." Ryburn also said, "To teach we must use experience already gained as starting point of work". Many expertises have defined lesson plan as;

1. "Daily Lesson Planning involves defining the objectives, selecting and arranging the subject-matter and determining the method and procedure." **Bining and Biwin**
2. Lesson Plan is the title given to a statement of the objectives to be realized and the specific means by which these are attained as a result of activities during the period. **Bossing, N.L.**
3. We may define lesson plan as a draft of the lesson put upon paper with all the important points whether of matter or method clearly marked. **Joseph London.**
4. Lesson must be prepared for, there is nothing so fatal to a teachers' progress as unpreparedness. **Davis**

Thus it is clear from the above definitions that lesson planning is the program of the teacher which indicates the objectives, subject-matter, learning experiences, audio-visual aids, methods and techniques etc. Generally, there are three levels of Planning.

- Weekly Planning
- Monthly Planning
- Annual Planning

These levels may be plans as—



Hence, the lesson plan reflects the intelligence, ability, capacity, resourcefulness and personality of the teacher. Lesson planning provides awareness to the structure and content with which teacher is involved in the direction to achieve the objectives.

Approaches to Lesson Planning:

There are various approaches to make a lesson plan. These are Herbartian approach, Unit approach, Evaluation approach and project approach.

- **Herbartian Approach** is based on understanding the mass theory of learning. All the knowledge and information is to be given from outside by the teacher because the student is considered to be similar to a clean slate. For the students, if an old knowledge makes a base for new knowledge (his previous knowledge or experiences), it may be acquired easily and retained for a longer period. Herbart has given five steps:- (i) Introduction (ii) Presentation (iii) Organization (iv) Comparison and (v) Evaluation. with the main focus being on content presentation.
- **Unit Approach of Morrison** consists on unit transaction and planning. The Morrison's lesson plan of teaching is cyclic; The five steps for 'cyclephase' of teaching are:- (i) Exploration, (ii) Presentation, (iii) Assimilation (iv) Organization and (v) recitation.
- **Evaluation Approach of B.S. Bloom:** Education is objective centered not content centered. The main aim in this approach is objective based teaching and testing. It takes into consideration the learning objectives and teaching methods on the basis of the objectives and to assess learning outcomes. Then a decision can be taken about objectives of learning are achieved or need to provide the revision.
- **Project Approach originated by Dewey and W.H. Kilpatrick,** stresses on group activity, social activity, self activity and related to real life experiences. It is pre-planned work completed by a person or group in social condition.

It is not necessary that a lesson plan made by a teacher will be successful at every place and time due to many reasons . Many factors influence lesson planning such as:

- i. Unavailability of teaching aids,

- ii. Strength of students in class,
- iii. Composition of the Class (Age wise),
- iv. School Location (Area wise),
- v. The Nature of the topic (Medium wise) etc.

These factors are not enough, other factors from own experiences can be added in it.

3.6 PRINCIPLES OF LESSON PLANNING

Some principles of planning are as follows—

1. ***Plan should be Flexible.*** The plan should be concrete and specific as well as it should also be flexible. Each child is different in his ability, experiences, needs, his interests and each has his own attitudes. Necessary extension, revision and reorganization of plan is a favorable index of the quality of teaching.
2. ***Plan Should be Specific in Nature.*** A plan should give specific information as to how the teaching-learning process is expected to move forward. There should be specific provision for the anticipated procedures.
3. ***Plan Should be Realistic***
 - a. The mathematics teacher should do the task of planning in the real sense.
 - b. The level of educational development must be taken into account.
 - c. The amount of time anticipated should be examined carefully.
 - d. The mathematics teacher should assess his teaching resources carefully.

It should be linked with the previous knowledge of the child. It should contain suitable subject matter.

Main Features

1. The lesson plan's subject matter should be at the disposal of the teacher according to the time for teaching
2. Homework related to the taught subject matter should be given.
3. It should provide maximum participation of the child in teaching and learning progress.
4. There should be proper provision in the lesson plan for the teaching aids and good illustrations.
5. There should be proper provision in the lesson plan of recap to have view of evaluation of the subject-matter taught to the students.
6. In the lesson plan there should be provision of summary of the whole subject matter.
7. Lesson plan should be child centered.
8. Example quoted to teach and explain the subject matter should be according to the age and the mental level of the students.
9. Subject matter arranged in the lesson plan should be related to the previous knowledge of the child.
10. Selection and organization of the subject matter should be to the point and systematic.

11. It should be written clearly and brightly.
12. Vague and irrelevant material should be avoided.
13. It should clearly state the aims and objectives of the subject matter.
14. It should be in written form, in good hand and bold letters.

3.7 MODELS OF LESSON PLANNING

Many written lesson plans are used by practicing teachers. Before attempting to write a lesson plan, a teacher needs some information about the class, students and their background. In general, a Model lesson plan is divided into many steps/ stages. One can write a macro lesson plan based on the different approaches stated above.

Format of Presentation: Effective teaching needs proper planning, transaction in the classroom and feedback. Practically there are three stages of a planned lesson:

- i. Pre- active stage is a planning before going to classroom.
- ii. Interactive stage is a interaction between teacher and students in the real classroom situation.
- iii. Post active stage is a self evaluation of our teaching work.

There is not a single format for writing a lesson plan because it varies teacher to teacher and subject to subject, It is suggested that it should be a well-organized structure, it follows basic fundamental parts of a lesson.. Some of the formats based on different approaches to the lesson planning. Teachers' are free to make changes according to the objectives you plan to achieve and nature of the subject etc.

i. Herbartian Lesson Plan Format

Subject: ----- Date: -----
 Unit: ----- School: -----
 Topic: ----- Class: -----
 Duration: ----- Period: -----

1. General Objectives:
2. Specific Objectives:
3. Teaching Aids:
4. Method
5. Previous Knowledge:
6. Introduction:
7. Statement of the Topic:
8. Presentation or Development of the lesson:

Teaching Points	Teacher's Activity	Student's Activity
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OR

Content	Objective	Teaching-Learning Activity	Evaluation
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9. Recapitulation:
10. Black Board Summary:
11. Home Assignment
12. Reference:

Note:

1. Steps of S. No. 1 to 5 may be changed according to your needs.
2. In presentation/ development objectives, teaching – learning activities and evaluation are in relation to particular teaching content.
3. If evaluation of students learning is done after each teaching step, then you can ignore recapitulation at the end of the lesson.
4. Black board summary should be developed as the lesson progresses.

ii. Bloom's or Evaluation Lesson Plan Format

Subject: ----- Date: -----
 Unit: ----- School: -----
 Lesson: ----- Class: -----
 Duration: -----

1. Specific Objectives:
2. Previous knowledge:
3. Introduction:
4. Statement of the Topic:
5. Presentation:

Objective	Teaching Method & Aids	Teacher's Activity	Student's Activity
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6. Evaluation:
7. Home Work:
8. References:

Note:

1. The objectives should be written in behavioural terms.
2. Teaching activity should be related to the learning structures.
3. Black board summary should be developed as the lesson progresses.

iii. RCE Lesson Plan Format

Date: -----
 Subject: ----- School: -----
 Unit: ----- Class: -----
 Topic/ Lesson: ----- Duration: -----

1. Concepts:
2. Teaching Aids:
3. Previous Knowledge:
4. Introduction:
5. Statement of Topic/ Aim:

6. Presentation:

Expected Behavioural Outcomes (EBOs)	Sequential Learning Experiences/ Activities (LEs)	Real Learning Outcomes(RLOs)
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7. Black board Summary:

8. Home Assignment:

9. References:

Note:

1. RCE is Regional College of Education.
2. Concepts are to be identified through content analysis.
3. Black Board summary should be developed as the lesson starts.

3.7.1 Model Lesson Plan for Teaching Algebra

Subject: Mathematics

Date:

Unit: 3

Class: 8

Topic: Algebraic Expressions and Identities

Sub Topic: Algebraic Identities

Duration: 30 minutes

Sr. #	Teacher's Activity/Points of Lesson Plan	Student's Activity &Board Writing
1.	General Objectives: The students of grade 7 will be able to know about: i. Algebraic Expressions. ii. Algebraic Identities. iii. Its usage in daily life.	
2.	Specific Objectives: The students will understand the: i. Relationship between algebraic expressions ii. Relationship between algebra, geometry and arithmetic.	
3.	Teaching Aids: White board, marker, pencil, geometry box, notebook, pen glaze paper, scissors, pasting material.	
4.	Teaching Method: Art Integrated Learning(AIL), This method involves learning through integrating various subjects with different models of art such as painting, drawing, clay modeling, paper-cutting	

	etc... also called Interactive method Presentation based Learning .	
5.	Previous Knowledge: Students have studied mathematics up to grade 7 and they know about variables, constants and algebraic expressions.	
6.	Preview: Teacher will address the students in a very happy mood. Before starting the today's' topic, teacher will give review about Algebraic expression and ask the following questions; Q1. Students tell me have you ever learnt Algebra? But the teacher will say yes you learnt algebra when you were a kid and your mother started calling you with names chikku, guddu etc. Q2. You know from where these names came? Teacher will tell that you were unnamed that time and your mother wanted to interact with you she needed some variable to address you and started calling by these names. You learnt your name and started giving response to your mother. So you became a variable named guddu, chikku etc. Teacher will revise that; similarly in mathematics the numbers 2, 3, 4 represent with some other names like a, b, c these are called variables. Variables and constants form a term and terms are added to form algebraic expressions. Your name is a variable which you learnt when you were a kid.	Ans. Some students will say Yes and some will say No. Ans. Student will reply, from Muma.
7.	Introduction: Teacher will tell the students; Today we will represent different numbers and algebraic expressions geometrically to find Algebraic Identity.	
8.	Presentation or Development of the lesson: Teacher will tell the students that we call algebraic expressions are represented as geometrical expressions and ask some questions;	Ans. Because, these numbers represent squares.

	Q1. Why 4, 9, and 16 are called squared numbers?							
	Teacher will tell students that we can also represent squared algebraic terms with squares. X^2 is the area of a square with side length X. Area of square of side X = X^2	ACTIVITY 1 <table><tr><td>a</td><td>b</td></tr><tr><td>a^2</td><td>ab</td></tr><tr><td>ab</td><td>b^2</td></tr></table> $a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$ $(a + b)^2 = a^2 + 2ab + b^2$	a	b	a^2	ab	ab	b^2
a	b							
a^2	ab							
ab	b^2							
	Teacher will tell that students can apply algebraic identity to work out arithmetic problems as;	Identity Taken; $(a + b)^2 = a^2 + 2ab + b^2$ Example; $(17)^2 = (10 + 7)^2$ $= (10)^2 + 2(10)(7) + (7)^2$ $= 100 + 140 + 49 = 289$ ACTIVITY 2 $(a - b)(a - b) = a(a - b) - b(a - b)$ $= a^2 - ab - ab + b^2$ $= a^2 - 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$						
	Teacher will tell that students can apply algebraic identity to work out arithmetic problems as;	Identity Taken; $(a - b)^2 = a^2 - 2ab + b^2$ Example; $(17)^2 = (20 - 3)^2 = (20)^2 - 2(20)(3) + (3)^2$ $= 400 - 120 + 9 = 289$						
		ACTIVITY 3 $(a - b)(a + b) = a(a + b) - b(a + b)$ $= a^2 + ab - ab - b^2$ $= a^2 - b^2$ $(a - b)(a + b) = a^2 - b^2$						
	Teacher will tell that students can apply algebraic identity to work out arithmetic problems as;	Identity Taken; $(a - b)(a + b) = a^2 - b^2$ Example; $(20-3)(20+3) = (20)^2 - (3)^2$ $17 \times 23 = 400 - 9$ $391 = 391$						

	Teacher will ask the question; Q1. What is identity? Teacher will explain; Consider the equality $(a + 1) (a + 2) = a^2 + 3a + 2$ Put $a = 2$ in above we can show that L.H.S = R.H.S, $(2+1)(2+2) = 2^2 + 3(2) + 2$ $(3)(4) = 4 + 6 + 2$ $12 = 12$ Also put $a = -3$ we can see $(-3+1)(-3+2) = (-3)^2 + 3(-3) + 2$ $(-2)(-1) = 9 - 9 + 2$ $2 = 2$ Teacher will told that such an equality which is true for every value of variable in it is called IDENTITY.	
9.	Revision; Teacher will ask , what we have learnt? An identity is equality, which is true for all values of the variables in the equality. The following are standard identities: $(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$ $(a - b)(a + b) = a^2 - b^2$ The above identities are useful in carrying out squares and products of algebraic expressions. They also allow easy alternative method to calculate products of numbers'. i. Using identity 1 find $(2x + 2y)^2$ ii. Using identity 2 find $(4p - 3q)^2$ iii. Using identity 3 find $(\frac{2}{3}m + \frac{1}{5}n)(\frac{2}{3}m - \frac{1}{5}n)$ iv. 194×206	
10.	Assignment; Try at home; 1. Represent the identity 1 by taking the value of $a = 3$ and $b = 4$ geometrically. 2. Using identity 2 find the arithmetic problem $(204)^2$ 3. Find the product of 104×96 by using the algebraic identity.	

3.7.2 Model Lesson Plan for Teaching Geometry

Subject: Mathematics

Date:

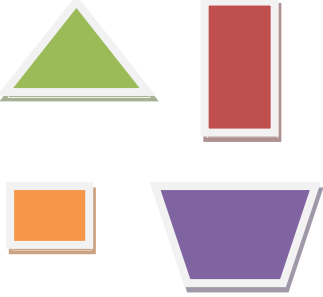
Unit: 8

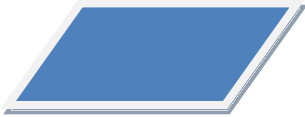
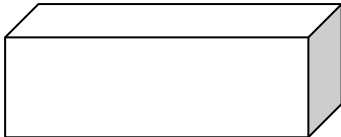
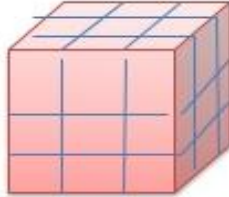
Class: 6

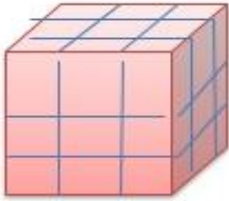
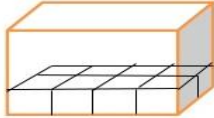
Topic: Volume of a cuboid or rectangular prism

Sub Topic: Formula of finding volume of cuboid.

Duration: 30 minutes

Sr.#	Points of lesson plan /Teacher's Activity	Student's Activity& Board writing
1.	General Objectives: The students of grade 6 will be able to know about; • Formulae of Volume of cuboid • Solution for finding rectangular prism • Work cooperatively to achieve result.	
2.	Specific Objectives: The students will understand the; • Volume of geometrical shapes • Difference of cube and cuboid	
3.	Teaching Aids: Transparent rectangular containers, small cubes, Rubik's cube, worksheets, pictures of plan figures, paper, ball pen	
4.	Teaching Method: Inquiry &Activity methods	
5.	Previous Knowledge: • Students know about the geometrical shapes like triangle, square, rectangle, parallelogram. • Know the how to take length & width. • Also know the about the area and perimeter of square and rectangle. • Know the unit of area of geometrical shapes.	

6.	Review: - The teacher will show the class geometrical shapes of square, rectangle, triangle trapezium with different measurement of the sides mentioned on it. - Ask the students some questions about areas of these figures.	
7.	Introduction; - Before introducing topic teachers will divide the class into groups for preparatory activity. - Teacher will assign the students group A, B, and C and give them shapes of different length and width. Ask questions about areas of these shapes. - Teacher will told students the difference of cube and cuboid by showing cube, Rubik's cube and a transparent rectangular container. - Today we will find the volume of rectangular container called volume of cuboid.	Group A. Ans. Area of square of length 10 unit is 100sq.units.  
8.	Presentation / Development of the lesson: - Teacher will show a solid cube and ask question about it; Q1. What it is ? Q2 Do you know about its length, width and height? Teacher will show an object related to cube for motivation and ask the questions; Q1. What do you call it? Teacher will tell the students that a Rubik's cube is 3 x 3 x3 cube that can be manipulated so that each face of the cube will have same design. Q2. Do you know how to play it? Q3. How many small cubes does this Rubik's cube have? - Teacher will told that number of small cubes that make up Rubik's cube is 27.	 Ans.1 Group A, it is a cube Ans.2 Group B, yes, it has equal length, width and height.  Ans.1 Group C, it is a Rubik's cube. Ans.2 Whole class, Yes teacher. Ans3. Some students may give estimate or exact answer.

- Teacher will show the container and ask; Q. What kind of solid figure is this container? Now the teacher will put cubes in the container and ask: Q1. How many cubes in a layer did I put inside the container? - Teacher will tell that 4 cubes are length and 2 are width. - Teacher will fill the container with small cubes until its upper portion and ask; Q1. How many cubes are in height? Q2. How can you find the number of cubes in the container without counting Q3. How many cubes in all? Q4 Teacher will ask when we get the total number of cubes that the container has, what have you looked for? Q5. What kind of polygon is the base of the container? Q6. What are the dimensions of container? Q7. How many cubes fit the length? The width? Q8. Tell me , can you give the volume of the rectangular container by just using the dimensions(length, width, height)? How? Q9 What is the formula in finding the volume of rectangular container? Q10. What is its unit? Q11. What is the area of rectangular base? Q12. What is volume? Q13. How can we write volume? - Teacher will ask the students did you understand grade 6. - Teacher told the students about the formula of volume of cuboid.	 Ans. Group C, rectangular prism called cuboid. Ans.1 Group B, 4 and 2  Ans.1 Group A, 3 cubes Ans.2 By counting the number in one layer then multiply into other horizontal layer. Ans3. There are 24 cubes.. Ans.4 students looked for its volume. Ans.5 Group A, rectangle Ans.6 Group B, length , width and height.=L, W,H. Ans.7 Group C, 4, and 2. Ans.8, yes teacher, by multiplying the length, width and height. Ans.9 Volume = $L \times W \times H$ Ans.10, cubic Unit. Ans 11. $L \times W$ Ans12. Area of base X Height Ans13. $V = B \times H$ $= L \times W \times H.$
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9.	Revision: For developing skill of finding the volume of the given cubic units. Teacher will assign each group to find the missing number. Students you have an announcement!!!! Teacher will make comparison among the groups on the bases of their participation and announced that students of group A have excellent participation for doing activity 1, group B efficiently participated in activity 2 while group C shown good co-operation. Well done students!!!!	A. L=3.5 cm, W=2.4cm, H=4 cm, V= ? B. L=9dm, W=4dm, H=8.5 dm, V= ? C. V =375cu. cm, W=21cm, H=4 cm, L= ? D. V=1232cm³, W=8cm, L=12 cm, H= ? Thank you Teacher.																									
10.	Assignment: Teacher will give home work related to the text book exercise .and complete the table; <table><tr><td></td><td>L</td><td>W</td><td>H</td><td>V</td></tr><tr><td>1</td><td>9dm</td><td>8.6dm</td><td>5dm</td><td></td></tr><tr><td>2</td><td>1.4m</td><td>1.5m</td><td>1.8m</td><td></td></tr><tr><td>3</td><td>40cm</td><td>15m</td><td>24cm</td><td></td></tr><tr><td>4</td><td>18.5cm</td><td>9.4cm</td><td>11cm</td><td></td></tr></table>		L	W	H	V	1	9dm	8.6dm	5dm		2	1.4m	1.5m	1.8m		3	40cm	15m	24cm		4	18.5cm	9.4cm	11cm		
	L	W	H	V																							
1	9dm	8.6dm	5dm																								
2	1.4m	1.5m	1.8m																								
3	40cm	15m	24cm																								
4	18.5cm	9.4cm	11cm																								

3.7.3 Model Lesson Plans for teaching Theorem

Subject: Mathematics

Date:

Unit: 7

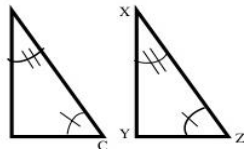
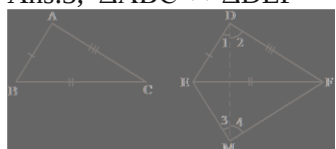
Class: 9

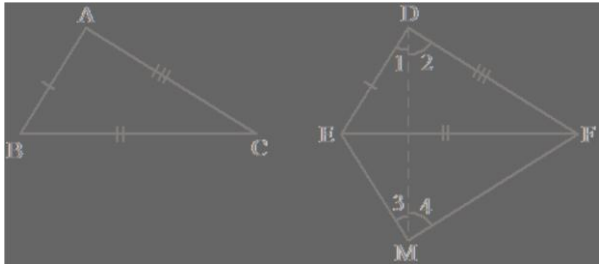
Topic: Congruent Triangles

Sub Topic: Postulate of congruent triangles (SSS $\approx$ SSS)

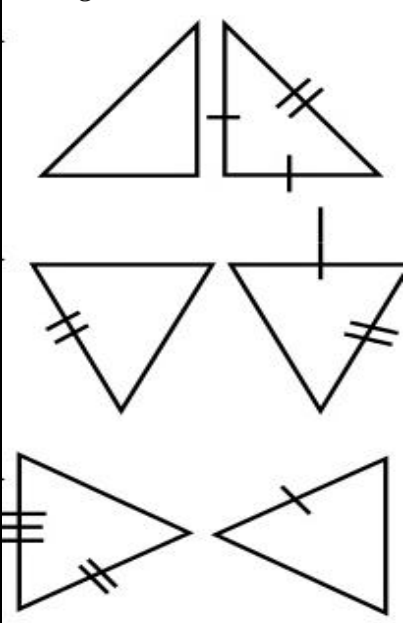
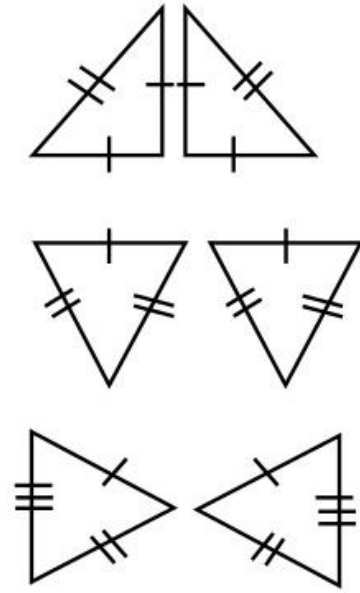
Duration: 30 minutes

Sr. #	Points of lesson plan /Teacher's Activity	Student's Activity& Board writing
1.	General Objective: Students of grade 9 will be able to know about: - Congruency of two triangles - With respect to their side and angles.	
2.	Specific Objectives: Student will understand SSS postulate of two congruent triangles.	

3.	Teaching Aids: White board, marker, pencil, geometry box, cardboard, paper.	
4.	Teaching Method: Inquiry & Deductive method	
5.	Previous knowledge: Students of grade 9 know about; 1-1 Correspondence of triangles - Angles and sides of triangles	
6.	Review: Teacher will ask question about yesterday's lesson; Q1 What you have done yesterday? Q2 What is a postulate? Q3 Which postulates you studied yesterday? Q4 Who will draw it on board? Q5. What is the 2 nd postulate of congruent triangles?	Ans.1, We have studied about the postulates of triangles Ans.2, A postulate in geometry is a truth without formal proof. Ans.3, ASA postulates of congruent triangle .  Ans.5. SAS postulate of congruent triangle
7.	Introduction: Teacher will tell that Today we will deduce 3 rd postulate of congruent triangles (SSS). The statement of the theorem is " In a correspondence of two triangles, if three sides of one triangle are congruent to the corresponding three sides of the other, then the two triangles are congruent" (S.S.S $\cong$ S.S.S).	
8.	Presentation/ Development of Lesson: - Teacher will divide the class into two groups for answering the questions; - Questions for group A Q. 1 State the first line of the theorem? Q. 2 What is given in it? Q. 3 How we write the correspondence in triangle? Q. 4 Draw diagram. - Questions for group B	Ans.1, Group A, In a correspondence of two triangles Ans.2, We are given two triangles ABC and DEF. Ans.3, $\triangle ABC \leftrightarrow \triangle DEF$ 

Q. 5 Read the next line of the statement. Q. 6 Which triangles? Q. 7 What is given in it? Q. 8 Sum up the given statement. - Questions for group A Q. 9 Read the next part of the statement? Q. 10 Which triangles? Q. 11 What is to prove? - Questions for group B Q. 12 Write what is given and what is to prove? Q. 13 What we do next?(Hint of previous Theorem) - Questions for group A Q. 14 Draw and steps of construction. Teacher will draw and students will write steps.  - Teacher will tell the students now think logically then answer with reasoning.	Ans.5, Group B, if three sides of one triangle are congruent to the corresponding three sides of the other. Ans.6, $\triangle ABC$ and $\triangle DEF$. Ans.7, $AB \cong DE$, $BC \cong EF$ and $CA \cong FD$. Given $\triangle ABC \leftrightarrow \triangle DEF$ $AB \cong DE$, $BC \cong EF$ and $CA \cong FD$ Ans.9, Group A then the two triangles are congruent" Ans.10. $\triangle ABC$ and $\triangle DEF$ Ans.11. $\triangle ABC \cong \triangle DEF$ Ans.12. Given: $\triangle ABC \leftrightarrow \triangle DEF$ $AB \cong DE$, $BC \cong EF$ and $CA \cong FD$ To Prove: $\triangle ABC \cong \triangle DEF$ Ans.13. We construct another triangle on $\triangle DEF$ Construction; Suppose that in $\triangle DEF$ the side EF is not smaller than any of the remaining two sides. On EF construct a $\triangle MEF$ in which, $\angle FEM \cong \angle B$ and $ME \cong AB$. Join D and M. As shown in the above figures we label some of the angles as $\angle 1$, $\angle 2$, $\angle 3$ and $\angle 4$.
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	Q. From figure how many correspondence of triangles exist? Teacher will write the statement and students will give the reasoning as; <u>STATEMENTS</u> In $\triangle ABC \leftrightarrow \triangle MEF$ $BC \cong EF$ $\angle B = \angle FEM$ $AB \cong ME$ $\therefore \triangle ABC \cong \triangle MEF$ and $CA \cong FM$ (i) Also $CA \cong FD$..... (ii) $\therefore FM \cong FD$ In $\triangle FDM$ $\angle 2 \cong \angle 4$..... (iii) Similarly $\angle 1 \cong \angle 3$..... (iv) $\therefore m\angle 2 + m\angle 1 = m\angle 4 + m\angle 3$ $\therefore m\angle EDF = m\angle EMF$ Now, in $\triangle ADB \leftrightarrow \triangle EDC$ $FD \cong FM$ And $m\angle EDF \cong m\angle EMF$ $DE \cong ME$ $\therefore \triangle DEF \cong \triangle MEF$ Also $\triangle ABC \cong \triangle MEF$ Hence $\triangle ABC \cong \triangle DEF$	Ans. Correspondences of three triangles. <u>REASONS</u> Given Construction Construction S.A.S postulate (corresponding sides of congruent triangles) Given {From (i) and (ii)} $FM \cong FD$ (proved) {from (iii) and (iv)} Proved Proved Each one $\cong AB$ S.A.S. postulate Proved Each $\triangle \cong \triangle MEF$ (Proved)
9.	Revision: For revising the theorem teacher will call students of group A & B on the board one by one for illustrating this theorem. Teacher will make comparison among the groups on the bases of their participation and announced that students of group A have excellent participation for introducing the theorem while the group B efficiently participated in steps of construction. Well done students!!!!	Students happily participated in it Thank you Teacher...

10.	Assignment: Teacher will provide sheets to students, to complete which pair of triangles congruent through SSS. 	
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SUMMARY

- Laboratory is a motivation for student teachers.
- Objectives of teaching mathematics can be achieved through laboratory learning.
- In mathematical laboratory, teachers make the learning of mathematics 'Learner Centered'.
- Mathematical laboratory can develop 21st- Century skill in students.
- Mathematical laboratory is a place where the students can learn mathematics by 'making', 'doing' and 'discovering'.
- Low- cost material for practical activities can be developed in mathematical laboratory.
- Concrete materials, pictures, charts models are the main equipment of school mathematical laboratory.
- Slide rulers and calculators are computer devices used for computational purpose.
- A plan is a draft of the lesson, put upon a paper with all the important points.
- Lesson planning constitutes essential learning experiences for student teachers.
- Lesson plan reflects the intelligence, ability, capacity, resourcefulness and personality of the teacher.

- Herbartian approach of lesson planning includes five main steps; (i) Introduction (ii) Presentation (iii) Organization (iv) Comparison and (v) Evaluation.
- Unit approach consists of five steps for 'cyclephase' of teaching, i.e. (i) Exploration, (ii) Presentation, (iii) Assimilation (iv) Organization and (v) recitation.
- The main aim of Evaluation approach is objective based teaching and testing.
- Project approach stresses on group activity, social activity, self- activity and related to real life experiences.
- Practically there are three stages of a planned lesson:
- Pre- active stage is a planning before going to classroom.
- Interactive stage is a interaction between teacher and students in the real classroom situation.
- Post active stage is a self evaluation of our teaching work.

EXERCISE

1. Suggest five recommendations for Mathematical Laboratory in schools.
2. State some demerits of laboratory learning in schools.
3. Design a creative activity for grade 7 students to teach geometry.
4. Illustrate a list of individual activities for grade 10 students to teach algebra.
5. Develop group activities for grade 6 students to teach them a new concept in arithmetic.
6. Design a project for grade 9 students to teach them 'Real and Complex Numbers'.
7. Design a Laboratory Manual for Practical Activities of Mathematics in the table given:

Class	Lesson #	Type of Activity	Name of Activity	Topic of Activity Book Page #	Remarks of teacher
V1	Unit 1	Group/ individual/ project	Game/ puzzle/ doing work	3--10	Interesting, Excellent/ Very Good.
	Unit2				
	Unit3				
	Unit4				
	Unit5				
	Unit6				
	Unit7				

8. Draw this manual for class VI - X.

FURTHER READINGS

- Daniel O. Apronti, Joyce Asante Afful (2001). Teaching Mathematics in Basic Schools. Accra: Teacher Education Division.
- GES-JICA Science, Technology and Mathematics Project (2004). Manual for the Teaching of Science and Mathematics in Basic School. GES-JICA STM Project.
- Juliet C. Donkor, Daniel O. Apronti, Michael E. Ampiah (2002). Primary Mathematics Pupil's Book 1-6. Unimax Macmillan.
- Al-Qirim N, Mesmari A, Mazroeei K, Khatri S, Kaabi Z. Developing teaching scenarios in the classroom using interactive smart board ecosystem, Digital Ecosystems and Technologies (DEST), 2010 4th IEEE International Conference 13-16 April 2010; 525-530. doi: 10.1109/DEST.2010.5610596

Unit - 04

**INSTRUCTIONAL MATERIALS/
AV AIDS**

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INTRODUCTION

Teaching and learning is not only telling and listening, it is a process of creating a clear and concrete picture of any concept which is being taught. Especially while teaching mathematics it is very difficult for teacher to teach and learner to learn mathematical concepts. Therefore teachers need some helping material to augment their teachings. Instructional materials or aids are the objects or devices which may help the teacher to make the learning meaningful, interesting and easy. In this unit we will learn about instructional aids and their appropriate use for teaching mathematics.

OBJECTIVES

After grasping the theory of this unit you will be able to understand:

1. The Importance of instructional materials
2. Principles of Effective Use of Instructional Materials/Aids
3. Various Aids in Teaching Mathematics
4. Projective Aids in teaching mathematics
5. Assistance of computer in Teaching Mathematics

4.1 INTRODUCTION TO INSTRUCTIONAL MATERIALS/AIDS

Instructional material refers to the material that helps a teacher in conveying his/her teaching outcomes to the students. These Instructional material also known as **teaching/learning materials**. Instructional material is a collection of objects that may include animated, unanimated or human and non-human resources that a teacher might use for delivering his teaching objectives. The basic purpose of using these instructional material is to enhance the learning outcomes of students and to facilitate their achievements.

Instructional materials are important aspect of teaching because these are used to enhance the students' capabilities, skills, their overall development and upbringing. Instructional material increase the communication between teacher and student. It helps to store and shares the human experiences of knowledge with the students. With effective use of instructional material the interest of students could be maximize, their retention to the accurate ideas would increase. It aids teaching experience and enhance the teachers' competence and also making learning outcomes more expressive for students.

With increase in the learning outcomes the instructional material also make the teaching process so easy and feasible. As it helps the students to attain the learning outcomes more effectively similarly it helps the teachers to deliver their ideas with more efficiency.

There are different types of instructional materials that are being used in mathematical instructions. First type of instructional materials is graphic materials which include graphics, charts, posters, diagrams, maps, comics, cartoon and globes. Usually these are drawn on a piece of cloth or a cardboard to make the students understand what a teacher is trying so hard to explain them verbally. This material belongs to two dimensional material. Graphic materials are basically used to captivate attention, to focus and as an encouragement to recall.

Second type of instructional materials is three-dimensional materials, it is different from two dimensional materials because it helps to present an idea in a replica form of the real things. Different types of three-dimensional materials include models, mock-ups, specimen, kits, and dioramas which is the formation of a scene in an event.

Third type is still pictures that belongs to the group of two dimensional materials. These are the pictures taken from journals and magazines and are placed on a surface or we hold them in our hands and we do not need the help of a projector for presenting these pictures.

Another type of instructional materials includes still projected pictures or motion pictures and audio materials. To present the idea with the help of these pictures we need a projector. A projector throw the light on the image and the image is finally projected to the wall or on a screen. There is a more helpful type of instructional materials that is audio materials. It refers to tape recordings and discs. It creates audio sounds, and students can understand the concepts by listening to a recording.

Further details of instructional materials are given in this unit for more clarity and understanding. This unit is helpful for teachers to grasp the importance of instructional materials in their teaching process.

4.2 IMPORTANCE OF INSTRUCTIONAL MATERIALS/AIDS

The main importance of instructional materials is to retain the interest and motive the students. It creates a high level of interest for the students because it represents a reality of experience that motivates the students for self-activity.

Instructional materials increase the efficiency of learners and contribute in the development of thoughts. As it provides tangible materials that develop a continuity of critical thinking.

Instructional materials make learning more permanent for the students because they provide a clarity of concepts, ideas, and learning outcomes. It reduces the meaningless work of the students by providing them a concrete basis for concepts.

Another important aspect of instructional materials is feasibility and essence of producing instructional materials. So it facilitates the teaching learning process. The production and selection of instructional materials should be on sound principles. In the process of developing an instructional material a teacher and student both get benefits by getting physical concept basics.

An important contribution of instructional materials in the process of learning is that it provides primary experiences to the students that enriches their learning outcomes.

The simplicity of instructional materials to represent a concept is also an important contribution in teaching learning process. It enhances the effectiveness of learners in grasping the concepts and efficiency of teachers in delivering the concepts.

Another important benefit of instructional materials is that it provides centralized study facilities that is very effective for enhancement of students' learning. When learning comes for the average students, instructional materials play a vital role in creating a real picture of the concept due to which the average students can also easily grasp the complexity of lessons. For example if a teacher is trying to give the concept of cylinder to a student it would be more difficult to convey the message verbally instead of showing the students an object that is cylindrical. So the visual things remain stick to the memory of the students. Today's learner expects to see short sessions of details and they want to learn in shorter time period that is only possible with instructional materials.

Instructional materials are also important in a way it helps to fully fill the state standards. Because they are usually aligned with standards set by government. The combination of good teachers and materials can have a huge impact on learners' outcomes.

4.3 PRINCIPLES OF EFFECTIVE USE OF INSTRUCTIONAL MATERIALS/AIDS

There are some principles for the effective use of instructional materials/aids.

Instructional materials should be easy to understand and the coverage of the lesson should be straightforward and away from complexity. The more it is near to facts the more realistic outcomes would be generated.

The instructional materials should be designed or organized in useful bits or portions for effective coding and rehearsal. Beautiful designs are important to attract the students so charts, graphs, pictures and visual aids should be effectively designed to reach the attention of the students.

The theory based learning is generic but now in the era of technology some visual aids or instructional materials should be added to the content so it becomes interesting for the learners, as the today's learners are more attracted towards technology.

An important principle for the use of instructional materials is that it should stick to the key points and contents. The more it is relevant to the topic the more it would give productive results.

The instructional materials should be used properly. When it is used more accurately it would help the students to remember the information. Before delivering the lessons proper working of instructional aids must be tested. If it would not work properly it would be frustrating for students.

Selection of words, terms, pictures and diagram or figures in instructional materials should be sensibly designed to deliver the same message to the learner as they do for the teachers.

Another important principle for the use of instructional aids is that it should be free from language barrier problems. Considering the continued expansion of technical terminology in daily usage and diverse background of students, makes it imperative for instructors to be very precise in their selection of terminology.

Similarly while using the instructional materials it is important to clarify the relationship between the materials objects and concepts. The visually of relationship would be more effective in terms of final results.

While using material aids maximum variety of tools must be used to make the lesson more interesting. Using different videos, charts, models, overhead, chalkboard, and visual aids cover the need of every receiver.

The instructor should be very expert of using the material aids so he would be able to explain the complexity of lesson with the help of these materials. A teacher must learn how to use the instructional materials.

The principle under consideration is to align the instructional materials with the lesson objectives. It would help to remain student centered.

A smart principle for the construction of learning aids is that these should be appropriate for the age, level of ability, social and emotional development of the students. If the instructional material would be according to the level of the students then it would be easy for them to understand the lesson and concept. The selection of instructional materials should be according to the students' objective rather than on teachers' personal preference or availability of materials.

4.4 VARIOUS AIDS IN TEACHING MATHEMATICS.

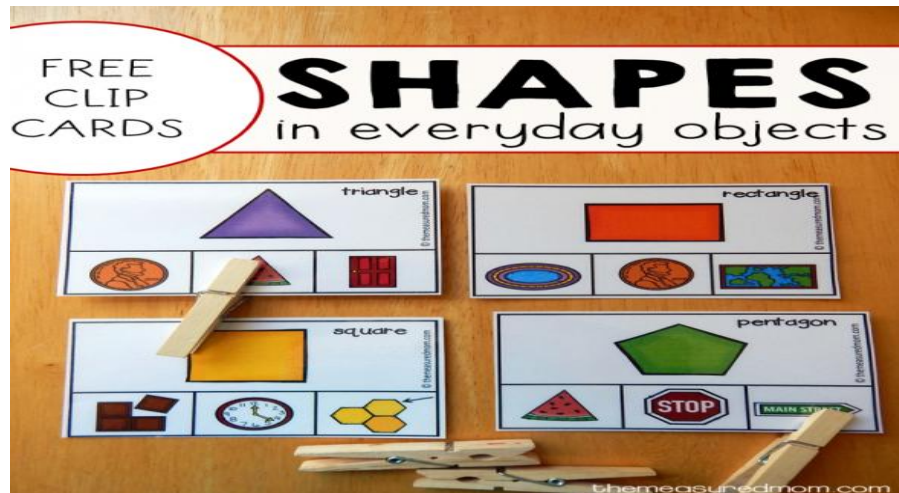
There are different aids that are being used in teaching mathematics including:

- Real Objects
- Models
- Charts
- Black Board
- Flannel Board
- Excursions and Collections
- Text Books and Work Books
- Radio and T.V.

Now we will discuss all of these aids one by one in detail so one can get familiar with helping side of them.

4.4.1 Real objects:

Any object about which we talk and write while we are doing maths is mathematical object. There are many examples for real objects some of them are numbers, functions, shapes, matrices, groups and vector spaces and infinite series. The objects of maths are abstract objects they are not physical objects basically. Like number 42 is an example of math object. Rectangle is another example of maths object. Similarly when we are talking about set (2, 7, 6, 8, 7, 8). This is a math object and its number are the elements of set. We can also show these objects in the form of picture like if we want to show the shapes we can have the visual form of these shapes as are shown in the figure:



Maths is applicable in every field of life, we can easily find the example of maths in our daily life if the kids are plying with blocks we can count the number of blocks. If someone is going to buy ice-cream we can count the money for which we are buying that ice-cream.

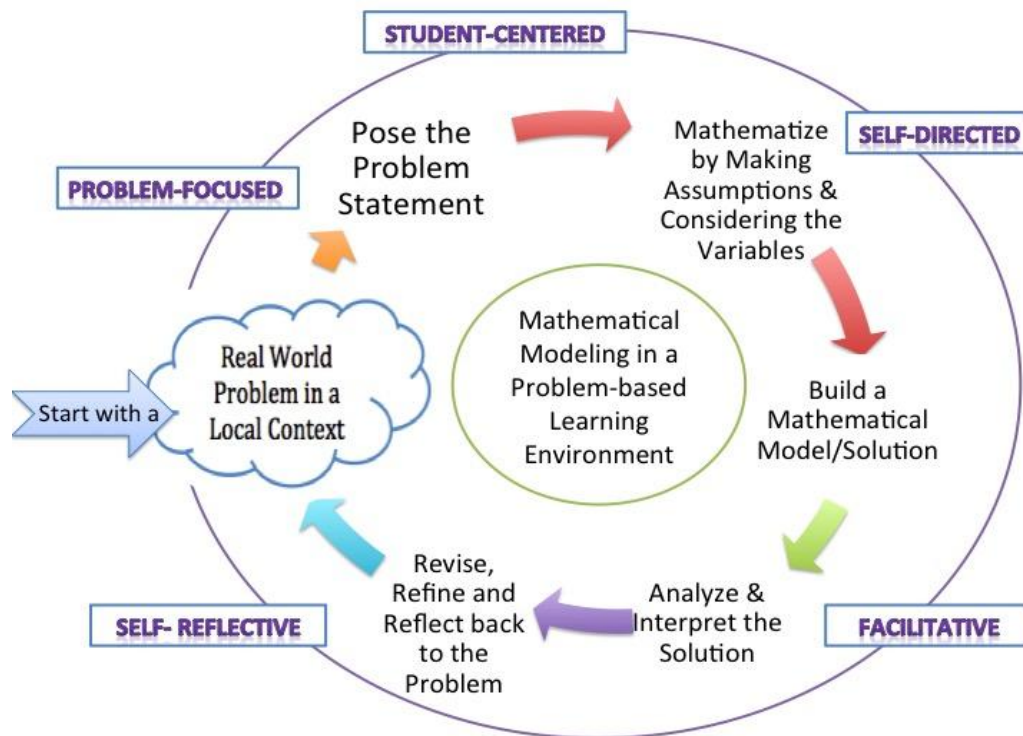


This picture is showing us a 3d shape due to this visual presentation of the shape a learner can easily understand its structure, sides, edges and vertices. For teachers it is easier to explain the students what are vertices instead of telling them verbally. By showing such real life example the concept of math would stick to the minds of students.

4.4.2 Models

Models are construction of a system related to real life world with perspectives of mathematics including its concepts. Everything we do in our daily routine is dependent on mathematical concept. For example if we have to serve the dinner at 7 pm so we have to decide the preparation time. Mathematical models are being used in every field of life like in physics, economics, biology, chemistry, sociology, computer sciences, electrical engineering, political sciences etc.

The following diagram showing what is a model and how it is being constructed.



To make a model we have to start with a real world problem in a local context. After considering the problem from real life a problem statement is to be formed. To continue the model assumptions and variables are created by using concepts of mathematics. After that a model or the solution is created. The process continue with analyzation and interpretation of the solution. The last step of the process includes revision, reflection of the problem.

Mathematical models are mostly composed of relationship and variables. Relationship are operators like algebraic operators, functions, differential operators etc. and variables are parameters of interest. Numerous models are defined by their structures:

Linear vs. non-linear:

If all the operators have linearity the model is considered as linear and if the operators don't have linearity then the model would be non-linear.

Statics vs. dynamic:

A static model is time –invariant means it calculates the system in equilibrium while a dynamic model calculates for time dependent changes in the state of the system.

Explicit vs. implicit:

If the input parameters are given then output parameters can be calculated after a series of calculation this type of model is called explicit model. Sometimes output parameters are given and input parameters are solved by an iterative procedure such models are called implicit models.

Discrete vs. continuous:

In discrete model the objects are being treated as discrete like states in a statistical model, while the objects are continuous in continuous model such as temperature and stresses in a solid.

Deterministic vs. probabilistic:

In deterministic model the variable states is determined by the parameters given in the model and by sets of previous states of these variables. While on the other hand the probabilistic model uses probability distributions, randomness exists due to which variables states are not defined by unique values.

Deductive, inductive or floating:

Deductive models are based on theory while inductive models are based on observation, it is formed by empirical judgements and generalization from them and floating model is only the entirety of expected structure.

Activity:

Ask the student to calculate the space inside a card box by using linear model.

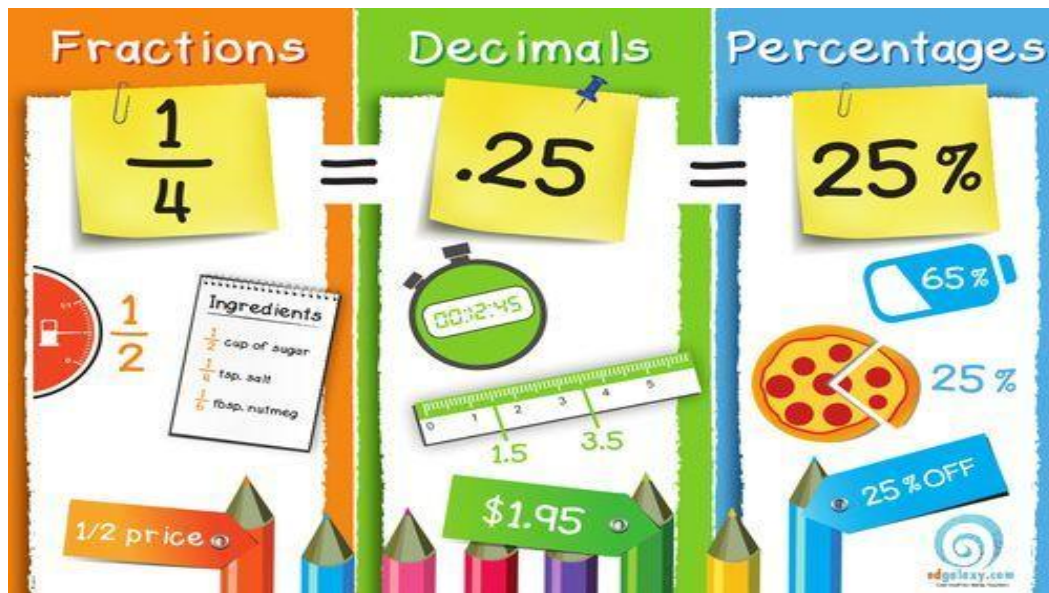
4.4.3 Charts

Mathematical charts are traditional tools that aid students to keep the concepts in mind. Usually charts are pieces of paper that are different from simple papers, on charts concepts, models of mathematics are drawn and decorated on the walls. Charts and graphs are a helpful way to present and convey the scientific concepts. For example weather patterns can be presented through bar graphs that show the experimental findings.

Activity:

Ask the students to prepare a chart for their favourite concept of mathematics and the best chart would be displayed in the classroom.

Some examples of charts are shown in given picture.



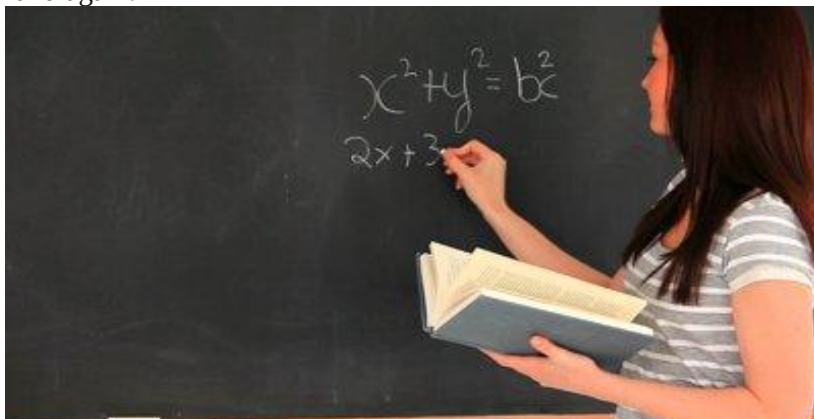
This chart is displaying the difference between fractions, decimals and percentages. A beautiful visual of 3 concept that will definitely stick to the minds of the students.

Activity:

The teacher can ask the students to create a pie chart for total number of students in their class. If there are 30 students in class. The students are asked to tell their favourite colours. For this information a pie chart creation is a best presentation.

4.4.4 Blackboard

Blackboard is a black coloured surface used as a teaching aids in schools and colleges, universities. It is painted with matte black colour and sometime it can be of dark green colour. On it teachers write to deliver their lectures with the help of chalks and it can be erase again and again.



The use of blackboard is a traditional aid now in most of the places blackboard is replaced with white boards. On blackboard one can write with chalks while on white board a marker is used to write. But still the advantages of this tool is irreplaceable. Some of best advantages are:

It is Easy to present, teachers can easily use this tool they don't need any special training for using this tool.

Students get bored with long verbal lectures to keep their attention or focus on the lesson, the writing on board is helpful for teacher and students both. In this way teacher would be able to keep them in classroom and students would be able to understand the mathematical concepts without any difficulty while doing maths at home. Because maths is a subject that is totally based on writing and practice there is nothing involve verbal things unlike theoretical subjects. The more students write maths more they would learn.



It is helpful for delivering the lecture of maths. Maths problem can easily be explained with it.

Formulas can be written on it that students can easily memorize. It would be helpful for students to take notes. Taking notes for students is big help to keep things in mind for exams.

4.4.5 Flannel board

It is a board display that is made up of wood, cardboard or strawboard. It is covered with woollen cloth or coloured flannel. Different objects like cut-outs, drawings, pictures will stick to the flannel board. The objects and materials used or stick to this board can be removed and relevant material can be used for the next lesson.

Usually a flannel board of size 1.5* 1.5m is used. It can be fixed next to the chalkboard or can be placed on a stand. Where the pictures are adhere and detach from the flannel board for subsequent lesson along with telling the story or description.

The biggest advantage of this medium is that it is dynamic tool that is easy to use and change, according to the requirements. Changes can be done by adding or removing materials or objects from the board.

Instead of this medium is not only useful for primary grades it can be used with equal benefits for all level of ages and all level of languages. This tool is so helpful for teachers to keep the students in focus.

Activity:

Teacher can increase the motivation of the students by involving them in creating a flannel board. Ask the student to prepare pictures or objects to display on the flannel board for the concept of fraction.

Following is a flannel board that is displaying the concepts of shapes. Black colour flannel board is having the shape of triangle while other different shapes are given below that can be attached or remove to describe the shapes. A teacher can even ask the student to come and stick a specific shape to the flannel board. This activity will increase their focus and attention.



4.4.6.Excursions and Collections

Excursions are short trips that are arranged by a teacher for correlating the lesson to the real life experiences. The purpose of excursions is to enhance the subject relevance to the real world as well as to increase the knowledge. Field trips should be an integral parts of lesson instructions. While collection means the materials that is being collected at excursions. Students should be encouraged by the teachers to take notes or collect the regarding materials while they are on trip.

Significance of excursions and collections can be understand by the following postulates.

Field trips provide first-hand experience.



Enhance the interest of students and motivate them to learn maths.

Providing a better meaning of learning in mathematics and interrelationships.

Enhance the observational, perceptual and analytical skills.

Develop the personal and social developments and also help to develop scientific attitude.

Provide a platform for self-expression.

Support the students to establish contact with outer world.

Increase the chance for the selection of the projects.

Activity:

Plan an excursion for the students that exhibit the applications and implications of mathematical concepts so well. So the students may get more of understanding from this trip.

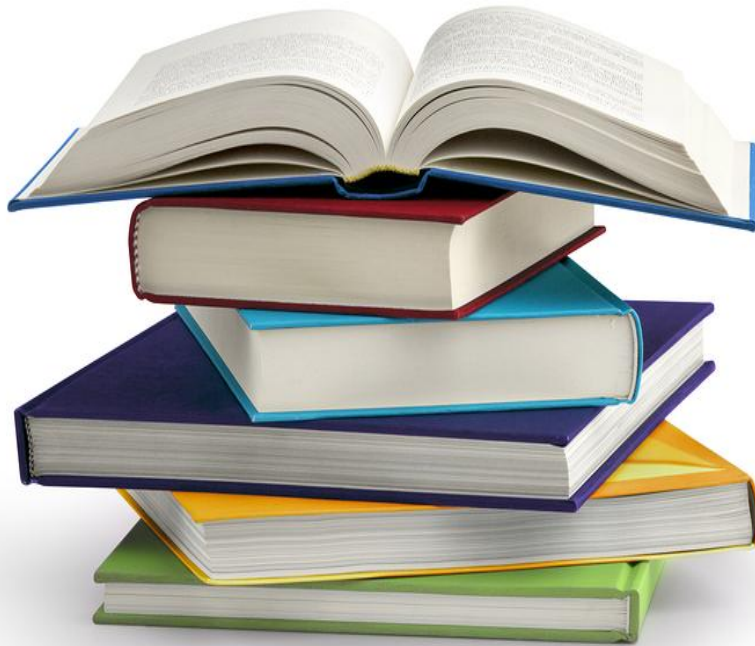
The selection of place for excursions should be according to the relevance of lessons. For example visit of a bank for explaining to the students how maths is implemented

everywhere in real life. After excursions the experience of the students should be discussed in class, their queries and confusions must be answered by the teachers.

4.4.7 Textbook and workbook

Textbooks are compilation of theory based study and are helpful in every field of life. It is an assemblage of knowledge, concepts and principles. These are helpful for the need of students to study the lesson in a sequence. In schools, colleges and universities textbooks are most commonly used for student learning. These books can be in digital or printed form.

When these books are used for solving maths it helped a lot to the teachers as well as the students for learning mathematics. This tool gives step by step instructions for solving mathematics



Textbooks and workbooks provide numerous advantages like:

Textbooks are helpful for beginning teachers because it is a guideline which follows the sequence from simple to complex concepts of mathematics.

Textbooks of mathematics provide unit, and exercises for practice as well as the details that are required for covering a lesson.

Everything is well explained in a balance, a teacher can use it as a helping tool to have guidance for what to do and when to do. Latest editions are updated according to recent research.

While using textbook or a workbook a teacher should keep some points in mind.

Teachers should use it as a helping tool but not the only resource. It is a helping guide teacher should not take it as mandate for teaching. Teachers should feel free to add some

additional data in textbooks for teaching mathematics. Outside topics and reading can be introduced by the teachers to help the students to get the concepts. For further reading go to www.teachervision.com.

4.4.8. Radio and T.V.

Radio is a scientific device that is helpful in audio communication. It helps for more and more learning. As we can hear weather forecast, similarly it covers a wide variety of areas. In developing countries like Pakistan radio and T.V are playing vital role for enhancing the interest of people for education where education is inaccessible. In such countries radio has massive values where limitations of finance, appropriate equipment and appliances adversely disturb educational planning and administration.

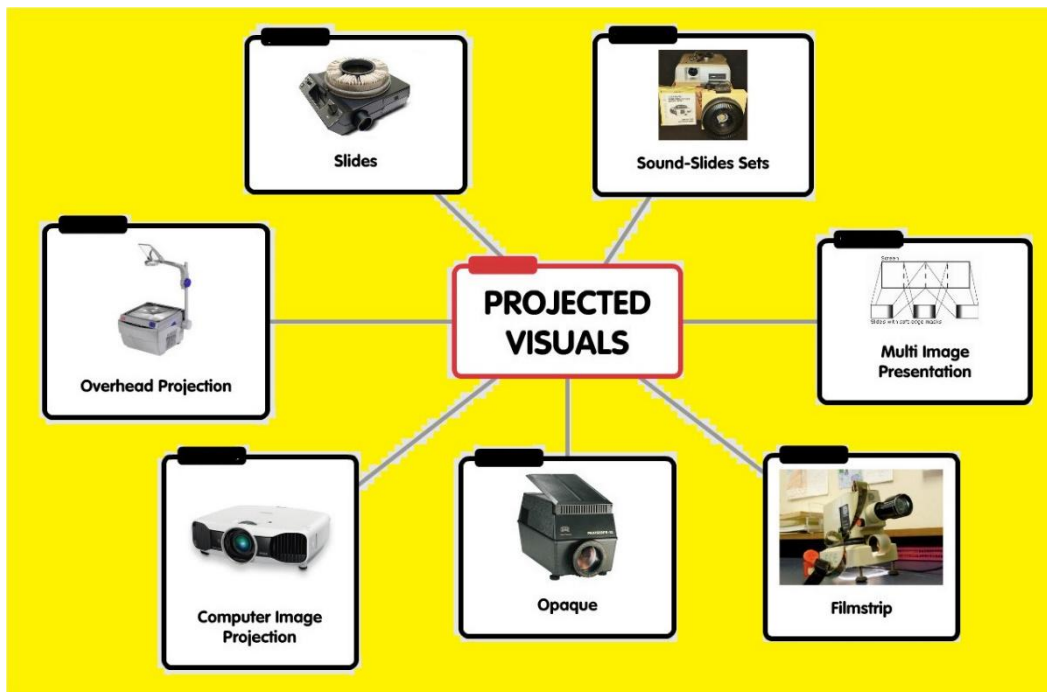


Radio helps to improve the qualitative enhancement of education. In inaccessible areas it promotes the facility of education. It provides growing awareness for the formal system of education and also improves the standards of education. Radio and T.V. gives a sense of unity and integrity where people bring issues to discuss and make arguments regarding every issue of life. Both of these things provide mathematical data regarding stock exchange, broadcasting, weather casting. Now a days there are special channels that give education of maths by delivering lectures online. So students can get benefits for their mathematical subject through these traditional devices. These help to compare our country standards with international standards. Another important role is that these devices are inexpensive and all types of people can reach this. Educational broadcasting improves the educational purposes.

While telecasting of T.V. helps to improve the training of people, inculcation of socio-political and cultural values. T.V. also helps to understand the implication of mathematics in different fields of life. It gives a sense how it is implemented in different structures etc.

4.5. PROJECTIVE AIDS

Projective aids are material used to show a picture on a screen by a machine. These machine are of various types that helps to get the lesson with more realistic basis. Some of projective aids are:



Some of these projective aids are discuss below:

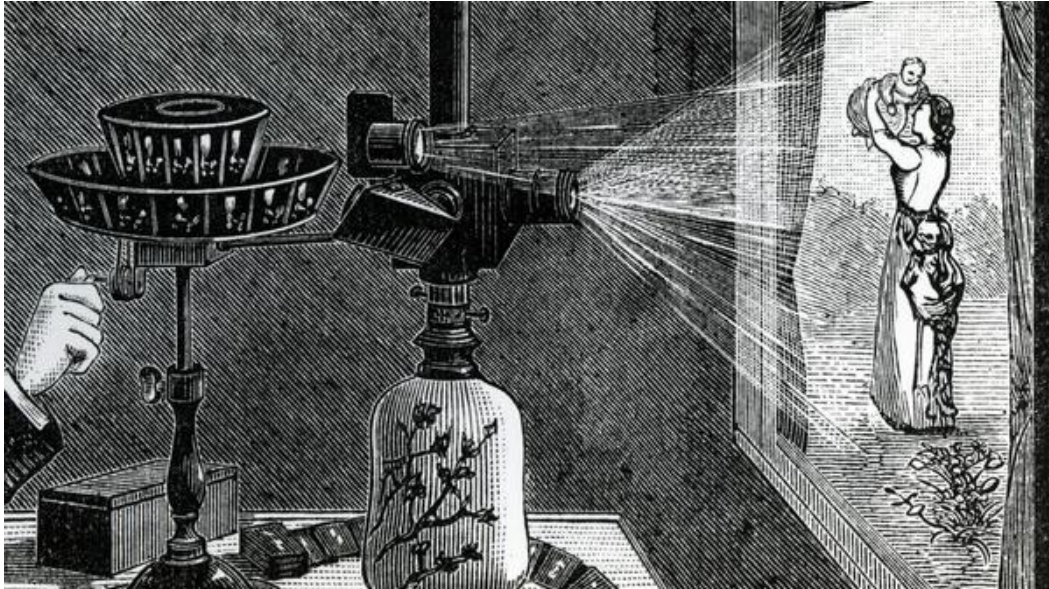
4.5.1 Magic Lanterns

It is a very traditional type of projector that was used massively for educational and entertainment purpose back in 17th century. It was used to show the printed pictures on a big surface with help of light thrown on it. The use of it remain heavily beneficial till mid of 18th century and it was later replaced by slide projectors.

In early century it was illuminated by candles but with the passage of time it was lighten with a brighter source of light. For further details check www.luikerwaal.com.

In early years handmade slide were used in magic lanterns. As the technology increases there was advancement in the slide also. Due to which the process was speed up and use of lanterns was also increased. Artist create different pictures on the glass and the operator use to move that glass slowly that means to create movements in the pictures. In biunial or triunial lantern that use two and three lens respectively, one picture could keep stationary while other one and two pictures could keep moving. And more advancement was found when the pictures were drawn on the discs, which spun around and show continuous movements in the film.

The following picture shows a magic lanterns.



There are different types of magic lanterns including children's lanterns, plural lanterns, ceramic lanterns, sciopiconsetc all of these are little different from each other for their characteristics. Some of them are having more than one lens that shows the picture.

4.5.2 Films strip projectors



A kind of projector with which the lanterns were replaced. A FILM STRIP PROJECTOR use a roll of 35mm films on which a fixed sequence is arranged. Teachers can use it to cover a large variety of lessons. It can also use a roll of 8mm, 16mm that consist of picture with descriptive titles. It is composed of metal case having concave reflector and a set of concave lens that are used to illuminate the film roll. It also have an exhaust fan to control the heat of the projectors.

Teaching mathematics with help of film strip projector seems to be successful. It is helpful for showing students the step wise techniques in solving mathematical problems.

A teacher gives the life to classroom and tools like film strip projector are helpful tool for giving mathematical lesson to the students.

Film strip projector having several advantages like:

Require simple operating skills

Economical medium

Small and weightless in size

Moveable from one place to another place

Suitable source for teaching and convenient for attaining the interests of students.

Images can be stopped on screen to elaborate

Free from mess because it gives you a sequential arrangement of data.

Help to save time and allow students to participate.

Can use rechargeable batteries plus AC/DC power.

4.5.3 Epidiascope



This projector is helpful in showing both slides that are transparent and opaque objects. This function is unique from other projector so it is best for schools. A prism like system including projection lens, mirrors is used in this projector to show an image. A brighter source of light like a bulb is used in this projector to display opaque objects by shining light onto the objects. For maths instructors it is supportive in showing books, maps, charts and graphs etc. Important benefits of epidiascope are:
It is supportive in showing flat objects like coins, cloths, charts.

It is useful for teaching mathematic because it is most effective device in creating graphic aids.

Teachers don't need to clip pictures from books and materials of various size can be used by teachers.

4.5.4 Motions pictures

Motion pictures means a film or a movie, it is being generated by collaborating different pictures or clips of picture to display a movie. Due to its advancement and perfection of work it is most influential media of visual aids. With progression in technology motion pictures have been changed, now these are available in different styles that also vary from country to country. The excellence of motion pictures enhance the interest of students

because it is not made by humans rather these are made scientifically. It gives the students an exact clarity of real life because in it real people and things are involved.

Some appealing advantages of motion pictures are:

It helps to display a process at once in projected form that is not sometimes possible to see with human eye in real.

Major advantage is it helps to get the understanding of things, ideas and concepts and are those correlated with real life.

There are three different types of motion pictures that are:

Documentary: based on special events in the world.

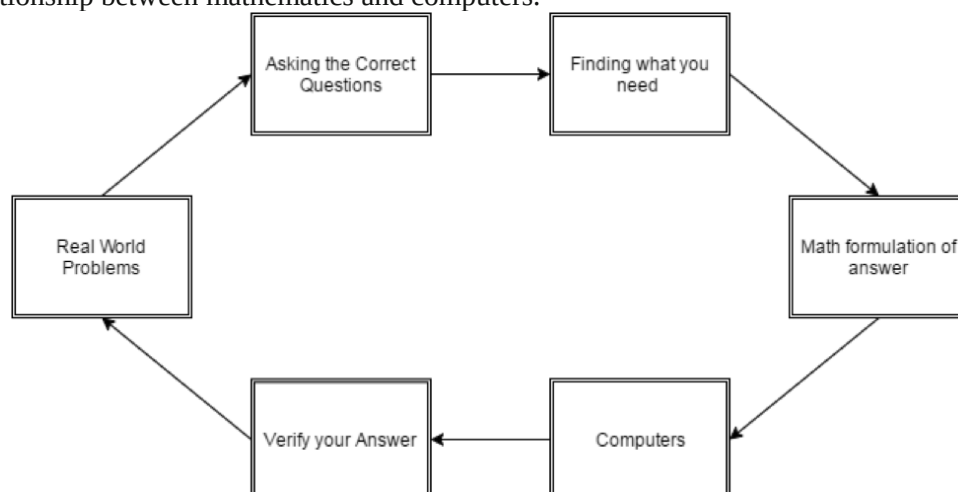
Experimental: based on variety of approaches.

Fictional: based on imagination of the writer.

4.6. COMPUTER IN TEACHING MATHEMATICS.

Computer is a technology full device used for programming and now a days it is being used in every field of life. Computers bring new areas of study in more advance form and helps to bring new approaches to old problems. That can change the student's point of view for mathematics. When the solution of a problem is computerize in front of students it gives them same understanding as a scientist get by doing experiments in laboratory. The process of solving a problem on a computer gives a sense of experimental science. Although all branches of mathematics need computation, such as combinatorial mathematics or mathematical physics. In such fields calculations on computers helps to overcome the constraints of tedious routine manipulation.

In mathematical education the notion of an algorithm is very important. It start when students learn to carry out arithmetic operations. When students starts to learn algorithm it is the best time to give them a sense of programming that means how the algorithm can be performed on a computer. That will give them an understanding of what is the relationship between mathematics and computers.



This above figure is showing us the process how we can incorporate computers in teaching mathematics. When students formulate their own questions using computers, forming sub-questions, solving the calculation and implementing their answers in the form of verified applications in the real world, it will give them real experience of dealing mathematics and they will learn more interestingly.

As every teacher has a limited time for delivering his lesson in a class period, sometimes it might not be possible to give and solve interesting examples in limited time period. The experimental nature of mathematics can be combined with computer to carry out large computations, including formation of guesses, carrying experiments to find out the evidence and reformulation of data. For further details visit www.sciencedirectassests.com.

4.6.1 Computer assisted instructions

CAI is a term in which computers are used to teach concepts, skills and different subject areas. This is supportive aid in this era of technology. As the technology is arising day by day so there is a need to adopt new methods as well as new medium to teach the students. First computer for this purpose was developed in 1960s. In these computers different applications and software are used to deliver the concepts of mathematics.

Learning management system is an application used for educating the students. Electronic books are another source of computer assisted instructions. In which actually books are available for students in computer where they can learn the respective lessons. Teacher can get benefits with this tool a lot they can customize the curriculum text. Assignments and homework could be forwarded on a server. As now a days student are more interested in computer based activities they will get more learning through such medium. Electronic performance support system is another example of computer assisted instruction.

Advantages of computer assisted instructions

CAI assistances the teachers to provide a wide range of experience. So teachers can easily make their student understand the concepts of mathematics.

With aids of CAI teachers are able to deliver their concept in a symmetrical manner and students get reinforce when they give right answers. It helps the students to learn mathematics with fun because it shows the graphics and pictures.

It enhance the teaching learning process by providing individualized instructions to each and every student. Some student are slow learner they need more time for getting concept, CAI provide the facility to select the topic of their own interest.

It offers the opportunity of interactive learning, as it provide instant feedback to the students and student react continuously. Due to which they grasp the concept more effectively.

4.6.2 Using computer graphics in mathematics

Computer graphic shows moving pictures and it is also possible to generate animations. With it teachers can control the speed of animation, portion of view, the geometric relationship of the objects.

Computer graphics are used in subject of mathematic in order to understand the lessons conceptually. Computer graphics are used in mathematics to show the pictures and films created by computer for the results of mathematical formulation. This approach is very much helpful for giving students an instant visual understanding even of abstract, synthetic data such as mathematical surface in 4D. This application has a great impact on teaching learning process.

There are important fields of mathematics where computer graphics play a vital involvement.

- Algebra and trigonometry
- Linear algebra
- Calculus
- Differential geometry
- Numerical methods
- Sampling theory and signal process
- Matrix equations
- Numerical solutions of differential equations
- Optimization
- Probability and statistics
- Computational geometry

The computer graphics also provide a feature of update dynamics. With this facility the teachers can update the changes the shape, colour and other properties of the objects.

There is another feature of computer dynamic i.e. motion dynamic, with its application the objects can be moved with respect to a stationary observer or vice versa.

4.6.3 Mathematics in computer

As computer is a big helping tool in field of mathematics similarly mathematics is a giant help for computer subjects. The combination of these two subjects offers a better understanding of mathematical reasoning with computing. Mathematics is a basic helping aid in computing while on the other hand computing is a key constituent in mathematics. Combination of these two areas gives a better understanding of bridge relationship between theory and practice. Standard arithmetic is being used in programming functions. Almost every program of computer use addition, subtraction, multiplication and division function of mathematics. A lot of algebra is used by programmers to solve simple problems. Similarly some other advance areas of mathematics like single variable calculus, differential equations, linear algebra, multivariate calculus, probability and basic combinatorics and graph theory are helpful in computer programming. So

mathematical teachers should use the combination of these two things to give a broad perspective of mathematics in computing.

SUMMARY

This unit explains the importance of instructional materials and the principles of effective use of instructional materials. This unit also summarises various aids in teaching that include real objects, models, charts, black board, flannel board, excursions and collections, text books and work books, radio and T.V. It also illustrate the projective aids that include magic lanterns, film strips projector, epidiascope, motion pictures. One can easily grasp the concept of computer aid in teaching, by studying this unit as it elaborates the computer assisted instructions, use of computer graphics in mathematics and mathematics in computer.

Instructional materials are important aspect of teaching because these are used to enhance the students' capabilities, skills, their overall development and upbringing. Instructional material increase the communication between teacher and student. It helps to store and shares the human experiences of knowledge with the students. With effective use of instructional material the interest of students could be maximize, their retention to the accurate ideas would increase. It aids teaching experience and enhance the teachers' competence and also making learning outcomes more expressive for students. So all types of instructional materials are having important impact on teaching learning process that must be included by teachers in their teaching process.

SELF-ASSESSMENT QUESTIONS

Subjective Questions

1. What are instructional materials? How are they helpful in teaching learning process?
2. Elaborate the principles for effective use of instructional materials.
3. Make a list of various aids in teaching mathematics and elaborate the difference between black board and flannel board.
4. Discuss the advantages of motion pictures and limitations of magic lanterns.
5. Two terms are different from each other "using computer graphics in mathematics" and "mathematics in computer" elaborate?

Objective Questions

Identify true and false statements.

1. Magic lanterns were invented back in 14th century.
2. Radio comes in the category of visual aids in teaching learning process.
3. A lot of algebra is used by programmers to solve simple problems of computer programming.

4. Computer assisted instructions enhance the teaching learning process by providing individualized instructions to each and every student.
5. With the help of epidiascope students cannot watch flat objects like coins etc.

Answers

1. False
2. False
3. True
4. True
5. False

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Unit - 05

ASSESEEMENT AND EVALUATION

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INTRODUCTION

When somebody says the sentences such as “I like it”, “This looks good”, “Well done” or the opposite, “I should have done it better”, “I did not do so well” or “I hate you” then actually the assessment is being taken (Brno, 2012). Every person is assessing the things (intentionally or non- intentionally) almost all the time. It means that assessment is used in every aspect of life.

In teaching and learning process, assessment is one of the essential elements. Assessment practices of teachers connect the curriculum, instructional mechanism and students learning outcomes. Consequently these practices enable the students to practice learning contents, develop thinking patterns, activate their neurons and enhance their confidence on acquired skills and knowledge (Hussain, 2008).

Due to the dominant importance of classroom assessment practices, it is essential for teachers to equip with competencies in assessment practices. Teacher understanding and proper implementation of assessment strategies are directly linked to assessment results and instructional decisions. Therefore, teacher’s familiarity with assessment tools, principles, strategies and procedure are obligatory. Similarly, they need to be skillful in designing assessment tools and aware of: purpose of assessment, quality in Assessment, types of test and its significance, teacher made tests and its role in learning and types of test items and its utility.

OBJECTIVES

After grasping the theory of this unit you will be able to:

1. Define Assessment and Evaluation
2. Elaborate the Purposes of Assessment
3. Explain the Types of test and its significance.
4. Explain the Types of Test items and its utility
5. Apply different techniques of Assessment for mathematics teaching.

5.1 DEFINITION OF ASSESSMENT AND EVALUATION

The term “assessment” has been commonly used by educationist in meaning of evaluation, measurement, and documentation of the academic readiness, learning progress, and skill acquisition of students throughout in their life through lifelong learning. Commonly the terminologies used for are there for assessment and evaluation are Measurement, Tests, Examination and Appraisal.

Assessment is the part of evaluation which is used for gathering of information systematically. It is very important for planning the next steps, tracing the progress, involving the parents, children and young people in the process of learning. It is carried out to see that whether the students know, understand and are able to perform. Following are the some definitions of assessment in different perspective of educationists.

Erwin 1991 has defined the assessment as systematic root for making implications about the development and learning of students. According to him the assessment is the process of collecting, designing, defining, selecting, interpreting, analyzing, and using evidence to upsurge the learning of students.

According to Black and Wiliam (1998) the assessment comprise of all the activities undertaken by teachers and students to acquire information which can be used to modify the teaching and learning process.

Palomba and Banta (1999) consider the assessment as a tool for reviewing, collecting and using of information about educational programs developed for the improvement of students’ learning and development.

Huba and Freed 2000 has defined the term assessment as the process by which information are discussed and gathered from different sources for development of deep understanding about the knowledge and understandings of students resulting due to their educational experiences.

In the words of Allen (2004) assessment is he process of refining programs and improving student learning by using empirical data on student learning.

5.2 THE PURPOSE OF ASSESSMENT

Assessment is an essential part of instruction, it determines whether the goals of education are being met or not. Assessment affects decisions about instructional needs, grades, curriculum, placement, advancement, and, in some cases, funding. Assessment stimulate us to ask these hard questions: "Are teachers teaching what they are supposed to be teaching?" "Are students learning what they are expected to be learning?" "Is there any other way to teach the subject matter for promoting better learning?"

In 21st century, students need not to know only the basic reading skills, but also the skills which will allow them to face a dynamic world that is frequently changing. They must be able to: think critically, analyze, and make conclusions. A paradigm shift to the skills based knowledge require new learning goals for our students. These new learning goals has changed the relationship between assessment and instruction in present era. Teachers are needed to be active in making decisions about the purpose of assessment and the content that is being assessed.

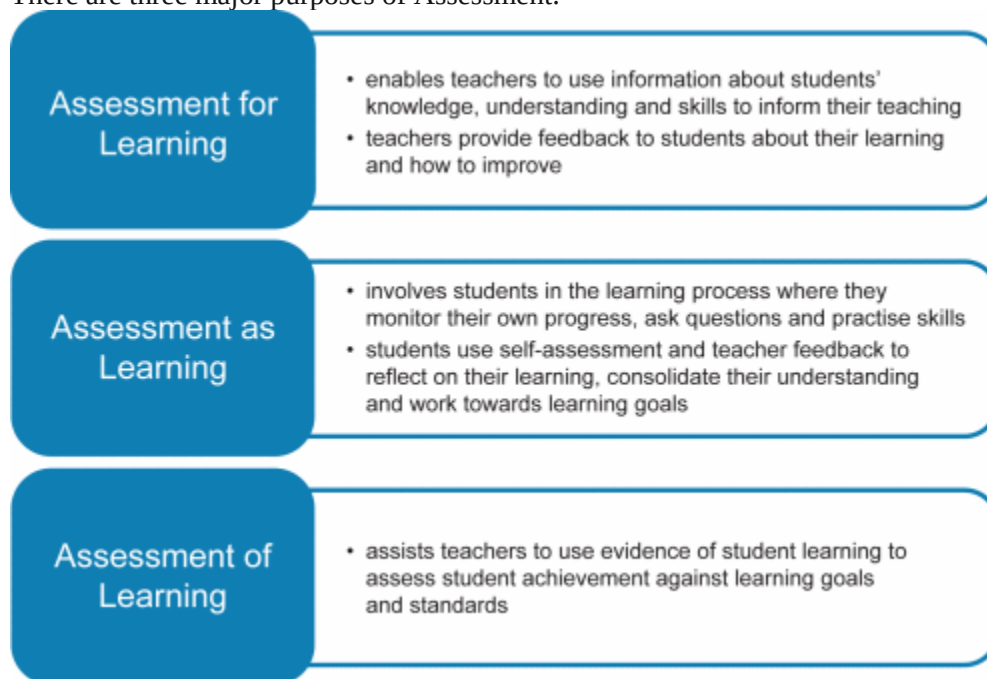
Assessment can be used for various purposes, such as improving teaching and learning, student placement, providing student grades, national accountability, monitoring and determining interventions, and providing individual feedback to students and their parents (Newton 2007). Claims that are made may be different, depending on the goals and design of the assessment activity. For example classroom and large-scale assessment have different purposes and goals. Classroom assessment gathers information and provides feedback to support individual student learning (De Lange 2007).

In classroom assessment, teacher-selected or teacher-made assessments are usually used which are most effective particularly, when they are closely aligned with “what” and “how” the students are learning. Uses of a range of assessment tools, strategies, and formats, providing manifold opportunities for students to reveal their learning, making strong use of formative feedback on a timely and regular basis, and participation of students in the process of assessment are Current perspectives in classroom assessment.

In large-scale assessment, information of systems, monitoring of system, evaluation of programs, and student placements is made. In many parts of the world, students are assessed in mathematics, using large-scale assessment in the form of provincial, national, and state level assessments. It can be taken in the form of international assessments also. Large-scale assessment of mathematics is used to monitor educational systems for accountability purposes. It plays a vital role in the lives of students and teachers since grade promotion often depends upon students’ test results. Teachers are evaluated on the basis that how well their students perform on such assessments (Wilson and Kenney 2003).

Either we are talking about classroom or large-scale assessment, it is unavoidable to pay attention towards the purpose of the assessment or to the way the results of the assessment are interpreted and used appropriately for particular purpose. The results of the assessment should be used for the purposes for which the assessment was designed and the inferences need to be appropriate to the assessment design. However, it is recognized that sometimes the purpose of assessment is indistinct. For example sometime, teachers use tasks from large-scale assessments for instructional purposes. In some parts of the world data from large-scale assessments are provided to teachers for better understanding of teaching and learning in their own classrooms. And sometime they use summative classroom assessments for formative purposes when they use the summative results for understanding the misconceptions of students and then design their future instruction accordingly.

There are three major purposes of Assessment:



<https://abdao.wordpress.com/2015/07/18/purposes-of-assessment/>

5.2.1 Assessment for learning

It is a continuing process that monitors students' learning and teachers' teaching. It helps students to improve their learning and teachers to improve their teaching. It constantly informs instruction and helps students to manage their own learning. The assessment information is used to decide the next teaching paces and learning ladders to improve the teaching and learning process continuously. Formative assessments serve the purpose of assessment for learning.

Some examples of assessment for learning are given below.

- concept maps
- progress/monitoring reports
- checklists/surveys
- interviews
- observations
- anecdotal records
- research proposal (for feedback)
- quizzes
- home works
- worksheets
- performance tasks
- essays
- questioning strategies

- projects
- graphic organizers
- self-assessments/ peer assessment
- collaborative activities
- portfolios (collection of student work)

5.2.2 Assessment as learning

Formative assessments serve the purpose of assessment as learning too. It is also a continuing process which is helpful for students for self-reflection, monitoring of their own learning, and adjusting their learning strategies to achieve their goals. This helps them to become more independent, successful learners, self-directed, and metacognitive as well.

Some examples of assessment as learning are given below.

- self-assessment
- peer-assessment
- personal learning logs

5.2.3 Assessment of learning

Summative assessments fulfill the purpose of assessment of learning. It measures what and how well the students had learned at the end of instruction. It states the status of learning and measures overall achievement of students. It finds out that learning outcomes and goals have been achieved or not.

Some examples of assessment of learning are given below.

final performance tasks

- final papers/written outputs
- final oral presentations
- standardized tests
- end of unit tests or projects
- recitals
- long exams
- periodical tests
- final exams

5.3 QUALITY IN ASSESSMENT

Quality in assessment depends upon the level of reliability and validity. Approaches to reliability and validity are explained in following lines.

5.3.1 Reliability

Reliability is related to the consistency of an assessment. Assessment is reliable if consistently achieves the same results with the same (or similar) cohort of students.

Reliability is effected by several factors such as: too many options within a question paper, ambiguous questions, poorly trained markers and vague marking instructions etc.

Usually, reliability of an assessment is based on the following features:

Temporal stability: Temporal stability means comparison of test performance on two or more than two different occasions.

Form equivalence: It means that there should be equivalence of performance among examinees on different forms of a test based on the same content.

Internal consistency: Responses on a test are consistent across questions. For example: In a survey that asks respondents to rate attitudes toward technology, consistency would be expected in responses to the following questions:

"I feel very negative about computers in general."

"I enjoy using computers".

5.3.2 Validity

Valid assessment measures exactly that, which is intended to measure. For example, it would not be valid to assess driving skills through a written test alone. A more valid way of assessing driving skills would be through a combination of tests that help determine what a driver knows, such as through a written test of driving knowledge, and what a driver is able to do, such as through a performance assessment of actual driving. Teachers frequently complain that some examinations do not properly assess the syllabus upon which the examination is based; they are, effectively, questioning the validity of the exam.

Validity of an assessment is generally gauged through examination of evidence in the following categories:

Content – Does the content of the test measure stated objectives? Content validity is concerned with a test's ability to include or represent all of the content of a particular construct. The question " $1 + 1 = \underline{\quad}$ " may be a valid basic addition question. Would it represent all of the content that makes up the study of mathematics? It may be included on a scale of intelligence, but does it represent all of intelligence? The answer to these questions is obviously no. To develop a valid test of intelligence, not only must there be questions on math, but also questions on verbal reasoning, analytical ability, and every other aspect of the construct we call intelligence. There is no easy way to determine content validity aside from expert opinion.

Criterion – Do scores correlate to an outside reference? (Do high scores on a 4th grade reading test accurately predict reading skill in future grades?)

Construct – Does the assessment correspond to other significant variables? (Do ESL students consistently perform differently on a writing exam than native English speakers?)

A good assessment has both validity and reliability, plus the other quality attributes noted above for a specific context and purpose. In practice, an assessment is rarely totally valid or totally reliable. A ruler which is marked wrongly will always give the same (wrong)

measurements. It is very reliable, but not very valid. Asking random individuals to tell the time without looking at a clock or watch is sometimes used as an example of an assessment which is valid, but not reliable. The answers will vary between individuals, but the average answer is probably close to the actual time. In many fields, such as medical research, educational testing, and psychology, there will often be a trade-off between reliability and validity. A history test written for high validity will have many essay and fill-in-the-blank questions. It will be a good measure of mastery of the subject, but difficult to score completely accurately. A history test written for high reliability will be entirely multiple choice. It isn't as good at measuring knowledge of history, but can easily be scored with great precision. We may generalize from this. The more reliable our estimate is of what we purport to measure, the less certain we are that we are actually measuring that aspect of attainment.

It is well to distinguish between "subject-matter" validity and "predictive" validity. The former, used widely in education, predicts the score a student would get on a similar test but with different questions. The latter, used widely in the workplace, predicts performance. Thus, a subject-matter-valid test of knowledge of driving rules is appropriate while a predictively valid test would assess whether the potential driver could follow those rules.

5.4 TYPES OF TEST AND ITS SIGNIFICANCE

Whether students like them or not, tests are a way of checking their knowledge and comprehension. Tests are the main instrument which are used to evaluate the learning of students. According to research studies. Another benefit of tests is that, they make students to learn and remember more than they might have otherwise. Different types of tests are used and each has a different style and purpose.

Diagnostic Tests

These types of tests are used to diagnose that how much the students know and what do they know exactly. These types of tests can help the teacher to know that what was needed to be reviewed or reinforced in class. These tests are beneficial for students also as they enable them to identify their areas of development.

Placement Tests

These tests are used to place students in the appropriate class or level. For example, in language schools, placement tests are used to check a student's language level through grammar, vocabulary, reading comprehension, writing, and speaking questions. After establishing the students level, the student is placed in the appropriate class to suit his/her needs.

Progress or Achievement Tests

Achievement or progress tests measure the students, improvement in relation to their syllabus. These tests only contain items which the students have been taught in class.

There are two types of progress tests: short-term and long-term.

Short-term progress tests check how well students have understood or learned material covered in specific units or chapters. They enable the teacher to decide if remedial or consolidation work is required.

Long-term progress tests are also called Course Tests because they check the learners' progress over the entire course. They enable the students to judge how well they have progressed. Administratively, they are often the sole basis of decisions to promote to a higher level.

Progress tests can also be structured as quizzes, rather than as tests. They can be answered by teams of students, rather than individuals. They can be formulated as presentations, posters, assignments, or research projects. Structuring progress tests in this way takes into account the multiple intelligences and differing learning styles of the students. Yet many students still expect a regular test as a part of normal learning.

Proficiency Tests

These tests check learner levels in relation to general standards. They provide a broad picture of knowledge and ability. In English language learning, examples are the TOEFL and IELTS exams, which are mandatory for foreign-language speakers seeking admission to English-speaking universities. In addition, the TOEIC (Test of English for International Communication) checks students' knowledge of Business English, as a prerequisite for employment.

Internal Tests

Internal tests are those given by the institution where the learner is taking the course. They are often given at the end of a course in the form of a final exam.

External Tests

External tests are those given by an outside body. Examples are the TOEFL, TOEIC, IELTS, SAT, ACT, LSAT, GRE and GMAT. The exams themselves are the basis for admission to university, job recruitment, or promotion.

Objective Tests

Objective tests are those that have clear right or wrong answers. Multiple-choice tests fall into this group. Students have to select a pre-determined correct answer from three or four possibilities.

Subjective Tests

Subjective tests require the marker or examiner to make a subjective judgment regarding the marks deserved. Examples are long questions and oral interviews. For such tests, it is especially important that both examiner and student are aware of the grading criteria in order to increase their validity.

Combination Tests

Many tests are a combination of objective and subjective styles. For example, on the

TOEFL iBT, the Test of English as a Foreign Language, the reading and listening sections are objective, and the writing and speaking sections are subjective.

Standardized tests

A standardized test is a test that is administered and scored in a consistent, or "standard", manner. Standardized tests are designed in such a way that the questions, conditions for administering, scoring procedures, and interpretations are consistent and are administered and scored in a predetermined, standard manner. Any test in which the same test is given in the same manner to all test takers, and graded in the same manner for everyone, is a standardized test.

Steps for constructing a standardized Test

Step # 1. Planning the Test:

Planning of the test is the first important step in the test construction. The main goal of evaluation process is to collect valid, reliable and useful data about the student.

Therefore before going to prepare any test we must keep in mind that:

1. What is to be measured?
2. What content areas should be included and
3. What types of test items are to be included.

Step # 2. Preparing the Test:

After planning preparation is the next important step in the test construction. In this step the test items are constructed in accordance with the table of specification. Each type of test item need special care for construction.

The preparation stage includes the following three functions:

- i. Preparing test items.
- ii. Preparing instruction for the test.
- iii. Preparing the scoring key.

Step # 3. Try Out of the Test:

Once the test is prepared now it is time to be confirming the validity, reliability and usability of the test. Try out helps us to identify defective and ambiguous items, to determine the difficulty level of the test and to determine the discriminating power of the items.

Try out involves two important functions:

- a. Administration of the test.
- b. Scoring the test.

Step # 4. Evaluating the Test:

Evaluating the test is most important step in the test construction process. Evaluation is necessary to determine the quality of the test and the quality of the responses. Quality of

the test implies that how good and dependable the test is? (Validity and reliability). Quality of the responses means which items are misfit in the test. It also enables us to evaluate the usability of the test in general class-room situation.

Evaluating the test involves following functions:

- a. Item analysis.
- b. Determining validity of the test.
- c. Determining reliability of the test.
- d. Determining usability of the test.

5.5 TEACHER MADE TESTS AND ITS ROLE IN LEARNING

5.5.1 Meaning of Teacher Made Test

Teacher-made tests are prepared and used to: test the students' achievement in classroom, evaluate the method of teaching adopted by the teacher and curricular program of the school. It is prepared to measure the learning outcomes and content of curriculum in local perspective. It is too much flexible to be adopted for any procedure and material. No sophisticated technique is needed for its preparation.

Teacher-made test is one of the most valuable instrument in the hands of the teachers to elucidate their purpose. It is designed to solve the problem and requirements of the class for which it is being prepared.

Teacher-made tests and standardized tests are similar in many ways. Both are constructed carefully and planned on the basis of table of specifications, both provide clear directions to the students and consisting of the same type of test items. However they differ in: the quality of test items, the procedures for scoring and administering, the interpretation of scores and the reliability of test measures. No doubt, standardized tests are good in nature, more reliable, valid and better in quality.

But a classroom teacher cannot always depend on standardized tests. Because following requirements are needed to be addressed in this regards: They must suit to his local needs, be readily available, not be costly, and must not have different objectives. In order to fulfill the above mentioned requirements, the teacher has to prepare his own tests Taylor has extremely recommended for the use of these teacher-made tests. Teacher made tests do not require rigorous processes of standardization nor need the all the four steps of standardized tests. Only the first two steps planning and preparation are sufficient for their construction.

5.5.2 Features of Teacher-Made Tests

Some of the features of teacher-made test are as follow.

1. A teacher-made test are prepared by the teachers which can be used for prognosis and diagnosis purposes.
2. These tests do not cover all the steps of a standardized test.

3. Order of difficulty is under consideration while arranging the test items.
4. The preparation of the items follows the blueprint.
5. Test construction is not a single man's business, rather it is a co-operative endeavour.
6. The test includes a large number of items and covers the whole content area.
7. Teacher-made tests may also be used as a tool for formative evaluation.
8. Preparation and administration of these tests are cost-effective.
9. The test is developed by the teacher to determine the student's proficiency and achievement in a given subject.
10. Teacher-made tests are not used for research purposes usually.
11. Teacher-made tests do not have norms whereas these are quite essential for standardized tests.

5.5.3 Principles of Construction of Teacher-made Test:

In order to make teacher-made test more efficient and effective tool of evaluation, careful considerations are needed to be given while constructing such tests.

The following steps may be followed for the preparation of teacher-made test:

1. Planning:

Planning of a teacher-made test includes:

- a. Decision about the portion of the syllabus to be covered and length of test.
- b. Determination of the objectives and the purpose of the test, 'as why to measure and what to measure'.
- c. Decision about incorporation of the number and forms of items (questions) according to blueprint.
- d. Specification of the objectives in behavioral terms. A table may be prepared for specifications and weightage given to the objectives to be measured if needed.
- e. Giving enough time to teachers for test preparation and administration.
- f. Having a clear understanding and knowledge of the principles of constructing objective type questions, short answer questions and long questions.
- g. Seeking the co-operation and suggestion of test experts, co-teachers, and experienced teachers of other schools as well.

2. Preparation of the Test:

Planning is the philosophical aspect and preparation is the practical aspect of test construction. All the practical aspects will have to be taken into consideration while teacher constructs the tests. Test preparation is said to be an art or technique which is to be acquired by teachers. It requires reading, much thinking and rethinking during constructing test items.

When construction of test items is done, it should be given to the other experts of the concerning subject for review and opinion. Suggestions and views may be sought even from others on languages, statements given, modalities of the items, options of correct answers supplied and on other possible mistakes expected. These suggestions and views

will be helpful for a test constructor in modifying and verifying the items to make the test more usable and acceptable.

After construction of the test, items should be arranged in a simple to complex order. For arranging the items, a teacher can adopt the methods like topic wise, unit-wise and group-wise etc. In order to avoid delay in scoring, a scoring key may also be prepared on spot.

Another important aspect of a test construction is the “direction” or “instruction”. Without giving a proper direction or instruction, there will be a possibility to create a misunderstanding on the part of students. It may also lose the authenticity of the test reliability.

Thus, the direction or instruction should be simple and suitable to enable the students to know:

- i. The marks allotted to each item
- ii. The time for completion of test
- iii. How and where to record the answer?
- iv. Required number of items to be attempted
- v. The materials (like logarithmic table or graph papers) to be used.

5.5.4 Uses of Teacher-Made Tests

1. Teacher-made test helps the teachers to assess the level (normal, average, above average or below average) of the class.
2. It can assess whether specified instructional objectives have been achieved or not.
3. It may be used as an achievement test also.
4. It helps him for incorporating new strategies for the purpose of teaching and learning.
5. It measures students' academic achievement in a given course.
6. It helps to know the efficacy of learning experiences.
7. Such type of test may be exchanged among neighboring schools.
8. It helps to classify and certify the students on the basis of scores achieved.
9. It helps to diagnose students learning problems and to suggest necessary corrective methods of teaching.
10. It helps the teacher to render counseling and guidance.
11. It helps to measure growth of pupils in particular area.
12. It can be used as a tool for diagnostic, summative and formative evaluation.
13. It can be served as the purpose of standardized test also.

Teacher Made Test

- Basically teacher made tests are used to evaluate the progress of the students in school. However , the specific use of tests may vary from school to school and teacher or teacher.
- The test results can be used for students, teachers, and for other administrative purposes.
- These tests are very simple to use.
- Easy for the students.
- Teachers can assess the strengths and weaknesses of students.
- Tests are conducted continuously and children get immediate feedback.

5.6 TYPES OF TEST ITEMS AND ITS UTILITY

Examinations are a very common tools used in schools, colleges and universities for assessment and evaluation purpose. There are many types of examination questions but seven important types of them are: 1) multiple choice, 2) true/false, 3) matching, 4) short answer, 5) essay, 6) oral, and 7) computational. In following lines each of them is being discussed one by one.

5.6.1 Multiple choice

Multiple choice questions are composed of one question which is called “stem” and multiple possible answers (choices) are provided including either only the one correct answer and several incorrect answers (distractors) or one distractors and several correct answer. Students are directed to select the correct answer by circling the associated number or letter, or filling in the associated circle on the machine-readable response sheet.

Example: Distractors are:

- A) Incorrect but plausible choices used in multiple choice questions
 - B) Unnecessary clauses included in the stem of multiple choice questions
 - C) Elements of the exam layout that distract attention from the questions
- Answer: A

Multiple type questions have great significance for students. They can respond to these type of questions quite swiftly and as a result, these test are often used to test student's knowledge of a broad range of content. Although it is a time consuming task to develop such type of questions because it is difficult to generate several plausible distractors. However, they can be checked easily and very quickly.

Following are some tips for writing effective multiple choice items:

<i>Avoid</i>	<i>Do use</i>
In the stem: • Long / complex sentences • Trivial statements • Negatives and double-negatives • Ambiguity or indefinite terms, absolute statements, and broad generalization • Extraneous material • Item characteristics that provide a clue to the answer misconceptions In the choices: • Statements too close to the correct answer • Completely implausible responses • 'All of the above,' 'none of the above' • Overlapping responses (e.g., if 'A' is true)	In the stem: • Your own words – not statements straight out of the textbook • Single, clearly formulated problems In the choices: • Plausible and homogeneous distractors • Statements based on common student misconceptions • True statements that do not answer the questions • Short options – and all same length • Correct options evenly distributed over A, B, C, etc. • Alternatives that are in logical or numerical then 'C' is also true) order • At least 3 alternatives

Activity:

Keeping in mind the characteristics of effective multiple type questions mentioned above, jot down five multiple choice questions based on the Pythagoras theorem from 9th grade mathematics.

5.6.2 True/false

True/false questions consist of a statement only. Students are asked to respond about the questions whether it is false or true. For example: True/false questions have only two possible answers (Answer: False). Usefulness of true/false questions can be increased by asking students to correct the false statements.

True/false questions have also great significance as:

- They let students to respond quickly hence these types of tests are frequently used in exams to test knowledge of a comprehensive range of content.

- Misconceptions are checked and familiarity with course content is assessed through True/false questions.
- with the help of True/false questions, It is quick and easy to grade but still time consuming to develop.
- These tests provide 50% chance of guessing the right answer. Hence, multiple choice questions should be used instead of true/false questions. Here are some Tips for writing good and effective true/false items:

<i>Avoid</i>	<i>Do use</i>
• Negatives and double-negatives • Long / complex sentences • Trivial material • Broad generalizations • Ambiguous or indefinite terms	• Your own words • The same number of true and false statements (50 / 50) or slightly more false statements than true (60/40) – students are more likely to answer true • One central idea in each item

Activity:

Keeping in mind, the above mentioned strategies develop five True/false questions for “Matrices” from 9th grade mathematics.

5.6.3 Matching

Students are asked to match the questions by pairing each of a set of item or stem in one column with one of the choices provided in other column. Such types of questions are usually used to assess recall and recognition skills of the students. They are often used for acquisition of detailed knowledge in courses. They are generally quick and easy to develop and mark, but on the part of students, more time is required to respond the number of such items as compared to the similar number of multiple choice or true/false items. In order to reduce the effects of guessing, teachers can use some choices more than once in the same matching exercise.

Example: Match each question type with one attribute:

1. Multiple Choice a) Only two possible answers
2. True/False b) Equal number of stems and choices
3. Matching c) Only one correct answer but at least three choices

- Following are the some tips for writing good matching items:

<i>Avoid</i>	<i>Do use</i>
• Long stems and options • Heterogeneous content (e.g., dates mixed with people) • Implausible responses	• Short responses 10-15 items on only one page • Clear directions • Logically ordered choices (chronological, alphabetical, etc.)

5.6.4 Short answer

These types of questions are composed of a brief prompt which demands a written answer, varies in length from few words to one or two sentences. They are most often used in order to test the basic knowledge of key terms and facts. Following is an example of short question.

“Write down the name of the exam format in which students associate a set of prompts with a set of options?” Answer: Matching questions

Alternatively, this could be written as a fill-in-the-blank short answer question:

“Type of exam question in which the students have to associate prompts and options is called a question.” Answer: Matching.

Short questions are also used to test higher thinking skills, including evaluation or analysis. For example:

“Would you include short questions format in your next exam? Justify in few sentences by explaining the factors which have influenced your decision.”

There are many advantages of short answer questions. Short answer questions can be developed relatively easily and faster as compare to matching, true/false, and multiple choice questions. However they make it difficult for students to guess the answer. In this type of questions, students have provision for more flexibility in explanation and demonstration of their understanding and creativity than the multiple choice questions. Hence the scoring as compare to multiple choice questions is relatively hard for teachers and could be quite subjective but easy and faster to mark and often test a broader range of the course content than full essay questions.

Tips for writing good short answer items:

<i>Type of question</i>	<i>Avoid</i>	<i>Do use</i>
All short-answer	• Trivia • Long / complex sentences	• Your own words • Specific problems • Direct questions
Fill-in-the-blank	• Taking out so many words that the sentence is meaningless	• Prompts that omit only one or two key words at the end of the sentence

5.6.5 Long questions

Long questions provide a complex prompt which requires written answers. They can vary in length from few paragraphs to many pages. Like short answer questions, they provide students an opportunity to explain and demonstrate their understanding creativity, but it is

hard for students to arrive at a standard answer by guessing. It is easy and take less time to construct such type of question but hard and time consuming in marking.

Long questions are different as they are more structured. This broadness allows students to demonstrate the course material in creative ways. Due to this reason, long questions are a preferred to test higher order thinking skills like analysis, synthesis and evaluation.

Tips for writing good long question items:

<i>Avoid</i>	<i>Do use</i>
• Complex, ambiguous wording • Questions that are too broad to allow time for an in-depth response	• Your own words • Words like ‘compare’ or ‘contrast’ at the beginning of the question • Clear and unambiguous wording • A breakdown of marks to make expectations clear • Time limits for thinking and writing

SELF ASSESSMENT

Fill in the Blanks

1. Type of assessment information used to decide the next teaching paces and learning ladders to improve the teaching and learning process continuously is called.....
2. Quality in assessment depends upon the level of.....
3. self-assessment, peer-assessment and personal learning logs are example of.....
4. A **test** is a **test** that is administered and scored in a consistent, or "standard" manner.
5. Incorrect but plausible choices used in multiple choice questions are called.....

Answers

1. Assessment for learning
2. reliability and validity
3. Assessment as learning
4. Standardized
5. Distractors

EXERCISE

1. What are the three major Purposes of Assessment explain?
2. How quality in Assessment depends upon validity and reliability?

3. Describe types of test and its significance
4. How teacher made test plays a vital role in learning of students?
5. Write down the major types of Test items and its utility in teaching of mathematics.

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Unit - 06

DEVELOPING THE CONCEPTS OF SETS

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INTRODUCTION

Teaching of mathematics is not a simple art. It is an art of knowing how to teach effectively and efficiently to ensure meaningful learning for obtaining the desired outcomes. It is wise to say that if two people are trying to communicate about some topic, they must have the same understanding of its technical terms. Teachers should be aware about the curriculum of mathematics at his teaching level then he will be able to think critically about the complex issues and solve the problems of various kinds. Teacher should be aware to both what the student is capable of and how he can achieve high outcome. So that teacher should improve his pedagogical skills as well as knowledge that is why he will be able to explain the required topic to his students with examples, pictures and charts. In this unit we discussed sets, and relations. Self-assessment questions and activities are also the part of the unit. These activities and self-assessments into the classroom can awaken student's passion and interest which can make better understanding of the material provided.

OBJECTIVES

After studying this unit, prospective teachers will be able to describe how to teach and explain

1. Set
2. Some important sets
3. Complex numbers
4. Types of sets
5. Different operation on set i.e union, intersection, difference and complement.
6. The basic laws of Set Operations
 - Commutative Law
 - Associative Law
 - Distributive Property
 - De-Morgan's Law
7. An order pairs, Cartesian Product
8. Binary Relation
9. Function
 - into Function
 - one-one Function
 - into and one-one Function
 - onto Function
 - one-one and onto Function

Developing the Concepts of Sets

George Cantor was a German mathematician who is regarded as a founder of set theory and he was the one who introduced this theory of sets in the world of mathematics. Nowadays, this theory is a foundation of many advance mathematical research. Mathematicians use a term ‘set’ to describe a collection of objects. For example the image below shows the set of shoes, the set of color pencils, the set of jewelry and a Set of English Novels.



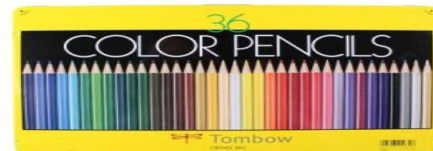
Set of English Novels



Set of Shoes



Set of Jewelry



Set of colored Pencil

6.1 INTRODUCTION OF A SET

In our daily life we use many collective nouns to narrate collection of things / objects. For example a fleet of ships, a set of stamps issued after the Pakistan's independence. So the set is a simply a collection of things which are distinct and clearly identified. The objects consisting in a set are called the elements or the member of the set. We represent the set by capital letters A, B, C,... and list the elements of the set by the small letters a, b, c, ... separate all elements by the comma and enclosed within braces. For example,

$$A = \{a, b, c, d\}$$

The letters a, b, c and d are the members of Set A. we can write it as: $a \in A, b \in A, c \in A, d \in A$. Where the symbol ' $\in$ ' means "belongs to", 'is a member of' or "is an element of". On the other hand e is not the member of set A. so we write it as $e \notin A$. There are two methods for representing a set.

- **Tabular/ Roster method**
In this method a set is represented by listing all its entries within braces and separating them by commas. For example set A containing elements p, r, t, s and z can be written as $A = \{p, r, t, s, z\}$
- **Property Method**
In this method a set is represented by stating all the properties which are satisfied by the elements of the set but not by the elements outside of the set. For example set Y containing all values of y for which the condition $P(y)$ is true, then we write it as: $Y = \{y: P(y)\}$

Some Fundamental Sets

Let us recall the following sets before giving the concept of different sets.

- **Set of Natural number**
Counting numbers are said to be the Natural numbers i.e 1,2,3 and so on. We know the natural numbers are denoted by N.
So $N = \{1, 2, 3, 4, \dots\}$
- **Set of Whole Numbers**
We include 0 in the set of natural numbers, and whole numbers are denoted by W.
So $W = \{0, 1, 2, 3, 4, \dots\}$
- **Set of Integers**
Set of integers made up of positive integers, negative integers and 0. We denote it by Z, Such that
$$Z = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$
- **Set of Even Numbers**
We know that any integer which is exactly divided by 2 and leaves no remainder is called even number. Set of even numbers is denoted by E i.e. $E = \{0, 2, 4, 6, 8, \dots\}$
- **Set of Prime Numbers**
We know that the integer which is divisible by only itself is called a prime number. Set of prime numbers is denoted by P. i.e. $P = \{2, 3, 5, 7, 11, 13, \dots\}$
- **Set of Rational Numbers**
We know that a rational number is formed by dividing two integers p, q such that $\frac{p}{q}$ such that $q \neq 0$. Rational numbers is denoted by Q.
The set of rational number is $Q = \{\frac{p}{q}: p, q \in Z \text{ and } q \neq 0\}$
- **Set of Irrational Numbers**
We know that the numbers which are not rational are called irrational numbers such that $\sqrt{5}, \sqrt{2}, \pi$ and e are the examples of irrational numbers. We denote irrational numbers by Q' .
So set of irrational numbers is $Q' = \{x; x \neq \frac{p}{q}, q \neq 0, \wedge p, q \in Z\}$
- **Set of Real Numbers**
We know that if positive, negative, large, small, decimal numbers and whole numbers together are called real numbers hence we can say that the union of

rational and irrational numbers is known as real numbers. The set of real number is denoted by R such that $R = \{x: x \in Q \cup Q'\}$

Remember:

- Q and Q' are the subset of R .
- Q and Q' are the disjoint sets.

- **Set of Complex Numbers**

Complex numbers were developed after the development of the real numbers, because complex numbers contain real numbers and every number whose square is a negative number.

A number which can be presented in the form of $z = a + ib$ is called a complex number. Here a and b are real numbers and $i = \sqrt{-1} \Rightarrow i^2 = -1$. Leonard Euler (1707-1783) was a swiss mathematician who first use the symbol i (iota) for the number $\sqrt{-1}$. Set of complex numbers are denoted by C . The set of complex number is $C = \{a + ib: a, b \in R\}$

Remember:

- In $z = a + ib$ the number a is called the real part of z and b is called the imaginary part of z .
- If the real part of $z = a + ib$ is zero, then z will be reduced in purely imaginary number ib . The set of purely imaginary numbers is also contained in C .
- Every real number is a complex number but every complex number is not a real number. For example for a complex number $z = a + ib$, if $b = 0$ then $z = a$

Activity:

If i^{12} then we can write it $(i^2)^6 = (-1)^6 = 1$

6.2 TYPES OF SETS

- **Finite and infinite sets**

The set of all natural number less than 9 is $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. The number of elements in a set A is 8 and we write $n(A) = 8$. The set A is called a finite set. On the other hand the set of natural number consist of infinite elements and is called infinite set as it is much difficult to list all natural numbers in the set. Hence we write it as $N = \{1, 2, 3, \dots\}$ we used “...” to indicate the remaining elements. In this case value of $n(A)$ is infinite.

Example 1:

List all the elements, if possible, of each of the following sets and also write down the number of elements for each finite set.

- L, the set of positive natural numbers between 7 and 13.

- b. P, the set of Prime numbers less than 14.
- c. M, the set of positive integers which are also multiple of 9.
- d. H, the set of integers x defined by $-5 < x < 4$.

Solution:

- a. $L = \{8, 9, 10, 11, 12\}$ and $n(L) = 5$
- b. $P = \{2, 3, 5, 7, 11, 13\}$ and $n(P) = 6$
- c. $M = \{9, 18, 27, 36, 45, \dots\}$, Here M is infinite set.
- d. $H = \{-4, -3, -2, -1, 0, 1, 2, 3\}$ and $n(P) = 8$

- **Null set / Empty Set**

If there is no object present in a set, then set is called empty set or a null set.

We write it as $\{ \}$ or $\varnothing$.

Example 2:

D = Set of days in a week which starts with the letter Y

Solution:

Set of days in a week = {Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}

We observed that there does not exist any day which starts with Y. So $P = \{ \}$

Example 3:

F = Set of positive integers which are both even as well as odd.

Solution:

Any positive integer is either even or odd. Hence $F = \{ \}$.

Example 4:

C = Set of months with 27 days.

Solution:

There does not exist any month with 27 days, hence $C = \{ \}$.

- **Equal Sets**

Set A and set B will be said equal if they have the identical list of elements. We can write $A = B$.

For example, $L = \{\alpha, \beta, \gamma\}$, $M = \{\alpha, \beta, \gamma\}$, $R = \{\alpha, \beta, \gamma, \delta, \theta\}$ here $L = M$ but $L \neq R$.

- **Equivalent Sets**

Set A and set B will be said equivalent if they have equal number of elements. For example, $L = \{\alpha, \beta, \gamma, \omega\}$, $M = \{n, x, y, z\}$

You can observe that there are four elements in each of the two sets L and M. therefore, the set L and M are called equivalent sets.

Remember

- Equivalent sets may not be Equal sets, because in equivalent sets all elements must be identical in the sets whereas in Equal sets, the number of elements must be the same.

Activity:

Explain the difference between equal set and equivalent set with examples.

- **Subset**

If each element of set B is also an element of set A, then set B is called the subset of set A. we use the notation $B \subseteq A$ to denote B is a subset of A. we can also say that A contains B or B includes A such that $A \supseteq B$.

Consider the sets $A = \{\alpha, \beta, \gamma, \delta, \theta\}$

$$B = \{\alpha, \beta, \gamma\}$$

$$C = \{\beta, \gamma, \alpha\}$$

Observed from above that each element of B is also an element of set A, such that $B \subseteq A$ or $B \subset A$, read as B is a **proper subset** of A.

Each element of set B is also an element of set C, such that $B \subseteq C$ or $C \subseteq B$ hence $B = C$ so we say that B and C are **improper subset** of each other.

Example 5:

If $L = \{\text{cat}, c, a, r\}$, $M = \{t, c, a, r\}$, is $L \subseteq M$?

Solution:

The word cat is an element of set L but not of M so L is not the subset of M, mathematically we write it as $L \not\subseteq M$.

Remember

- Every set is an improper subset of itself.
- Null set/ Empty set is a proper subset of every set.
- Empty set does not have any proper subset.
- Singleton set have only one proper subset.
- All the subsets of the set except the set itself are proper subsets of the set.

Example 6:

List the subsets of set $\{\alpha, \beta\}$ also classify the proper and improper subsets.

Solution:

The subsets are $\{\ }, \{\alpha\}, \{\beta\}, \{\alpha, \beta\}$

Proper subsets: $\{\ }, \{\alpha\}, \{\beta\}$

Improper subset: $\{\alpha, \beta\}$

Example 7:

List the subsets of set $\{\alpha, \beta, \gamma\}$ also classify the proper and improper subsets.

Solution:

The subsets are $\{\ }, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}, \{\alpha, \beta, \gamma\}$

We observed that $\{\ }, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}$ are the proper subsets and $\{\alpha, \beta, \gamma\}$ is the improper subset of the set $\{\alpha, \beta, \gamma\}$.

- **Power set**

If A is a set, then the set consisting of all the subsets of the set A is said to be the

Power Set of the set A. we represent it by $P(A)$. If we have a set $C = \{م, ل, گ\}$ so the $P(C) = \{\{\}, \{م\}, \{ل\}, \{گ\}, \{م, ل\}, \{م, گ\}, \{ل, گ\}, \{م, ل, گ\}\}$

Observe that a set $C = \{م, ل, گ\}$ which contains 3 elements and the number of subsets of set C will be 8 because $2^n = 2^3 = 8$ similarly if the set contains 5 elements the the number of subsets in the power set will be $2^n = 2^5 = 32$.

Think...!

Power set of an empty set is empty or not?

Universal Set

A universal set is defined as the set which consists of all elements under consideration. It is usually represented by U. if the student of class B.ed of Allama Iqbal Open University is under consideration, then a set consisting of all the students of class B.ed shall be the universal set.

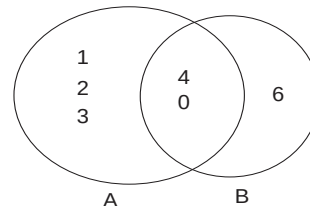
• Venn Diagrams

John Venn (1834-1923) was an English mathematician who introduced the concept of Venn diagram, so that Venn diagram is named after him. Venn diagram consists of circles or ovals. The elements of the sets are labeled within the circles. We can organize information visually that what relation is defined between the sets. We can easily compare and differentiate the sets to generate the result. Overlapping circles in the Venn diagram shows that they have some elements in common, similarly disjoint circles shows that there is nothing in common between circles.

Activity:

See the corresponding Venn diagram and write sets.

- Set A
- Set B
- Set of common elements between set A and set B.



6.3 UNION AND INTERSECTION OF SETS

• Union of Sets

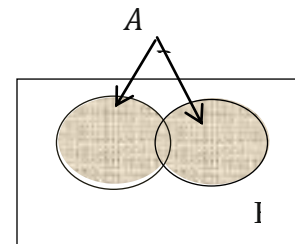
The union of two sets A and B is a set consisting of all elements which are either in the set A or in set B or in both. For union we use the symbol " $\cup$ ". For "A union B" we write " $A \cup B$ ".

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

Clearly $A \subseteq A \cup B$

$$B \subseteq A \cup B$$

So, $A \cup B = B \cup A$



Example 8:

If $A = \{1, 2, 4, 5, 6, 8\}$ and $B = \{1, 2, 3, 8\}$ then prove that $A \cup B = B \cup A$

Solution:

$$\text{L.H.S} = A \cup B$$

$$A \cup B = \{1, 2, 4, 5, 6, 8\} \cup \{1, 2, 3, 8\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8\} \dots (1)$$

$$\text{R.H.S} = B \cup A$$

$$B \cup A = \{1, 2, 3, 8\} \cup \{1, 2, 4, 5, 6, 8\}$$

$$B \cup A = \{1, 2, 3, 4, 5, 6, 8\} \dots (2)$$

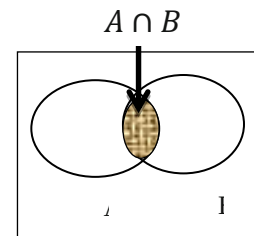
From (1) and (2), L.H.S=R.H.S

- Intersection of Sets**

The intersection of two sets A and B is a set consisting of all the common elements of the sets A and B. For intersection we use the symbol $\cap$. For “A intersection B” we write “ $A \cap B$ ”

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

Trivially, $A \cap B \subseteq A$, $A \cap B \subseteq B$, $A \cap B = B \cap A$

**Example 9:**

If $A = \{1, 2, 4, 5, 6, 8\}$ and $B = \{1, 2, 3, 8\}$ then prove that $A \cap B = B \cap A$

Solution:

$$\text{L.H.S} = A \cap B$$

$$A \cap B = \{1, 2, 4, 5, 6, 8\} \cap \{1, 2, 3, 8\}$$

$$A \cap B = \{1, 2, 8\} \dots (1)$$

$$\text{R.H.S} = B \cap A$$

$$B \cap A = \{1, 2, 3, 8\} \cap \{1, 2, 4, 5, 6, 8\}$$

$$B \cap A = \{1, 2, 8\} \dots (2)$$

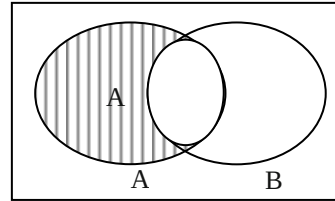
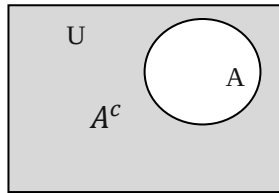
From (1) and (2), L.H.S=R.H.S

Remember

- Two sets A and B are supposed to be **disjoint sets** if they have no identical element $\Leftrightarrow A \cap B = \varnothing$
- Two sets A and B are supposed to be **overlapping sets** if they have at least one identical element $\Leftrightarrow A \cap B \neq \varnothing$
- Two sets A and B are supposed to be **overlapping sets** if they have at least one identical element.

- Complement of A Set**

Complement of a set A can be defined as $\cup - A = \{x: x \in U, x \notin A\}$, where $\cup$ is a universal set and $A \subseteq \cup$. For “complement of A” we write “ A' ” or A^c . So it is clear by definition that A^c contains the elements which do not exist in the set A.



$U - A = A^c$ is shown in colored segment

Remember

- $A \cup A^c = U$, $A \cap A^c = \varnothing$, $U^c = \varnothing$

Example 9:

If $U = \{\alpha, \beta, \gamma, \delta, \theta, v\}$, and $A = \{\alpha, \beta, \eta, \omega, \theta\}$ then find A^c .

Solution:

$$A^c = U - A = \{\alpha, \beta, \gamma, \delta, \theta, v\} - \{\alpha, \beta, \eta, \omega, \theta\} = \{\gamma, \delta, v\}$$

Here we can observe that $A^c \neq A$

Remember: $(A')' = U$

Activity:

If $U = \{x: 3 \leq x \leq 28\}$

$A = \{x: x \text{ is a factor of } 28\}$

Then Find A^c and $n(A^c)$.

6.4 BASIC LAWS OF SET OPERATIONS

6.4.1 Commutative Law

- **Commutative law with respect to union**

State and prove the commutative law with respect to union.

For non-empty sets A and B, it is $A \cup B = B \cup A$

Proof:

Let A and B are two non-empty sets.

Suppose $a \in A \cup B$

$\Rightarrow a \in A$, or $a \in B$, by definition of union of sets

$\Rightarrow a \in B$, or $a \in A$

$\Rightarrow a \in B \cup A$

$\Rightarrow A \cup B \subseteq B \cup A \dots (i)$

Now suppose $b \in B \cup A$

$\Rightarrow b \in A$, or $b \in B$, by definition of union of sets

$\Rightarrow b \in B$, or $b \in A$

$$\Rightarrow b \in B \cup A$$

$$\Rightarrow B \cup A \subseteq A \cup B \dots (ii)$$

From (i) and (ii) we have $A \cup B = B \cup A$, by definition of equal sets

• **Commutative law with respect to intersection**

State and prove the commutative law with respect to intersection.

For non-empty sets A and B, it is $A \cap B = B \cap A$

Proof:

Let A and B are two non-empty sets.

Suppose $a \in A \cap B$

$\Rightarrow a \in A$, and $a \in B$, by definition of intersection of sets

$\Rightarrow a \in B$, and $a \in A$

$\Rightarrow a \in B \cap A$

$\Rightarrow A \cap B \subseteq B \cap A \dots (i)$

Now suppose $b \in B \cap A$

$\Rightarrow b \in A$, and $b \in B$, by definition of union of sets

$\Rightarrow b \in B$, and $b \in A$

$\Rightarrow b \in B \cap A$

$\Rightarrow B \cap A \subseteq A \cap B \dots (ii)$

From (i) and (ii) we have $A \cap B = B \cap A$, by definition of equal sets

Example 10:

If $A = \{\alpha, \beta, \gamma, \rho\}$ and $B = \{\gamma, \rho, \phi, \phi\}$ then verify the commutative law of union as well as commutative law of intersection.

Solution:

Commutative law of union for set A and B such that $A \cup B = B \cup A$

L.H.S = $A \cup B$

$$= \{\alpha, \beta, \gamma, \rho\} \cup \{\gamma, \rho, \phi, \phi\}$$

$$= \{\alpha, \beta, \gamma, \rho, \gamma, \rho, \phi, \phi\} \dots (i)$$

R.H.S = $B \cup A$

$$= \{\gamma, \rho, \phi, \phi\} \cup \{\alpha, \beta, \gamma, \rho\}$$

$$= \{\alpha, \beta, \gamma, \rho, \gamma, \rho, \phi, \phi\} \dots (ii)$$

From equation (i) and equation (ii)

$$A \cup B = B \cup A$$

Now, Commutative law of intersection for set A and B such that $A \cap B = B \cap A$

L.H.S = $A \cap B$

$$= \{\alpha, \beta, \gamma, \rho\} \cap \{\gamma, \rho, \phi, \phi\}$$

$$= \{\gamma, \rho\} \dots (1)$$

R.H.S = $B \cap A$

$$= \{\gamma, \rho, \phi, \phi\} \cap \{\alpha, \beta, \gamma, \rho\}$$

$$= \{\gamma, \rho\} \dots (2)$$

From equation (1) and equation (2)

$$A \cap B = B \cap A$$

Activity

Let X, Y and Z are non-empty sets

Such that $X \cap Y = \varnothing, Y \cap Z = \varnothing$

Then which statement is true:

- i. $X = Z$
- ii. $X \subset Z$
- iii. $Z \subset X$
- iv. None of these

6.4.2 Associative Law

• Associative law with respect to Union

State and prove the Associative law with respect to union.

For three non-empty sets A, B and C, it is $(A \cup B) \cup C = A \cup (B \cup C)$

Proof:

$$\text{L.H.S} = (A \cup B) \cup C$$

$$\Rightarrow a \in (A \cup B) \cup C$$

Then $a \in (A \cup B)$ or $a \in C$, by definition of union of sets

So it follows that $a \in A$, or $a \in B$, or $a \in C$

However, that means $a \in (A \cup B) \cup C$

$$\Rightarrow (A \cup B) \cup C \subseteq A \cup (B \cup C) \dots (i)$$

$$\text{Similarly } A \cup (B \cup C) \subseteq (A \cup B) \cup C \dots (ii)$$

From (i) and (ii), we have $(A \cup B) \cup C = A \cup (B \cup C)$

• Associative law with respect to Intersection

State and prove the Associative law with respect to Intersection.

For three non-empty sets A, B and C, it is $(A \cap B) \cap C = A \cap (B \cap C)$

Proof:

$$\text{L.H.S} = (A \cap B) \cap C$$

$$\Rightarrow a \in (A \cap B) \cap C$$

$\Rightarrow a \in (A \cap B)$ and $a \in C$, by definition of intersection of sets

$\Rightarrow a \in A$, and $a \in B$, and $a \in C$

$$\Rightarrow a \in (A \cap B) \cap C$$

$$\Rightarrow (A \cap B) \cap C \subseteq A \cap (B \cap C) \dots (i)$$

$$\text{Similarly } A \cap (B \cap C) \subseteq (A \cap B) \cap C \dots (ii)$$

From (i) and (ii), we have $(A \cap B) \cap C = A \cap (B \cap C)$

Example 11:

If $A = \{\alpha, \beta, \gamma, \rho\}$ and $B = \{\gamma, \rho, \varphi, \phi\}$, $C = \{\delta, \theta, \rho\}$ then verify the associative law with respect to Union as well as Intersection.

Solution:

Associative law of union for set A, B and C such that $(A \cup B) \cup C = A \cup (B \cup C)$

$$\text{L.H.S} = (A \cup B) \cup C$$

$$\begin{aligned}
&= (\{\alpha, \beta, \gamma, \rho\} \cup \{\gamma, \rho, \varphi, \phi\}) \cup \{\delta, \theta, \rho\} \\
&= \{\alpha, \beta, \gamma, \rho, \gamma, \rho, \varphi, \phi\} \cup \{\delta, \theta, \rho\} \\
&= \{\alpha, \beta, \gamma, \delta, \rho, \gamma, \rho, \theta, \varphi, \phi\} \dots (i) \\
&\text{R.H.S} = A \cup (B \cup C) \\
&= \{\alpha, \beta, \gamma, \rho\} \cup (\{\gamma, \rho, \varphi, \phi\} \cup \{\delta, \theta, \rho\}) \\
&= \{\alpha, \beta, \gamma, \rho\} \cup \{\gamma, \rho, \delta, \theta, \varphi, \phi\} \\
&= \{\alpha, \beta, \gamma, \delta, \rho, \gamma, \rho, \theta, \varphi, \phi\} \dots (ii)
\end{aligned}$$

From equation (i) and equation (ii)

$$(A \cup B) \cup C = A \cup (B \cup C)$$

Now, associative law with respect to intersection for set A, B and C we will verify that

$$(A \cap B) \cap C = A \cap (B \cap C)$$

$$\text{L.H.S} = (A \cap B) \cap C$$

$$\begin{aligned}
&= (\{\alpha, \beta, \gamma, \rho\} \cap \{\gamma, \rho, \varphi, \phi\}) \cap \{\delta, \theta, \rho\} \\
&= \{\gamma, \rho\} \cap \{\delta, \theta, \rho\} \\
&= \{\rho\} \dots (i)
\end{aligned}$$

$$\text{R.H.S} = A \cap (B \cap C)$$

$$\begin{aligned}
&= \{\alpha, \beta, \gamma, \rho\} \cap (\{\gamma, \rho, \varphi, \phi\} \cap \{\delta, \theta, \rho\}) \\
&= \{\alpha, \beta, \gamma, \rho\} \cap \{\rho\} \\
&= \{\rho\} \dots (ii)
\end{aligned}$$

From equation (i) and equation (ii) it is proved that $(A \cap B) \cap C = A \cap (B \cap C)$

6.4.3 Distributive Property

- **Distributive property of union over intersection**

State and prove the Distributive property of union over intersection.

For three non-empty sets A, B and C, it is $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof:

Suppose $a \in A \cup (B \cap C)$

$$\Rightarrow a \in A \text{ or } a \in B \cap C$$

$$\Rightarrow a \in A \text{ or } a \in B \cap C$$

$$\Rightarrow a \in A \text{ or } (a \in B \text{ and } a \in C)$$

$$\Rightarrow [a \in A \text{ or } a \in B] \text{ and } [a \in A \text{ or } a \in C]$$

$$\Rightarrow a \in (A \cup B) \text{ and } a \in (A \cup C)$$

$$\Rightarrow a \in (A \cup B) \cap (A \cup C)$$

$$\text{Thus } A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \dots (i)$$

Similarly now suppose $b \in (A \cup B) \cap (A \cup C)$

$$\Rightarrow b \in (A \cup B) \text{ and } b \in (A \cup C)$$

$$\Rightarrow [b \in A \text{ or } b \in B] \text{ and } [b \in A \text{ or } b \in C]$$

$$\Rightarrow b \in A \text{ or } (b \in B \text{ and } b \in C)$$

$$\Rightarrow b \in A \text{ or } b \in B \cap C$$

$$\Rightarrow b \in A \cup (B \cap C)$$

$$\Rightarrow (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \dots (ii)$$

So from (i) and (ii) we can say that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

• **Distributive property of intersection over union**

State and prove the Distributive property of intersection over union.

For three non-empty sets A, B and C, it is $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof:

Suppose $a \in A \cap (B \cup C)$

$\Rightarrow a \in A$ and $a \in (B \cup C)$

$\Rightarrow a \in A$ and $(a \in B \text{ or } a \in C)$

$\Rightarrow (a \in A \text{ and } a \in B) \text{ or } (a \in A \text{ and } a \in C)$

$\Rightarrow (a \in A \cap B) \text{ or } (a \in A \cap C)$

$\Rightarrow a \in (A \cap B) \cup (A \cap C)$

$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C) \dots (i)$

Similarly $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C) \dots (ii)$

So from (i) and (ii) we can say that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Example 11:

If $K = \{a, b, c\}$, $L = \{b, c, d, e\}$, $M = \{c, d, e, f, j\}$ then prove that

i. $K \cup (L \cap M) = (K \cup L) \cap (K \cup M)$

ii. $K \cap (L \cup M) = (K \cap L) \cup (K \cap M)$

Solution (i):

L.H.S = $K \cup (L \cap M)$

$= \{a, b, c\} \cup (\{b, c, d, e\} \cap \{c, d, e, f, j\})$

$= \{a, b, c\} \cup \{c, d, e\}$

$= \{a, b, c, d, e\} \dots (i)$

R.H.S = $(K \cup L) \cap (K \cup M)$

$= (\{a, b, c\} \cup \{b, c, d, e\}) \cap (\{a, b, c\} \cup \{c, d, e, f, j\})$

$= \{a, b, c, d, e\} \cap \{a, b, c, d, e, f, j\}$

$= \{a, b, c, d, e\} \dots (ii)$

From (i) and (ii) we have, $K \cup (L \cap M) = (K \cup L) \cap (K \cup M)$

Solution (ii):

L.H.S = $K \cap (L \cup M)$

$= \{a, b, c\} \cap (\{b, c, d, e\} \cup \{c, d, e, f, j\})$

$= \{a, b, c\} \cap \{b, c, d, e, f, j\}$

$= \{b, c\} \dots (i)$

R.H.S = $(K \cap L) \cup (K \cap M)$

$= (\{a, b, c\} \cap \{b, c, d, e\}) \cup (\{a, b, c\} \cap \{c, d, e, f, j\})$

$= \{b, c\} \cup \{c\}$

$= \{b, c\} \dots (ii)$

From (i) and (ii) we have $K \cap (L \cup M) = (K \cap L) \cup (K \cap M)$

6.4.4 De-Morgan's Law

Augustus De Morgan was an English logician and mathematics. He introduced De Morgan's laws, so this famous formula is named as De Morgan's law. De Morgan's law stated algebraically as:



1806 to 1871

- i. $(A \cup B)' = A' \cap B'$
 i. $(A \cap B)' = A' \cup B'$

i. Proof: Suppose $a \in (A \cup B)'$
 $\Rightarrow a \notin A \cup B$
 because $a \notin A$ and $a \notin B$
 $\Rightarrow a \in A'$ and $a \in B'$

by definition of
complement of sets

$\Rightarrow a \in A' \cap B'$

by definition of intersection
of sets

$(A \cup B)' \subseteq A' \cap B'$

(i)

Similarly $A' \cap B' \subseteq (A \cup B)'$

(ii)

From (i) and (ii), we have proved that $(A \cup B)' = A' \cap B'$

ii. Proof: Suppose $a \in (A \cap B)'$
 then $a \notin A \cap B$
 because $a \notin A$ or $a \notin B$
 $\Rightarrow a \in A'$ or $a \in B'$
 $\Rightarrow a \in A' \cup B'$

by definition of
complement of sets

$(A \cap B)' \subseteq A' \cup B'$

(i)

Similarly $A' \cup B' \subseteq (A \cap B)'$

(ii)

From (i) & (ii), it is $(A \cap B)' = A' \cup B'$

Example 12:

If $U = \{1, 2, 3, \dots, 16\}$, $A = \{1, 3, 5, \dots, 9\}$, $B = \{2, 4, 6, \dots, 10\}$ then verify De Morgan's law.

Solution:

$A = \{1, 3, 5, \dots, 9\}$

So $A' = U - A$

$\Rightarrow A' = \{1, 2, 3, \dots, 16\} - \{1, 3, 5, \dots, 9\}$

$\Rightarrow A' = \{2, 4, 6, 8, 10, 11, 12, 13, 14, 15, 16\}$

Now $B' = U - B$

$\Rightarrow B' = \{1, 2, 3, \dots, 16\} - \{2, 4, 6, \dots, 10\}$

Thus $B' = \{1, 3, 5, 7, 9, 11, 12, 13, 14, 15, 16\}$

Consider $A \cup B = \{1, 3, 5, \dots, 9\} \cup \{2, 4, 6, \dots, 10\}$

$= \{1, 2, 3, \dots, 10\}$

L.H.S = $(A \cup B)' = U - A \cup B$

$= \{1, 2, 3, \dots, 16\} - \{1, 2, 3, \dots, 10\}$

$= \{11, 12, 13, \dots, 16\}$ (i)

R.H.S = $A' \cap B'$

$= \{2, 4, 6, 8, 10, 11, 12, 13, 14, 15, 16\} \cap \{1, 3, 5, 7, 9, 11, 12, 13, 14, 15, 16\}$

$= \{11, 12, 13, 14, 15, 16\}$

(ii)

From (i) and (ii) it is proved that $(A \cup B)' = A' \cap B'$

Now consider $\cup = \{1, 2, 3, \dots, 16\}$

$A = \{1, 3, 5, \dots, 9\}$; $A' = \{2, 4, 6, 8, 10, 11, 12, 13, 14, 15, 16\}$

$B = \{2, 4, 6, \dots, 10\}$; $B' = \{1, 3, 5, 7, 9, 11, 12, 13, 14, 15, 16\}$

$A \cap B = \{1, 3, 5, \dots, 9\} \cap \{2, 4, 6, \dots, 10\}$

$A \cap B = \{ \}$

L.H.S = $(A \cap B)' = \cup - A \cap B$

$= \{1, 2, 3, \dots, 16\} - \{ \}$

$= \{1, 2, 3, \dots, 16\}$ (i)

R.H.S = $A' \cup B'$

$= \{2, 4, 6, 8, 10, 11, 12, \dots, 16\} \cup \{1, 3, 5, 7, 9, 11, 12, \dots, 16\}$

$= \{1, 2, 3, \dots, 16\}$ (ii)

From (i) and (ii) it is proved that $(A \cap B)' = A' \cup B'$

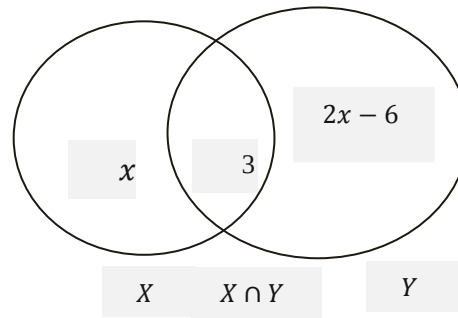
Activity:

State and prove the De Morgan's Laws.

Activity: X and Y are two non-empty sets and the number of element in each set is given in the corresponding Venn diagram. It is given that $n(X) = n(X' \cap Y)$.

Then Calculate

- the value of x .
- $n(X)$ and $n(Y)$



6.5 CARTESIAN PRODUCT, BINARY RELATION AND FUNCTION

• Order Pairs

Order Pair is the pair of elements that exist in a particular order within bracket are called ordered pairs. We denote order pair in the form (x, y) whose first element is x and the second element is y . Obviously, $x \neq y$, therefore $(x, y) \neq (y, x)$

So remember that $(x, y) = (y, x)$ if and only if $x = y$

Example 13:

What will be the value of x , if $(x - 2, 1) = (-4, 1)$

Solution:

we have $(x - 2, 1) = (-4, 1)$

Comparing the corresponding elements

$$x - 2 = -4$$

$$x = -4 + 2$$

Hence, $x = -2$

- **Cartesian Product of Sets**

The Cartesian product of two non-empty sets A and B can be stated as $A \times B$ (A cross B)

$$A \times B = \{(x, y) \mid x \in A, y \in B\}$$

Example 14:

If $A = \{7, 9\}$ and $B = \{3, 1, 0\}$ then find $A \times B$ and $B \times A$.

Solution:

$$A \times B = \{(7, 3), (7, 1), (7, 0), (9, 3), (9, 1), (9, 0)\}$$

It is obvious that set A has 2 elements and set B has 3 elements so that the set $A \times B$ has 6 ordered pairs because $2 \times 3 = 6$.

$$\text{Now } B \times A = \{(3, 7), (3, 9), (1, 7), (1, 9), (0, 7), (0, 9)\}$$

We observed that $A \neq B$ therefore, $A \times B \neq B \times A$.

Remember

- $A \times \varnothing = \varnothing \times A = \varnothing$

- **Binary Relation**

In mathematics, every subset R of $A \times B$ (Cartesian product) is called a binary relation from set A to set B. Remember that if $A \neq B$ then the binary relation R will not be the same for $A \times B$ and $B \times A$ Because $A \times B \neq B \times A$.

Activity:

If $K = \{l, m\}$ then write the number of binary relations in $K \times K$.

- **Domain and Range of Relation**

In a binary relation R, the set of first number in each ordered pair is called the **domain** of R. Similarly the set of second number in each ordered pair is called the **range** of R.

Think...!

- What will be the domain of $\{(1, 2), (3, 4)\}$?
- What will be the range of $\{(2, 1), (1, 3)\}$?

Example 15:

If $L = \{2\}$ and $M = \{a, b\}$ then write the binary relations in $L \times M$ also write the domain and the range of binary relations.

Solution:

$$L = \{2\}, \text{ and } M = \{a, b\}$$

To find binary relations in $L \times M$, first we will find the Cartesian product $L \times M$.

$$L \times M = \{(2, a), (2, b)\}$$

We know that binary relation R is a sub set of $L \times M$.

So the subsets of $L \times M$ are: φ , $\{(2, a)\}$, $\{(2, b)\}$ and $\{(2, a), (2, b)\}$

Hence, the binary relations in $L \times M$ are

$$R_1 = \varphi$$

$$R_2 = \{(2, a)\}$$

$$R_3 = \{(2, b)\}$$

$$R_4 = \{(2, a), (2, b)\}$$

We have observed that the number of binary relations in $L \times M$ is equal to the number of subsets of $L \times M$.

Now we shall find the Domain and Range for the above relation.

$$R_1 = \varphi$$

Here, the empty relation has no such ordered pairs so its domain and range will also be empty.

For $R_2 = \{(2, a)\}$, Domain (R_2) = $\{2\}$ and Range (R_2) = $\{a\}$

For $R_3 = \{(2, b)\}$, Domain (R_3) = $\{2\}$ and Range (R_3) = $\{b\}$

For $R_4 = \{(2, a), (2, b)\}$, Domain(R_4) = $\{2\}$, Range (R_4) = $\{a, b\}$

Activity:

If set $A = \{1, 3, 5, 7\}$ and $B = \{1, 2, 4\}$, then write a binary relation R for $B \times A$ when $R = \{(x, y)/x \in B \wedge y \in A \wedge x < y\}$

Think...?

If the number of element in set A is 5 and set B is 4 then how many number of elements in set $A \times B$?

• Function

Technically, function is a special relation or a mapping where each input has a single output. We write it $f: A \longrightarrow B$ and express $(x, y) \in f$ by $f(x) = y$ where $x \in A$ and $y \in B$.

Remember $\forall x \in A$ there is no repetition in any two ordered pairs of f .

If $f: A \longrightarrow B$ then domain of f is the set A and set B is said to be a codomain and range of f .

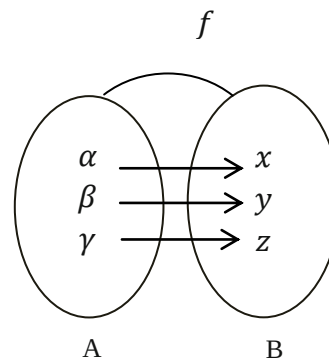
Example 15:

If $A = \{\alpha, \beta, \gamma\}$, $B = \{x, y, z\}$, then a binary relation $f = \{(\alpha, x), (\beta, y), (\gamma, z)\}$ is a function from A to B . because Domain of the function is equal to set A and range of function is equal to the set B .

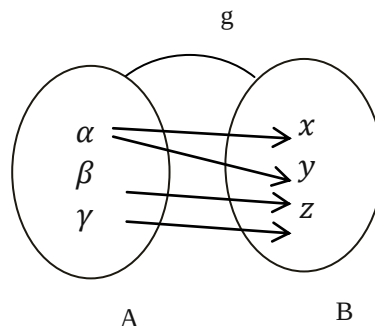
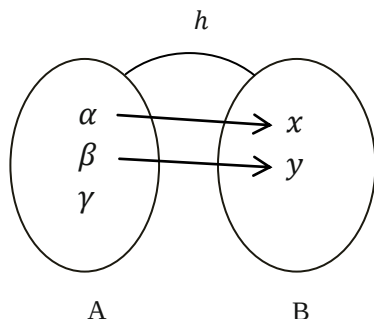
We observed that $\forall x \in A$ there is no repetition and every element of set A is associated with one and only one element of set B .

The given below sets are the examples of relation but not function.

h is not a function because $\gamma \in A$ has no image in set B , as



well as g is also not a function because $\alpha \in A$ has two image in set B.



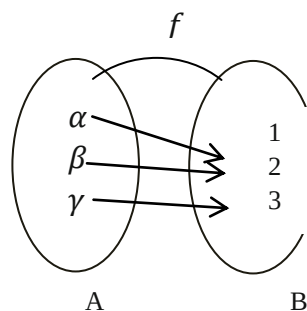
- **Into Function**

If $f: A \longrightarrow B$ and $\text{range}(f) \subset B$ means $\text{range}(f) \neq B$ then f is called into function from set A to set B.

$$f = \{(\alpha, 2), (\beta, 2), (\gamma, 3)\}$$

$$\text{Range}(f) = \{2, 3\} \subset B.$$

So it is clear that f is an in-to function.



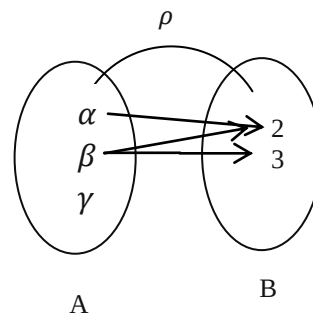
- **Onto Function**

If $\rho: A \longrightarrow B$ and $\text{range}(\rho) = B$ then ρ is called onto function from set A to set B.

$$\rho = \{(\alpha, 2), (\beta, 2), (\gamma, 3)\}$$

$$\text{Range}(\rho) = \{2, 3\} = B.$$

So the ρ is a onto function.



- **One to One Function**

If $\rho: A \longrightarrow B$ and $\text{range}(\rho) = B$ then ρ is called one to one function if $\forall x \in A$ there exist a distinct image in set B.

$$\text{i.e., } \rho(x_1) = \rho(x_2)$$

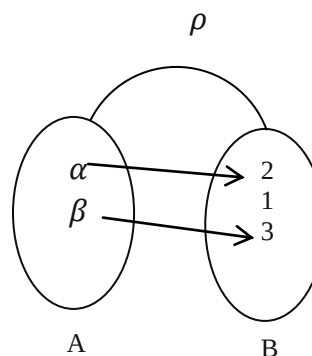
$$\implies x_1 = x_2 \in A$$

$$\text{Or } \forall x_1 \neq x_2 \in A \implies \rho(x_1) \neq \rho(x_2)$$

So it is obvious that there is no repetition in the second element of any of its two ordered pairs.

In the given figure, $\rho: A \longrightarrow B$, ρ is a one to one function.

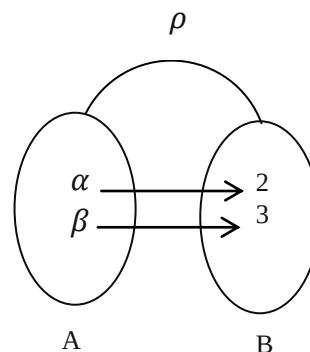
Binary relation $\rho = \{(\alpha, 2), (\beta, 3)\}$ because there is one to one correspondence between set A and set B.



- **Bijjective Function**

If $\rho: A \longrightarrow B$, then ρ is said to be abijective function if ρ is onto and one to one .

In the adjoining figure, we can see that ρ is one to one function because distinct element of A have the distinct image in set B. ρ is also onto function because every element of set B is the image of at least one element of set A.



Think...!

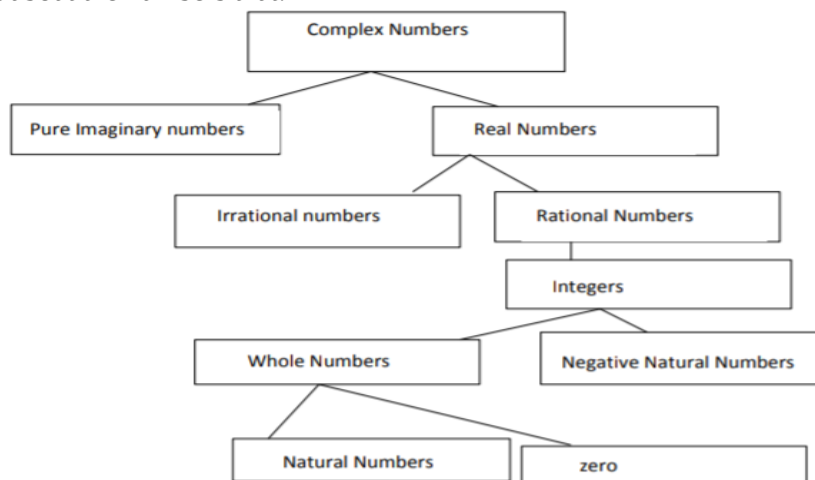
- Every function is onto or not?
- Every function is one to one or not?
- One to one function is also into function or not?

SUMMARY

In this unit we have discussed the concept of set that the collection of well-defined set is called the set. We name the sets by capital alphabets A, B, C,... Z and the element of sets are denoted by small alphabets enclosed within braces. We discussed some important sets and their notations. For example

Set	Notation
Set of natural number	N
Set of whole numbers	W
Set of even numbers	E
Set of odd numbers	O
Set of prime number	P
Set of rational numbers	Q
Set of irrational number	Q'
Set of complex number	C
Set of real numbers	R

We learnt about the numbers that:



We learnt that if set contains uncountable number of element then set is called infinite set and if set have countable number of elements then it is called finite. We have studied another beautiful set which have no single element that is called an empty set similarly set with a single element is called a singleton element. We discussed subsets and then we described a set which consist all subsets of any set we named that set a power set. We saw difference between equal and equivalent sets that if there is equal number of elements in two sets then they are said to be equivalent sets, but if the elements are same in both sets then they will be said to be equal sets.

In this unit we learnt an interesting concept “Complement of set A” is a set which do not have any element which belongs to set A. Such that $A' = U - A$

We learnt about some important operation of union and intersection on two and three sets. For example:

- Commutative property with respect to Union
 $A \cup B = B \cup A$
- Commutative property w.r.t Intersection
 $A \cap B = B \cap A$
- Associative Property w.r.t Union
 $A \cup (B \cup C) = (A \cup B) \cup C$
- Associative property w.r.t Intersection
 $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive property of union over intersection
 $(A \cap (B \cap C) = A \cap (B \cap C)$
- De Morgan’s Laws
- $(A \cup B)' = A' \cap B'$
- $(A \cap B)' = A' \cup B'$

We learnt that if A and B are two non-empty subsets then any subset of Cartesian product $A \times B$ is called a binary relation from A to B.

We learnt about the function that f is a function from set A to set B ($f: A \rightarrow B$)

- i. $\text{DOM}(f) = A$
- ii. There is no repetition in the first elements of any two ordered pairs of f .

We also used some symbol in this unit given below:

=	Is equal to
≠	Is not equal to
∀	For all
⇒	Implies that
∈	Belongs to
∉	Does not belongs to
⊆	Subset
⊄	Is not a subset

$\subset$	Is a proper subset
$\cup$	Union
$\cap$	Intersection
$>$	Greater than
$\geq$	Is greater than and equal
$<$	Less than
$\leq$	Is less than and equal
$:$ or $/$	Such that
Iff	Iff and only if
$\wedge$	And
$\vee$	Or
$\varnothing$	Empty set
U	Universal set
A'	Complement of A
GREEK LETTERS	
α	Alpha
β	Beta
γ	Gamma
δ	Delta
ε	Epsilon
η	Eta
θ	Theta
ι	Iota
κ	Kappa
λ	Lambda
μ	Mu
ν	Nu
φ	Psi
ξ	Xi
$\omicron$	Omicron
π	Pi
Ω	Omega
ρ	Rho
σ	Sigma
ϕ	Phi
τ	Tau

SELF-ASSESSMENT EXERCISE

- List the elements of the following sets.
 $A = \{y: y \text{ is a positive integer and } 5y \leq 17\}$
 $B = \{y: y \text{ is a positive integer and } y^2 \leq 21\}$
 $C = \{y: y \text{ is a positive integer and } \sqrt{y} \leq 2\}$

2. If $U = \{y: y \text{ is a positive integer } 5y > 15\}, A = \{1, 2, 3\}$ then find A' .
3. Let X, Y and Z are the sets of letters needed to spell the word PROGRESS, STUDENT and CONGRUENT, respectively then find $n(A \cup (B \cap C))$.
4. If $L = \{\alpha, \beta, \gamma, \delta\}, M = \{\beta, \gamma, \alpha, \delta\}$, Is $L \subseteq M$? Is $M \subseteq L$? Is $\varphi \subseteq L$? Explain with reasons.
5. List the subset of $\{\alpha, \beta, \gamma, \delta\}$ and find all proper and improper subsets.
6. Find power sets of the following sets.
 - a. $\{\text{Pakistan, India}\}$ c. $\{0\}$ d. φ .
7. If $U = N, A = P, B = \emptyset$, then verify De Morgan's Laws.
8. If $A = \{ \}, B = \{0, 1\}, C = \{1\}$ then verify the following
 - Associative property of union of sets
 - Associative property of intersection of sets
 - Distributive property of union over intersection
9. If set $X = \{\alpha, \nabla, \Delta, \delta\}, Y = \{\alpha, \beta, \Delta\}$ then find $(X \cup Y) \times (X \cap Y)$
10. If $R = \{(x, y): x, y \in N \wedge x + y = 7\}$, then write domain and the range of R .
11. If $X = \{4, 8, 16\}, Y = \{0, 6, 10\}$, then find the function from the following relations. Also state the type of the function.
 - $R_1 = \{(4, 0), (4, 6), (4, 10)\}$
 - $R_2 = \{(8, 0), (4, 6), (16, 0)\}$
 - $R_3 = \{(16, 6), (8, 10), (4, 0)\}$

ANSWERS

1. $A = \{1, 2, 3\}, B = \{1, 2, 3, 4\}, C = \{1, 2, 3, 4\}$
2. $\{4, 5, 6, \dots\}$
3. 94. YES, Every element of L is also an element of M and vice versa and empty set is a subset of every set therefore empty set is a subset of set
5. $\varphi, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\delta\}, \{\alpha, \beta\}, \{\alpha, \gamma\}, \{\alpha, \delta\}, \{\beta, \gamma\}, \{\beta, \delta\}, \{\gamma, \delta\}, \{\alpha, \beta, \gamma\}, \{\alpha, \beta, \delta\}, \{\alpha, \gamma, \delta\}, \{\beta, \gamma, \delta\}, \{\alpha, \beta, \gamma, \delta\}$
6. a. $\{\{ \}, \{\text{Pakistan}\}, \{\text{India}\}, \{\text{Pakistan, India}\}\}$ b. $\{\{ \}, \{0\}\}$ c. $\{ \}$ or φ
9. $\{(\alpha, \alpha), (\nabla, \alpha), (\alpha, \alpha), (\Delta, \alpha), (\delta, \alpha)\}$
10. Domain of $R = \{2, 4, 5, 6\}$, Range of $R = \{2, 4, 5, 6\}$
11. R_1 is not a function, R_2 is an into function, R_3 is a bijective function.

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Unit - 07

**DEVELOPING THE CONCEPTS
OF ALGEBRA, MATRICES AND
PARTIAL FRACTION**

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INTRODUCTION

Teaching of mathematics is not a simple art or just knowing how to teach. Teaching mathematics is an art of knowing how to teach effectively and efficiently to ensure meaningful learning for obtaining the desired outcomes. This unit consists of seven sections. Ist section is about Factorization, second is about Linear equations, third is about Linear Homogenous equations. Fourth is about application of Linear Homogenous equations, fifth is about Solution of Quadratic equations, Sixth is about Introduction of matrices and basic operations on 2×2 matrices, seventh is about Introduction of 3×3 matrices, Eight is about partial fractions of different types, Ninth is about Resolution of partial fraction.

Teaching mathematics by Enquiry and by problem solving because advances in cognitive research and developmental psychology, combined with today's urgency to educate all students in an increasingly diverse and technological society, have transformed the way we think about teaching mathematics and science.

OBJECTIVES

In teaching of Factorization you will be able to:

1. Concept of factorization.
2. Explain Factorize different type of expressions.
3. State and apply remainder theorem.
4. Find remainder when a polynomial is divided by a linear polynomial.
5. Explain concept of zeros of a polynomial.
6. Apply factor theorem.
7. Use factor theorem to factorize a cubic polynomial.

7.1 FACTORIZATION

Factors of an expression are the expressions whose product is the given expression.

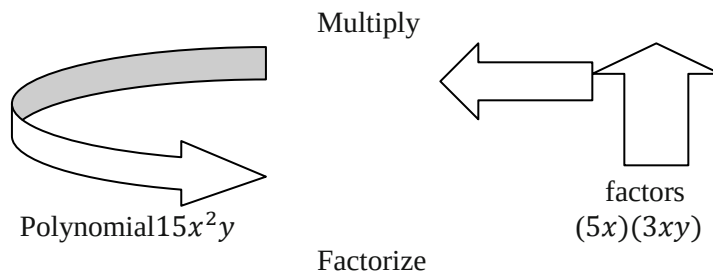
The process of expressing the given expressions as a product of its factors is called Factorization or factoring.

“Factorization is the reverse of process of multiplication”

For example $15x^2y = (5x)(3xy)$

Here $5x, 3xy$ are the factor of $15x^2y$

Hence, the factorization process which converts expression like $15x^2y$ into $(5x)(3xy)$ is the opposite of the multiplication process.



Factoring different type of algebraic expression.

Type 1 $\implies Ka + Kb + Kc$

Factorization rule for the expression of type –I is

Consider $Ka + kb + Kc = K(a + b + c)$

Example 1: factorize $2x - 4y + 6z$

Solution: $2x - 4y + 6z$
 $= 2(x - y + z)$

Example 2: Factorize $x^2 - xy + xz$

Solution: $x(xy - y + z)$

Example 3: $6x^3 - 4x^2 + 2x$

Solution: $= 2x[3x^2 - 2x + 1]$

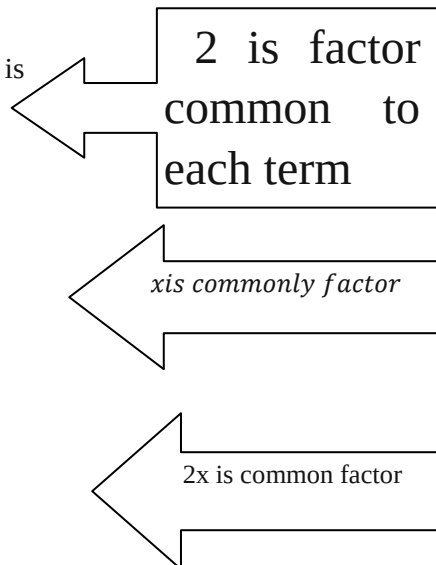
Example 4: Factorize $x^2y^2 + x^2z^2 + x^4$

Solution: $x^2y^2 + x^2z^2 + x^4$
 $= x^2y^2 + x^2z^2 + (x^2)(x^2)$
 $= x^2(y^2 + z^2 + x^2)$

TYPE 2 $ax + ay + bx + by$

Factorization of the expression of Type II is

$ax + ay + bx + by = (a + b)(x + y)$



Example5: $6xy - 16x + 9y - 24$

$$= 6xy - 16x + (9y - 24)$$

$$= (6xy - 16x) + (9y - 24)$$

$$= 2x(3y - 8) + 3(3y - 8)$$

$$= (2x + 3)(3y - 8)$$

Factorize the expressions

$$2ax + bx + 6ay + 3by$$

$$= x(2a + b) + 3y(2a + b)$$

$$= x(2a + b) + 3y(2a + b)$$

$$= (2a + b)(x + 3y)$$

Now check: $(2a + b)(x + 3y) = 2ax + bx + 6ay + 3by$

Example2: $2yx + 18y^2 + 3zx + 27zy$

Solution: $2yx + 18y^2 + 3zx + 27zy$

$$= 2y(x + 9y) + 3z(x + 9y)$$

$$(2y + 3z)(x + 9y)$$

Type 3: Factorize of the expression of the form

i. $a^2 + 2ab + b^2 = (a + b)^2$

ii. $a^2 - 2ab + b^2 = (a - b)^2$

Expression which have the pattern of the left side of eq(1) and eq(2) are called perfect squares. These identities are useful in helping us to factorize certain expressions.

Example1: factorize the following

$$1: x^2 + 6x + 9$$

Solution: $x^2 + 6x + 9$

$$= x^2 + 2(x)(3) + (3)^2$$

$$= (x + 3)^2$$

Type 4: Factorization of the expression of the form $a^2 - b^2$.

This is called difference of two squares $a^2 - b^2 = (a - b)(a + b)$

Example 1: Factorize the following.

$$k^2 - 81$$

Solution: $k^2 - 81$

$$1) : k^2 - 81 = k^2 - 9^2$$

Example 2: $9a^2 - (b + c)^2$

$$= (3a)^2 - (b + c)^2$$

$$= [3a + b + c][3a - (b + c)]$$

$$= [3a + b + c][3a - b - c]$$

Example 3: factorize $36d^2 - 1$

Solution: $36d^2 - 1$

$$= (6d)^2 - (1)^2$$

$$= (6d - 1)(6d + 1)$$

Self Assessment Exercise: 1**Factorize:**

1. $3a(x + y) - 7b(x + y)$
2. $ax + ay - xy - x$
3. $a^3 + a - 3a^2 - 3$
4. $3ax + 6ay - 8by - 4$
5. $1 - 14x + 49$
6. $x^2 + x + \frac{1}{4}$
7. $x^2 + \frac{1}{x^2}$
8. $a^2 + b^2 + 2ab + 2bc + 2ac$

Activity 1: Factorize:

1. $x^2 + y - xy - x$
2. $8 - 4a - 2a^3 + a^4$
3. $20x^2 + 5 - 20x$
4. $2a^3b + 2ab^3 - 4a^2b^2$

Type V: Factorization of the expression of the form

$$x^2 \pm px + q$$

For factorizing the expression of type $x^2 \pm px + q$ the following steps may be following.

- i. Find two vectors, whose sum or difference is p.
- ii. Split up the middle term px into two term writing the factors of q as coefficient of x (or the variable used)
- iii. Group the term into two expressions
- iv. Factor out the common factors of the groups.

Example 1: Factorize $x^2 + 15x - 100$ **Solution:** $= x^2 + 15x - 100$ Step I $= x^2 + 20x - 5x - 100$ Step II $= x(x + 20) - 5(x + 20)$ Step III $= (x + 20)(x - 5)$ We get $100 x^2$ if we recollect the two middle term at step I $20x - 5x = 15x$ **Example 2:** $x^2 + 3x + 4x + 12$ $= x^2 + 3x + 4x + 12$ $= x(x + 3) + 4(x + 3)$ $(x + 4)(x + 3)$ **Example 3:** $x^2 - 5x - 6$ **Solution:** $x^2 - 5x - 6$ $x^2 - 6x + x - 6$ $= x(x - 6) + 1(x - 6)$ $= (x - 6)(x + 1)$

Example 4: $x^2 - 5x - 14$

Solution: $x^2 - 7x + 2x - 14$

$$=x(x-7)+2(x-7)$$

$$=(x+2)(x-7)$$

Type VI: Factorization of expression of form $ax^2 + bx + c, a \neq 0$

To factorize the expression of the form $ax^2 + bx + c$, we find number find q such p that $p+q=b$,

$Pq=ac$ in the given expression where a, b, c are constants, $a \neq 0$

Example 1: Factorize:

1. $6x^2 + 7x - 3$

Solution: Compare given expression with $ax^2 + bx + c$

$$ac = (6) \times (-3) = -18$$

$$6x^2 + 7x - 3$$

$$= 6x^2 + 9x - 2x - 3$$

$$= (3x - 1) - 1(2x + 3)$$

$$= (3x - 1)(2x + 3)$$

Example 2: $\sqrt{3}x^2 + 11x + 6\sqrt{3}$

Solution: $ac = \sqrt{3} \times 6\sqrt{3}$

$$= 6(3) = 18$$

Clearly $9+2=11$

$$\begin{aligned} \therefore \sqrt{3}x^2 + 11x + 6\sqrt{3} &= \sqrt{3}x^2 + 9x + 2x + 6\sqrt{3} \\ &= \sqrt{3}x(x + 3\sqrt{3}) + 2(x + 3\sqrt{3}) \\ &= (\sqrt{3}x + 2)(x + 3\sqrt{3}) \end{aligned}$$

Self-Assessment Exercise 2:

FACTORIZE:

1. $x^2 + 9x + 20$

2. $x^2 - 7x + 12$

3. $x^2 - x - 156$

4. $x^2 - x - 2$

5. $y^2 - 11y - 15$

Activity 2: Factorize:

1. $3x^2 - 10x + 8$

2. $5x^2 - 32x + 12$

3. $10x^2 - 7x - 12$

4. $4\sqrt{3}x^2 + 5x - 2\sqrt{3}$

Type VII: Factorization of expression of the form

$$a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3$$

$$a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3$$

We know that

- i. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- ii. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Example 1: Factorize

- i. $x^3 + 6x^2 + 12x + 8$

Solution: $(x)^2 + 3(2)(x)^2 + 3(2)^2 + (2)^3 = (x + 2)^3$

7.1.2 Remainder Theorem and Factor theorem:

A function defined by the equation

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} \dots \dots \dots + a_1 x + a_0$$

$$a_n \neq 0$$

Where 'n' is non-negative integer and the coefficients are constants is called a polynomial function of degree, 'n'.

For Example:

- i. $p(x) = a_1 x + a_0$ (is a polynomial function of degree one) $a_1 \neq 0$
- ii. $p(x) = 3x^2 + 5x + 11$ (is a polynomial function of degree two)
- iii. $p(x) = 7x^5 + 2x^4 + 4x^3 + 7x^2 + 5x + 6$ (is a polynomial function of degree 5)
- iv. $p(x) = 5x^5 + \frac{7}{x} + 6 = 5x^5 + 7x^{-1} + 6$
(is not a polynomial function because power of x is negative integer)

Remember that

Divided = divisor $\times$ quotient + remainder

7.1.3 The Remainder Theorem

If R is the remainder after dividing the polynomial $p(x)$ by $x - a$, then $p(a) = R$ (or)

If a polynomial $p(x)$ is divided by polynomial $x - a$,

where a is any constant, then remainder is $p(a)$, provided there is no term left containing 'x'

Example 1:

If $p(x) = 4x^4 + 10x^3 + 19x^2 + 5x + 5$ is divided by $x + 3$ then find the remainder.

Solution: $p(x) = 4x^4 + 10x^3 + 19x^2 + 5x + 5$
 $x - a = x + 3 \Rightarrow a = -3$

- ii. $x^3 - 6x^2 + 12x - 8$
 $= (x)^3 - 3(2)(x)^2 + 3(2)^2 x - (2)^3 = (x - 2)^3$

Factorization of the expression of the form $a^3 \pm b^3$

We know that

- i. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- ii. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Example 1:**Factorize:**

$$\begin{aligned}
 \text{i)} \quad & x^3 + 27 \\
 & = (x)^3 + (3)^3 \\
 & = (x + 3)(x^2 - 3x + 9) \\
 \text{ii)} \quad & a^3 - b^3 - a + b \\
 & = (a^3 - b^3) - (a - b) \\
 & = (a^3 - b)(a^2 + ab + b^2) - (a - b) \\
 & = (a - b)[(a^2 + ab + b^2) - 1]
 \end{aligned}$$

Self Assessment Exercise 3: Factorize:

1. $8x^3 - y^3$
2. $27x^3 - 1$
3. $a^3b^3 + 512$
4. $27x^3 - 64y^3$
5. $a - b - a^3 + b^3$
6. $1 - \frac{64p^3}{q^3}$

Activity 3: Factorize:

1. $m^6 - n^6$
2. $x^3 - 27a^3$
3. $8x^3 - 6x - 9y + 27y^3$

Example 2: IF $P(x) = 5x^4 + 14x^3 + 3x^2 - 5x - 3$

Is divided by $x - 1$? Find the Remainder.

Solution: $p(x) = 5x^4 + 14x^3 + 3x^2 - 5x - 3$
 $x - a = x - 1 \implies a = 1$
 $p(1) = 5(1)^4 + 14(1)^3 + 3(1)^2 - 5(1) - 3$
 $= 5 + 14 + 3 - 5 - 3$
 $= 14$
 $P(1) = 14$

$R = 14$

7.1.4 Finding Remainder without dividing.

In the following examples, we learn to find the remainder without Division when polynomial is divided by a linear polynomial.

Example 1:

Use the remainder theorem to find the remainder when the first polynomial is divided by the second polynomial

$$x^2 + 3x + 7, x + 1$$

Solution: i. Let $p(x) = x^2 + 3x + 7$

Since the divisor $= x + 1$

Therefore $x - a = x + 1$



$$a = -1$$

By the remainder theorem

$$R = p(-1)$$

$$P(-1) = (-1)^2 + 3(-1) + 7$$

$$= 1 - 3 + 7$$

$$R = 5$$

Activity 4:

Q. Use remainder theorem to find

Remainder, when first polynomial is divided by the second polynomial.

$$x^3 - 2x^2 + 3x + 3, x - 3$$

Example 2: when $x^4 + 2x^3 + kx^2 + 3$ is divided by $x - 2$ the Remainder is 1. Find the value of K .

Solution: Let $p(x) = x^4 + 2x^3 + kx^2 + 3$

Since the divisor $= x - 2$

Therefore $x - a = x - 2$

$a = 2$

$$P(2) = (2)^4 + 2(2)^3 + k(2)^2 + 3$$

$$= 16 + 16 + 4k + 3$$

$$= 35 + 4k$$

$$P(2) = 1 \text{ (Given)}$$

$$1 = 35 + 4k \implies 4k = -34$$

$$K = -17/2$$

7.1.5 Zero of a polynomial

If $p(x) = x - a_1$, $Q(x) = x - a_2$ are any first degree polynomials such that $P(a_1) = 0$ and $Q(a_2) = 0$ for Polynomials $p(x)$ and $Q(x)$. Then a_1, a_2 are called zero of $p(x)$ and $Q(x)$.

7.1.6 The factor theorem

If a polynomial $p(x)$ is divided by $x - a$ such that $p(a) = 0$ then $x - a$ is a factor of $p(x)$.

Conversely if $x - a$ is a factor of $p(x)$, Then ' a ' is zero of $p(x)$.

Example 1: Use the factor theorem to determine if the first polynomial is a factor of the second polynomial.

$$x - 1, x^2 + 4x - 5$$

Solution: let $p(x) = x^2 + 4x - 5$

$$x - a = x - 1$$

$a = 1$

$$p(1) = (1)^2 + 4(1) - 5$$

$$=1+4(1)-5$$

$$=1+4-5$$

$$=5-5$$

$$=0$$

Since $p(1)=0$

Thus by factor theorem, $x-1$ is a factor of $x^2 + 4x - 5$.

Example 2: Use the factor theorem to show that $x=1$ is a factor of $p(x)=x^{25} + 1$

Solution: by direct substitution we see that -1 is zero of $p(x)$

$$P(x) = x^{25} + 1$$

$$P(-1)=(-1)^{25} + 1$$

$$P(-1) = -1+1$$

$$=0$$

Since -1 is a zero of $p(x) =$

$x^{25} + 1$ linear polynomial $x - (-1) =$

$x + 1$ is by factor theorem, a factor of $x^{25} + 1$.

Activity 5:

Q. 1 Use factor theorem to show that $x-1$ is not a factor of $4x^7 - 2x^6 + x^2 + 2x + 5$?

Q. 2 Use the factor theorem to show that $x+1$ is not a factor of $2x^5 - 5x^2 - x + 4$.

7.1.7 Factorizing a cubic polynomial

In order to factorize a cubic polynomial, we see the following examples:

Example 1: Factorize the following :

$$x^3 - x^2 - 10x + 10, x - 1.$$

$$\textbf{Solution: } p(x) = x^3 - x^2 - 10x + 10, x - 1$$

$$x - a = x - 1$$

$$\xrightarrow{\hspace{1cm}} a = 1$$

$$p(1) = 1^3 - 1^2 - 10 + 10$$

$$=0,$$

therefore $x - 1$ is a factor of $p(x)$.

$$\begin{array}{r} x^2 - 10 \\ x - 1 \overline{) x^3 - x^2 - 10x + 10} \\ \underline{x^3 \mp x^2} \\ -10x + 10 \\ \underline{\mp 10x \pm 10} \\ 0 \end{array}$$

$$P(x) = \text{quotient} \times \text{divisor}$$

$$= (x - 1)(x^2 - 10)$$

Activity 6: Factorize the following

$$x^3 - 8, x - 2$$

Self Assessment Exercise 4:

Evaluate each of the polynomials for the value indicated.

Q. 1 $p(x) = 2x^3 - 5x^2 + 7x - 2; p(2)$

Q. 2 $p(x) = x^4 + 5x^3 - 13x^2 - 30; p(-1)$

Q. 3 $p(x) = x^5 - 10x^3 + 7x + 6; p(3)$

Determine whether the second polynomial is a factor of the first polynomial without dividing.

Hint (evaluate directly and use the factor theorem).

Q. 4 $x^{18} - 1; x + 1.$

Q. 5 $x^9 - 2^9; x - 2.$

Q. 6 $3x^4 - 2x^3 + 5x - 1; x - 1$

Q. 7 Solve if $p(x) = x^3 - kx^2 + 3x + 5$ is divided by $x-1$, find k , if remainder is 8

Activity 7: Evaluate each of the polynomials for the value indicated.

Q. 1 $p(x) = x^4 - 10x^2 + 25x - 2, p(-4)$

Q. 2 $p(x) = x^4 + 4x^3 - 9x^2 + 19x + 6, p(-2)$

Q. 3 Solve

If $p(x) = 3x^3 + kx - 26$ is divided by $x - 2$, find k , if remainder is 0.

7.2 LINEAR EQUATION AND THEIR SOLUTIONS

7.2.1 Concept of Symbolic Expressions

The concept of symbolic expression may be understood with the following statement say seven added to five times a number given thirty. Taking $x = 5$ $5x + 7 = 30$. This algebraic statement is known as an equation. Similarly, $ax - b = c$ is an equation in which x connected by a sign of equality constitute an algebraic equation. The expressions to the left and to the right of the sign of equality are two sides of the equation. The symbol for which a value is to be determined is formed as the unknown quantity of the equation. Other symbols is used in an algebraic equation are taken as constant, such as the letter a, b, c etc.

The value of the unknown quantity satisfying the equality of both the sides of the equation is known as a root of the equation. Simply equation may be off three different forms. For example:

1. $ax = b$ 2. $ax + b = c$ 3. $ax + b = cx + d$

Linear Equations

As equation consisting of polynomial of degree one is called the linear equation. For example: $2x + 7y = 9$

7.2.2 Formation of Linear Equation: In One Variable

A linear equation is a statement of equality in which there is at least one variable involved.

For example: $5x+7=13$ is a linear equation in variable x with degree one.

Solution of linear Equations

To solve linear equation means to find the value of variable or unknown quantity in the given equation. To find the value of variable or to solve linear equations, remember the following values.

1. When equal equation quantities are added to both sides of equation, the results are also equal.

For example: $x-5=6$

Add 5 to both sides of equation.

$$x-5+5=6+5$$

$$x=11$$

2. When equal quantity subtracted from both sides of equation, the results are also equal.

For example: $x+7=11$

Subtract 7 from both sides of equation.

$$x+7-7=11-7$$

$$x=4$$

3. When equal quantities are multiplied with both sides of equation, the products are same.

For example

$$\frac{1}{5}x = 2$$

Multiply both sides of equation by 5

$$5\left(\frac{1}{5}\right)x = 5 \times 2$$

$$x=10$$

4. When both sides of equation are divided by a non Zero quantity ,the quotients are also equal For example: $5x=15$

Dividing both sides by 5

$$\frac{5x}{5} = \frac{15}{5}$$

5. Collect together the like terms on both sides and simplify

For example: $5x=6-x$

$$5x+x=6$$

$$6x=6$$

$$x=1$$

Example 1: Solve the equation

$$3x + 6 = 9$$

$$3x = 9 - 6$$

$$3x = 3$$

$$\frac{3x}{3} = \frac{3}{3}$$

$$x=1$$

All the above equations are equivalent equations.

Example 2: Solve

Solution:

$$\frac{x}{2} - 6 = 8 - \frac{2x}{3}$$

$$\frac{x}{2} - 6 = 8 - \frac{2x}{3}$$

$$\frac{x}{2} + \frac{2x}{3} = 8 + 6$$

$$\frac{3x+4x}{6} = 14$$

$$\frac{7x}{6} = 14$$

$$7x = 84$$

$$\frac{7x}{7} = \frac{84}{7}$$

$$X = 12$$

Solution set={12}

7.2.3 Concept of Linear Equations in Two Variables

Many statements can be written in the form of algebraic expressions or algebraic sentences.

Example 1: Price of 2 chair and 3 tables is =Rs40 let the price of one chair b=x and the price of one table=y

Example 2: Sum of two numbers is 45 let the number are a and b then we can write a+b=45

Example 3: Six times a number subtracted from twice of another number is 80 it can be expressed as
 $2x - 6y = 80$

All equations formed in example 1,2, 3 are linear equations in two variables

In general a linear equation in two variables may be written as $ax + by = c$

7.2.4 Equivalent Linear Equations in Two Variables

We know that a linear equation in one variable can be converted to an equivalent by

1. Adding the same expressions to both sides of an equation
2. Subtracting the same expressions from both sides of an equations
3. Multiplying dividing both sides of the equation by a non zero quantity.

7.2.5 Simultaneous Linear Equations

If two or more equations consisting of two variables are simultaneously satisfied by the same values of the variable then these are Simultaneous linear equations.

The pair of these values is the solution of these equations. In general Simultaneous Linear Equations are written in the form :

$$a_1x + b_1y = c_1 \dots \dots \dots (1)$$

$$a_2x + b_2y = c_2 \dots \dots \dots (2)$$

Remember that:

The Simultaneous linear equations (1) and (2) have a unique solution if and only if $a_1b_2 - b_1a_2 \neq 0$

Do you know?

If $a_1b_2 - b_1a_2 \neq 0$ then the two equations (1) and (2) are either equivalent linear equations or they have no solutions.

7.2.6 Solving Simultaneous Linear Equations

There are different methods for solving Simultaneous linear equations. Two of these are explained here by examples.

Example1: Solve $3x+4y=20$

$$2x+y=15$$

Solution: $3x + 4y = 20$ (1)

$$2x + y = 15$$
 (2)

$$6x + 8y = 40$$

$$\pm 6x \pm 3y = \pm 45$$

$$5y = -5$$

$$Y = -1$$

Placing this value of y in (1) or in equation (2)

$$3x + 4(-1) = 20$$

$$3x - 4 = 20$$

$$3x = 20 + 4$$

$$3x = 24$$

$$X = \frac{24}{3}$$

$$X = 8$$

So the solution of these equations

$$X = 8 \quad Y = -1$$

It can also be written as a pair (8,-1)

Example2: Solve

$$3x + 5y = 5$$

$$X + 2y = 1$$

Solution: $3x + 5y = 5$ (1)

$$X + 2y = 1$$
 (2)

From 2 $x = 1 - 2y$ (3)

$$3(1 - 2y) + 5y = 5$$
 (putting value of x from eq (3) in eq (1))

$$3 - 6y + 5y = 5$$

$$-y + 3 = 5$$

$$-y = 2$$

$$Y = -2$$

Placing this value of y in (3)

$$X = 1 - 2(-2)$$

$$= 1 + 4$$

$$x = 5$$

Self Assessment Exercise 5:

From the linear equations in two variables from the following sentences

1. Price of 6 candies and 2 Chocolates is Rs .1230
2. Price of 5 books is equal to Rs 3 less than the price of 7 note books.

Activity 8:

Q. 1 $3x - 4y = -14$

$$y - 5x + 5 = 0$$

Q. 2 $\frac{1}{5}x + \frac{1}{4}y = \frac{7}{5}$

$$3(x + y) = 16 - y$$

Solve the following equations:

Q. 4 $8\left(\frac{x}{2} - 3\right) = 7x + 2$

Q. 5 $5x + 2(2x - 1) = 3x - 14$

Q. 6 $\frac{x}{2} + \frac{x}{3} = \frac{x}{4} + \frac{x}{5}$

7.2.7 Solving Problems with the Help of Linear Equations

Many arithmetic problems can be solved with the help of algebraic equations. By using this method, the variable may be assigned to unknown quantity and is made an equation with the known quantity, Finally the equation framed in this way is solved.

Before dealing with this Kind of problems the students should have sufficient practice in forming symbolic expressions.

For unknown quantity, the variable X, Y, Z may be assigned. Keep the following points in mind to solve the word problem.

1. Read to equations in mind to solve the word problem.
2. Make sure particularly about
 - i. What is required?
 - ii. What is given?
3. Unknown quantity may be represented by X, Y or Z.
Form the equation according to given condition.
4. Solve the equation.
5. Solution the equation.

Example 1: The sum of two consecutive integers is 23. Find the integers.

Solution: Let X be the first integer. Then the next consecutive integer will be $x + 1$. Sum of two consecutive numbers $= x + (x + 1)$ according to the conditions

$$X + x + 1 = 23$$

$$2x = 23 - 1$$

$$2X = 22$$

$$X = 11$$

First integer = 11

Second integer $= x + 1$

$$=11+1$$

Second integer =12

Example 2: A number is 5 more than the twice another, then find the number.

Solution: Let the number be x

Twice of number=2x

2nd number is 5 more than twice of the first number.

Second number=2x+5

According to the condition

$$X + (2x+5) = 35$$

$$3x = 35 - 5$$

$$3x = 30$$

$$\frac{3x}{3} = \frac{30}{3}$$

$$X = 10$$

Second number = 2x+5

$$= 2(10) + 5$$

$$= 20 + 5$$

Second number = 25

Example 3: Two sisters are given Rs 72 for buying some sweets. One of them gets Rs 14 more than the other. How much does each get?

Solution: Let the one brother gets Rs x

Then the other sister gets Rs (x+14)

Then according to the given condition

$$X + x + 14 = 72$$

$$2x + 14 = 72$$

$$2x = 72 - 14$$

$$2x = 58$$

$$X = 29$$

One sister gets =29

Other sister gets=x+14

$$= 29 + 14$$

Then other sister gets=43

Activity 9:

Q. 1 The sum of numbers is 215 and their difference is 53. Find the number?

Q. 2 The age of a man 7 years ago was seven times the age of his son. Father's

Age after 3 years will be three times the age of the son. Find their present Ages.

Q. 3 The sum of three consecutive integers is 27. Find the smallest integer?

Q. 4 Ali walks by a rope 27 meters long .He cuts in into two pieces in such a way That one rope is 9 meters longer than other what are the lengths of two Ropes?

Answers of Self Assessment Exercise

Self Assessment Exercise 5

Q. 1 (1)- $6x + 2y = 1230$

$$(2)- 5x=7y-3$$

Activity8:

Q. 1 (2, 5)

Q. 2 (32,-20)

Q. 3 (28, 2)

Q. 4 $-\frac{26}{3}$

Q. 5 $\{-2\}$

Q. 6 0

7.3 LINEAR HOMOGENOUSE EQUATIONS

An equation of the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b$$

Is called a linear equation in the variables $x_1, x_2, x_3, \dots, x_n$

Here $a_1, a_2, \dots, a_n, b$ are constants, usually real numbers.

A collection of several such equations is called system of linear equations.

Consider the m linear equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_n \quad (1)$$

In n unknowns $x_1, x_2, x_3, \dots, x_n$, where a_{ij}, b_i are scalars $i=1, 2, \dots, m$, $j=1, 2, \dots, n$.

Using the matrix notation, the system (1) can be written as

$$Ax=b$$

Where A, x, b are the matrices as given below:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The matrix A is called the matrix of coefficients of the system of equations (1), the column of constants b_i forms a column vector b and the unknowns x_j form the column vector x.

Definition: The equations of the type given in (1) above with $b \neq 0$, are called a system of non homogenous linear equations. If $b=0$, then the system of equations (1) is known as a homogenous linear system.

Definition: An ordered n-tuple $y = (y_1, y_2, \dots, y_n)$ is called a solution of the system (1) if, where $x_1, x_2, \dots, x_n$ are replaced by $y_1, y_2, \dots, y_n$ respectively in each equation of (1), the resulting statements are true statements. The vector y is then also said to satisfy the system of equations (1).

For example, the system of linear equations

$$x_1 + 2x_2 = 1$$

$$2x_1 + x_2 = 2$$

has the ordered pair (1,0) as a solution. Here $y = (1,0)$.

Example 1: Solve the system of equation:

$$x_1 - x_2 + 2x_3 = 0 \quad (1)$$

$$4x_1 + x_2 + 2x_3 = 1 \quad (2)$$

$$x_1 + x_2 + x_3 = -1 \quad (3)$$

Solution The coefficient of x_1 in the (1,1) position is already 1. Following the procedure described above, multiplying equation (1) by 4 the resulting equation from equation (2) and subtracting equation (1) from equation (3) respectively, we have

$$x_1 - x_2 + 2x_3 = 0 \quad (1)$$

$$5x_2 - 6x_3 = 1 \quad (4)$$

$$2x_2 - x_3 = 1 \quad (5)$$

Multiply equation (5) by 2 and subtract from equation (4) the set of equations becomes

$$x_1 - x_2 + 2x_3 = 0 \quad (1)$$

$$x_2 - 4x_3 = 3 \quad (6)$$

$$2x_2 - x_3 = -1 \quad (5)$$

Multiply equation (6) by 2 and subtract from equation (5) we have

$$x_1 - x_2 + 2x_3 = 0 \quad (1)$$

$$x_2 - 4x_3 = 3 \quad (6)$$

$$7x_3 = -7 \quad (7)$$

Now (7) gives

$$x_3 = -1, \quad x_1 = 1.$$

By backward substitution,

$$x_2 = -1, \quad x_1 = 1.$$

7.3.1 Gauss-Jordan Elimination Method

While reducing the given system of linear equations into the equivalent system (II), we eliminated x_k , $k = 1, 2, \dots, n - k$ from the $(k+1)$ th, $(k+2)$ th... nth equation only. If we remove x_k from the first $(k-1)$ equations also then x_k will appear only the k th equation. With this modification we get another equivalent system from the given system expressing each explicitly and we no more require any backward substitution of these variables for finding the values of other unknowns. This process is called the **Gauss-Jordan Elimination Method**. The following example explains the working of this method.

Example 4: Find the solution of the following system of linear equations gauss-Jordan elimination method:

$$\begin{aligned} 2x_1 - x_2 - x_3 &= 4 \\ 3x_1 - 4x_2 - 2x_3 &= 11 \\ 3x_1 - 2x_2 - 4x_3 &= 11. \end{aligned}$$

Solution: We reduce the augmented matrix A_b into the reduced echelon form as follows:

$$\begin{aligned} A_b &= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 4 \\ 3 & 4 & -2 & 11 \\ 3 & -2 & 4 & 11 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 3 & 4 & -2 & 11 \\ 2 & -1 & -1 & 4 \\ 3 & -2 & 4 & 11 \end{array} \right] \text{ by } R_{12} \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 2 & -1 & -1 & 4 \\ 3 & -2 & 4 & 11 \end{array} \right] \text{ by } R_1 - R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & -11 & 1 & -10 \\ 0 & -17 & 7 & -10 \end{array} \right] \text{ by } R_2 - 2R_1 \text{ and } R_3 - 2R_1 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & -11 & 1 & -10 \\ 0 & -6 & 6 & 0 \end{array} \right] \text{ by } R_3 - R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & -11 & 1 & -10 \\ 0 & -1 & 1 & 0 \end{array} \right] \text{ by } R_3 - R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & -10 & 0 & -10 \\ 0 & -1 & 1 & 0 \end{array} \right] \text{ by } R_2 - R_3 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{array} \right] \text{ by } R_3 - R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & 1 & 0 & 1 \\ 0 & -6 & 6 & 0 \end{array} \right] \text{ by } R_3 + R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & -1 & 7 \\ 0 & -11 & 1 & -10 \\ 0 & 0 & 1 & 1 \end{array} \right] \text{ by } R_3 - R_2 \\ &R \left[\begin{array}{ccc|c} 1 & 5 & 0 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & -6 & 6 & 0 \end{array} \right] \text{ by } R_1 - R_3 \\ &R \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right] \text{ by } R_1 - 5R_2 \end{aligned}$$

So, $x_1 = 3$, $x_2 = 1$, $x_3 = 1$.

Example 6: The housing department of the government plans to undertake four housing projects and lists material requirements for the houses in each of the projects as follows:

	Project 1	Project 2	Project 3	Project 4
Paint (in 100 gallons)	1	2	1	1.5

Wood (kn 10,000 cu. ft)	3	4	2.5	2.5
Bricks (in millions)	1	2	1.5	1
Labour (in 1000 hours)	10	10	9	8

If the supplier delivers 6,800 gallons of paint, 14,200,200 cubic feet of wood, 64 million bricks and 448,000 hours of labor, find the number of houses built for each project.

Solution: let x_1, x_2, x_3 and x_4 be the number of houses built under project 1, project 2, project 3, and project 4 respectively. Then according to the conditions of the problem:

$$1 \times x_1 + 2 \times x_2 + 1 \times x_3 + 1.5 \times x_4 = 68 \quad (\text{in 100 gallons})$$

$$3 \times x_1 + 4 \times x_2 + 2.5 \times x_3 + 2.5 \times x_4 = 142 \quad (\text{in 10,000 cu.ft})$$

$$1 \times x_1 + 2 \times x_2 + 1.5 \times x_3 + 1 \times x_4 = 64 \quad (\text{in million})$$

$$10 \times x_1 + 10 \times x_2 + 9 \times x_3 + 8 \times x_4 = 448 \quad (\text{in 1000 hours})$$

The augmented matrix is

$$\begin{aligned}
 A_b &= \left[\begin{array}{cccc|c} 1 & 2 & 1 & 1.5 & 68 \\ 3 & 4 & -2.5 & 2.5 & 142 \\ 1 & 2 & 1.5 & 1 & 64 \\ 10 & 10 & 9 & 8 & 448 \end{array} \right] \\
 \sim R &\left[\begin{array}{cccc|c} 1 & 2 & 1 & 1.5 & 68 \\ 0 & -2 & -0.5 & -2 & -62 \\ 0 & 0 & 0.5 & -0.5 & -4 \\ 0 & -10 & -1 & -7 & -232 \end{array} \right] \text{ by } R_2 - 3R_1, R_3 - R_1 \text{ and } R_4 - 10R_1 \\
 R &\left[\begin{array}{cccc|c} 1 & -1 & 2 & 2 & 2 \\ 0 & 0 & 0 & -1 & -1 \\ 0 & 5 & -6 & -8 & -7 \\ 0 & 2 & -1 & -1 & -2 \end{array} \right] \text{ by } R_1 + R_2, 2R_3 + R_4 \text{ and } (-5)R_2 \\
 R &\left[\begin{array}{cccc|c} 1 & 0 & 0.5 & -0.5 & 6 \\ 0 & 1 & 0.25 & -1 & 31 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & 1.5 & 3 & 78 \end{array} \right] \text{ by } -\frac{1}{2}R_2 \\
 R &\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 1.25 & 33 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & 0 & 4.5 & 90 \end{array} \right] \text{ by } R_1 - (0.5)R_3; R_2 + (-0.25)R_3 \text{ and } R_4 + (-1.58)R_3 \\
 R &\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 1.25 & 33 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & 0 & 1 & 20 \end{array} \right] \text{ by } \frac{1}{4.5}R_4 \\
 R &\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 12 \\ 0 & 0 & 0 & 1 & 20 \end{array} \right] \text{ by } R_2 + (-1.25)R_4 \text{ and } R_3 + R_4
 \end{aligned}$$

Hence $x_1 = 10$, $x_2 = 8$, $x_3 = 12$, and $x_4 = 20$ are the number of houses built for each project.

7.3.3 Consistency Criterion

(Another Method of Solving A Consistent System of Equations)

A general system

$$Ax = b \quad (I)$$

Of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ may or may not have a solution. If the system $Ax = b$ has a solution $y = (y_1, y_2, \dots, y_n)$ say, then it is called a **consistent system**. If the system has no solution, that is, if there does not exist a vector y which satisfies the equation $Ax = b$, then the system is termed as **inconsistent**. In the simple $m = n$, that is, the number of equations equals the number of unknowns and the matrix A of coefficients is nonsingular the following theorem shows that the system (I) is consistent.

7.4 APPLICATION OF LINEAR HOMOGENOUS EQUATIONS

7.4.1 Theorem

A system of m homogeneous equation in n unknowns

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n = 0$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n = 0$$

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n = 0$$

$$\vdots \quad \vdots \quad \dots \quad \vdots \quad \vdots$$

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n = 0$$

has a nontrivial solution if $m < n$.

Proof

It can shown that , in a system of m linear equations in n unknowns, if $\text{rank } A = \text{rank } A_b = k < m$, then any x which satisfies k of the equations for which the corresponding rows of A in echelon form are nonzero satisfies all the equations of the set. Moreover, if $k < n$, then $n - k$ of the variables can be assigned arbitrary values and the remaining k variables can be found provided that the columns of A in echelon form associated with the k variables are nonzero. Naturally, therefore, in this case the solution set is infinite. For a homogeneous system of linear equations, we have the following necessary and sufficient condition for a nontrivial solution.

7.4.2 Theorem

A system of m homogeneous linear equations $Ax = 0$ in n variables has a nontrivial solution if and only if the rank of A is less than n . Hence a system $Ax = 0$ of n equations in n variables has a nontrivial solution if and only if A is singular.

Example 7: Examine the following homogeneous system for nontrivial solution.

$$x_1 - x_2 + 2x_3 + x_4 = 0$$

$$3x_1 - 2x_2 + x_4 = 0$$

$$4x_1 + x_2 + 2x_3 + 2x_4 = 0$$

Solution. The matrix of coefficients is

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 0 & 1 \\ 4 & 1 & 2 & 2 \end{bmatrix}$$

We bring A into the echelon form by the indicated elementary row operations.

$$\begin{array}{lcl}
A & R & \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 5 & -6 & -2 \\ 0 & 5 & -6 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_2 - 3R_1 \text{ and } R_1 - 4R_3 \\
R & & \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 5 & -6 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_3 - R_2 \\
R & & \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -\frac{6}{5} & -\frac{2}{5} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } \frac{1}{5}R_2 \\
R & & \begin{bmatrix} 1 & 0 & \frac{4}{5} & \frac{3}{5} \\ 0 & 1 & -\frac{6}{5} & -\frac{2}{5} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_1 + R_2
\end{array}$$

The rank of the matrix is $2 < 4$ hence, by theorem 7.4.1, the equations have a nontrivial solution. The first two rows of the above matrix give the following relations:

$$x_1 + \frac{4}{5}x_3 + \frac{3}{5}x_4 = 0$$

$$x_2 + \frac{6}{5}x_3 - \frac{2}{5}x_4 = 0$$

$$\text{i.e., } x_1 = -\frac{4}{5}x_3 - \frac{3}{5}x_4$$

$$x_2 = -\frac{6}{5}x_3 + \frac{2}{5}x_4$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{4}{5}x_3 - \frac{3}{5}x_4 \\ -\frac{6}{5}x_3 + \frac{2}{5}x_4 \\ x_3 \\ x_4 \end{bmatrix}$$

Assigning arbitrary values to x_3, x_4 , we find the corresponding values of x_1, x_2 in particular, when $x_3 = a, x_4 = b$,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{4}{5}x_3 - \frac{3}{5}x_4 \\ -\frac{6}{5}x_3 + \frac{2}{5}x_4 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ -\frac{6}{5} \\ 1 \\ 0 \end{bmatrix} a + \begin{bmatrix} -\frac{3}{5} \\ \frac{2}{5} \\ 0 \\ 1 \end{bmatrix} b$$

Which give solutions for arbitrary values of a and b .

Example 8. For what value of λ the equation:

$$(5 - \lambda)x_1 + 4x_2 + 2x_3 = 0$$

$$4x_1 + (5 - \lambda)x_2 + 2x_3 = 0$$

$$2x_1 + 2x_2 + (2 - \lambda)x_3 = 0$$

Have nontrivial solutions. Find these solutions.

Solution: The matrix of coefficients is

$$A = \begin{bmatrix} 5 - \lambda & 4 & 2 \\ 4 & 5 - \lambda & 2 \\ 2 & 2 & 2 - \lambda \end{bmatrix}$$

We reduce this matrix to the echelon form by applying elementary row operations:

$$\begin{aligned}
 \text{A} \quad \text{R} & \begin{bmatrix} 2 & 2 & 2-\lambda \\ 4 & 5-\lambda & 2 \\ 5-\lambda & 4 & 2 \end{bmatrix} \text{ by } R_{13} \\
 \text{R} & \begin{bmatrix} 1 & 1 & 1-\lambda/2 \\ 4 & 5-\lambda & 2 \\ 5-\lambda & 4 & 2 \end{bmatrix} \text{ by } \frac{1}{2}R_1 \\
 \text{R} & \begin{bmatrix} 1 & 1 & 1-\lambda/2 \\ 0 & 1-\lambda & -2+2\lambda \\ 0 & \lambda-1 & \frac{-6+7\lambda-\lambda^2}{2} \end{bmatrix} \text{ by } R_2 - 4R_1 \text{ and } R_3 - (5-\lambda)R_1 \\
 \text{R} & \begin{bmatrix} 1 & 1 & \frac{2-\lambda}{2} \\ 0 & 1-\lambda & -2(1-\lambda) \\ 0 & -(1-\lambda) & \frac{-(1-\lambda)(6-\lambda)}{2} \end{bmatrix} \quad (i) \\
 \text{R} & \begin{bmatrix} 1 & 1 & \frac{2-\lambda}{2} \\ 0 & 1 & -2 \\ 0 & -1 & \frac{(6-\lambda)}{2} \end{bmatrix} \text{ by } \frac{1}{1-\lambda}R_2 \text{ and } \frac{1}{1-\lambda}R_3 \text{ provide } 1-\lambda \neq 0 \\
 \text{R} & \begin{bmatrix} 1 & 0 & \frac{6-\lambda}{2} \\ 0 & 1 & -2 \\ 0 & 0 & \frac{(\lambda-10)}{2} \end{bmatrix} \text{ by } R_1 - R_1 R_3 \text{ and } R_3 + R_2 \quad (2)
 \end{aligned}$$

If $\lambda = 10$ in (2), we have

$$\text{A} \quad \text{R} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \text{ with } 1-\lambda \neq 0$$

The given system of equations reduces to

$$\begin{aligned}
 x_1 - 2x_2 &= 0 \\
 x_2 - 2x_3 &= 0
 \end{aligned}$$

So $x_1 = x_2 = 2x_3$

Take $x_3 = a, a \in R$. then

$$x_1 = 2a, x_2 = 2a, x_3 = a$$

$$(x_1, x_2, x_3) = (2a, 2a, a) =$$

$a(2, 2, 1)$ gives all the solutions of the system for $a \in$

R. If, however $\lambda \neq 10$ with $1-\lambda = 0$, then (1) gives

$$\text{A} \quad \text{R} \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The given system reduces to the single equation

$$x_1 + x_2 + \frac{1}{2}x_3 = 0$$

Take $x_3 = 2a, x_2 = b$. then $x_1 = -x_2 - \frac{1}{2}x_3 = -b - a$

Hence the solution vector is

$$(x_1, x_2, x_3) = (-b - a, b, 2a)$$

$$= b - (1, 1, 0) + a(-1, 0, 2), a, b \in R$$

For $\lambda \neq 1$ and $\lambda \neq 10$ the given system has only the trivial solution $(0,0,0)$ because, from (2) above, we see that A is invertible.

Example 9: Three species of bacteria co-exist in a test tube and feed in three foods f_1, f_2, f_3 . Suppose that a bacterium of the i th species consumes, in the average, an amount a_{ij} of the j th food per day where.

$$a_{11} = 1, a_{12} = 1, a_{13} = 1, a_{21} = 1, \\ a_{22} = 2, a_{23} = 3, a_{31} = 1, a_{32} = 3, a_{33} = 5.$$

Further suppose that there are 1500, 4500 units of food of types f_1, f_2, f_3 respectively. Assuming that all food is consumed, what are the populations of the three species that can co-exist in this environment?

Solution: Let the population of the three species of bacteria be x_1, x_2, x_3 . Since the i th species consumes A_{ij} units of food of type J , we have

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15000 \\ 30000 \\ 45000 \end{bmatrix}$$

That is $x_1 + x_2 + x_3 = 15,000$

$$x_1 + 2x_2 + 3x_3 = 30,000$$

$$x_1 + 3x_2 + 5x_3 = 45,000$$

Matrix of coefficient is $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$

The augmented matrix is

$$A_b = \begin{bmatrix} 1 & 1 & 1 & 15000 \\ 1 & 2 & 3 & 30000 \\ 1 & 3 & 5 & 45000 \end{bmatrix}$$

$$R \begin{bmatrix} 1 & 1 & 1 & 15000 \\ 0 & 1 & 2 & 15000 \\ 1 & 2 & 4 & 30000 \end{bmatrix} \text{ by } R_2 - R_1 \text{ and } R_3 - R_1$$

$$R \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 15000 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } R_1 - R_2 \text{ and } R_3 - 2R_2$$

Here we note that $\text{Rank } A = \text{Rank } A_b$

Thus we may choose x_3 arbitrarily, then

$$x_1 = x_3 = a, \quad x_2 = 1500 - 2x_3 = 15,000 - 2a$$

since x_1, x_2, x_3 are positive,

$$15,000 - 2x_3 \geq 0 \text{ that is, } 0 < x_3 \leq 7,500.$$

From the first equation we see that total population that can co-exist is 15,000

If we take $x_3 = a = 3,000$, then $x_1 = 3,000$ and $x_2 = 9,000$.

Self Assessment Exercise 6:

Solve the following system of linear equations, the field of scalar being R :

Q. 1 $2x_1 + x_3 = 1$

$$2x_1 + 4x_2 - x_3 = -2$$

$$x_1 - 8x_2 - 3x_3 = 2$$

$$\begin{aligned}\text{Q. 2 } x_1 + x_2 + x_3 &= a \\ x_1 + (1 + a)x_2 + x_3 &= 2a \\ x_1 + x_2 + (1 + a)x_3 &= 3a\end{aligned}$$

Activity10:

Solve the system of equations having the given matrices as their augmented matrices:

$$\begin{aligned}\text{Q. 1 (i) } & \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right] \\ \text{(ii) } & \left[\begin{array}{ccc|c} 4 & 2 & -1 & 0 \\ 3 & 3 & 6 & 3 \\ 5 & 1 & -8 & -1 \end{array} \right]\end{aligned}$$

Self Assessment Exercise 7:

For what value of

λ do the following homogeneous equations have nontrivial solutions?

Find these solutions:

$$\begin{aligned}\text{Q. 1 } (1-\lambda)x_1 + x_2 &= 0 \\ x_1 + (1-\lambda)x_3 &= 0 \\ \text{Q. 2 } (3-\lambda)x_1 + x_2 + x_3 &= 0 \\ x_1 - (1-\lambda)x_2 + x_3 &= 0 \\ x_1 - x_2 + (1-\lambda)x_3 &= 0\end{aligned}$$

In each of the following use Gauss-Jordan method to reduce the given system to reduced echelon form, indicating the operations performed and determine the solution if any:

$$\begin{aligned}\text{Q. 1 } 6x_1 - 6x_2 + 6x_3 &= 6 \\ 2x_1 + 4x_2 + 6x_3 &= 12 \\ 10x_1 - 5x_2 + 5x_3 &= 30 \\ \text{Q. 2 } 5x_1 - 5x_2 - x_3 &= 0 \\ 10x_1 + 5x_2 + 2x_3 &= 0 \\ 5x_1 + 15x_2 - 9x_3 &= 0\end{aligned}$$

7.5 QUADRATIC EQUATIONS

A quadratic equation in one variable is an equation that can be written in the form:

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

Where x is a variable and a, b and c are real numbers. We refer to this form as the **standard form** of the quadratic equation. A quadratic equation is also a polynomial equation in which the highest power of the unknown variable is two.

7.5.1 Solution of A Quadratic Equation

We can solve a quadratic equation by the following two methods:

(i) Factorization (ii) Completing the Square (iii) The Quadratic Formula

7.5.2. Solution of a Quadratic Equation by Factorization

The general form of a quadratic equation is $ax^2 + bx + c = 0$, $a \neq 0$. we can solve this equation algebraically to find x by using Null Factor Law.

If $a \times b = 0$ then $a=0$ or $b=0$ (or both a and b equal zero).

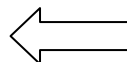
The Null Factor Law works only for expressions in factor form.

Example 1: Solve $x^2 + 4x - 77 = 0$

Solution: $x^2 + 4x - 77 = 0$

$(x - 7)(x + 11) = 0$

$x - 7 = 0$ or $x + 11 = 0$



Write the equation and check that the right hand side equals zero. The left hand side is factorized, so use the Null Factor Law to find two linear equations.

Equation that is not in factor form will need to be factorized first before the Null Factor Law can be applied.

Remember that the right hand side of the equation must be zero.

" A second degree polynomial $ax^2 + bx + c$ with integral coefficients has linear factors if and only if " ac " integral factors whose sum is " b ."

Example 2: Solve $6x^2 - 19x - 7 = 0$ using factorization.

Solution: Compare with standard form

$$ax^2 + bx + c = 0, a = 6, b = -19, c = -7$$

$$ac = 6(-7) = -42$$

$$-42 = (-21)2 \text{ and } -21 + 2 = -19 = b$$

Thus

$$6x^2 - 19x - 7 = 0$$

$$6x^2 - 21x + 2x - 7 = 0$$

$$3x(2x - 7) + 1(2x - 7) = 0$$

$$(2x - 7)(3x + 1) = 0$$

Either $2x - 7 = 0$ or $3x + 1 = 0$

$$x = \frac{7}{2} \quad \text{or} \quad x = -\frac{1}{3}$$

$$\text{Solution set} = \left\{ -\frac{1}{3}, \frac{7}{2} \right\}$$

Example 3: Solve $2x^2 = 3x$

Solution: $2x^2 = 3x$

$$2x^2 - 3x = 0$$

$$x(2x - 3) = 0$$

Either $x = 0$ or $2x - 3 = 0$

$$x = \frac{3}{2}$$

$$\text{Solution set} = \left\{ 0, \frac{3}{2} \right\}$$

Note: $x = 0, \frac{3}{2}$ are also called roots of the quadratic equation of $2x^2 = 3x$.

Example 4: If $x = 3$ is a solution of the equation $x^2 + kx + 15 = 0$. Find the value of 'k'. Also find the other solution of the equation.

Solution: Substitute $x = 3$ in $x^2 + kx + 15 = 0$

$$3^2 + 3k + 15 = 0$$

$$3k + 24 = 0 \quad \implies \quad k = -8$$

Now consider $x^2 - 8x + 15 = 0$

$$15 = (-5) \times (-3) \quad \text{And} \quad (-5) + (-3) = -8 = b$$

$$x^2 - 5x - 3x + 15 = 0$$

$$x(x - 5) - 3(x - 5) = 0$$

$$(x - 3)(x - 5) = 0$$

$$x - 3 = 0$$

$$\text{Or} \quad x - 5 = 0$$

$$x = 3$$

$$\text{Or} \quad x = 5$$

Solution set = $\{3, 5\}$

7.5.3 Solution of a Quadratic Equation by Completing the Square Method

The method of completing the square is based on the process of transforming the standard quadratic equation into the form

$$ax^2 + bx + c = 0 \quad \text{--- (1)}$$

$$(x + a)^2 = b \quad \text{--- (2), where } a \text{ and } b \text{ are constants.}$$

Equation (2) can easily be solved by completing the square method.

But how do we transform equation (1) into the form of equation (2)?

To complete the square of a quadratic equation of the form $x^2 + bx$, add the square of one-half of the coefficient of x that is $\left(\frac{b}{2}\right)^2$.

It is important to note that the rule stated above applies only to quadratic forms where the coefficients of the second degree term are 1.

Important forms used in completing the square are:

$$(i) \quad (x + m)^2 = x^2 + 2mx + m^2$$

$$(ii) \quad (x - m)^2 = x^2 - 2mx + m^2$$

Example 1: Solve $x^2 + 6x - 2 = 0$ by completing the square method.

Solution: $x^2 + 6x - 2 = 0$

$$x^2 + 6x - 2 + 2 = 2$$

Adding 2 to both sides

$$x^2 + 6x = 2$$

$$x^2 + 6x + (3)^2 = 2 + 3^2$$

$$(x + 3)^2 = 11$$

$$x + 3 = \pm\sqrt{11}$$

$$x = -3 \pm \sqrt{11}$$

$$\text{Solution set} = \{-3 + \sqrt{11}, -3 - \sqrt{11}\}$$

To complete the square of the left side, add the square of one half of the coefficient of x to each side of the equation.

Example 2:

Solve $(x - 3)^2 = 4$

Solution: $(x - 3)^2 = 4$,

$$\begin{aligned}
x^2 - 6x + 9 &= 4 \\
x^2 - 6x &= 5 \\
x^2 - 6x + (3)^2 &= -5 + 9 \\
(x - 3)^2 &= 4 \\
x &= 3 \pm 2 \\
\text{Either } x &= 5 \\
\text{Or } x &= 1 \\
\text{Solution set} &= \{1, 5\}
\end{aligned}$$

Example 3: Solve $3(x - 2)^2 = x(x - 2)$ by completing the square method.

Solution: $3(x^2 - 4x + 4) = x^2 - 2x$

$$3x^2 - 12x + 12 = x^2 - 2x$$

$$3x^2 - x^2 - 12x + 2x = -12$$

$$2x^2 - 10x = -12$$

$$x^2 - 5x = -6$$

$$x^2 - 5x + \left(\frac{5}{2}\right)^2 = -6 + \left(\frac{5}{2}\right)^2$$

$$\left(x - \frac{5}{2}\right)^2 = -6 + \frac{25}{4} = \left(x - \frac{5}{2}\right)^2 = \frac{1}{4}$$

$$x - \frac{5}{2} = \pm \frac{1}{2}$$

$$x = \frac{5}{2} \pm \frac{1}{2}$$

Either $x = \frac{5}{2} + \frac{1}{2} = \frac{6}{2} = 3$

Or $x = \frac{5}{2} - \frac{1}{2} = \frac{4}{2} = 2$

Solution set $\{2, 3\}$

Example 4:

Solve $10x^2 - 12x = 15$ by completing the square method.

Solution: $10x^2 - 12x = 15$

$$x^2 - \frac{12}{10}x = \frac{15}{10} \quad \leftarrow \boxed{\text{Dividing by '10'}}$$

$$x^2 - \frac{6}{5}x = \frac{3}{2} \Rightarrow x^2 - \frac{6}{5}x + \left(\frac{3}{5}\right)^2 = \left(\frac{3}{5}\right)^2 + \frac{3}{2}$$

Adding $\left(\frac{3}{5}\right)^2$ on the both sides

$$\left(x - \frac{3}{5}\right)^2 = \frac{9}{25} + \frac{3}{2} = \left(x - \frac{3}{5}\right)^2 = \frac{18+75}{50} \Rightarrow \left(x - \frac{3}{5}\right)^2 = \frac{93}{50}$$

$$x - \frac{3}{5} = \pm \sqrt{\frac{93}{50}} \quad \text{or} \quad x = \frac{3}{5} \pm \frac{\sqrt{93}}{5\sqrt{2}}$$

$$x = \frac{3\sqrt{2} \pm \sqrt{93}}{5\sqrt{2}} \implies x = \frac{3\sqrt{2} + \sqrt{93}}{5\sqrt{2}} \text{ or } x = \frac{3\sqrt{2} - \sqrt{93}}{5\sqrt{2}}$$

$$\text{Solution set} = \left\{ \frac{3\sqrt{2} + \sqrt{93}}{5\sqrt{2}}, \frac{3\sqrt{2} - \sqrt{93}}{5\sqrt{2}} \right\}$$

Example 5:

Solve $\frac{1}{x} + \frac{1}{x+8} = \frac{1}{3}$ by using factorization.

Solution: $\frac{1}{x} + \frac{1}{x+8} = \frac{1}{3}$

$$\frac{x+8+x}{x(x+8)} = \frac{1}{3}$$

$$\frac{2x+8}{x^2+8x} = \frac{1}{3}$$

$$x^2+8x = 6x+24$$

$$x^2 = 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$\text{Either } x+6=0 \text{ or } x-4=0$$

$$x = -6 \text{ or } x = 4$$

$$\text{Solution set} = \{4, -6\}$$

Self Assessment Exercise 8:

i. Solve by Using Factorization Method:

Q. 1 $x^2 - 4x - 12 = 0$,

Q. 2 $x^2 - 6x = 5 = 0$

Q. 3 $\frac{x}{4}(x+1) = 3$,

Q. 4 $2x = \frac{2}{x} + 3$

Q. 5 $5x^2 - 6x - 8 = 0$,

ii. Solve by Completing the Square Method:

Q. 7 $x^2 - 10x - 3 = 0$,

Q. 8 $x^2 - 6x - 3 = 0$

Q. 9 $2x^2 - 6x + 3 = 0$,

Q. 10 $x^2 + mx + n = 0$

7.5.4 The Quadratic Formula:

Quadratic formula is one of the techniques to solve a quadratic equation. Usually this formula is used when the factorization is not possible or seems to be too difficult.

Derivation of Quadratic Formula:

The general form of a quadratic equation is

$$ax^2 + bx + c = 0, \quad a \neq 0$$

where a, b, c are real numbers.

Now, we use the method of completing the square to derive a formula for the solution of all quadratic equations,

$$ax^2 + bx + c = 0, \quad a \neq 0$$

To make the leading coefficient that is of x^2 as 1, divide by a.

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \text{or} \quad x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Add the square of one half of the coefficient of x, which is $\left(\frac{b}{2a}\right)$, to each side to complete the square of the side.

$$\begin{aligned} x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 &= \frac{b^2}{4a^2} - \frac{c}{a} \quad \rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \\ x + \frac{b}{2a} &= \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ x &= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

The last equation is called the **quadratic formula**.

Example 1

Solve $2x + \frac{3}{2} = x^2$ by using the quadratic formula

Solution: $2x + \frac{3}{2} = x^2$

$$2x^2 - 4x - 3 = 0$$

Here $a = 2$, $b = -4$, $C = -3$

We have,

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ X &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(-3)}}{2(2)} \\ &= \frac{4 \pm \sqrt{16 - 24}}{4} = \frac{4 \pm \sqrt{40}}{4} \\ &= \frac{4 \pm 2\sqrt{10}}{4} \\ X &= \frac{2 \pm \sqrt{10}}{2} \end{aligned}$$

$$\text{Solution set} = \left\{ \frac{2 + \sqrt{10}}{2}, \frac{2 - \sqrt{10}}{2} \right\}$$

Example 2 : Solve $(x + 5)^2 + (2x - 1)^2 - 67 = (x + 5)^2 + (2x - 1)^2$

By using Quadratic formula.

Solution: $(x + 5)^2 + (2x - 1)^2 - 67 = (x + 5)^2 + (2x - 1)^2$

$$x^2 + 10x + 25 + 4x^2 - 4x + 1 - 67 = 2x^2 + 10x - x - 5$$

$$5x^2 + 6x - 41 = 2x^2 + 9x - 5$$

$$x^2 - x - 12 = 0 \quad \leftarrow \text{divide by 3}$$

Here $a = 1$, $b = -1$, $c = -12$

We have, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Therefore $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-12)}}{2(1)}$

$$\begin{aligned}
&= \frac{1 \pm \sqrt{1+48}}{2} \\
&= \frac{1 \pm \sqrt{49}}{2} = \frac{1 \pm 7}{2} \\
x &= \frac{1-7}{2}, \quad x = \frac{1+7}{2} \\
x &= -3, \quad x = 4
\end{aligned}$$

Solution set = $\{4, -3\}$

Self Assessment Exercise 9:

Solve using quadratic formula.

Q. 1 $x^2 - 5x + 6 = 0$

Q. 2 $\frac{x+1}{x+4} = \frac{2x-1}{x-6}$ where $x \neq -4, -6$.

Q. 3 $\frac{x}{6} + \frac{6}{x} = \frac{4}{x} + \frac{x}{4}$ where $x \neq 0$

Activity 11:

Q. 1 $3x^2 + x - 6 = 0$

Q. 2 $10x^2 - 5x = 15$

Q. 3 $\frac{x+4}{x-4} + \frac{x-4}{x+6} = \frac{10}{3}$ where $x \neq -4$

7.5.5 Problems involving quadratic equations

Example 1: Find two consecutive positive odd number such that the sum of their square is equal to 130.

Solution: Let one add number be x and the other number be $(x+2)$

$$x^2 + (x+2)^2 = 130$$

$$x^2 + x^2 + 4x + 4 = 130$$

$$2x^2 + 4x - 126 = 0$$

$$x^2 + 2x - 63 = 0$$

$$x^2 + 9x - 7x - 63 = 0 \leftarrow \text{dividing by 2}$$

$$x(x+9) - 7x(x+9) = 0$$

$$(x+9)(x-7) = 0$$

$$x+9=0 \text{ or } x-7=0$$

$$x=-9 \text{ or } x=7$$

$x=-9$ is not a solution because it is a negative number,

Where $x=7$

$$x+2 = 7+2 = 9$$

The two consecutive positive add number are 7 and 9.

Self Assessment Exercise 10

Q. 1 Find two consecutive positive odd numbers such that the sum of their square is 74.

Q. 2 Find two consecutive positive even numbers such that the sum of their square is 164.

Q. 3 The base and height of a triangle are $(x+3)$ cm and $(2x-5)$ cm respectively. If the area of the triangle is 20 cm^2 , find x .

- Q. 4 Find two consecutive positive odd numbers given that the difference between their reciprocals is $\frac{2}{63}$.
- Q. 5 The sum of the positive numbers is 12 and the sum of whose squares is 80. Find the numbers.

Activity 12:

- Q. 1 The difference of two numbers is 9 and the product of the numbers is 162. Find the two numbers.
- Q. 2 The perimeter and area of a rectangle are 22cm and 30cm^2 respectively. Find the length and breadth of the rectangle.
- Q. 3 The product of two consecutive positive numbers is 156. Find the numbers.

Answers of Self Assessment Exercise

Self Assessment Exercise 8

- 1: -2,6 2: 1,5 3: 3,-4 4: $2, -\frac{1}{2}$ 5: $2, -\frac{4}{5}$
- 7: $5 \pm 2\sqrt{7}$ 8: $3 \pm 2\sqrt{3}$ 9: $\frac{3 \pm \sqrt{3}}{2}$ 10: $\frac{-m \pm \sqrt{m^2 - 4n}}{2}$

Self Assessment Exercise 9

- 1: 2, 3 2: $\pm\sqrt{10}$ 3: $\pm 2\sqrt{6}$

Activity 11:

- 1: $-1, \frac{2}{3}$ 2: $-1, \frac{3}{2}$ 3: ± 8

Self Assessment Exercise 10

- 1: 5,7 2: 8,10 3: 5 4: 7,9 5: 4,8 Or 8,4.

Activity 12

- 1: 9, 8 2: 5, 6 3: 12,13

7.6 INTRODUCTION OF MATRICES AND BASIC OPERATION ON 2X2 MATRICES

7.6.1 Introduction

Here we will introduce a new mathematical form, called a matrix that will enable us to represent a number of different quantities as a single unit.

The idea of matrices was introduced by a famous mathematician Arther Cayley in 1857. Matrices are widely used in both the physical and the social sciences.

A matrix is a square or a rectangular array of numbers written within square brackets or parentheses in a definite order, in rows and columns.

Generally, the matrices (plural of the matrix) are denoted by capital letters A, B, Cetc. while the elements of a matrix are denoted by small letters a, b, c..... and numbers 1, 2, 3 For example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 7 \\ 2 \end{bmatrix}, \quad C = [4 \quad 5], \quad D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Look at another example:

A company that manufactures shirts makes a standard model and a competition model. The labor (in hours) required for each model is conveniently represented by the 2 x 3 matrix.

$$M = \begin{bmatrix} \text{Fabricating} & \text{Finishing} & \text{Packing \& Handling} \\ 5 & 1 & 0.2 \\ 7 & 2 & 0.2 \end{bmatrix} \text{Standard Shirt}$$

The Weekly production can be represented by the row matrix

$$\begin{bmatrix} \text{Standard Shirts} & \text{Competition Shirts} \\ 100 & 10 \end{bmatrix}$$

Each matrix consists of horizontally and vertically arranged elements.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

Vertically arranged elements

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

Horizontally arranged elements

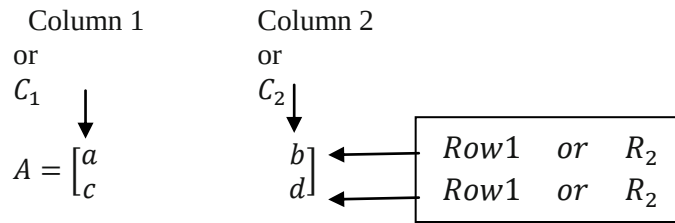
Rows:	Horizontally arranged elements are said to form rows.
Columns:	Vertically arranged elements are said to form columns.

The number of rows and columns in matrices may be equal or different. However, the number of elements in different rows are same and similar is the case in the columns of a matrix that remains the same.

Generally, rows and columns are denoted by R and C respectively.

For

example:



Matrix A has two, rows and two columns whereas a, b, c, d are its elements, The number of rows and the number of columns are denoted by m and n respectively.

In the above example $m = 2$ and $n = 2$

Order of a Matrix

If a matrix A has 'm' number of rows and 'n' number of columns, then order of the matrix A is $m \times n$ (read as "m-by-n matrix")

Example 1: Find the order of $P = [3]$

Solution: Matrix P has only one row and one column.

So order of P is 1-by-1 matrix.

Example 2: Find the order of $Q = [4 \ 7]$

Solution: Matrix Q has one row and two columns.

So order of Q is 1-by-2 matrix.

Example 4: Find the order of $A = \begin{bmatrix} 3 & 4 & 9 \\ 5 & 7 & 2 \\ 1 & 2 & 5 \end{bmatrix}$

Solution: In matrix A, the number of rows is three i.e. $m = 3$ and the number of columns is three, i.e. $n = 3$.

The order of A is 3-by-3 matrix

Remember that:

- We denote order of a matrix in-by-n instead of $m \times n$
- It is important to remember that in the order of a matrix, the number of rows are mentioned first.

Matrix Equality:

Two matrices A and B are said to be equal if and only if they have the same order and their corresponding elements are equal.

Their equality is denoted by $A = B$.

For example:

$$\begin{bmatrix} w & x \\ y & z \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Leftrightarrow \begin{cases} w = a \\ x = b \\ y = c \\ z = d \end{cases}$$

Example

Which of the following matrices are equal and which of them are not equal?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 3-2 & \frac{4}{2} \\ 3 & 2 \times 2 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & \frac{6}{2} & 7 \\ \frac{6}{3} & 1 & 3 \\ 2 \times 2 & 2 & \frac{10}{2} \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$$

Solution: Matrix B, can be written as

$$B = \begin{bmatrix} 3-2 & \frac{4}{2} \\ 3 & 2 \times 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A$$

- i. Order of A and B is same 2×2 , and corresponding elements are equal, so $A = B$
- ii. Order of A, B and C is same 2×2 , but corresponding elements are not equal, so $A = B \neq C$.
- iii. Matrix E can be written as:
Order of D and E is same, i.e. 3×3 and corresponding elements are equal, so $D = E$.

$$E = \begin{bmatrix} 1 & \frac{6}{2} & 7 \\ \frac{6}{3} & 1 & 3 \\ 2 \times 2 & 2 & \frac{10}{2} \end{bmatrix} = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{bmatrix}$$

- iv. Order of the matrix F is 2×3 so $F \neq D$ and $E \neq F$.

Self-Assessment Exercise 11

With help of the given matrices answer the following questions from Q. 1 to Q. 3

$$A = \begin{bmatrix} 2 & -2 \\ -5 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & -2 \\ 0 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix}$$

$$D = \begin{bmatrix} -3 & 2 & 0 \\ 0 & 1 & 5 \\ 4 & -2 & 2 \end{bmatrix}, \quad E = \begin{bmatrix} -2 & 4 \\ 0 & 5 \\ 3 & -1 \end{bmatrix}$$

- Q. 1 What are the order of matrices A, C and F ?
- Q. 2 What are the orders of matrices?
- Q. 3 What element is in the second row and third column of matrix D ?

7.6.2 Basic Operations on matrices of Same Order

Two matrices A and B are said to be conformable for addition $A + B$, if they are of the same order and their sum is obtained by adding their corresponding elements. Order of $A + B$ will be the same as the order of matrices A and B .

Addition of Matrices:

When two matrices are conformable for addition, we find addition by adding their corresponding elements.

For example:

$$\text{i.} \quad \begin{bmatrix} w & x \\ y & z \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} w+a & x+b \\ y+c & z+d \end{bmatrix}$$

$$\text{ii.} \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & -3 & 5 \\ 3 & 4 & 7 \end{bmatrix} + \begin{bmatrix} 3 & -2 & -5 \\ -5 & 2 & 1 \\ -1 & -3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1+3 & 2+(-2) & 4+(-5) \\ 2+(-5) & -3+2 & 5+1 \\ 3+(-1) & 4+(-3) & 7+(-2) \end{bmatrix} = \begin{bmatrix} 4 & 0 & -1 \\ -3 & -1 & 6 \\ 2 & 1 & 5 \end{bmatrix}$$

Subtraction of Matrices:

If two matrices A and B are of the same order then their difference can be written as $A - B$.

The difference $A - B$ is obtained by subtracting the elements of B from the corresponding elements of matrix A .

Example 1: If $A = \begin{bmatrix} 1 & x \\ y & 4 \end{bmatrix}$, $B = \begin{bmatrix} a & 2 \\ 3 & b \end{bmatrix}$, then find $A - B$

Solution: $A - B = \begin{bmatrix} 1 & x \\ y & 4 \end{bmatrix} - \begin{bmatrix} a & 2 \\ 3 & b \end{bmatrix}$

$$A - B = \begin{bmatrix} 1-a & x-2 \\ y-3 & 4-b \end{bmatrix}$$

Example-2 $A = \begin{bmatrix} 2 & 7 & 3 \\ -1 & 3 & 4 \\ 0 & 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 6 \\ -1 & 8 & 3 \end{bmatrix}$, find $A - B$

Solution: $A - B = \begin{bmatrix} 2 & 7 & 3 \\ -1 & 3 & 4 \\ 0 & 4 & -2 \end{bmatrix} - \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 6 \\ -1 & 8 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 2-1 & 7-3 & 3-5 \\ -1-2 & 3-1 & 4-6 \\ 0-(-1) & 4-8 & -2-3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & -2 \\ -3 & 2 & -2 \\ 1 & -4 & -5 \end{bmatrix}$$

Example-3 Add the matrices A, B and C where

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -6 \\ -4 & 1 \end{bmatrix}$$

Solution: Since A, B and C matrices have the same order, so they are conformable for addition

$$\begin{aligned} A + B + C &= \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} + \begin{bmatrix} 2 & -6 \\ -4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2+1+2 & 1+3-6 \\ 3-2-4 & 4+5+1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & -2 \\ -3 & 10 \end{bmatrix} \end{aligned}$$

A Scalar Multiplication:

Any element from the set of real numbers is also called a scalar. We define the product of a matrix A and a scalar k , denoted by kA , to be the matrix formed by multiplying each element of A by k .

For example:

i. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$

ii. If $A = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 6 & -1 \\ -3 & 4 & 7 \end{bmatrix}$, then $3A = \begin{bmatrix} 12 & 9 & 6 \\ 6 & 18 & -3 \\ -9 & 12 & 21 \end{bmatrix}$

7.6.3 Laws of Addition of Matrices

Commutative Law:

For any two matrices A and B of the same order

$$A + B = B + A$$

This law is called commutative law of matrices with respect to addition.

Example 1: If $A = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -6 & 1 \end{bmatrix}$

then show that $A + B$

Solution: Matrices A, B have same order, so they are conformable for addition

$$A + B = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ -6 & 1 \end{bmatrix} = \begin{bmatrix} 1+4 & 3+2 \\ 4-6 & 5+1 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 5 & 5 \\ -2 & 6 \end{bmatrix}$$

$$\text{And } B + A = \begin{bmatrix} 4 & 2 \\ -6 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 4+1 & 2+3 \\ -6+4 & 1+5 \end{bmatrix}$$

$$B + A = \begin{bmatrix} 5 & 5 \\ -2 & 6 \end{bmatrix}$$

$$\text{Thus } A + B = B + A$$

7.6.4 Additive Identity of Matrices

In real numbers, zero is the additive identity i.e. the sum of a real number and zero is equal to the real number e.g. $5 + 0$

Similarly, a zero matrix O of order $m \times n$ is called the additive identity matrix such that

$$A + O = O + A = A$$

$$\text{For example: } A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$$

$$\text{Consider, } A + O = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A + O = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix} = A$$

$$\text{And } O + A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix} = A$$

Thus $A + O = O + A = A$ so ' O ' is additive identity of matrix A .

Remember that: The order of ' A ' and ' O ' is same.

7.6.5 Additive Inverse of a Matrix

If two matrices A and B are such that their sum ($A + B$) is a zero matrix, then A and B are called additive inverse of each other.

$$\text{For example: } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } B = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$$

$$\text{Consider } A + B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Therefore A and B are inverse of each other.

Self-Assessment Exercise 12

$$\text{Q. No.1. If } A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 5 \\ 4 & 9 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 5 \\ 2 & 3 & 6 \\ 1 & 4 & -2 \end{bmatrix}$$

$$\text{Find } i). A + B \quad ii). B - A \quad iii). 2A + 3B$$

$$\text{Q. No.2. If } A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 7 \\ 4 & 6 \end{bmatrix} \text{ then}$$

Show that, $3B - 3A = 3(B - A)$

Q. No. 3. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$, Verify that $(A + B)^t = A^t + B^t$

Activity-14

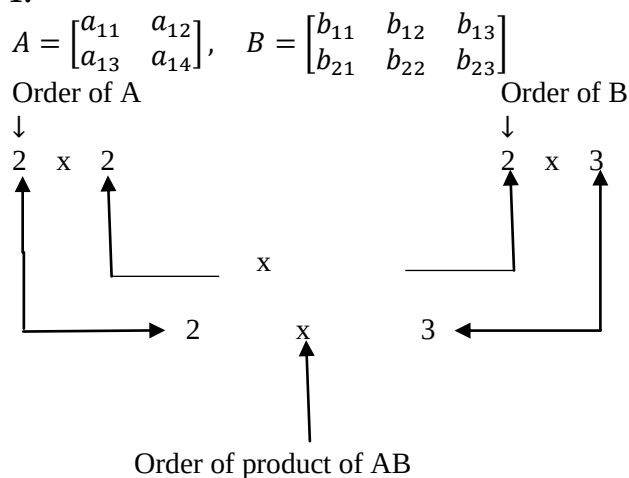
Q. No. 1. Find, x and y if, $\begin{bmatrix} x+3 & 1 \\ -3 & 3y-4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$

Q. No. 2. Find Additive Inverse of the Matrices. $A = \begin{bmatrix} 4 & 3 \\ 2 & 6 \end{bmatrix}$, $C = \begin{bmatrix} 1 \\ -7 \\ 4 \end{bmatrix}$

7.6.6 Multiplication of matrices

Two matrices A and B are said to be comfortable for the product AB if the number of columns in A are equal to the number of rows of B.

Example 1:



The product of AB shall contain the elements like

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix} \text{ where}$$

$$c_{11} = a_{11}b_{11} + a_{12}b_{21}$$

(Multiplication of the elements of 1st row of A with elements of 1st column of B)

$$c_{12} = a_{11}b_{12} + a_{12}b_{22}$$

(Multiplication of the elements of 1st row of A with elements of 2nd column of B)

$$c_{13} = a_{11}b_{13} + a_{12}b_{23}$$

(Multiplication of the elements of 1st row of A with elements of 3rd column of B)

$$c_{21} = a_{21}b_{11} + a_{22}b_{21}$$

$$c_{22} = a_{21}b_{12} + a_{22}b_{22}$$

$$c_{23} = a_{21}b_{13} + a_{22}b_{23} \quad \text{thus,}$$

$$AB = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \end{bmatrix}$$

Example 2: $A = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, then find AB

Solution: order of $A = 2 - by - 2$

order of $B = 2 - by - 1$

order

of

$$AB = 2 - by - 1$$

Because number of columns in $A = \text{number of rows in } B = 2$

$$AB = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \times 2 & 2 \times 3 \\ 3 \times 2 & 4 \times 3 \end{bmatrix} = \begin{bmatrix} 10 + 6 \\ 6 + 12 \end{bmatrix} = \begin{bmatrix} 16 \\ 18 \end{bmatrix}$$

Result:

If A is a square matrix then $A^2 = A.A$

$$A^3 = A.A.A = AA^2 = A^2A$$

Finally

$$A^n = A.A.A \dots \dots \dots n \text{ times}$$

7.6.7 Commutative Law

Commutative law does not hold in multiplication of matrices in general
i.e. $AB \neq BA$

Example 1: $A = \begin{bmatrix} 1 & 2 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}$, verify $AB \neq BA$

Solution: Given matrices A and B are conformable for multiplication AB and BA .

$$\text{Consider } AB = \begin{bmatrix} 1 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 8 & 3 + 4 \\ -3 - 8 & 9 - 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 7 & 7 \\ -11 & 5 \end{bmatrix} \dots \dots \dots (i)$$

$$\text{Consider } BA = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 9 & -2 - 6 \\ 4 + 6 & 8 - 4 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -8 \\ 10 & 4 \end{bmatrix} \dots \dots \dots (ii)$$

From equations (i) and (ii)

$$AB \neq BA$$

Hence commutative law does not hold in multiplication of matrices, in general,

Example 2: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, verify $I_2A = AI_2 = A$

Solution:

$$\begin{aligned} \text{Consider } I_2A &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ &= \begin{bmatrix} 1 \times a + 0 \times c & 1 \times b + 0 \times d \\ 0 \times a + 1 \times c & 0 \times b + 1 \times d \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

$$I_2A = A \dots \dots \dots (i)$$

$$\begin{aligned} \text{Consider } AI_2 &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} a \times 1 + b \times 0 & a \times 0 + b \times 1 \\ c \times 1 + d \times 0 & c \times 0 + d \times 1 \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

$$AI_2 = A \dots \dots \dots (ii)$$

From equations (i) and (ii)

$$I_2A = AI_2 = A$$

Self-Assessment Exercise 13

Q. 1 Verify statement, using

$$A = \begin{bmatrix} 4 & 2 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -2 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 2 \\ 4 & 2 \end{bmatrix}$$

- i). $(AB)C = A(BC)$
- ii). $(B + C)A = BA + CA$

Q. 2 Find the Matrix products.

$$1. \quad \begin{bmatrix} 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \quad 2. \quad \begin{bmatrix} 3 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Q. 3 If $\begin{bmatrix} 1 & 5 \\ 3 & a \end{bmatrix} \begin{bmatrix} b \\ 7 \end{bmatrix} = \begin{bmatrix} 35 \\ 10 \end{bmatrix}$, then find the values of a and b .

Activity 15

Q. 1 Find the Matrix Products.

$$\begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & -2 \end{bmatrix}$$

Q. 2 If $A = \begin{bmatrix} 2 & 6 \\ 7 & 8 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -3 \\ 2 & 0 \end{bmatrix}$ then
 verify $(AB)^t = B^t A^t$

7.6.8 Multiplicative Inverse of A Matrix

Determinant Function

In this section, we are going to define a new function, called a determinant of a square matrix. Its domain is the set of all square matrices with real elements, and its range is the set of all real numbers.

If A is a square matrix, then $\det A$ or $|A|$ read "The determinant of A " is used to denote the unique real number.

The determinant of a matrix of order 2 is defined as follows.

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$= ad - bc$$

Evaluate Determinant of a Matrix

Example-1 If $A = \begin{bmatrix} -1 & 2 \\ -3 & -4 \end{bmatrix}$, then evaluate $\det A$

Solution: $|A| = \begin{vmatrix} -1 & 2 \\ -3 & -4 \end{vmatrix} = (-1) \times (-4) - (-3) \times 2$
 $= 4 + 6 = 10$

7.6.9 Singular and Non-Singular Matrices

Singular Matrix

A square matrix A is called a singular matrix. If $\det A = 0$

EXAMPLE If $A = \begin{bmatrix} 12 & 6 \\ 6 & 3 \end{bmatrix}$, then evaluate $\det A$

$$|A| = \begin{vmatrix} 12 & 6 \\ 6 & 3 \end{vmatrix} = 36 - 36 = 0$$

$\det A = 0$, Hence matrix A is singular

Non-Singular Matrix:

A square matrix A is called non-singular matrix, if $\det A \neq 0$.

EXAMPLE If $A = \begin{bmatrix} 2 & 5 \\ 6 & 8 \end{bmatrix}$, then evaluate $\det A$

$$|A| = \begin{vmatrix} 2 & 5 \\ 6 & 8 \end{vmatrix} = 16 - 30 = -14 \neq 0$$

$\det A \neq 0$, Hence matrix A is non – singular

Adjoint of a Matrix

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a square matrix of order 2 by 2. Then the matrix obtained by interchanging the elements of the diagonal (i.e. a and d) and by changing the signs of the other elements b and c is called the adjoint of the matrix A .

The adjoint of the matrix A is denoted by $\text{adj } A$. For example:

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Look at another example:

$$\text{If } P = \begin{bmatrix} -6 & -2 \\ 3 & 4 \end{bmatrix}, \text{ then } \text{adj } P = \begin{bmatrix} 4 & 2 \\ -3 & -6 \end{bmatrix}$$

7.6.10 Multiplicative Inverse

In the set of real numbers, we know that for each real number a (except zero) there exists a real number a^{-1} such that $aa^{-1} = 1$.

The number a^{-1} is called the multiplicative inverse of a .

Similarly, each square matrix A has a multiplicative inverse A^{-1}

such that, $AA^{-1} = A^{-1}A = I$, provided $\det A \neq 0$.

Multiplicative inverse A^{-1} of any non-singular matrix A is given by

$$A^{-1} = \frac{\text{adj } A}{|A|}, \quad |A| \neq 0$$

If A is a singular matrix then the multiplicative inverse of A does not exist.

Remember That:

Inverse of square matrix A is denoted by A^{-1}

- i. Only non-singular matrices have inverses.
- ii. Inverse of square matrix A is always unique.
- iii. Non-square matrices cannot possess inverses.

7.6.11 Inverse of a Non- Singular Matrix

The matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has the inverse $\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, provide $ad - bc \neq 0$

Example 1: If $A = \begin{bmatrix} 7 & 3 \\ 14 & 9 \end{bmatrix}$, then find inverse of matrix A

Solution: $A = \begin{bmatrix} 7 & 3 \\ 14 & 9 \end{bmatrix}$, then $\text{adj } A = \begin{bmatrix} 9 & -3 \\ -14 & 7 \end{bmatrix}$

$$|A| = \begin{vmatrix} 7 & 3 \\ 14 & 9 \end{vmatrix}$$

$$|A| = 63 - 42 = 21 \neq 0$$

We know that $A^{-1} = \frac{1}{|A|} \text{adj } A$

$$A^{-1} = \frac{1}{21} \begin{bmatrix} 9 & -3 \\ -14 & 7 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{9}{21} & \frac{-3}{21} \\ \frac{-14}{21} & \frac{7}{21} \end{bmatrix} = \begin{bmatrix} \frac{3}{7} & \frac{-1}{7} \\ \frac{-2}{3} & \frac{1}{3} \end{bmatrix}$$

Example 3: If $P = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$, then find inverse of matrix P .

Solution:

$$P = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}, \quad \text{then } \text{adj } P = \begin{bmatrix} 4 & -2 \\ -6 & 3 \end{bmatrix}$$

$$|P| = \begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix}$$

$$|P| = 3 \times 4 - 2 \times 6 = 12 - 12 = 0$$

Since $|P| = 0$

The inverse of P is not defined

Because $\frac{1}{0}$ is not defined.

Self-Assessment Exercise 14

Q. 1 Find the determinants of matrices.

$$i). \begin{bmatrix} u & v \\ x & y \end{bmatrix} \quad ii). \begin{bmatrix} -8 & -4 \\ -4 & -2 \end{bmatrix}$$

Q. 2 Find the inverse of each matrix and show that $A^{-1}A = I$, if inverse does not exist give reasons.

$$i). \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \quad ii). \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

7.6.13 Solution of Simultaneous Linear Equations

To determine the value of two variables, we need a pair of equations. Such a pair of equations is called a system of simultaneous linear equations.

1. The technique of solving a pair of simultaneous equations by
 - a. *Matrix Inversion Method*
 - b. *Cramer's Rule*
2. To apply the technique to solve some practical problems,

Matrix Inversion Method

Let $a_1x + a_2y = b_1$ (i)

and $a_3x + a_4y = b_2$ (ii)

be the two simultaneous linear equations. These equations can be written in matrix form as:

$$\begin{bmatrix} a_1x + a_2y \\ a_3x + a_4y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$i.e. \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Or $AX = B$ (iii)

$$\text{Where } A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

To find values of the variable x and y , the equation (iii) is solved by the following method,

$$AX = B$$

If A has an inverse A^{-1} ,

Then $A^{-1}AX = A^{-1}B$ ($\because A^{-1}A = I$)

$$IX = A^{-1}B \quad (\because IX = X)$$

$$X = \frac{\text{adj } A}{|A|} B \quad \text{provided } |A| \neq 0$$

In case A is singular (OAI 0), then it is not possible to find the solution of the given equations.

Example

Solve the following set of equations using the matrix inversion method.

$$3x - 4y = 7, \quad 5x - 7y = 12$$

Solution:

The given simultaneous equations may be written in matrix form as.

$$\begin{bmatrix} 3 & -4 \\ 5 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

$$AX = B$$

Here

$$A = \begin{bmatrix} 3 & -4 \\ 5 & -7 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & -4 \\ 5 & -7 \end{vmatrix} = -21 + 20 = -1$$

$$|A| = -1 \neq 0$$

As A is non-singular matrix, so the equations can be solved.

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{-1} \begin{bmatrix} -7 & 4 \\ -5 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 7 & -4 \\ 5 & -3 \end{bmatrix}$$

But

$$X = A^{-1}B$$

$$X = \begin{bmatrix} 7 & -4 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

$$X = \begin{bmatrix} 49 - 48 \\ 35 - 36 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Thus

$$X = 1 \quad \text{and} \quad y = -1$$

$$\text{Solution Set} = \{(1, -1)\}$$

Cramer's Rule

Simultaneous linear equations can be solved by Cramer's rule. The method to solve linear equations by Cramer's rule is explained below.

Consider the linear equations.

$$a_1x + a_2y = b_1$$

$$a_3x + a_4y = b_2$$

In matrix form

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$AX = B$$

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix} \text{ provided } |A| \neq 0$$

$$|D_1| = \begin{vmatrix} b_1 & a_2 \\ b_2 & a_4 \end{vmatrix}$$

$$|D_2| = \begin{vmatrix} a_1 & b_1 \\ a_3 & b_2 \end{vmatrix}$$

$$\text{Now, } x = \frac{|D_1|}{|A|} \quad \text{and} \quad y = \frac{|D_2|}{|A|}$$

Example 1:

Use Cramer's rule to solve the following linear equations,

$$x + 3y = 6, \quad 2x + y = 4$$

Solution: $x + 3y = 6$, $2x + y = 4$ in matrix form:

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$AX = B$$

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = 1 - 6 = -5 \neq 0$$

$$\text{Consider } |D_1| = \begin{vmatrix} 6 & 3 \\ 4 & 1 \end{vmatrix} = 6 - 12 = -6$$

$$|D_2| = \begin{vmatrix} 1 & 6 \\ 2 & 4 \end{vmatrix} = 4 - 12 = -8$$

$$x = \frac{|D_1|}{|A|} = \frac{-6}{-5} \quad \text{and} \quad y = \frac{|D_2|}{|A|} = \frac{-8}{-5}$$

$$x = \frac{6}{5}, \quad y = \frac{8}{5}$$

Example 2:

7 apples and 4 pears cost Rs. 11 while the 5 apples and 2 pears cost Rs. 7. How much each apple and pear cost?

Solution: We denote apple by 'x' and pear by 'y'

$$\text{then } 7x + 4y = 11$$

$$5x + 2y = 7$$

In matrix form

$$\begin{bmatrix} 7 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \end{bmatrix}$$

$$AX = B$$

$$|A| = \begin{vmatrix} 7 & 4 \\ 5 & 2 \end{vmatrix} = 14 - 20 = -6 \neq 0$$

Consider $|D_1| = \begin{vmatrix} 11 & 4 \\ 7 & 2 \end{vmatrix} = 22 - 28 = -6$

$$|D_2| = \begin{vmatrix} 7 & 11 \\ 5 & 7 \end{vmatrix} = 49 - 55 = -6$$

$$x = \frac{|D_1|}{|A|} = \frac{-6}{-6} \quad \text{and} \quad y = \frac{|D_2|}{|A|} = \frac{-6}{-6}$$

$$x = 1 \quad \text{one apple costs Rs. 1}$$

$$y = 1 \quad \text{one pear costs Rs. 1}$$

Self-Assessment Exercise 15

Q. 1 Solve the simultaneous equation by matrix inverse method. Where possible, where there is no solution explain why?

$$\begin{array}{ll} \text{i). } 2x - 5y = 1 & \text{ii). } x + y = 2 \\ 3x - 7y = 5 & y = 2 + x \end{array}$$

Q. 2 Use Cramer's rule to solve the simultaneous equations. Give the reason where solution is not possible.

$$\begin{array}{ll} \text{i). } x + 2y = 3 & \text{ii). } -2x + 6y = 5 \\ x + 3y = 5 & x - 3y = -7 \end{array}$$

Activity 16

Q. 1 Write the equation $2x + ky = 7$ and $4x - 9y = 4$ in matrix form. Also find value of k if possible.

7.6.14 Determinants

The determinants of square matrices of order $n \geq 3$, can be written by following the same pattern as already discussed. For example, if $n = 4$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}, \quad \text{then} \quad |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

Now our aim is to compute the determinants of various orders. But before Describing a method for computation of determinants of order $n \geq 3$, we introduce the following definitions.

7.7 INTRODUCTION OF MATRICES AND BASIC OPERATION ON 3X3 MATRICES

7.7.1 Minor and Cofactor of an Element of a Matrix or its Determinant

Number of Element: Let us consider a square matrix A of order 3. Then the minor element a_{ij} , denoted by M_{ij} , is the determinant of the $(3-1) \times (3-1)$ matrix by deleting the i th row and the j th column of A (or $|A|$).

For example, if

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ then the matrix obtained by deleting the first row}$$

$$\text{and the second column of } A \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} \text{ (see adjoining figure) } \begin{bmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{and its determinant is the minor of } a_{12} \text{ that is, } M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

Cofactor of an Element: The cofactor of an element a_{ij} , denoted by A_{ij} is defined by,

$$A_{ij} = (-1)^{i+j} \times M_{ij}$$

Where M_{ij} is the minor of the element a_{ij} of A or $|A|$.

$$\text{For example, } A_{12} = (-1)^{1+2} \times M_{12} = (-1)^3 \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

Determinant of a Square Matrix of Order ≥ 3

The determinant of a square matrix of order n is the sum of the products of each element of row (or column) and its cofactor.

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2j} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3j} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & & a_{nj} & & a_{nm} \end{bmatrix}$$

$$\begin{aligned} |A| &= a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3} + \dots + a_{in}A_{in} \text{ for } i = 1, 2, 3, \dots, n \\ \text{or } |A| &= a_{1j}A_{1j} + a_{2j}A_{2j} + a_{3j}A_{3j} + \dots + a_{nj}A_{nj} \text{ for } j = 1, 2, 3, \dots, n \end{aligned}$$

Putting $i = 1$, we where

$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + \dots + a_{1n}A_{1n}$ which is called the expansion of $|A|$ by the first row

If A is a matrix of order 3, that is, $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then

$$|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3} \quad \text{for } i = 1, 2, 3 \quad (1)$$

$$\text{Or } |A| = a_{1j}A_{1j} + a_{2j}A_{2j} + a_{3j}A_{3j} \quad \text{for } j = 1, 2, 3 \quad (2)$$

For example, for $i = 1$, $j = 1$ and $j = 2$, we have

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} \quad (i)$$

$$\text{Or } |A| = a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31} \quad (ii)$$

$$\text{Or } |A| = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32} \quad (iii)$$

(iii) can be written as:

$$|A| = a_{12}(-1)^{1+2} \times M_{12} + a_{22}(-1)^{2+2} \times M_{22} + a_{32}(-1)^{3+2} \times M_{32}$$

$$\text{i.e. } |A| = -a_{12}M_{12} + a_{22}M_{22} - a_{32}M_{32} \quad (iv)$$

$$\text{Similarly (i) can be written as } |A| = a_{12} - a_{12}M_{12} + a_{13}M_{13} \quad (v)$$

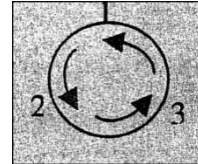
Putting the values of M_{12} , M_{12} and M_{13} in (v), we obtain

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\text{Or } |A| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{23}a_{31}) \quad (vi)$$

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{23}a_{31} \quad (vii)$$

The second scripts of positive terms are in circular order of anti-clockwise direction i.e., these are as 123, 231, 312 adjoining figure) while the second scripts of negative terms are 132 213, 321,



Example 1: Evaluate the determinant of $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & 2 \end{bmatrix}$

$$\text{Solution: } |A| = \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & 2 \end{vmatrix}$$

Using the result (v), that is,

$$|A| = a_{12} - a_{12}M_{12} + a_{13}M_{13}$$

$$\begin{aligned} |A| &= 1 \begin{vmatrix} 3 & 1 \\ -3 & 2 \end{vmatrix} - (-2) \begin{vmatrix} -2 & 1 \\ 4 & 2 \end{vmatrix} + 3 \begin{vmatrix} -2 & 3 \\ 4 & -3 \end{vmatrix} \\ &= 1[6 - 1(-3)] + 2[(-2).2 - 1.4] + 3[(-2)(-3) - 12] \\ &= (6 + 3) + 2(-4 - 4) + 3(6 - 12) \\ &= 9 - 16 - 18 = -25 \end{aligned}$$

Example 2: Find the cofactors A_{11}, A_{22} and A_{32}

$$\text{If } A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & 2 \end{bmatrix} \text{ and find } |A|.$$

Solution: We first find M_{12}, M_{12} and M_{13} ,

$$M_{12} = \begin{vmatrix} -2 & 1 \\ 4 & 2 \end{vmatrix} = -4 - 4 = -8; \quad M_{22} = \begin{vmatrix} 1 & 3 \\ 4 & 2 \end{vmatrix} = 2 - 12 = -10$$

$$\text{And } M_{32} = \begin{vmatrix} 1 & 3 \\ -2 & 1 \end{vmatrix} = 1 - (-6) = 7$$

$$\begin{aligned} \text{Thus } A_{12} &= (-1)^{1+2} M_{12} = (-1)(-8) = 8 \\ A_{22} &= (-1)^{2+2} M_{22} = (1)(-10) = -10 \\ A_{32} &= (-1)^{3+2} M_{32} = (-1)(8) = -7 \end{aligned}$$

$$\begin{aligned} \text{and } |A| &= a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32} = (-2)8 + 3(-10) + (-3)(-7) \\ &= -16 - 30 + 21 = -25 \end{aligned}$$

$$\begin{aligned} \text{Note that } a_{11}A_{12} + a_{21}A_{22} + a_{31}A_{32} &= 1(8) + (-2)(-10) + 4(-7) \\ &= 8 - 20 - 28 = 0 \end{aligned}$$

$$\begin{aligned} \text{And } a_{13}A_{12} + a_{23}A_{22} + a_{33}A_{32} &= 3(8) + 1(-10) + 2(-7) \\ &= 24 - 10 + 14 = 0 \end{aligned}$$

$$\begin{aligned} \text{Similarly we can show that } a_{11}A_{13} + a_{21}A_{23} + a_{31}A_{33} &= 0 \\ a_{11}A_{21} + a_{12}A_{23} + a_{13}A_{23} &= 0 \quad \text{and} \quad a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33} = 0 \end{aligned}$$

7.7.4 Properties of Determinants which Help in their Evaluation

1. For a square matrix A , $|A| = |A^t|$
2. If in a square matrix A , two rows or two columns are interchanged, the determinant of the resulting matrix is $-|A|$
3. If a square matrix A has two identical rows or two identical columns, then $|A| = 0$
4. If all the entries of a row (or a column) of a square matrix A are zero, then $|A| = 0$
5. If the entries of a row (or a column) in a square matrix A are multiplied by a number $k \in \mathcal{R}$, then the determinant of the resulting matrix is $k|A|$.
6. If each entry of a row (or a column) of a square matrix consists of two terms, then its determinant can be written as the sum of two determinants, i.e., if

$$B = \begin{bmatrix} a_{11} + b_{11} & a_{12} & a_{13} \\ a_{21} + b_{21} & a_{22} & a_{23} \\ a_{31} + b_{31} & a_{32} & a_{33} \end{bmatrix}, \text{ then}$$

$$|B| = \begin{vmatrix} a_{11} + b_{11} & a_{12} & a_{13} \\ a_{21} + b_{21} & a_{22} & a_{23} \\ a_{31} + b_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} b_{11} & a_{12} & a_{13} \\ b_{21} & a_{22} & a_{23} \\ b_{31} & a_{32} & a_{33} \end{vmatrix}$$

7. If to each entry of a row (or a column) of a square matrix A is added a non-zero multiple of the corresponding entry of another row (or column) the determinant of the resulting matrix is $|A|$.
8. If a matrix is in triangular form, then the value of its determinant is the product of the entries on its main diagonal.

Self-Assessment Exercise 16

Q. 1 Evaluate the following determinants.

$$i). \begin{vmatrix} 5 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 1 & 2 \end{vmatrix} \quad ii). \begin{vmatrix} 2a & a & a \\ b & 2b & b \\ c & c & 2c \end{vmatrix}$$

Q. 2 Without expansion show that,

$$\begin{vmatrix} 6 & 7 & 8 \\ 3 & 4 & 5 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

Q. 3 Show that, $\begin{vmatrix} a & b+c & a+b \\ b & c+a & b+c \\ c & a+b & c+a \end{vmatrix} = a^3 + b^3 + c^3 + 3abc$

Q. 4 Show that, $\begin{vmatrix} a+l & a & a \\ a & a+l & a \\ a & a & a+l \end{vmatrix} = l^2(3a+l)$

7.8 PARTIAL FRACTIONS OF DIFFERENT TYPES

7.8.1 Introduction

We have learnt in previous classes how to add two or more fractions into a single rational fraction. For example,

$$1 \frac{1}{x-1} + \frac{2}{x+2} = \frac{3}{(x-1)(x+2)}$$

$$2 \frac{2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x-2} = \frac{5x^2+5x-3}{(x+1)^2(x-2)}$$

In this chapter we shall learn how to reverse the order in 1 and 2 that is to express a single rational function as a sum of two or more single rational functions which are called Partial fractions.

7.8.2 Fraction

The quotient of two numbers or algebraic expressions is called a Fraction. The quotient is indicated by a bar (—).we write , the dividend above the bar and the divisor below the bar. For example $\frac{x^2+2}{x-2}$ is a fraction with $x-2$ is denominator. $x-2 \neq 0$.if $x-2=0$,then the fraction is not defined because $x-2=0, x=2$ which makes the denominator of the fraction zero.'

Rational Fraction

An expression of the form $\frac{N(x)}{D(x)}$, where $N(x)$ and $D(x)$ are polynomials in x with real coefficients and $D(x) \neq 0$, is called a rational fraction. For example, $\frac{3}{x+1}$, $\frac{2x-5}{x^2+4}$ and $\frac{9x^2}{x^3-1}$ are proper rational fractions or proper fractions.

7.8.3 Improper Rational Fraction

A rational fraction $\frac{p(x)}{Q(x)}$ is called an Improper Rational Fraction if the degree of polynomial $p(x)$ in the numerator is equal to or greater than the degree of the polynomial $Q(x)$ in the denominator

For example, $\frac{x}{2x-3}$, $\frac{(x-2)(x+1)}{(x-1)(x+4)}$, $\frac{x^2-3}{3x+1}$ and $\frac{x^3-x^2+x+1}{x^2+5}$ are Improper Rational Fractions or Improper Fractions.

Any improper rational fraction can be reduced by division to a mixed form, consisting

Of the sum of polynomial and a proper rational fraction.

For example, $\frac{3x^2+1}{x-2}$ is an improper rational fraction? By long division we obtain

$$\frac{3x^2+1}{x-2} = 3x+6 + \frac{13}{x-2}$$

i.e., an improper rational fraction $\frac{3x^2+1}{x-2}$ has been reduced to the sum of a polynomial

$3x+6$ and a proper rational fraction $\frac{13}{x-2}$

When a rational fraction is separated into of partial fractions, the result is an identity; i.e., it is true for all values of the variable.

The evaluation of the coefficients of the partial fractions is based on the following theorem :

“If two polynomials are equal for all values of the variable, then the polynomials have same degree and the coefficients of like powers of the variable in both the polynomials must be equal.”

For example ,

$$\text{If } px^3 + qx^2 - ax + b = 2x^3 - 3x^2 - 4x + 5, \quad \forall x$$

Then $p=2, q=-3, a=4$ and $b=5$.

7.9 RESOLUTION OF A RATIONAL FRACTION $\frac{p(x)}{Q(x)}$ INTO PARTIAL FRACTIONS

Following are the main points of resolving a rational fraction $\frac{p(x)}{Q(x)}$ into Partial fractions:

1. The degree of $P(x)$ must be less than that of $Q(x)$. If not divide and work with the

- remainder theorem.
2. Clear the given equation of fractions.
 3. Equate the coefficients of like terms.
 4. Solve the resulting equations for the coefficients.

We now discuss the following cases of partial fractions resolution.

Case 1: Resolution of $\frac{p(x)}{Q(x)}$ into Partial fractions when $Q(x)$ has only non –repeated linear factors:

The polynomial $Q(x)$ may be written as :

$$Q(x) = (x-a_1)(x-a_2)\dots\dots\dots(x-a_n), \text{ where } a_1 \neq a_2 \neq \dots \dots \dots \neq a_n$$

$$\frac{p(x)}{Q(x)} = \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \dots\dots\dots + \frac{A_n}{x-a_n} \text{ is an identity.}$$

Where the coefficients $A_1, A_2, \dots \dots \dots, A_n$ are numbers to be found.

The method is explained by the following examples:

Example 1: Resolve $\frac{7x+25}{(x+3)(x+4)}$ into partial fractions.

Solution: Suppose $\frac{7x+25}{(x+3)(x+4)} = \frac{A}{(x+3)} + \frac{B}{(x+4)}$

Multiplying both sides by $(x+3)(x+4)$, we get

$$7x+25 = A(x+4) + B(x+3)$$

$$7x+25 = Ax+4A+Bx+3B$$

$$7x+25 = (A+B)x + 4A+3B$$

This is an identity in x .

So, equating the coefficients of like powers of x we have

$$7 = A+B \quad \text{and} \quad 25 = 4A+3B$$

Solving these equations, we get $A=4$, and $B=3$

Hence the partial fractions are $\frac{4}{(x+3)} + \frac{3}{(x+4)}$

Example 2: Resolve $\frac{x^2-10x+13}{(x-1)(x^2-5x+6)}$ into Partial Fraction

Solution: The Factor $x^2 - 5x + 6$ in the denominator can be factorized and its factors are $x - 3$ and $x - 2$.

$$\frac{x^2-10x+13}{(x-1)(x^2-5x+6)} = \frac{x^2-10x+13}{(x-1)(x-2)(x-3)}$$

Suppose $\frac{x^2-10x+13}{(x-1)(x-2)(x-3)} = \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}$

$$x^2 - 10x + 13 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

This is an identity in x .

Putting $x=1$ in the identity, we get

$$(1)^2 - 10(1) + 13 = A(1-2)(1-3) + B(1-1)(1-3) + C(1-1)(1-2)$$

$$1-10+13 = A(-1)(-2) + B(0)(-2) + C(0)(-1)$$

$$4 = 2A \quad A = 2$$

Putting $x=2$ in the identity, we get

$$(2)^2 - 10(2) + 13 = A(0)(2-3) + B(2-1)(2-3) + C(2-1)(0)$$

$$4-20+13=B(1)(-1)$$

$$-3=-B \quad B=3$$

Putting $x=3$ in the identity, we get

$$(3)^2-10(3)+13=A(3-2)(0)+B(3-1)(0)+C(3-1)(3-2)$$

$$9-30+13=C(2)(1)$$

$$-8=2C \quad C=-4$$

Hence partial fractions are: $\frac{2}{x-1} + \frac{3}{x-2} - \frac{4}{x-3}$

In example 2:

- The denominator of A is $x-1$, and the value of A has been found by putting $x-1=0$ i.e. $x=1$
- The denominator of B is $x-2$, and the value of B has been found by putting $x-2=0$ i.e. $x=2$ and
- The denominator of C is $x-3$, and the value of C has been found by putting $x-3=0$ i.e. $x=3$

Example 3: Resolve $\frac{2x^3+x^2-x-3}{x(2x+3)(x-1)}$ into partial Fractions.

Solution: $\frac{2x^3+x^2-x-3}{x(2x+3)(x-1)}$ is an improper fraction so, transform it into mixed form.

$$\text{Denominator} = x(2x+3)(x-1)$$

$$= 2x^3 + x^2 - 3x$$

Dividing $2x^3 + x^2 - x - 3$ by $2x^3 + x^2 - 3x$, we have

Quotient=1 and Remainder= $2x-3$

$$\frac{2x^3+x^2-x-3}{x(2x+3)(x-1)} = 1 + \frac{2x-3}{x(2x+3)(x-1)}$$

$$\text{Suppose } \frac{2x-3}{x(2x+3)(x-1)} = \frac{A}{x} + \frac{B}{2x+3} + \frac{C}{x-1}$$

$$2x-3 = A(2x+3)(x-1) + B(x)(x-1) + C(x)(2x+3)$$

Which is an identity in x .

Putting $x=0$ in the identity, we get $B = \frac{-8}{5}$

Putting $x-1=0$, $x=1$ in the identity, we get $C = \frac{-1}{5}$

Hence partial fractions are $1 + \frac{1}{x} - \frac{8}{5(2x+3)} - \frac{1}{5(x-1)}$

Self Assessment Exercise 17

Resolve the following into Partial Fractions.

Q. 1 $\frac{1}{x^2-1}$

Q. 2 $\frac{x^2+1}{(x+1)(x-1)}$

Q. 3 $\frac{1}{(x-1)(2x-1)(3x-1)}$

Q. 4 $\frac{(x-1)(x-3)(x-5)}{(x-2)(x-4)(x-6)}$

Activity 17:

Q. 1 $\frac{3x^2-4x-5}{(x-2)(x^2+7x+10)}$

Q. 2 $\frac{x}{(x-a)(x-b)(x-c)}$

Q. 3 $\frac{2x^3+x^2-5x+3}{2x^3+x^2-3x}$

Case 2: When $Q(x)$ has a factor $(x-a)^n, n \geq 2$ and n is a +ve integer, then $\frac{P(x)}{Q(x)}$ may be written as the following identity:

$$\frac{P(x)}{Q(x)} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)} + \dots + \frac{A_n}{(x-a_n)^n}$$

Where the coefficients $A_1, A_2, \dots, A_n$ are numbers to be found.

The method is explained by the following examples.

Example 1: Resolve, $\frac{x^2+x-1}{(x+2)^3}$

Solution: $\frac{x^2+x-1}{(x+2)^3} = \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{(x+2)^3}$

$$x^2 + x - 1 = A(x+2)^2 + B(x+2) + Cs$$

$$x^2 + x - 1 = A(x^2 + 4x + 4) + B(x+2) + C$$

Putting $x+2=0$ in (1), we get

$$(-2)^2 + (-2) - 1 = A(0) + B(0) + C$$

$$C=1$$

Equating the coefficients of x^2 and x in (2), we get $A=1$

And $1=4A+B$

$$1=4+B$$

$$B=-3$$

Hence the partial fractions are $\frac{1}{x+2} - \frac{3}{(x+2)^2} + \frac{1}{(x+2)^3}$

Example 2: Resolve, $\frac{1}{(x+1)^2(x^2-1)}$ into Partial Fractions.

Solution: Here denominator $= (x+1)^2(x^2-1)$

$$= (x+1)^2(x+1)(x-1) = (x+1)^3(x-1)$$

$$\frac{1}{(x+1)^2(x^2-1)} = \frac{1}{(x+1)^3(x-1)}$$

Suppose

$$\frac{1}{(x+1)^3(x-1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

$$1 = A(x+1)^3 + B(x+1)^2(x-1) + C(x-1)(x+1) + D(x-1) \quad (1)$$

$$\Rightarrow 1 = A(x^3 + 3x^2 + 3x + 1) + B(x^3 + x^2 - x - 1) + C(x^2 - 1) + D(x - 1)$$

$$\Rightarrow 1 = (A+B)x^3 + (3A+B+C)x^2 + (3A-B+D)x + (A-B-C-D) \quad (2)$$

Putting $x-1=0 \Rightarrow x=1$ in (1), we get,

$$1 = A(2)^3 \Rightarrow A = \frac{1}{8}$$

Putting $x+1=0 \Rightarrow x=-1$ in (1), we get

$$1 = D(-1-1) \Rightarrow D = \frac{-1}{2}$$

Equating the coefficients of x^3 and x^2 in (2), we get

$$0 = A + B$$

$$\Rightarrow B = -A \quad \Rightarrow B = \frac{-1}{8}$$

$$\text{And } 0 = 3A + B + C \quad \Rightarrow 0 = \frac{3}{8} - \frac{1}{8} + C \quad \Rightarrow C = \frac{-1}{4}$$

Hence the partial fractions are:

$$\frac{\frac{1}{8}}{x-1} + \frac{\frac{-1}{8}}{x+1} + \frac{\frac{-1}{4}}{(x+1)^2} + \frac{\frac{-1}{2}}{(x+1)^3} = \frac{1}{8(x-1)} - \frac{1}{8(x+1)} - \frac{1}{4(x+1)^2} - \frac{1}{2(x+1)^3}$$

Self Assessment Exercise 18:

Resolve the following into Partial Fractions.

Q. 1 $\frac{2x^2 - 3x + 4}{(x-1)^3}$

Q. 2 $\frac{4x}{(x+1)^2(x-1)}$

Q. 3 $\frac{1}{(x-3)^2(x+1)}$

Q. 4 $\frac{x^2}{(x-1)^3(x+1)}$

Q. 5 $\frac{2x^4}{(x-3)(x+2)^2}$

Activity 18:

Q. 1 $\frac{5x^2 - 2x + 3}{(x+2)^3}$

Q. 2 $\frac{x^2}{(x-2)(x-1)^2}$

Q. 3 $\frac{x-1}{(x-2)(x+1)^3}$

Q. 4 $\frac{4x^3}{(x^2-1)(x+1)^2}$

CASE 3: when $Q(x)$ contains non-repeated irreducible quadratic factor

Definition: A quadratic factor is irreducible if it cannot be written as the product of two linear factors with real coefficients. For example, $x^2 + x + 1$ and $x^2 + 3$ are irreducible quadratic factors.

If the polynomial $Q(x)$ contains non-irreducible quadratic factor then

$\frac{p(x)}{Q(x)}$ may be written as the identity having partial fractions of the form:

$\frac{Ax+B}{ax^2+bx+c}$ where A and B are the numbers to be found.

The method is explained by the following examples:

Example 1: Resolve $\frac{3x-11}{(x^2+1)(x+3)}$ into Partial Fractions.

Solution: Suppose $\frac{3x-11}{(x^2+1)(x+3)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+3}$

$$\Rightarrow 3x - 11 = (Ax + B)(x+3) + C(x^2 + 1) \quad (1)$$

$$\Rightarrow 3x - 11 = (A+C)x^2 + (3A+B)x + (3B+C) \quad (2)$$

Putting $x+3=0 \Rightarrow x=-3$ in (1), we get

$$-9-11=C(9+1) \Rightarrow C=-2$$

Equating the coefficients of x^2 and x in (2), we get

$$0=A+C \Rightarrow A=-C \Rightarrow A=2$$

$$\text{And } 3=3A+B \Rightarrow B=3-3A \Rightarrow B=3-6 \Rightarrow B=-3$$

Hence the Partial fractions are : $\frac{2x-3}{x^2+1} - \frac{2}{x+3}$

Example2: Resolve $\frac{4x^2+8x}{x^4+2x^2+9}$ into Partial Fractions.

Solution: Here, denominator $=x^4+2x^2+9=(x^2+2x+3)(x^2-2x+3)$

$$\therefore \frac{4x^2+8x}{x^4+2x^2+9} = \frac{4x^2+8x}{(x^2+2x+3)(x^2-2x+3)}$$

Suppose

$$\frac{4x^2+8x}{(x^2+2x+3)(x^2-2x+3)} = \frac{Ax+B}{x^2+2x+3} + \frac{Cx+D}{x^2-2x+3}$$

$$4x^2+8x = (Ax+B)(x^2-2x+3) + (Cx+D)(x^2+2x+3)$$

$$4x^2+8x = (A+C)x^3 + (-2A+B+2C+D)x^2 + (3A-2B+3C+2D)x + 3B+3D$$

This is an identity in x .

Equating the coefficients of x^3, x^2, x, x^0 in 1, we have

$$0=A+C$$

$$4=-2A+B+2C+D$$

$$8=3A-2B+3C+2D$$

$$0=3B+3D$$

Solving (1),(2),(3) and (4), we get

$$A=1, \quad B=2, \quad C=-1 \quad \text{and} \quad D=-2.$$

Hence the partial fractions are: $\frac{x+2}{x^2+2x+3} + \frac{-x-2}{x^2-2x+3}$

Self Assessment Exercise 19

Resolve the following into Partial Fractions:

Q. 1 $\frac{9x-7}{(x^2+1)(x+3)}$

Q. 2 $\frac{3x+7}{(x^2+4)(x+3)}$

Q. 3 $\frac{x^2+15}{(x^2+2x+5)(x-1)}$

Q. 4 $\frac{x^4}{1-x^4}$

Activity 1:

Q. 1 $\frac{1}{(x^2+1)(x+3)}$

Q. 2 $\frac{x^2}{(x^2+4)(x+2)}$

Q. 3 $\frac{1}{(x-1)^2(x^2+2)}$

Q. 4 $\frac{x^2-2x+3}{x^4+x^2+1}$

CASE4: When $Q(x)$ has repeated irreducible quadratic factors.
If the polynomial $Q(x)$ consists a repeated irreducible quadratic factors

$(a_n x^2 + bx + c), \quad , n \geq$

2 and n is +ve integer, then $\frac{p(x)}{Q(x)}$ may be written as the following identity

$$\frac{p(x)}{Q(x)} = \frac{A_1 x + B_1}{a_1 x^2 + bx + c} + \frac{A_2 x + B_2}{(a_1 x^2 + bx + c)^2} + \cdots + \frac{A_n x + B_n}{(a_n x^2 + bx + c)^n}$$

Where $A_1, B_1, A_2, B_2, \dots, A_n, B_n$ are numbers to be found. The method is explained through the following example:

Example 1: Resolve $\frac{4x^2}{(x^2+1)(x-1)}$ into partial fractions.

Solution: Let $\frac{4x^2}{(x^2+1)(x-1)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2} + \frac{E}{x-1}$

$$\Rightarrow 4x^2 = (Ax+B)(x^2+1)(x-1) + (Cx+D)(x-1) + E(x^2+1)^2$$

$$\Rightarrow 4x^2 = (A+E)x^4 + (-A+B)x^3 + (A-B+C+2E)x^2 + (-A+B-C+D)x + (-B-D+E)$$

Putting $x-1=0 \Rightarrow x=1$ in (1), we get

$$4=E((1+1)^2) \Rightarrow E=1$$

Equating the coefficients of x^4, x^3, x^2, x in (2), we get

$$0=A+E \Rightarrow A=-E \Rightarrow A=-1$$

$$0=-A+B \Rightarrow B=A \Rightarrow B=-1$$

$$4=A-B+C+2E$$

$$\Rightarrow C=4-A+B-2E=4+1-1-2 \Rightarrow C=2$$

$$0=-A+B-C+D$$

$$\Rightarrow D=A-B+C=-1+1+2=2 \Rightarrow D=2$$

$$\text{Hence partial fractions are: } \frac{-x-1}{x^2+1} + \frac{2x+2}{(x^2+1)^2} + \frac{1}{x-1}$$

Self Assessment Exercise 20:

Resolve into Partial Fractions.

Q. 1 $\frac{x^3+2x+2}{(x^2+x+1)^2}$

Q. 2 $\frac{x^2}{(x^2+1)^2(x-1)}$

Q. 3 $\frac{2x-5}{(x^2+2)^2(x-2)}$

Activity 20:

Q. 1 $\frac{8x^2}{(x^2+1)^2(1-x^2)}$

Q. 2 $\frac{4x^4+3x^3+6x^2+5x}{(x-1)(x^2+x+1)^2}$

Q. 3 $\frac{2x^4-3x^3-4x}{(x^2+2)^2(x+1)^2}$

Answer of Self Assessment Exercises

Answer of Self Assessment Exercise 1

1. $(x + y)(3a - 7b)$
2. $(a - x)(x + y)$
3. $(a - 3)(a^2 + 1)$
4. $(x + 2y)(3a - 4b)$
5. $(1 - 7x)^2$
6. $\left(x + \frac{1}{2}\right)^2$
7. $\left(x - \frac{1}{x}\right)^2$
8. $(a + b)(a + b + 2c)$

Activity 1:

1. $(x - 1)(x^2 + x - y)$
2. $(4 - a^3)(2 - a)$
3. $5(2x - 1)^2$
4. $2ab(a - b)^2$

Self assessment Exercise 2

1. $(x + 4)(x + 5)$
2. $(x - 3)(x - 4)$
3. $(x - 13)(x + 5)$
4. $\{(x - 2)(x + 1)\}$
5. $(y - 19)(y + 8)$

Activity 2:

1. $(4 - 2)(3x - 4)$
2. $(x - 6)(5x - 2)$
3. $(2x - 3)(5x + 4)$
4. $(4x - \sqrt{3})(\sqrt{3x} + 2)$

Self Assessment Exercise 3

1. $(2x - y)(4x^2 + 2xy + y^2)$
2. $(3x + 1)(9x^2 - 3x + 1)$
3. $(ab + 8)(a^2b^2 - 8ab + 64)$
4. $(3x - 4y)(9x^2 + 12xy + 16y^2)$
5. $(a - b)[1 - (a^2 + ab + b^2)]$
6. $\left(1 - \frac{4p}{q}\right)\left(1 - \frac{4p}{q} + \frac{16p^2}{q^2}\right)$

Activity 3:

1. $(m + n)(m - n)(m^2 + mn + n^2)$
2. $(x - 3a)(x^2 + 3ax + 9a^2)$
3. $(2x + 3y)(4x^2 + 9y^2 - 6xy - 3)$

Self Assessment Exercise 4

- 1) 3 2) -47 3) 0 4) yes 5) No 6) yes 7) K=1

Activity 4:

R=21

Activity 5:

1. $x - 1$ Is not factor of polynomial.
 2. $x + 1$ Is not factor of polynomial.

Activity 6:

$$x^3 - 8 = (x^2 - 2x + 4)$$

Activity 7:

1. -6 2. -84 3. K=1

Self Assessment Exercise 17

Q. 1 $\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$

Q. 2 $\frac{1}{(x+1)} + \frac{1}{(x-1)}$

Q. 3 $\frac{1}{2(x-1)} - \frac{4}{2x-1} + \frac{9}{2(3x-1)}$

Activity 17:

Q. 1 $\frac{-1}{28(x-2)} + \frac{30}{7(x+5)} - \frac{5}{4(x+2)}$

Q. 2 $\frac{a}{(a-b)(a-c)(x-a)} + \frac{b}{(b-a)(b-c)(x-b)}$

Q. 3 $1 - \frac{1}{x} + \frac{8}{5(2x+3)} + \frac{1}{5(x-1)}$

Self Assessment Exercise 18

Q. 1 $\frac{2}{x-1} + \frac{1}{(x-1)^2} + \frac{3}{(x-1)^3}$

Q. 2 $\frac{1}{x-1} + \frac{1}{x+1} + \frac{3}{(x+1)^2}$

Q. 3 $\frac{1}{16(x-1)} - \frac{1}{16(x-3)} + \frac{3}{4(x-3)^2}$

Q. 4 $\frac{-1}{8(x-1)} + \frac{1}{8(x-1)} + \frac{3}{4(x-1)^2} + \frac{1}{2(x-1)^3}$

Q. 5 $2x-2 + \frac{162}{25(x-3)} + \frac{288}{25(x+2)} - \frac{32}{5(x+2)^2}$

Activity 18:

Q. 1 $\frac{5}{x+2} + \frac{22}{(x+2)^2} + \frac{27}{(x+2)^3}$

Q. 2 $\frac{-3}{x-1} + \frac{1}{(x-1)^2} + \frac{4}{(x-2)}$

$$\begin{aligned} \text{Q. 3 } & \frac{1}{27(x-2)} + \frac{1}{27(x+1)} + \frac{1}{9(x-1)^2} + \frac{2}{3(x+1)^3} \\ \text{Q. 4 } & \frac{2}{(x+1)^3} - \frac{5}{(x+1)^2} + \frac{7}{2(x+1)} + \frac{1}{2(x-1)} \end{aligned}$$

Self Assessment Exercise 19

$$\begin{aligned} \text{Q. 1 } & \frac{17x-6}{5(x^2+1)} - \frac{17}{5(x+3)} \\ \text{Q. 2 } & \frac{-2}{13(x+3)} - \frac{2x+33}{5(x^2+4)} \\ \text{Q. 3 } & \frac{2}{x-1} - \frac{x+5}{(x^2+2x+5)} \\ \text{Q. 4 } & -1 + \frac{1}{4(1-x)} + \frac{1}{4(1+x)} + \frac{2}{2(1+x^2)} \end{aligned}$$

Activity 19:

$$\begin{aligned} \text{Q. 1 } & \frac{1}{2(1+x)} + \frac{1-x}{2(1+x^2)} \\ \text{Q. 2 } & \frac{1}{2(1+x)} + \frac{x-2}{2(x^2+4)} \\ \text{Q. 3 } & \frac{-1}{18(x-1)} + \frac{1}{3(x-1)^2} + \frac{x+4}{18(x^2+2)} \end{aligned}$$

Self Assessment Exercise 20

$$\begin{aligned} \text{Q. 1 } & \frac{x-1}{x^2+x+1} + \frac{2x+3}{(x^2+x+1)^2} \\ \text{Q. 2 } & \frac{1}{4(x-1)} - \frac{x+1}{4(x^2+1)} + \frac{x+1}{2(x^2+1)^2} \\ \text{Q. 3 } & \frac{-1}{36(x-1)} + \frac{x+2}{36(x^2+2)} + \frac{x+14}{6(x^2+2)^2} \end{aligned}$$

Activity 20:

$$\begin{aligned} \text{Q. 1 } & \frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{(x^2+1)} - \frac{4}{(x^2+1)^2} \\ \text{Q. 2 } & \frac{2}{(x-1)} + \frac{2x-1}{x^2+x+1} + \frac{3}{(x^2+x+1)^2} \\ \text{Q. 3 } & \frac{-1}{(x+1)} + \frac{1}{(x-1)^2} + \frac{x}{(x^2+2)} - \frac{2x}{(x^2+2)^2} \end{aligned}$$

FURTHER READINGS

Mathematics: a very short introduction Timothy Gowers (CUP, 2002)

Solving Mathematical Problems Terence Tao (OUP, 2006)

Calculus for the Ambitious T.W. Körner (CUP, 2014)

Mathematical Methods for Science Students G. Stephenson (Longman, 1973)

Probability and Statistics M.R. Spiegel (Schaum's outline series; McGraw-Hill, 1982)

Unit - 08

**DEVELOPING THE CONCEPTS
OF TRIGONOMETRY**

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INTRODUCTION

The word “trigonometry” is a Greek word. Trigonometry means “measurement of a triangle”. It helps us find distances and angles, and is used a lot in Social Sciences like video games, science like Physical, Surveying, Business, Navigation, Astronomy, Engineering, and more. Trigonometry is that branch of mathematics concerned with the measurement of sides and angle of a plane triangle and the investigations of the various relations which exist among them. Currently, the subject of trigonometry also includes another distinct branch and concerns itself with properties relations between and behavior of trigonometric functions.

The importance of trigonometry will be immediately realized when trigonometric applications in solving problem of astronomy, menstruation, surveying, mechanics and physics are encountered.

OBJECTIVES

After completion of the unit you will be able to:

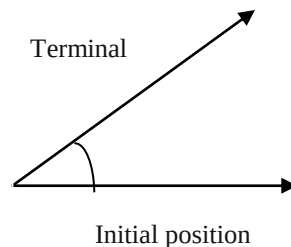
1. Differentiate between different types of trigonometry.
2. Convert angle measured in Degrees to radian and Vice Versa.
3. Define trigonometric functions.
4. Plot different trigonometric functions.
5. Prove trigonometric identities.
6. Apply trigonometry to solve real life problems.

8.1 TYPES OF TRIGONOMETRY

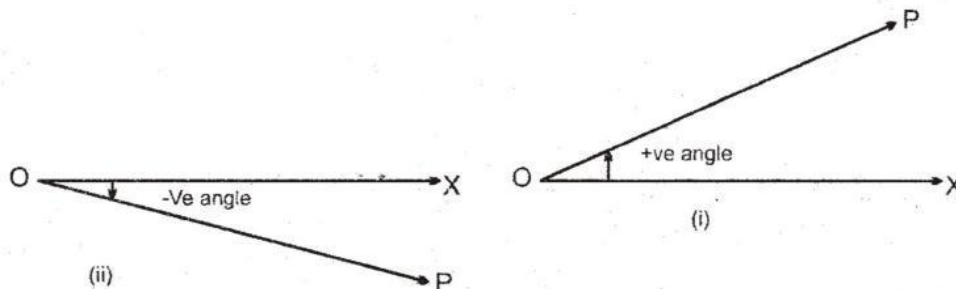
- Plane trigonometry** deals with triangles, angles and other figures which lie in a plane.
- Spherical Trigonometry** deals with the spherical triangles, that is, triangles lie within a sphere and sides of which are circular arcs.

8.2 ANGLE

An angle is defined as the union of two non-collinear rays that have common end-points. An angle measures the rotation of a line from one position to another about a fixed point on it.



In figure 8.1 the OX is called initial line (position) and OP is called terminal line or generating line (position) of $\angle XOP$. If the terminal side resolves in anticlockwise direction the angle described is positive. Which shown in figure (i) (if terminal side resolves in clockwise direction, then the angle described is negative as shown in figure (ii)).



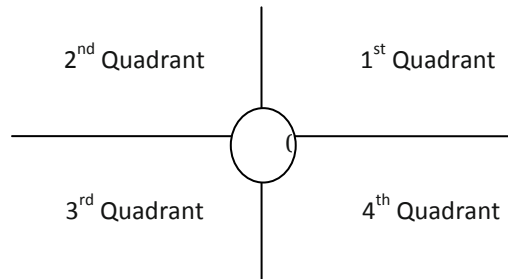
8.3 QUADRANTS

Two mutually perpendicular straight lines xox and yoy that divide the plane into four equal parts, each part is called quadrant. The XOY , $X'OY$, $X'OY'$ and XOY' are called the 1st, 2nd, 3rd and 4th quadrants respectively.

In first quadrant the angles vary from 0° to 90° in anti-clockwise direction and the angle

vary from -270° to -360° in clockwise direction.

In second quadrant the angle vary from 90° to 180° in anti-clockwise direction and angle vary -180° to -270° in clockwise direction.



In the third quadrant the angle vary from 180° to 270° in anti- clockwise direction and the angle vary from -90° to -180° in clockwise direction.

In the fourth quadrant the angle vary from 270° to 360° in anti-clockwise direction and the angle vary from -0° to -90° in clockwise direction.

8.4 MEASUREMENT OF ANGLES

The size of any angle is determined by the amount of rotations and in trigonometry two systems of measuring angles are used.

i. Sexagesimal or English System (Degree):

The sexagesimal system is the older more commonly used and the name derive from the Latin for “sixty”. The fundamental unit of angle measure in the sexagesimal system is the degree of arc when a circle is divided into 360 equal parts,

One degree = $\frac{1}{360}$ th part of a circle.

Then one full circle = 360 degrees. The symbol of degrees is denoted by $(^\circ)$.

Thus an angle of 20 degrees may be written as 20° therefore there are four right angles in a complete circle

$$\text{One right angle} = \frac{1}{4} \text{ circle} = \frac{1}{4} (360^\circ) = 90^\circ$$

The degree is subdivided in two ways that depending upon whether we work in the common sexagesimal system or in the decimal sexagesimal system. The most common sexagesimal system, the degree is subdivided into 60 equal parts which are called minutes, denoted by the symbol $(')$, and the minute is further subdivided into 60 equal parts which are called second and indicated by the symbol $('')$. Therefore

1 minute = 60 seconds

1 degree = 60 minutes = 3600 seconds

1 circle = 360 degrees = 21600 minutes = 12,96,000 sec.

ii. Circular measure system (Radian)

This system is comparatively recent use. The unit used in this system is called a “Radian”. The Radian is define as “The angle subtended at the centre of a circle by an arc equal in length to the radius of the circle” and shown in fig. The arc AB is equal in length to the radius OB of the circle. The angle subtended $\angle AOB$ is then one radian.

i.e. $m \angle AOB = 1$ radian.

8.5 RELATION BETWEEN DEGREE AND RADIAN MEASURE

Consider a circle of radius r and the circumference of the circle is $2\pi r$. By definition of radian, An arc of length ' r ' subtends an angle = 1 radian. Therefore an arc of length $2\pi r$ subtends an angle = 2π radian and an arc of length $2\pi r$ subtends an angle = 360°

Then 2π radians = 360°

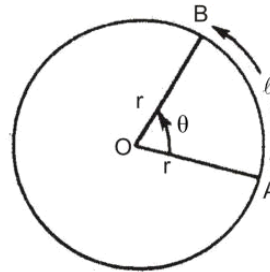
Or π radians = 180°

1 radians = $\frac{\pi}{180^\circ}$

1 radians = $\frac{180}{3.1416}$

Or 1 radians = 57.3°

Therefore to convert radians into degree and we multiply the number of radians by $\frac{180^\circ}{\pi}$ or 57.3.



Now we have to convert it into Radian measure before applying the formula.

Again, $360^\circ = 2\pi$ radians

$$1^\circ = \frac{2\pi r}{360^\circ} \text{ radians}$$

$$1^\circ = \frac{\pi}{180^\circ}$$

$$\text{Or } 1^\circ = \frac{3.1416}{180^\circ}$$

$$1^\circ = 0.01745 \text{ radians}$$

when to convert degree into radians and we multiply the number of degrees by

$$\frac{\pi}{180} \text{ or } 0.0175.$$

Note: One complete revolution = $360^\circ = 2\pi$ radius.

Self Assessment 8.1

Q. 1 Convert the following to Radian measure

- i. 210° ii. 540° iii. $42^\circ 36' 12''$ iv. $24^\circ 32' 30''$

Q. 2 Convert the following to degree measure:

- i. rad ii. rad iii. 5.52 rad iv. 1.30 rad $5\pi 4 2\pi 3$

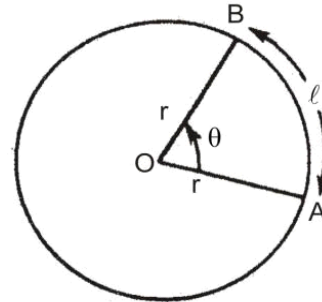
8.6 RELATION BETWEEN LENGTH OF A CIRCULAR ARC AND THE RADIAN MEASURE OF ITS CENTRAL ANGLE

Let “ l ” be the length of a circular arc AB of a circle of r radius, the θ be its central angle measure in radians and the ratio of l to the circumference $2\pi r$ of the circle is the same like the ratio of θ to 2π .

Therefore

$$l : 2\pi r = \theta : 2\pi$$

Note: If the angle will be given in degree measure then We have to convert it into Radian measure before applying the formula.



Example 1:

Convert 120° into Radian Measure.

Solution:

$$\begin{aligned} 120^\circ &= 120 \times \frac{\pi}{180} \\ &= \frac{2\pi}{3} = \frac{2(3.1416)}{3} = 2.09 \text{ rad.} \end{aligned}$$

Example 2:

Convert $37^\circ 25' 38''$ into Radian measure.

Solution:

$$\begin{aligned} 37^\circ 25' 28'' &= 37^\circ + \frac{25}{60} + \frac{28}{3600} \\ &= 37^\circ + \frac{5^\circ}{12} + \frac{7^\circ}{900} \\ &= 37^\circ + \frac{382^\circ}{900} \\ &= 37 + \frac{181^\circ}{450} = \frac{16831^\circ}{450} = \frac{16831}{450} \times \frac{\pi}{180} \\ &= \frac{16831(3.14160)}{81000} = \frac{52876.26}{81000} \\ 37^\circ 25' 28'' &= 0.65 \text{ radians} \end{aligned}$$

Example 3:

Express in Degrees:

- (i) $\frac{5\pi}{3}$ rad (ii) 2.5793 rad (iii) $\frac{\pi}{6}$ rad

Solution:

- (i) Since 1 rad = $\frac{180}{\pi}$ deg
 $\therefore \frac{5\pi}{3}$ rad = $\frac{5\pi}{3} \times \frac{180}{\pi}$ deg = $5(60)$ deg = 300°
- (ii) 2.5793 rad = $2.5793 (57^\circ.29578)$ $\therefore$ 1 rad = $57^\circ.295778$
 $= 147^\circ.78301 = 147.78$ (two decimal places)
- (iii) $\frac{\pi}{6}$ rad = $\frac{\pi}{6} \times \frac{180}{\pi}$ deg = 30°

Example 4:

A curve on a highway is laid out an arc of a circle of radius 620m. How long is the arc that subtends a central angle of 32° ?

Solution: $r = 620\text{m}$ $l = ?$ $\theta = 32^\circ = 32 \times \frac{\pi}{180} \text{ rad}$

$$l = 620 \times 32 \times \frac{\pi}{180} = 346.41 \text{ m}$$

Example 5:

The moon subtends an angle of 0.5° as observed from the Earth. Its distance from the earth is 384400 km. Find the length of the diameter of the Moon.

Solution:

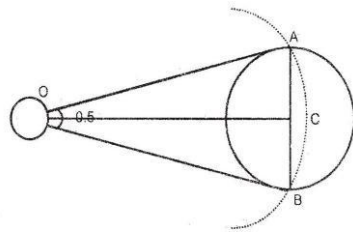


Fig. 4.8

$l = AB = \text{diameter of the Moon} = ?$ as angle 0.5 is very small.

i.e. AB (arc length) consider as a straight line AB

$$\theta = 0.5^\circ = 0.5 \times 0.01745 \text{ rad} = 0.008725 \text{ rad}$$

$r = OC = d = \text{distance between the earth and the moon}$

$$r = OC = 384400 \text{ km}$$

Since $l = r\theta$

$$l = 384400 \times 0.008725 = 3353.89 \text{ km}$$

Class Activity: Trigonometry in flight engineering

Flight engineers have to take in account their distance, direction and speed along with the speed and direction of the wind. The wind plays an very important role in how and when a plane will arrive where ever needed this is solved using vectors to create a triangle using trigonometry to solve. For example, if a plane is travelling at 234 mph, 45 degrees N of E, and there is a wind blowing due south at 20 mph. Then trigonometry will help to solve for that third side of your triangle that will lead the plane in the right direction and the plane will actually travel with the force of wind added on to its course.
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Self Assessment 8.2

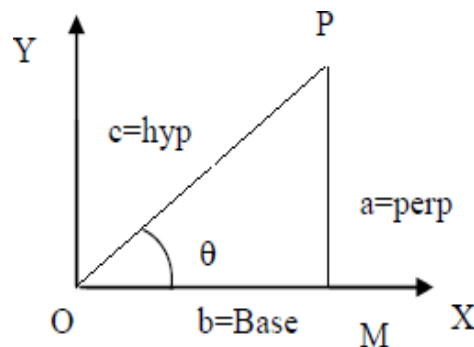
Q. 1 Find the missing element l , r , θ when:

- $l = 8.4 \text{ cm}$, $\theta = 2.8 \text{ rad}$
- $l = 12.2 \text{ cm}$, $r = 5 \text{ cm}$
- $r = 620 \text{ m}$, $\theta = 32^\circ$

- Q. 2 How far apart are two cities on the equator whose longitudes are 10°E and 50°W ? (Radius of the Earth is 6400km)
- Q. 3 A space man land on the moon and observes that the Earth's diameter subtends an angle of $1^\circ 54'$ at his place of landing. If the Earth's radius is 6400km, find the distance between the Earth and the Moon.

8.7 TRIGONOMETRIC FUNCTION AND RATIOS

Let the initial line OX revolves and trace out an angle θ and take a point P on the final line. Draw a perpendicular PM from P on OX: $\angle XOP = \theta$, where θ may be in a degree or radians.



Now OMP is a right angled triangle and We can form the six ratios as follows:

$$\frac{a}{c}, \frac{b}{c}, \frac{a}{b}, \frac{b}{a}, \frac{c}{b}, \frac{c}{a}$$

These ratios depend only on the size of the angle, not on the triangle formed. We called these ratios O trigonometric functions of angle θ or Trigonometric ratios. Therefore defined as below: θ

$$\sin \theta = \frac{a}{c} = \frac{MP}{OP} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{b}{c} = \frac{OM}{OP} = \frac{\text{Base}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{a}{b} = \frac{MP}{OM} = \frac{\text{Perpendicular}}{\text{Base}}$$

$$\cot \theta = \frac{b}{a} = \frac{OM}{MP} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\sec \theta = \frac{c}{b} = \frac{OP}{OM} = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

$$\operatorname{Cosec} \theta = \frac{c}{a} = \frac{OP}{MP} = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$$

8.8 RECIPROCAL FUNCTIONS

From the above definition of trigonometric functions then observe that

- (i) $\sin \theta = \frac{1}{\operatorname{Cosec} \theta}$ or, $\operatorname{Cosec} \theta = \frac{1}{\sin \theta}$ i.e. $\sin \theta$ and $\operatorname{Cosec} \theta$ are reciprocal of each other.
- (ii) $\cos \theta = \frac{1}{\sec \theta}$ or, $\sec \theta = \frac{1}{\cos \theta}$ i.e. $\cos \theta$ and $\sec \theta$ are reciprocals of each other.
- (iii) $\tan \theta = \frac{1}{\cot \theta}$ or, $\cot \theta = \frac{1}{\tan \theta}$ i.e. $\tan \theta$ and $\cot \theta$ are reciprocals of each other.

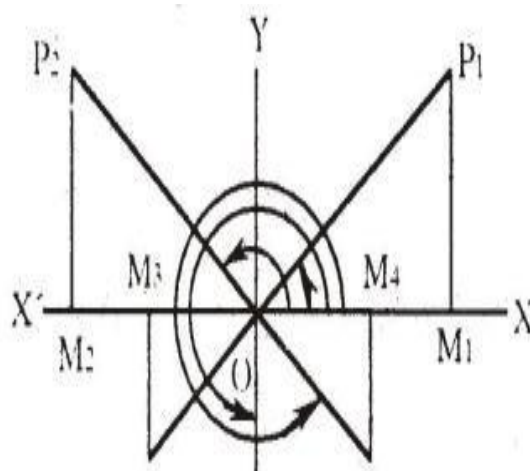
We can also see that;

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \text{and} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

8.9 RECTANGULAR CO-ORDINATES AND SIGN CONVENTION

In plane geometry the position of a point can be fixed by measuring its perpendicular distance from each of two perpendicular which is called co-ordinate axes and the horizontal line (x-axis) is called abscissa, the vertical line (y-axis) is called ordinate. The distance measured from the point O in the direction OX and OY are regarded positive, while in the direction of OX' and OY' are considered as negative.

Thus in the figure OM₁, OM₄, MP₁ and M₂P₂ are positive and OM₂, OM₃, M₃P₃ and M₄P₄ are negative. The terminal line OP₁, OP₂, OP₃, and OP₄ are positive in all the quadrants.

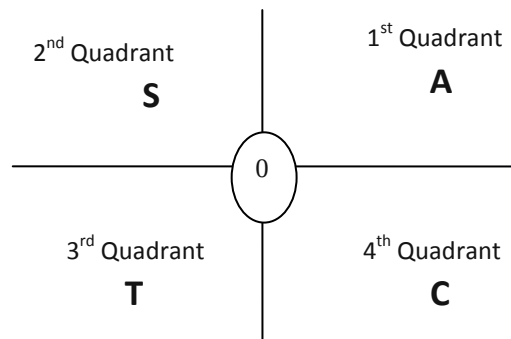


8.10 SIGNS OF TRIGONOMETRIC FUNCTIONS

The trigonometric ratios discussed above have different signs in different quadrants and in the above discussion see that OM and MP changes their sign in different quadrants. We can remember that the sign of trigonometric function by “CAST” or “ACTS” rule. The “CAST” C stands for cosine, A stands for All, S stands for Sine and T stands for Tangent.

First Quadrant:

In the first quadrant sign of all the trigonometric functions are all positive i.e., sin, cos, tan, Cot, Sec, Cosec.



Second Quadrant:

In the second quadrant sine and its inverse cosec are positive and the remaining four trigonometric function i.e., cos, tan, cot, sec are negative.

Third Quadrant:

In the third quadrant tan and its reciprocal cot are positive and the remaining four function i.e., Sin, cos, sec and cosec are negative.

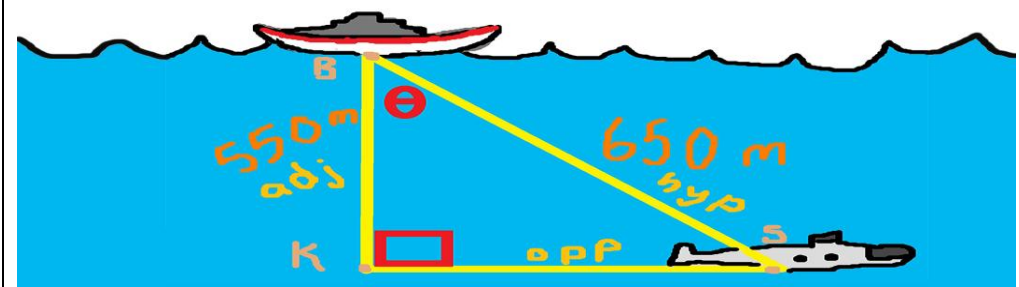
Fourth Quadrant:

In fourth quadrant cos and its reciprocal sec are positive. The remaining four functions i.e., sin, tan, cot and cosec are negative.

Trigonometric Ratios of Particular Angles:

Class Activity: Trigonometry in marine engineering:

In the marine engineering trigonometry is used to build and navigate marine vessels. To be more specific trigonometry is used to design the a Marine ramp which is a sloping surface to connect lower and higher level areas and it can be a slope or even a staircase depending on its application.



8.11 TRIGONOMETRIC RATIOS OF PARTICULAR ANGLES

8.11.1 Trigonometric Ratios of 30° or $\frac{\pi}{6}$

Let the initial line OX revolve and trace out an angle of 30° and take a point P on the final line then draw a perpendicular PQ from P on OX. The right angled triangle 30° and the side opposite to the 30° angle is one-half the length of the hypotenuse, i.e., if PQ = 1 unit and OP will be 2 units.

From fig. OPQ is a right angled triangle

By using Pythagorean theorem,

$$(OP)^2 = (OQ)^2 + (PQ)^2$$

$$(2)^2 = (OQ)^2 + (1)^2$$

$$4 = (OQ)^2 + 1$$

$$(OQ)^2 = 3$$

$$\sqrt{(OQ)} =$$

$$\sin 30^\circ = \frac{\text{Prep.}}{\text{Hyp}} = \frac{PQ}{OP} = \frac{1}{2}$$

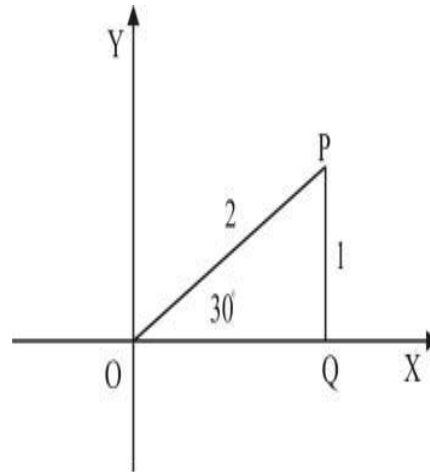
$$\cos 30^\circ = \frac{\text{Base}}{\text{Hyp}} = \frac{OQ}{OP} = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{\text{Prep.}}{\text{Base}} = \frac{PQ}{OQ} = \frac{1}{\sqrt{3}}$$

$$\cot 30^\circ = \frac{\text{Base}}{\text{Prep.}} = \frac{OQ}{PQ} = \frac{\sqrt{3}}{1} = \sqrt{3}$$

$$\sec 30^\circ = \frac{\text{Hyp.}}{\text{Base}} = \frac{OP}{OQ} = \frac{2}{\sqrt{3}}$$

$$\operatorname{cosec} 30^\circ = \frac{\text{Hyp.}}{\text{Prep.}} = \frac{OP}{PQ} = \frac{2}{1} = 2$$



8.11.2 Trigonometric Ratios of 45° or $\frac{\pi}{4}$

Let the initial line OX revolve, trace out an angle of 45° and take a point P on the final line. Draw a PQ perpendicular from P on OX. In the right angled triangle 45° and length of the perpendicular is equal to the length of the base i.e.,

if PQ = 1 unit. So OQ = 1 unit From figure by using Pythagorean theorem.

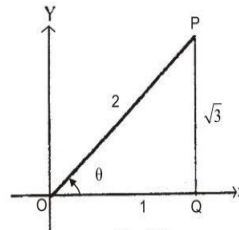
$$(OP)^2 = (OQ)^2 + (PQ)^2$$

$$(OP)^2 = (1)^2 + (1)^2 = 1 + 1$$

$$OP = \sqrt{2}$$

$$\text{Then } \sin 45^\circ = \frac{\text{Prep.}}{\text{Hyp}} = \frac{PQ}{OP} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{\text{Base}}{\text{Hyp}} = \frac{OQ}{OP} = \frac{1}{\sqrt{2}}$$



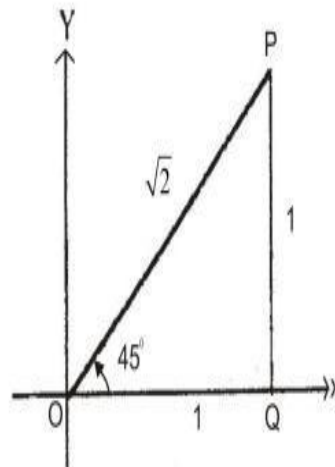
$$\begin{aligned}\tan 45^\circ &= \frac{\text{Prep.}}{\text{Base}} = \frac{PQ}{OQ} = \frac{1}{1} = 1 \\ \cot 45^\circ &= \frac{\text{Base}}{\text{Prep.}} = \frac{OQ}{PQ} = \frac{1}{1} = 1 \\ \sec 45^\circ &= \frac{\text{Hyp.}}{\text{Base}} = \frac{OP}{OQ} = \frac{\sqrt{2}}{1} = \sqrt{2} \\ \operatorname{cosec} 45^\circ &= \frac{\text{Hyp.}}{\text{Prep.}} = \frac{OP}{PQ} = \sqrt{2} = \sqrt{2}\end{aligned}$$

8.11.3 Trigonometric Ratios of 60° or $\frac{\pi}{3}$:

Let the initial line OX revolve, trace out an angle of 60° and take a point P on the final line. Then draw a PQ perpendicular from P on OX. In $^\circ$ right angle triangle 60 the length of the base is one-half of the Hypotenuse i.e., OQ = Base = 1 unit and , OP= Hyp = 2 units from figure by using Pythagorean Theorem:

$$\begin{aligned}(OP)^2 &= (OQ)^2 + (PQ)^2 \\ (2)^2 &= (1)^2 + (PQ)^2 \\ 4 &= 1 + (PQ)^2 \\ (PQ)^2 &= \sqrt{3} \quad PQ = \sqrt{3}\end{aligned}$$

$$\begin{aligned}\text{Then Sin } 60^\circ &= \frac{\text{Prep.}}{\text{Hyp}} = \frac{PQ}{OP} = \frac{\sqrt{3}}{2} \\ \text{Cos } 60^\circ &= \frac{\text{Base}}{\text{Hyp}} = \frac{OQ}{OP} = \frac{1}{2} \\ \tan 60^\circ &= \frac{\text{Prep.}}{\text{Base}} = \frac{PQ}{OQ} = \frac{\sqrt{3}}{1} = \sqrt{3} \\ \cot 60^\circ &= \frac{\text{Base}}{\text{Prep.}} = \frac{OQ}{PQ} = \frac{1}{\sqrt{3}} \\ \sec 60^\circ &= \frac{\text{Hyp.}}{\text{Base}} = \frac{OP}{OQ} = \frac{2}{1} = 2 \\ \operatorname{cosec} 60^\circ &= \frac{\text{Hyp.}}{\text{Prep.}} = \frac{OP}{PQ} = \frac{2}{\sqrt{3}}\end{aligned}$$

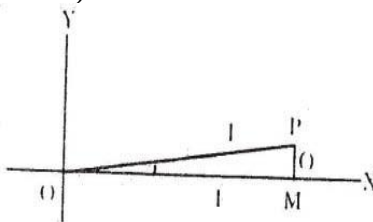


8.11.4 Trigonometric Ratios of 0° :

Let the initial line revolve, trace out a small angle nearly equal to zero 0° . Then take a point P on the final line and draw a PM perpendicular on OX i.e., PM = 0 , OP = 1 and OM = 1 (Because they just coincide x-axis)

Therefore from figure.

$$\begin{aligned}\sin 0^\circ &= \frac{\text{Prep.}}{\text{Hyp}} = \frac{PM}{OP} = \frac{0}{1} = 0 \\ \cos 0^\circ &= \frac{\text{Base}}{\text{Hyp}} = \frac{OM}{OP} = \frac{1}{1} = 1 \\ \tan 0^\circ &= \frac{\text{Prep.}}{\text{Base}} = \frac{PM}{OM} = \frac{0}{1} = 0\end{aligned}$$



$$\begin{aligned}\text{Cot } 0^\circ &= \frac{\text{Base}}{\text{Prep.}} = \frac{\text{OM}}{\text{PM}} = \frac{1}{0} = \infty \\ \text{Sec } 0^\circ &= \frac{\text{Hyp.}}{\text{Base}} = \frac{\text{OP}}{\text{OM}} = \frac{1}{1} = 1 \\ \text{Cosec } 0^\circ &= \frac{\text{Hyp.}}{\text{Prep.}} = \frac{\text{OP}}{\text{PM}} = \frac{1}{0} = \infty\end{aligned}$$

8.11.5 Trigonometric Ratio of 90° :

Let initial line revolve, trace out an angle nearly equal to 90° and take a point P on the final line. Then Draw a PQ perpendicular from P on OX i.e.,

OQ = 0, and OP = 1, PQ = 1 (Because they just coincide y-axis).

Therefore $\text{Sin } 90^\circ = \frac{\text{Prep.}}{\text{Hyp.}} = \frac{\text{PQ}}{\text{OP}} = \frac{1}{1} = 1$

$$\begin{aligned}\text{Cos } 90^\circ &= \frac{\text{Base}}{\text{Hyp.}} = \frac{\text{OQ}}{\text{OP}} = \frac{0}{1} = 0 \\ \tan 90^\circ &= \frac{\text{Prep.}}{\text{Base}} = \frac{\text{PQ}}{\text{OQ}} = \frac{1}{0} = \infty\end{aligned}$$

$$\text{Cot } 90^\circ = \frac{\text{Base}}{\text{Prep.}} = \frac{\text{OQ}}{\text{PQ}} = \frac{0}{1} = 0$$

$$\text{Sec } 90^\circ = \frac{\text{Hyp.}}{\text{Base}} = \frac{\text{OP}}{\text{OQ}} = \frac{1}{0} = \infty$$

$$\text{Cosec } 90^\circ = \frac{\text{Hyp.}}{\text{Prep.}} = \frac{\text{OP}}{\text{PQ}} = \frac{1}{1} = 1$$

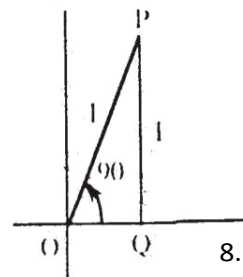


Table for Trigonometrically Ratios of Special angle

Angles Ratios	0°	30° Or $\frac{\pi}{6}$	45° Or $\frac{\pi}{4}$	60° Or $\frac{\pi}{3}$	90° Or $\frac{\pi}{2}$
Sin θ	$\sqrt{\frac{0}{4}} = 0$	$\sqrt{\frac{1}{4}} = \frac{1}{2}$	$\sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$	$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$	$\sqrt{\frac{4}{4}} = 1$
Cos θ	$\sqrt{\frac{4}{4}} = 1$	$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$	$\sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$	$\sqrt{\frac{1}{4}} = \frac{1}{2}$	$\sqrt{\frac{0}{4}} = 0$
Tan θ	$\sqrt{\frac{0}{4}} = 0$	$\sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$	$\sqrt{\frac{2}{2}} = 1$	$\sqrt{\frac{3}{1}} = \sqrt{3}$	$\sqrt{\frac{4}{0}} = \infty$

Example 1: If $\cos \theta = \frac{5}{13}$, the terminal side of the angle lies in the first

Quadrant find the values of the other five trigonometric ratio of θ .

Solution: In this cause $\cos \theta = \frac{5}{13}$

$$\cos \theta = \frac{\text{Base}}{\text{Hyp}} = \frac{5}{13}$$

Therefore from Fig.

$$(\text{OP})^2 = (\text{OQ})^2 + (\text{PQ})^2$$

$$(13)^2 = (5)^2 + (\text{PQ})^2$$

$$169 = 25 + (\text{PQ})^2$$

$$(\text{PQ})^2 = 169 - 25$$

$$= 144$$

$$\text{PQ} = \pm 12$$

Because θ lies in the first quadrant

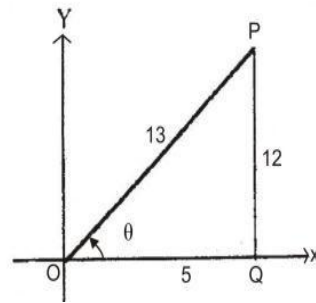
$$\text{i.e., } \sin \theta = \frac{12}{13}$$

All the trigonometric ratios will be positive in first quadrant.

$$\cos \theta = \frac{5}{13}$$

$$\tan \theta = \frac{12}{5}, \quad \sec \theta = \frac{13}{5}$$

$$\cot \theta = \frac{5}{12}, \quad \text{cosec} \theta = \frac{13}{12}$$



Example 3:

Verify that $\sin^2 30^\circ + \sin^2 60^\circ + \tan^2 45^\circ = 2$

Solution:

$$\begin{aligned} \text{L.H.S} &= \sin^2 30^\circ + \sin^2 60^\circ + \tan^2 45^\circ \\ &= \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2 \\ &= \frac{1}{4} + \frac{3}{4} + 1 \\ &= \frac{1+3+4}{4} \\ &= \frac{8}{4} \end{aligned}$$

Class Activity

What is the height of the tree on the left?

$$\tan(16^\circ) = 14 / \text{height}$$

$$\text{height} = 14 / \tan(16^\circ) = 48.8''$$

Self Assessment 8.3

Q. 1 If $\sin \theta = \frac{2}{3}$ the terminal side of the angle lies in the second quadrant, find out the remaining trigonometric ratios of θ .

Q. 2 If $\cos \theta = -\frac{\sqrt{3}}{2}$, the terminal side of the angle lies in the third quadrant, find out the remaining trigonometric ratios of θ .

Q. 3 Prove that:

i. $4 \tan 60^\circ \tan 30^\circ \tan 45^\circ \sin 30^\circ \cos 60^\circ = 1$

iii. $2 \sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ = \frac{3}{\sqrt{2}}$

Q. 4 Evaluate

i. $\cos 30^\circ \cos 60^\circ - \sin 30^\circ \sin 60^\circ$

ii. $\frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ}$

8.12 FUNDAMENTAL IDENTITIES

For any real number θ then we shall derive the following three fundamental identities

i. $\cos^2 \theta + \sin^2 \theta = 1$

ii. $\sec^2 \theta = 1 + \tan^2 \theta$

iii. $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

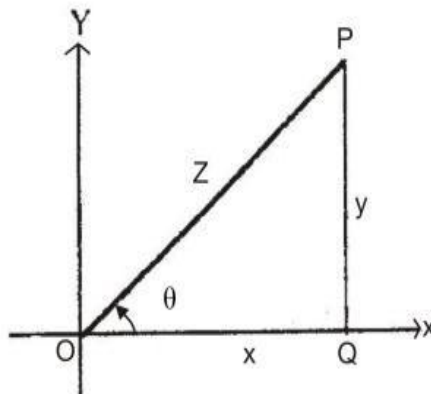
Proof :

Consider an angle $\angle XOP = \theta$ in the standard position and take a point P on the terminal line of the angle θ . Then Draw a PQ perpendicular from P on OX.

From the fig. $\triangle OPQ$ is a right angled triangle. By applying Pythagoras theorem

$$(OP)^2 = (OQ)^2 + (PQ)^2$$

$$\text{Or, } z^2 = x^2 + y^2$$



(i) Dividing both sides by z^2

$$\text{then } \frac{z^2}{z^2} = \frac{x^2}{z^2} + \frac{y^2}{z^2}$$

$$1 = \left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2$$

$$1 = (\cos \theta)^2 + (\sin \theta)^2$$

$$1 = \cos^2 \theta + \sin^2 \theta$$

or, $\cos^2 \theta + \sin^2 \theta = 1$

(ii) Dividing both sides of Eq. (i) by x^2 , we have

$$\frac{z^2}{x^2} = \frac{x^2}{x^2} + \frac{y^2}{x^2}$$

$$\left(\frac{z}{x}\right)^2 = 1 + \left(\frac{y}{x}\right)^2$$

$$(\sec \theta)^2 = 1 + (\tan \theta)^2$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

Example 1:

Prove that $\frac{\sin x}{\operatorname{cosec} x} + \frac{\cos x}{\sec x} = 1$

Solution:

$$\text{L.S.H.} = \frac{\sin x}{\operatorname{cosec} x} + \frac{\cos x}{\sec x}$$

$$= \sin x \cdot \frac{1}{\operatorname{cosec} x} + \cos x \cdot \frac{1}{\sec x} \because \frac{1}{\operatorname{cosec} x} = \sin x$$

$$= \sin x \cdot \sin x + \cos x \cdot \cos x \because \frac{1}{\sec x} = \cos x$$

$$= \sin^2 x + \cos^2 x$$

$$= 1$$

Q.E.D.

Example 2:

prove that $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sec \theta - \tan \theta$

Solution:

$$\begin{aligned}
 \text{L.H.S. } & \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} \\
 &= \sqrt{\frac{(1 - \sin \theta)(1 - \sin \theta)}{(1 + \sin \theta)(1 - \sin \theta)}} \\
 &= \sqrt{\frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta}} = \sqrt{\frac{(1 - \sin \theta)^2}{\cos^2 \theta}} \\
 &= \frac{(1 - \sin \theta)}{\cos \theta} = \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \\
 &= \sec \theta - \tan \theta = \text{R.H.S.}
 \end{aligned}$$

Self Assessment 8.4

Prove the following Identities:

Q. 1 $1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$

Q. 2 $\cos^4 \theta - \sin^4 \theta = 1 - 2 \sin^2 \theta$

Q. 3 $\frac{1}{\tan \theta + \cot \theta} = \sin \theta \cos \theta$

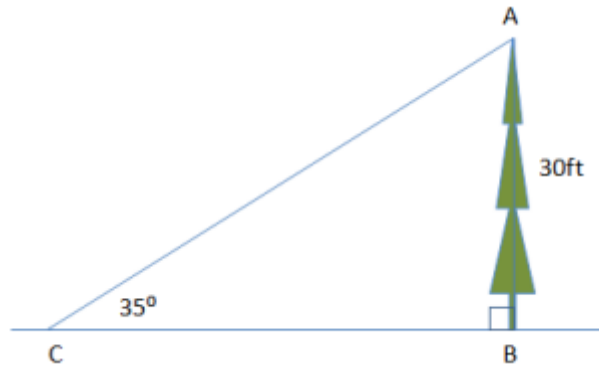
Q. 4 $(1 - \sin^2 \theta)(1 + \tan^2 \theta) = 1$

8.13 APPLICATION OF TRIGONOMETRY

The fundamental trigonometric functions i.e., sine and cosine are used to describe the light and sound waves. Trigonometry is used in oceanography to calculate heights of waves, tides in oceans. It is used in the creation of maps. Trigonometric is also used in satellite systems. The sine and cosine functions are fundamental to the theory of periodic functions which describe the sound and light waves and calculus is made up of Trigonometry and Algebra.

In its geometry application, it is mainly used to solve triangles like right triangles. That is given some angles and side lengths and we can find some or all the others.

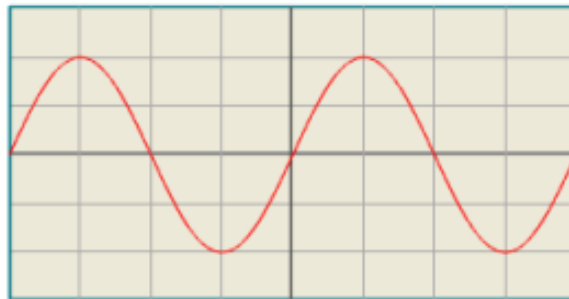
For example, in the figure below, knowing the height of the tree and the angle made and when we look up at its top so we can calculate how far away it is (CB). Using our full toolbox we can actually calculate all sides and all three angles of the right triangle ABC.



In a more advanced use of trigonometric ratios such as Sine and Tangent are used as functions in equations and is manipulated using algebra. In this way, we have many engineering applications such as electronic circuits and mechanical engineering. In this analytical application deals with angles drawn on a coordinate plane and can be used to analyze things like motion and waves.

For example a radio or sound wave can be described by an equation $y = \sin(ax+b)$

When graphed might look something like the wave below. The values of a,b and c determine as the wave's amplitude, frequency and phase shift.



SUMMARY

Trigonometry means as “measurement of triangles”.

1. Radian is an angle subtended at the center of a circle by an arc of the circle equal in length to its radius.
 i.e. π Radian = 180 degree
 $1 \text{ rad} = 57^\circ 17' 45''$
 $1 \text{ degree} = 0.01745 \text{ radian}$
2. Length of arc of the circle is , $l = s = r\theta$
3. Trigonometric functions are defined:
 $\sin\theta = \frac{AP}{OP}$, $\operatorname{Cosec}\theta = \frac{OP}{AP}$

$$\cos\theta = \frac{OA}{OP}, \quad \sec\theta = \frac{OP}{OA}$$

$$\tan\theta = \frac{AP}{OA}, \quad \cot\theta = \frac{OA}{AP}$$

4. Relation between trigonometric ratios:

$$\sec\theta = \frac{1}{\cos\theta}, \quad \operatorname{cosec}\theta = \frac{1}{\sin\theta}$$

$$\cot\theta = \frac{1}{\tan\theta}, \quad \cos\theta = \frac{1}{\sec\theta}$$

$$\sin\theta = \frac{1}{\operatorname{cosec}\theta}, \quad \tan\theta = \frac{1}{\cot\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\sec^2\theta = 1 + \tan^2\theta$$

$$\operatorname{cosec}^2\theta = 1 + \cot^2\theta$$

5. Signs of the trigonometric functions in the Four Quadrants.

Quadrant	I	II	III	IV
Positive	All +ve	$\sin\theta$, $\operatorname{cosec}\theta$	$\tan\theta$, $\cot\theta$	$\cos\theta$, $\sec\theta$
Negative	Nil	$\cos\theta$ $\sec\theta$ $\tan\theta$ $\cot\theta$	$\cos\theta$ $\sec\theta$ $\sin\theta$ $\operatorname{cosec}\theta$	$\sin\theta$ $\operatorname{cosec}\theta$ $\tan\theta$ $\cot\theta$

EXERCISE

Write the short answers of the following:

Q. 1 Define degree and radians measure

Q. 2 Convert into radius measure.

(a) 120° , (b) $22\frac{10}{2}$, (c) $12^\circ 40'$, (d) $42^\circ 36' 12''$

Q. 3 Convert into degree measure

(a) $\frac{\pi}{2}$ rad, (b) 0.726 rad. (c) $\frac{2\pi}{3}$ rad.

Q. 4 Prove that $\ell = r\theta$

Q. 5 What is the length of an arc of a circle of radius 5 cm whose central angle is 140° ?

Q. 6 Find the length of the equatorial arc subtending an angle 1° at the centre of the earth taking the radius of earth as 6400 KM.

Q. 7 Find the length of the arc cut off on a circle of radius 3 cm by a central angle of 2 radius.

Q. 8 Find the radius of the circle when $\ell = 8.4$ cm, $\theta = 2.8$ rad

Q. 9 If a minute hand of a clock is 10 cm long, how far does the tip of the hand move in 30 minutes?

Q. 10 Find x, if $\tan^2 45^\circ - \cos^2 60^\circ = x \sin 45^\circ \cos 45^\circ \tan 60^\circ$.

Q. 11 Find r when $\ell = 33$ cm. $\theta = 6$ radian

- Q. 12 Prove that $2 \sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ = \frac{3}{\sqrt{2}}$
- Q. 13 prove that $\cos 30^\circ \cos 60^\circ - \sin 30^\circ \sin 60^\circ = 0$
- Q. 14 Prove that $\cos 90^\circ - \cos 30^\circ = -2 \sin 60^\circ \sin 30^\circ$
- Q. 15 Prove that $\sin^2 \theta + \cos^2 \theta = 1$
- Q. 16 Prove that: $1 + \tan^2 \theta = \sec^2 \theta$
- Q. 17 Prove that $1 + \cot^2 \theta = \operatorname{Cosec}^2 \theta$
- Q. 18 Prove that: $(1 + \sin \theta)(1 - \sin \theta) = 1 \sec^2 \theta$
- Q. 19 Show that: $\cot^4 \theta + \cot^2 \theta = \operatorname{Cosec}^4 \theta - \operatorname{cosec}^2 \theta$
- Q. 20 Prove that: $\cos \theta + \tan \theta \sin \theta = \sec \theta$
- Q. 21 Prove that $1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$
- Q. 22 $\cos^4 \theta - \sin^4 \theta = 1 - 2 \sin^2 \theta$

ANSWERS

- | | | | |
|-------------------|--------------------------|----------------|-----------------|
| 2. (a) 2.09 rad | (b) 0.39 rad | (c) 0.22 rad | (d) 0.74 radius |
| 3. (a) 90° | (b) $41^\circ 35' 48''$ | (c) 120 degree | |
| 5. 12.21 cm. | 6. 111.7 Km | 7. 6 cm | 8. 3cm. |
| 9. 31.4 cm | 10. $\frac{\sqrt{3}}{2}$ | 11. 5.5 cm. | |

FURTHER READINGS

Trigonometry by Ron Larson

Algebra and Trigonometry by Robert F. Blitzer

Trigonometry by I.M. Gelfand, Mark Saul

Trigonometry for Dummies by Mary Jane Sterling

Unit - 09

**DEVELOPING THE CONCEPTS
OF GEOMETRY**

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INTRODUCTION

Geometry has always been a great source of interest and fascination. It has an immediate intuitive appeal at a simple level. From an early age children play with shapes, noting their obvious properties and seeing how they relate to each other.

The forth coming unit will equip the students about the fundamental concepts of geometry. In this unit the origin, types, history and need of geometry is also elaborated . Basic definitions like, point, line, ray, segment, angles and types of angles etc have been mentioned in depth .This unit also contains the applications of geometry in daily life and in advanced studies .

The concept of similar and congruent triangles and table of similarity versus congruence is included. Also test to prove a triangle is similar / congruent is the part of unit. Collinear points and ways to prove that given points are collinear and a comparison of collinear and non collinear points is there in the unit . As collinear points have a link with line and distance formula, therefore without mentioning distance formula the notion of collinear points would be meaningless. Use of distance formula to show given points are collinear are maintained in the unit.

A straight line and its midpoint have been explained with the help of mathematical formulae. Different forms of a straight line with examples have been presented. Solved examples have been added in each topic.

Without going to discuss about theorems and their proofs , geometry cannot be explained satisfactorily . So theorem, types of theorems and proofs and types of proofs will also equip the students.

Besides the above, each topic has been explained with the help of diagrams and mathematical expressions (where applicable) .Self-assessment Questions have been added at the end of each topic .

Teaching suggestions have also been added at the end of each topic mentioning the different techniques for teachers to teach the students. As geometry has no limit students may further explore the geometry with the help of list of books given at the end of the unit.

OBJECTIVES

After the completion of this unit you will be able to:

1. Define Geometry
2. Understand types of geometry
3. Know history of geometry
4. Know need of geometry
5. Explain application of geometry
6. Explain basic concepts of geometry
7. Find similarities between two triangles
8. Find congruency between two triangles
9. Define different types of angles
10. Define complementary, supplementary, obtuse and acute angles
11. Distinguish between different types of triangles
12. Differentiate between collinear and Non collinear points
13. Use distance formula to show that given three (or) more points are collinear
14. Find distance between two given points
15. Explain applications of distance formula
16. Use distance formula to show that the given three non collinear points form
 - An equilateral triangle
 - An isosceles triangle
 - A right angled triangle
 - A scalene triangle
17. Find midpoint of two given point
18. Explain equation of straight line
19. Distinguish between different forms of straight lines
20. Explain mathematical theorems
21. Describe types of mathematical theorems
22. Solve / teach different types of mathematical theorems
23. Interpret the geometrical proofs
24. Classify the different types of proof with example

9.1 INTRODUCTION TO GEOMETRY

The word geometry is derived from two Greek words namely Geo (earth) and Metron (measurement). Geometry is the branch of mathematics, that focuses on the study of shape, size relative configurations, and spatial properties, position of geometric figures.

Geometry is perhaps the most seasoned science. It was first officially composed by the Greek mathematician Euclid around 300 BC when he organized 465 geometric recommendations into 13 books, titled 'Components'. This, nonetheless, was not the first run through geometry had been used. Truly, there exists proof to accept that geometry dates right back to 3,000 BC in antiquated Mesopotamia, Egypt.

Geometry has been the subject of innumerable advancements. Thus, numerous sorts of geometry exist, including Euclidean geometry, non-Euclidean geometry, Riemannian geometry, mathematical geometry, and logical geometry.

This unit essentially centers around the properties of lines and points. We will likewise put accentuation on geometric estimations including lengths, regions, and volumes of different shapes. Before the finish of this unit it won't be difficult to see that geometry is surrounding us.

9.1.1 Need of Geometry

The expanded requirement for solid geometry getting the hang of starting in early tutoring presents issues for educators who are generally underprepared to decipher geometry substance and successfully convey basic ideas of geometry to them. It makes sense that if educators are underprepared to show geometry because of their own meager early learning and the nonappearance of basic geometry learning in instructor training programs, understudy learning will be unfavorably influenced. In this way, it is progressively imperative to successfully instruct or re-teach educators in the geometry content territory paying little respect to their understudies' ages. It is additionally essential to comprehend the extent of impact on satisfactory instructor geometry accomplishment. The motivation behind this unit is to comprehend both subjective and full of feeling factors that impact the geometry learning of educators who are in charge of helping understudies create solid geometry content information.

9.1.2 Branches of Geometry

Modern Geometry is divided into two branches :

- a. **Pure Geometry**
- b. **Coordinate Geometry**

Pure Geometry is that branch of Geometry in which problems are solved by geometric constructions or arguments.

In Coordinate Geometry, the subject matter is geometric but treatment is algebraic. This branch of geometry was invented by a French mathematician and philosopher Rene

Descarte (1596 -1650). He Showed in a plane by means of two numbers --- a point ; called Coordinates . As the point is shown by two coordinates, therefore, this branch of geometry is known as Coordinate Geometry.

9.1.3 The History of Geometry

Geometry's roots return to roughly 3,000 BC in antiquated Egypt. Old Egyptians utilized a beginning time of geometry in a few different ways, including the looking over of land, development of pyramids, and space science. Around 2,900 BC, antiquated Egyptians started utilizing their insight to develop pyramids with four triangular appearances and a square base.

9.1.4 Applications of Geometry

Notwithstanding the majority of the distinctive branches of knowledge of science that exist, maybe geometry has the most significant effect on our regular daily existences. Consider the earth you are in the present moment. Everything around you has a shape, volume, surface region, area, and other physical properties. Since its cause, geometry has fundamentally affected the manners in which individuals live.

9.1.5 Common Applications of Geometry

While we may not quickly think "geometry" when we perform ordinary undertakings, geometry is surrounding us. For example, stop signs have the state of an octagon, fish tanks must be deliberately filled in order to anticipate flooding, and endowments need a specific measure of wrapping paper to look pleasant, just to give some examples genuine applications. In this geometry segment, you will learn a lot more uses of geometry that you can use on a regular premise.

9.1.6 Advanced Applications of Geometry

As we wind up in a dynamic, mechanically determined society, geometry is turning into a subject of expanding significance. For instance, sub-atomic displaying is a developing field that requires a comprehension of different courses of action of circles just as the capacity to process sub-atomic properties like volume and topology. Design is another significant use of geometry. The development of a structure and the structure of its parts are essential to consider so as to expand building wellbeing. Robot movement arranging utilizes a subarea of computational geometry that spotlights on the control of robot development. Other propelled uses of geometry incorporate biochemical displaying, typography, timber handling, PC designs and material format.

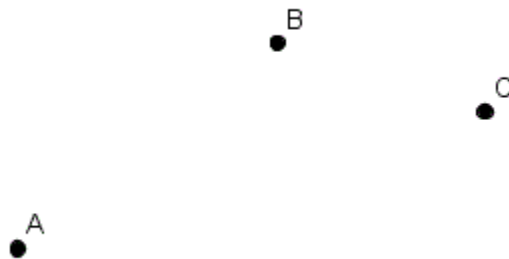
9.1.7 Basic Geometry Terms

The following are a portion of the key ideas and terms you should know so as to start your investigation of geometry.

i. Point

In geometry, we use points to determine accurate areas. They are by and large signified by a number or letter. Since points indicate a solitary, careful area, they are zero-

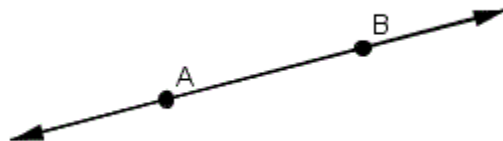
dimensional. As it were, points have no length, width, or tallness. It might be useful to think about a point as an infinitesimal "dab" on a bit of paper. A point is an exact location in space. They are shown as dots on a plane in 2 dimensions or a dot in space in 3 dimensions. It is labeled with capital letters. It does not take up any space.



Points A, B, and C

ii. Line

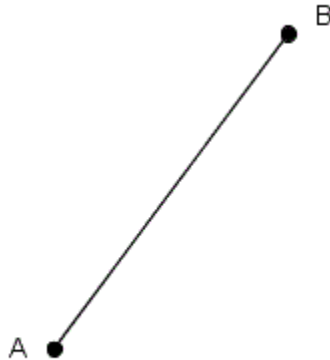
Line in geometry might be thought of as a "straight" line that can be drawn on paper with a pencil and ruler. In any case, rather than this line being limited by the elements of the paper, a line broadens boundlessly in the two headings. A line is one-dimensional, having length, however no width or tallness. Lines are extraordinarily dictated by two points. Consequently, we signify the name of a line going through the focuses A and B as AB , where the two-headed bolt connotes that the line goes through those remarkable focuses and broadens vastly in the two bearings. A line is a geometric assume that comprises of an unending number of focuses arranged straight that stretch out in the two bearings for ever (shown by the bolts toward the end). A line is recognized by a lower case letter or by two points that the line goes through. There is actually 1 line through two points. All points on a similar line are called collinear. Points not on a similar line are noncollinear.



iii. Line Segment

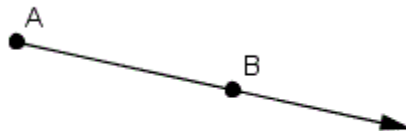
Consider the undertaking of drawing a "straight" line on a bit of paper (as we've done when pondering lines). What you've really done is make a line segment. Since our bit of paper has characterized measurements and we can't draw a line vastly toward any path, we have developed a section that starts some place and closures some place. We compose the name of a line segment with endpoints A and B as AB . Note that the documentation for lines and line segments vary on the grounds that a line segment has a characterized

length, though a line does not. A line segment is a piece of a line with two endpoints. A line segment begins and stops at two endpoints.



iv. Ray

A ray is a "straight" line that starts at one point and expands unendingly one way. A ray has one endpoint, which denotes the situation from where it starts. A ray starting at the point A that goes through point B is meant as $\overrightarrow{AB}$. This documentation demonstrates that the ray starts at point A and broadens endlessly toward point B. A ray is a piece of a line with one endpoint and reaches out one way until the end of time.



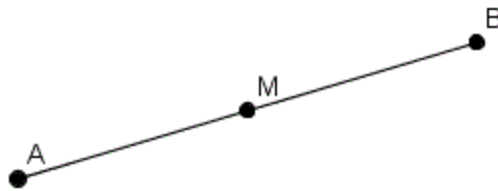
v. Endpoints

Endpoints mark the start or end of a line segment or ray. Line segments have two endpoints, giving them characterized lengths, though rays just have one endpoint, so the length of a ray can't be estimated.



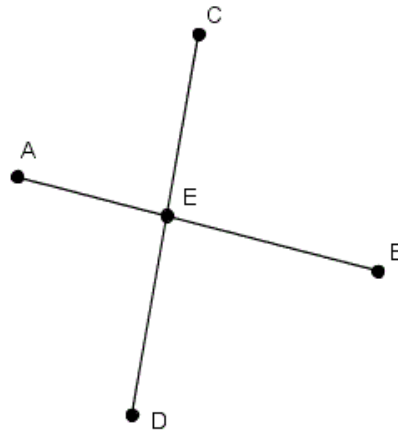
vi. Midpoint

The midpoint of a line segment denotes the position when the portion is separated into two equivalent fragments. At the end of the day, the lengths of the fragments from either endpoint to the midpoint are equivalent. For example, in the event that M is the midpoint of the portion, at that point. Note that neither lines nor rays can have midpoint since they expand endlessly in any event one bearing. It is difficult to locate the center of a line or beam that never closes.



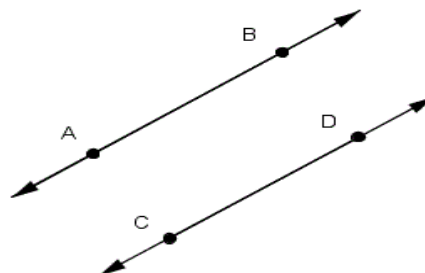
vii. Intersection

When we have lines, line segments or rays that meet, or cross at one point, we consider it as a point of intersection. As it were, those figures cross some place. In the following figure E is point of intersection of AB and CD.



viii. Parallel Lines

Two lines that will never cross are called parallel lines. On account of line segments and rays, we should consider the lines that they lie in. At the end of the day, we should consider the case that the line segments or rays were really lines that broaden limitlessly in the two bearings. In the event that the lines they lie on never intersect, they are called parallel. For example, the announcement " is parallel to ," is communicated

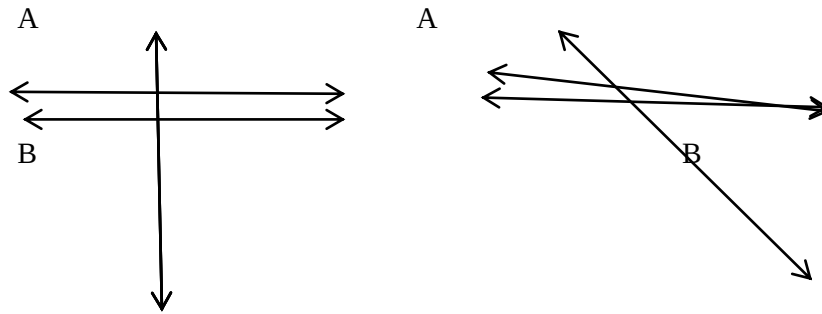


mathematically as $\parallel$.

If extended infinitely, the lines above will never meet.

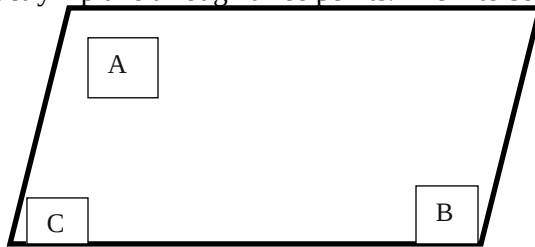
ix. Transversal

A transversal is a kind of line that crosses at any rate two different lines. The lines that a transversal crosses could and could not be parallel. In the following both cases AB is a transversal.



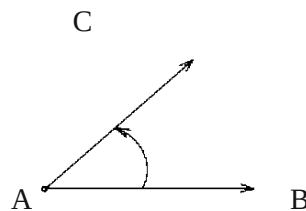
x. Plane

A plane can be thought of as a two-dimensional flat surface, having length and width, but no height. A plane extends indefinitely on all sides and is composed of an infinite number of points and lines. One way to think about a plane is as a sheet of paper with infinite length and width. A plane can be identified by 3 points in the plane or by a capital letter. There is exactly 1 plane through three points. The intersection of two planes is a line.



xi. Angle

An **angle** consists of two rays with a common endpoint. The two rays are called the sides of the angle and the common endpoint is the vertex of the angle.



An angle may be named as :

- By naming the Vertex, that is , $\angle A$.
- By naming the vertex and another point on each arm . In this case , the letter at the vertex is placed between the other two letters , that is $\angle CAB$

Each angle has a measure created by the turn about the vertex. The measure is dictated by the pivot of the terminal side about the underlying side. A counterclockwise turn produces a positive angle measure. A clockwise turn produces a negative angle measure. The units used to quantify an angle are either in degrees or radians.

xii. Types of angles

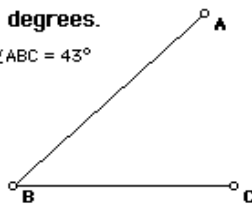
Angles can be arranged dependent on the measure: acute angle, right angle, obtuse angle, and straight angle. On the off chance that the total of proportions of two positive angles is 90° , the angles are called complementary. In the event that the total of proportions of two positive angles is 180° , the angles are called supplementary. In geometry, there are Four types of angles:

1. acute angle—an angle between 0 and 90 degrees
2. right angle—a 90 degree angle
3. obtuse angle—an angle between 90 and 180 degrees
4. straight angle—a 180 degree angle

Here are a few examples of angles:

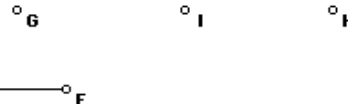
Angle ABC is an acute angle since it is between 0 and 90 degrees.

$$m\angle ABC = 43^\circ$$



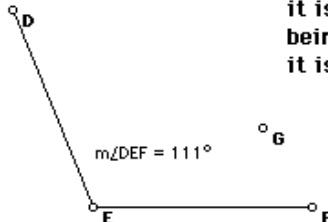
Angle GIH is a straight angle since it is a 180 degree angle. Notice that it is a line! Think about it being a straight line—so it is a straight angle

$$m\angle GIH = 180^\circ$$



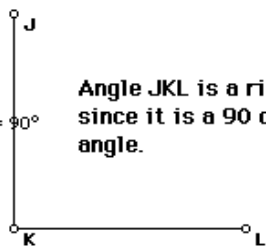
$$m\angle DEF = 111^\circ$$

Angle DEF is an obtuse angle since it is between 90 and 180 degrees



$$m\angle JKL = 90^\circ$$

Angle JKL is a right angle since it is a 90 degree angle.



xiii. Some pairs of Angles

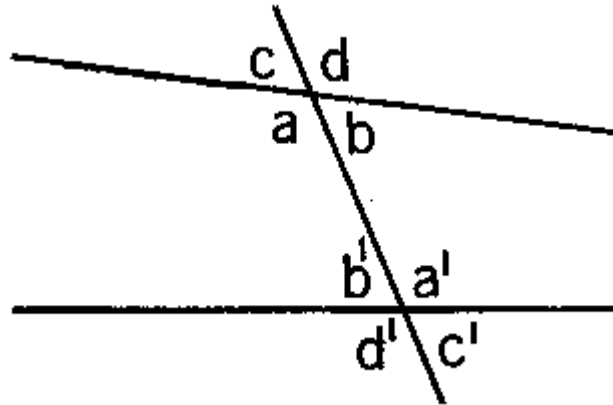
a. Alternate Angles

One of a pair of angles with different vertices and on opposite sides of a transversal at its intersection with two other lines:

A: Alternate interior angle one of a pair of angles inside the two intersected lines are called Alternate interior angles

B: Alternate exterior angle one of a pair of angles outside the two intersected lines

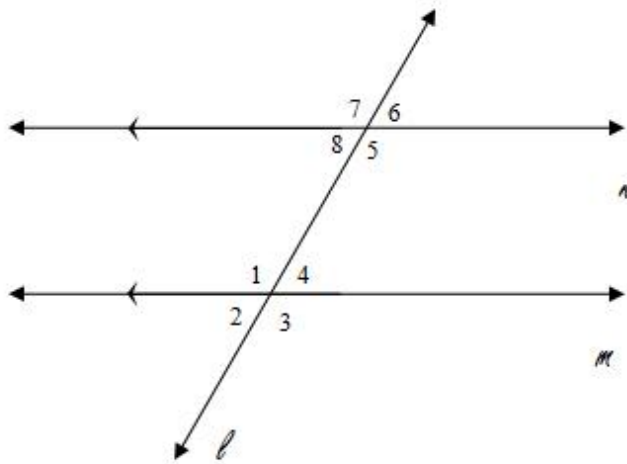
are called Alternate exterior angle .
Illustration of alternate angle



alternate interior angles a, a', b, b' ; alternate exterior angles c, c', d, d'

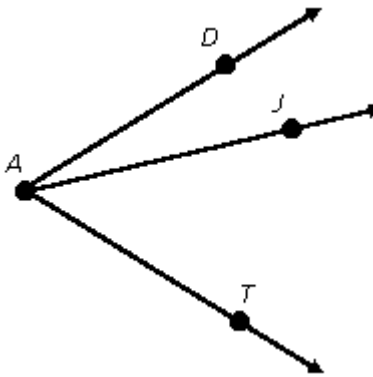
b. Corresponding angles

Corresponding angles are formed when a transversal passes through two lines. The angles that are formed in the same position, in terms of the transversal, are corresponding angles.



c. Adjacent Angles

Adjacent angles are two angles that have a typical vertex and a typical side. The vertex of a point is the endpoint of the rays that structure the sides of the edge. When we state basic vertex and normal side, we imply that the vertex point and the side are shared by the two angles. Here's a case of adjacent angles:

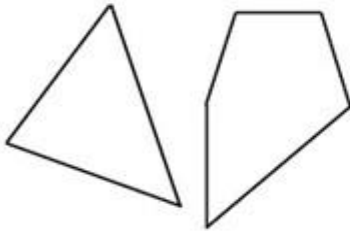


In this graph, point A is the vertex of the majority of the edges appeared. Angle DAJ and angle JAT are adjacent angles in light of the fact that the red beam AJ structures a side of every one of those angles.

xiv. Polygon

A polygon is a closed figure where the sides are all line segments. Each side must intersect exactly two others sides but only at their endpoints. The sides must be noncollinear and have a common endpoint.

Polygons

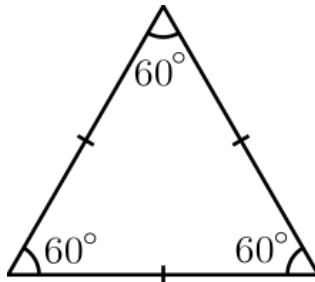


Not polygons



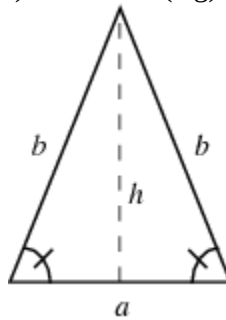
xv. Rectangular Polygon

In geometry, an equilateral triangle is a triangle in which all three sides are equal. An equilateral triangle is also equiangular; that is, all three internal angles are also congruent to each other and are each 60° . It is also a regular polygon, so it is also referred to as a **regular triangle**.



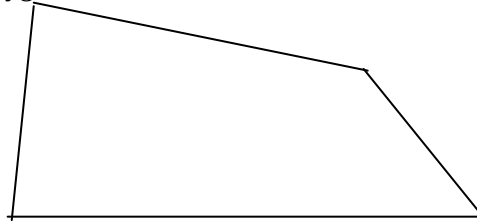
xvi. Isosceles Triangles

An isosceles triangle is a triangle with (at any rate) two equivalent sides. In the figure below, the two equivalent sides have length b and the staying side has length a . This property is proportionate to two edges of the triangle being equivalent. An isosceles triangle in this manner has both two equivalent sides and two equivalent angles. The name gets from the Greek iso (same) and skelos (leg).



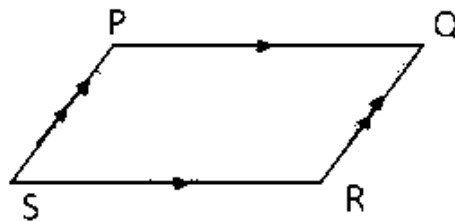
xvii. Quadrilateral

A quadrilateral is a polygon with four sides.



xviii. Parallelogram

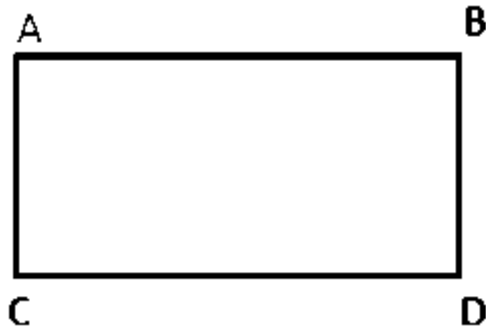
As the name suggests, a parallelogram is a quadrilateral with parallel opposite sides. In a parallelogram all the opposite sides are equal and the angles made by each opposite side is also equal. In the figure shown below, PQRS is a quadrilateral and since $PQ \parallel RS$ and $PS \parallel QR$ we call it a parallelogram



xix. Rectangle

The Rectangle is also a type of parallelogram with a little difference in the angles of the sides. In a rectangle, all the sides are at the right angle or to say perpendicular to each other. The figure below shows a quadrilateral ABCD and since all its sides are parallel to each other we may also call it a type of parallelogram with line segments placed at right

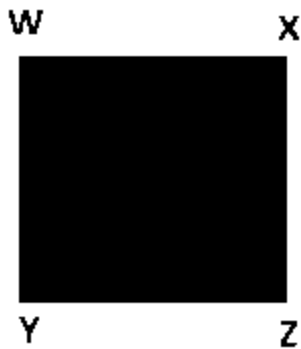
angle to each other. This can be written as $AB \parallel CD$ and $AC \parallel BD$. $\angle A = \angle B = \angle C = \angle D = 90^\circ$



xx. Square

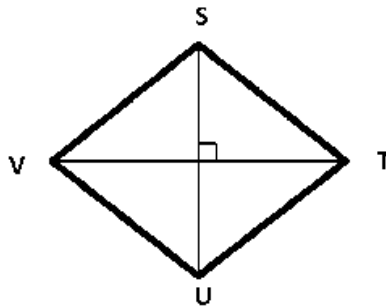
A square is a quadrilateral with all four sides equal. Every vertex of a square makes a right angle with its sides. A square is a quadrilateral which shows the following characteristic:

$WX \parallel YZ$ and $WY \parallel XZ$. $\angle W = \angle X = \angle Y = \angle Z = 90^\circ$



xxi. Rhombus

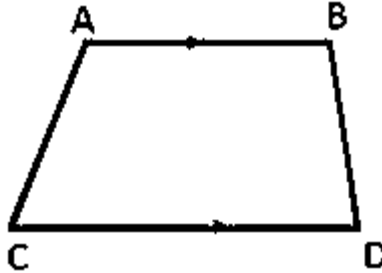
A rhombus is also a parallelogram but with all sides equal. In other words, a parallelogram that shows a resemblance to a square as regards its lengths of sides is called a rhombus. These are sometimes referred to as Diamonds also. Rhombus shows a very peculiar feature regarding their diagonals. The diagonals in a rhombus meet exactly in the middle at a right angle and are said to bisect with each other.



In the figure above we see that STUV is a quadrilateral with sides $ST \parallel UV$ and $SV \parallel TU$. All its sides are equal, thus $ST=TV=UV=VS$ which also happen to form equal angles.

xxii. Trapezium

Amongst the other types of quadrilaterals, a trapezium shows different characteristics, In a trapezium, only one pair of the four sides are parallel to each other. In a trapezium, all the sides may not be equal in measurement but one of the pairs of sides has to be parallel to each other.

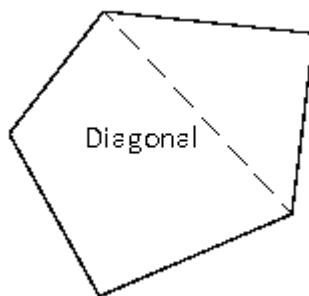


In the above figure, we can see a trapezium with sides $AB \parallel DC$. A trapezium does not show similarity in lengths and angles of sides hence is an irregular quadrilateral.

xxiii. Diagonal

A diagonal is made out of a straight line that is set at a point rather than straight up or over. On the off chance that you picture a square and draw a line associating the contrary corners, that is an askew line.

You'll discover inclining lines in geometry, and furthermore in your general surroundings. A quarterback may toss a corner to corner pass that edges over the field, or you may purchase a great new leotard with inclining stripes. Corner to corner can be either a thing or a descriptive word.



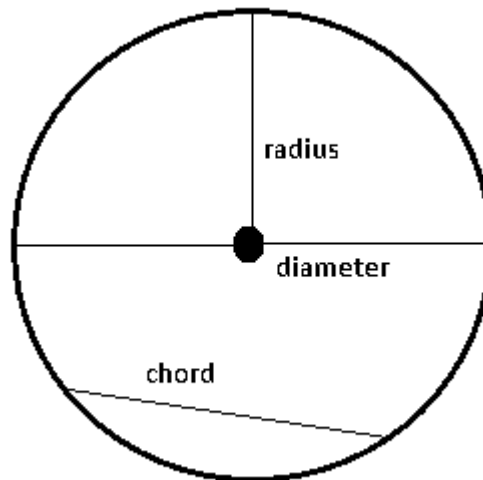
xxiv. Circle

A circle is all points in a similar plane that lie at an equal distance from an inside point. The circle is just made out of the points on the outskirts. You could think about a hula as a hula circle. It's just the points on the outskirts that are the circle. The points inside the hula band are not part of the circle and are called interior points.

Radius: The distance between the center and the circle outskirts is known as the radius.

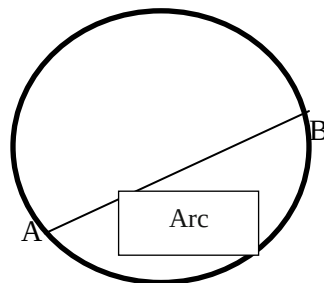
Diameter: A line portion that has the endpoints on the circle and goes through the center is known as the diameter. It is double the size of the radius.

Chord: A line fragment that has its endpoints on the circle however does not go through the center is known as a chord.



xxv. Arc

An arc is a portion of a circle consisting of two end points and the set of points on the circle between them. An arc is named by its end points. Arc AB ($\widehat{AB}$) is shown in the figure.



xxvi. Geometric Theorems

Theorems are carefully worded articulations about mathematical realities that have been demonstrated to be valid. Significant (and some not all that significant) thoughts in geometry and all of arithmetic are condensed as theorems.

Theorems have the form of **IF** one or more things are true (called the *hypothesis*), **THEN**

some other thing is true (called the *conclusion*).

Examples

- i. **IF** two straight lines intersect each other , **Then** the vertical angles are equal.
- ii. The Opposite Angle Theorem (OAT)
- iii. When two straight lines cross, opposite angles are equal
- iv. The Angle Sum of a Triangle Theorem
The interior angles of any triangle have a sum of 180°
- v. The Exterior Angle Theorem (EAT)
Any exterior angle of a triangle is equal to the sum of the opposite interior angles.
- vi. Parallel Lines Theorem (PLT).
Whenever a pair of parallel lines is cut by a transversal.
 - a. corresponding angles are equal.
 - b. alternate angles are equal .
 - c. interior angles have a sum of 180° .

xxvii. Space

Space is the arrangement of every single imaginable point on an endless number of planes. Along these lines, space covers each of the three measurements - length, width, and stature.

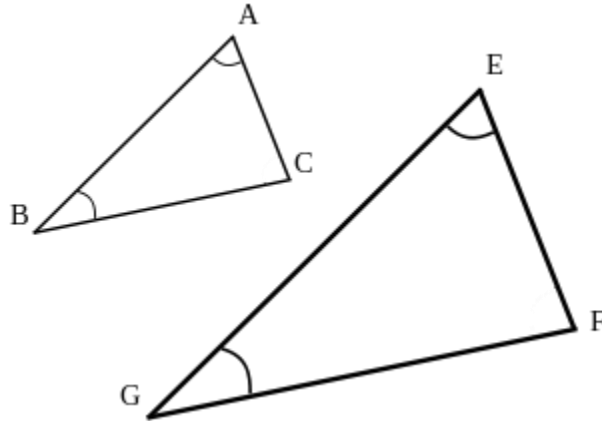
9.1.8 Teaching suggestions

Some of your students will come to class with a strong intuitive understanding of the terms point, line and, to a lesser extent, plan. For this reason, it may be difficult for them to accept the idea that these terms are undefined in a formal geometry system. The sophisticated notion that any logical system must be founded on certain undefined term is difficult for many student to appreciate, and it is suggested that you not belabor this concept. Instead, emphasize the concept of point, line, and plan as abstraction of position in physical space. The line we draw with over pencil and the flat surface of a student's desk are merely physical representations or model of these abstraction. It is worth noting that in many respects these physical representations are misleading, because they all possess a physical magnitude. It is quite difficult for some students to appreciate that a line has no thickness.

Because the concept of a plane has been studied less than the concepts of point and line in prior mathematics courses, we suggest that you spend more time discussion it. Two illustrate the concept, use examples of plane within your classroom , such as walls and book pages .Ask , would the plane represented by the flor of the classroom extend out side the building ? Down the street? How wide is this plane? Does it has Thickness?

9.2 SIMILARITY/ CONGRUENCY OF TRIANGLES

Two triangles are congruent if and only if their vertices can be matched up so that the corresponding parts (angles and sides) of the triangles are congruent .



In the above diagram, we see that triangle EFG is an enlarged version of triangle ABC i.e., they have the same shape. The same shape of the triangle depends on the angles of the triangles.

$$\angle ABC = \angle EGF, \angle BAC = \angle GEF, \angle EFG = \angle ACB$$

9.2.1 What is Congruent?

The word congruent originates from the Latin word 'congruo' which signifies 'I agree'. At the point when two items are harmonious they can be coordinated or mapped precisely to one another. They are a similar size and have a similar shape. They fit into the 's.s.s theorem' of side/side/side/every one of the three sides are the equivalent and each of the three of the angles are the equivalent. They can be superimposed on each other yet can be set distinctively regarding their direction on a plane or in a 3D space.

9.2.2 What is Similarity?

The word similarity originates from the Latin 'similis' which means like, looking like or comparative. Closeness in the mathematical world requires two items to have a similar shape however not really a similar size.

Two distinct circles for example are the two circles and along these lines similar however their size makes them extraordinary. They can be analyzed as similar shapes, however not mapped to each other. Two articles that are comparable will have a similar shape however the one could be a scaled up or a downsized rendition of the other. The direction of the shape could be unique, yet they will stay comparable. Mathematically items are similar in the event that they have a similar shape however not really a similar size. The region, height, and volume of Similar triangles are in a similar proportion as the proportion of the length of their sides.

9.2.3 Congruence Verses Similarity

Mathematical principle of exact size , shape and form – all sides and angle being equal.	Mathematical principle of similar shape but not exact in size.
No matter where the shape is set in space it can be turned and fitted into itself to be congruent.	Similar shapes match and look the same and match shape but will not be superimposed.
A very accurate mathematical and geometric term use to describe shapes.	A loss term to describe shapes and objects that may look alike or perform similar functions.
Can be used to define ideas that are specifically motivational or intrinsically linked by ideals and principles.	A means of comparing objects and experiences that are relatable to everyone who encounters similar objects and activities.
An adjective used specifically to describe objects or ideals that can be superimposed or coincidental.	An adjective used in everyday life to describe objects that could be linked due to their similar nature. Making Comparisons
Congruence uses precision and measurement in mathematical circles.	Similarity leads the reader to understand principles that are similar to be familiar with their meaning.
Congruent objects or experiences are more difficult to base on everyday human resources.	Similarities are part of life and easy to compare as we look at the world around as both mathematical and in art and literature.

9.2.4 Properties of Similar Triangles

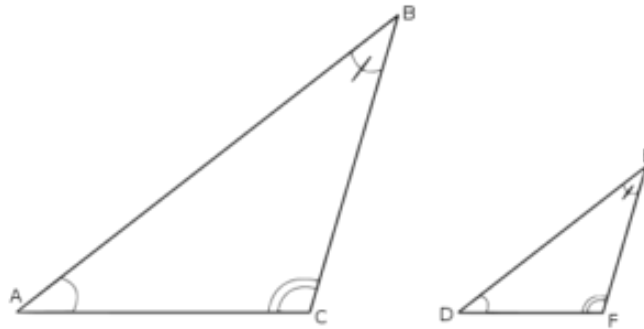
- Reflexivity: A triangle (Δ) is similar to itself
- Symmetry: If $\Delta ABC \sim \Delta DEF$, Then $\Delta DEF \sim \Delta ABC$
- Transitivity: If $\Delta ABC \sim \Delta DEF$ and $\Delta DEF \sim \Delta XYZ$, then $\Delta ABC \sim \Delta XYZ$

9.2.5 Tests to Prove that Triangles are Congruent

If two triangles are congruent, the six parts of one triangle are congruent to the six corresponding parts of the other triangle. If you are not sure whether two triangles are congruent, however, it is not necessary to compare all six parts. Sometimes three pairs of congruent corresponding parts will guarantee that two triangles are congruent. The following postulates give you three ways to show that two triangles are congruent by comparing only three pairs of corresponding parts.

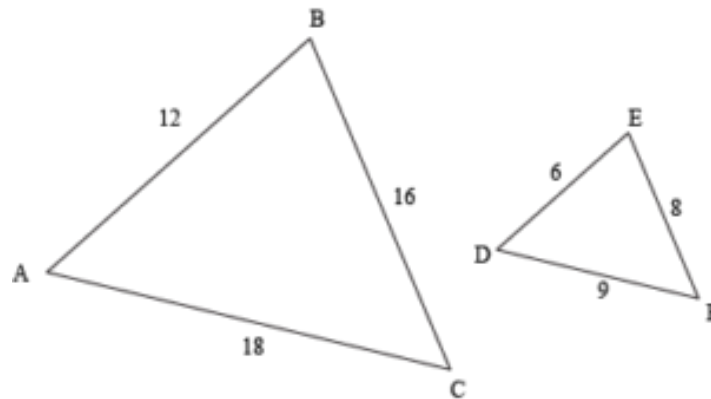
i. Angle-Side-Angle Similarity(ASA)

In any correspondence are two triangles , if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, then the triangle are congruent.



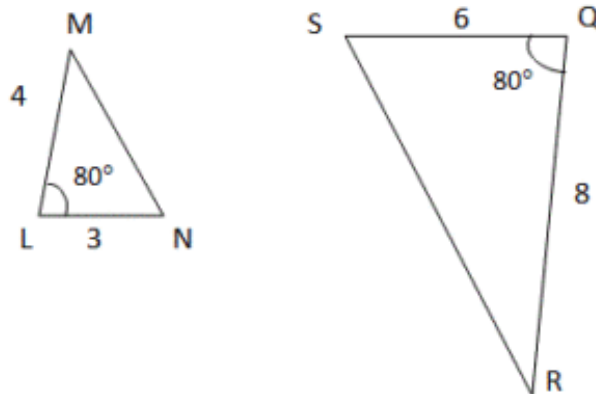
ii. Side-Side-Side Congruency(SSS)

In a correspondence of two triangles , if three sides of one triangle are congruent to the corresponding three side of the other , then the two triangles are congruent.



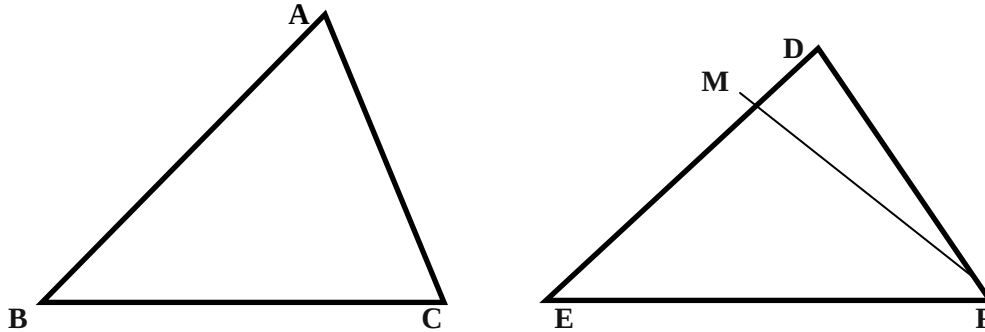
iii. Side Angle Side Congruency (SAS)

In any correspondence of two triangles , if one angle and any two sides of one triangle are congruent to the corresponding angle and sides of the other, then the triangle are congruent.



Theorem 1

In any correspondence of two triangles, if one angle and any two sides of one triangle are congruent to the corresponding angle and sides of the other, then the triangles are congruent.



Given

In $\triangle ABC \longleftrightarrow \triangle DEF$

$\angle B \cong \angle E$, $BC \cong EF$, $\angle C \cong \angle F$.

To Prove

$\triangle ABC \cong \triangle DEF$

Construction

Suppose $AB \neq DE$. Take a point M on DE such that $AB \cong ME$. Join M to F

Proof

Statements	Reasons
In $\triangle ABC \longleftrightarrow \triangle DEF$	
$\overline{AB} \cong \overline{ME}$(i)	Construction
$\overline{BC} \cong \overline{EF}$ (ii)	Given
$\angle B \cong \angle E$(iii)	Given
$\square \triangle ABC \cong \triangle MEF$	S.A.S. postulate
So, $\angle C \cong \angle MEF$	(Corresponding angles of congruent triangles)
But $\angle C \cong \angle DFE$	Given
$\square \angle DFE \cong \angle MFE$	Both congruent to $\angle C$
This is possible only if D and M are the same points, and $\overline{ME} \cong \overline{DE}$	
So, $\overline{AB} \cong \overline{DE}$(iv)	$\overline{AB} \cong \overline{ME}$ (construction) and $\overline{ME} \cong \overline{DE}$ (proved)
Thus from (ii),(iii) and (iv), we have	
$\triangle ABC \cong \triangle DEF$	S.A.S. postulate

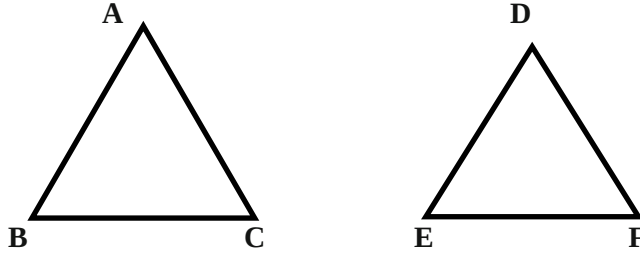
Theorem 2

In any correspondence of two triangles, if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, then the triangles are congruent. (S.A.A. $\cong$ S.A.A.)

Given

In $\triangle ABC \longleftrightarrow \triangle DEF$

$BC \cong EF, \angle A \cong \angle D, \angle B \cong \angle E$

**To Prove**

$\triangle ABC \cong \triangle DEF$

Proof

Statements	Reasons
In $\triangle ABC \longleftrightarrow \triangle DEF$	Given
$\angle B \cong \angle E$	Given
$BC \cong EF$	
$\angle C \cong \angle F$	$\angle A \cong \angle D, \angle B \cong \angle E, \text{ (Given)}$
$\square \triangle ABC \cong \triangle DEF$	A.S.A. $\cong$ A.S.A.

Theorem 3

If two angles of a triangle are congruent, then the sides opposite to them are also congruent.

Given

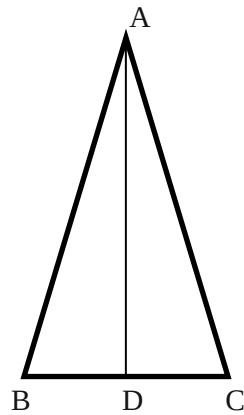
In $\triangle ABC, \angle B \cong \angle C$

To Prove

$\overline{AB} \cong \overline{AC}$

Construction

Draw the bisector of $\angle A$, meeting $\overline{BC}$ at the point D.



Proof

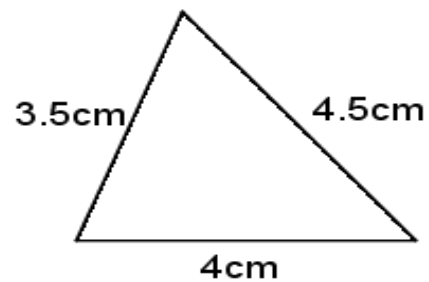
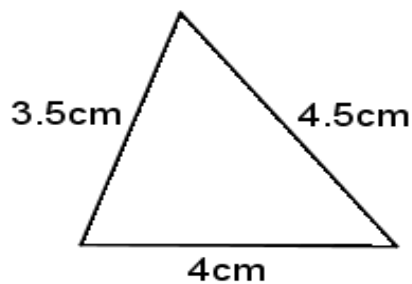
Statements	Reasons
In $\triangle ABD \leftrightarrow \triangle ACD$ $\overline{AD} \cong \overline{AD}$ $\angle B \cong \angle C$ $\angle BAD \cong \angle CAD$ $\square \triangle ABD \cong \triangle ACD$ Hence $\overline{AB} \cong \overline{AC}$	Common Given Construction S.A.A. $\cong$ S.A.A. (Corresponding sides of congruent triangles)

9.2.6 Self Assessment Questions

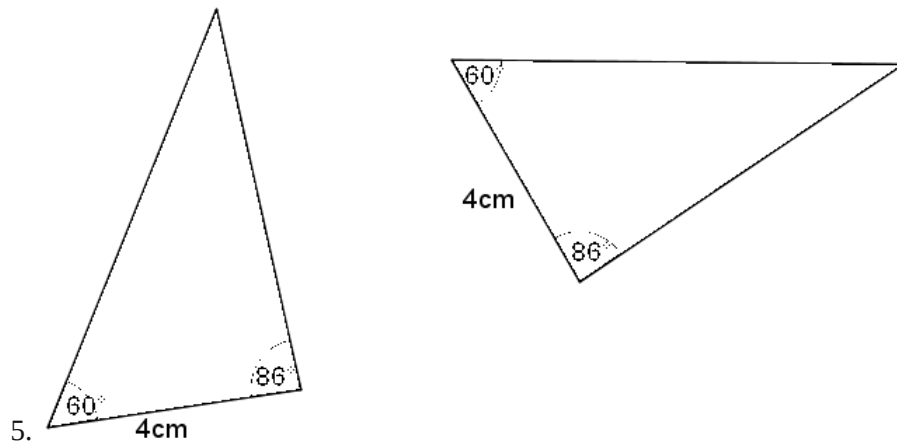
- Is every figure congruent to itself?
- Can an obtuse triangle and an acute triangle be congruent ?
- One triangle has vertices A, B, and C. Another has vertices T, R, and I. Are the two triangles similar? If so, state the similarity and the scale factor.

AB	BC	AC	TR	RI	TI
6	8	10	9	12	15
6	8	10	12	22	16
6	8	10	20	25	15
6	8	10	10	7.5	12.5

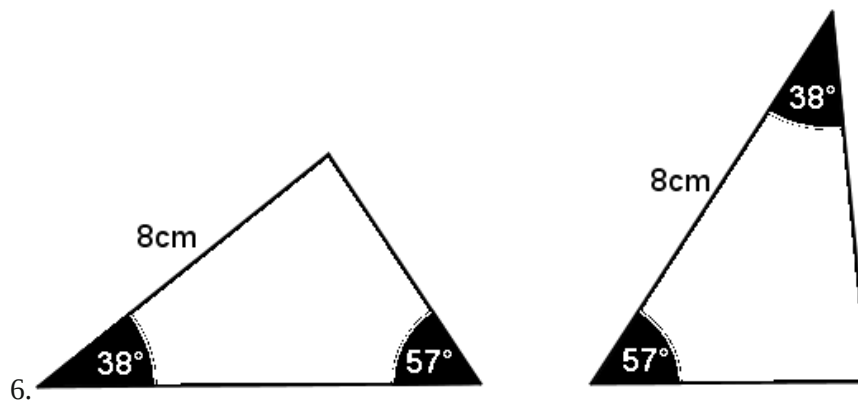
- For each pair of triangles select the correct description of congruency.



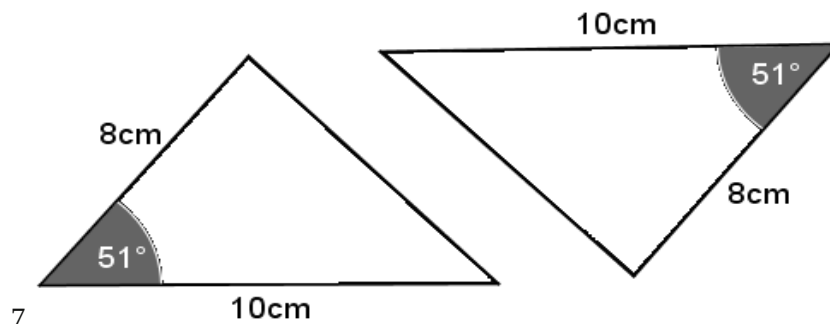
- ☐ A. SSS - three sides
- ☐ B. SAS - two sides and the included angle
- ☐ C. ASA or AAS - two angles and a corresponding side
- ☐ D. RHS - right angle, hypotenuse and another side
- ☒ E. Not (necessarily) congruent



- ☐ A. SSS - three sides
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- ☐ A. SSS - three sides
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- ☐ E. Not (necessarily) congruent



7.

- ☐ A. SSS - three sides
- ☐ B. SAS - two sides and the included angle
- ☐ C. ASA or AAS - two angles and a corresponding side
- ☐ D. RHS - right angle, hypotenuse and another side
- ☐ E. Not (necessarily) congruent

9.2.7 Teaching suggestions

To introduce the notion of similarity, you may draw two similar triangles on the white board and ask the students to compare them. Since you will want to elicit discussion about angle and side measure as well as comments about overall shape and size, encourage students to go to the board to measure your drawings. After the students have settled on an acceptable definition, present some examples which test their concept of similar triangles.

Ask students example for similar figures in their experience, such as photography slide , road side symbols , windows in classroom, desks ,doors, lawns of the school etc.

9.3 COLLINEAR POINTS

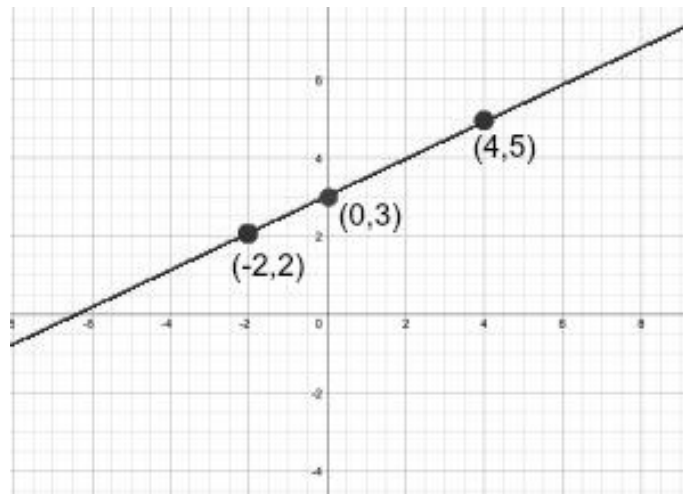
It is incredibly common to look up and see a bunch of birds lined up, perched on a group of power lines. Have you ever wondered how they do this without getting shocked? The fact is that birds don't get shocked while sitting on the power lines, because they are not

good conductors of electricity compared to the wire. For this reason, the electricity passes right by them and continues along the wire.

Another interesting fact about seeing birds on these power lines is that the set of birds on each individual power line can be said to be collinear. In mathematics, a set of points that all lie on the same line are collinear. When you see birds all lined up along a power line, you see that they all lie on the same line. Another example of collinear points would be musical notes on sheet music. We see that musical notes of the same key lie on the same bar line, making them collinear.

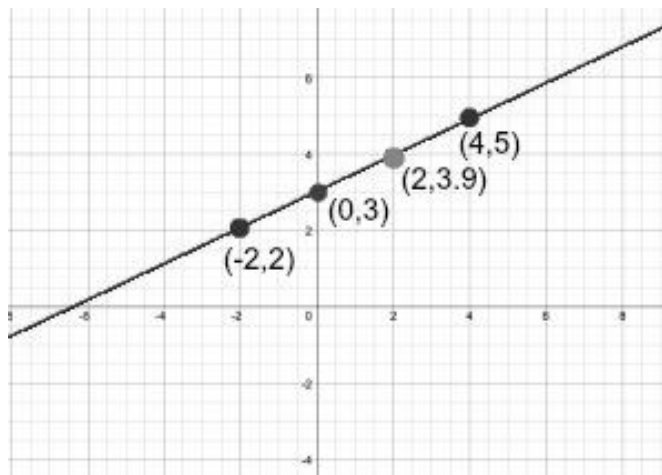
9.3.1 Determining Collinear Points

Mathematically speaking, it's easy to see that any two points are collinear. This is because you can draw a line through any two points, so they must lie on the same line. But how do we tell if a set of points are collinear if there are more than two of them? For instance, if I gave you the points $(0,3)$, $(4,5)$, and $(-2, 2)$, how can we determine if they all lie on the same line? One way to do this would be to graph them and see if they all lie on the same line.



Three Collinear Points

In this example, it appears that the three points are collinear, and in fact, they are. However, the downside to this is that you can sometimes get a point that is very close to a line and looks like it lies on it, but it actually doesn't. For instance, in our example, the three points all lie on the line $y = \frac{1}{2}x + 3$. What if I added the point $(2, 3.9)$ to the set, and I wanted to know if the points $(0,3)$, $(4,5)$, $(-2,2)$, and $(2, 3.9)$ were all collinear. If we plotted this extra point on the graph, it would appear that it fell on this line.



Appears Collinear

However, it actually doesn't, because it doesn't satisfy the equation of the line. If we plug in 2 for x , then our y ends up being $(1/2) * 2 + 3 = 4$, not 3.9. Thus, in reality, the point doesn't

9.3.2 Non Collinear Points

When the points are positioned in a place, where there is no line then it is termed as "Non Collinear Points". It is not feasible to draw a straight line through the non collinear points as the points are situated at different places.

• E

• D

• F

9.3.3 Collinearity of Three Points

Example

Points A, B and C are on the number line at a distance of 1, 4 and 8 units respectively from the origin.

Find $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$, and show that $\overline{AB} + \overline{BC} = \overline{AC}$

Solution:

$$|\overline{AB}| = 4 - 1$$

$$|\overline{AB}| = 3$$

$$|\overline{BC}| = 8 - 4 = 4$$

$$|\overline{AC}| = 8 - 1 = 7$$

$$\text{Now } |\overline{AB}| + |\overline{BC}| = 3 + 4 = 7 = |\overline{AC}|$$

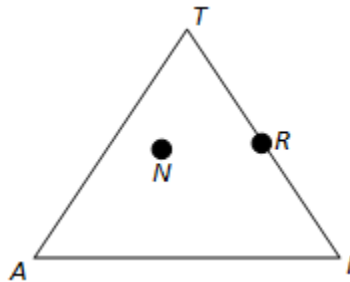
$$\text{Thus } |\overline{AB}| + |\overline{BC}| = |\overline{AC}|$$

9.3.4 Self Assessment Questions

1. Which is the correct term for points that DO NOT lie on the same line?

- ☐ Collinear
- ☐ Uncollinear
- ☐ Nonlinear
- ☐ Noncollinear

2. Name three collinear points in the diagram.



- ☐ T, R, and I
- ☐ T, R, and N
- ☐ T, A, and I
- ☒ A, R, and I

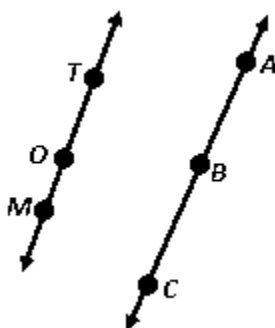
9.3.5 Teaching suggestions

A teacher should make realized the students by giving them an example from daily life like the concept of collinear points is to think about food on skewers, like in the following picture.



Each skewer represents part of a line, and the tomatoes are points. So, the tomatoes labeled A , B , and C are collinear because they have all been lined up on the same skewer. All of the tomatoes on the plate are not collinear because no single straight skewer can poke through all of them the way they are arranged.

If we take two of our tomato skewers and simplify them to a geometry diagram, it might look something like this:



We can see that one line contains points A , B , and C , so those three points are collinear. A different line contains points T , O , and M , so those three points are collinear, but they are not collinear to points A , B , and C . When points are not collinear, we call them **noncollinear**. So, for example, points A , T , and O are noncollinear because no line can pass through the three of them together.

The teacher should ask the student to locate the collinear subject in the classroom for example ceiling fan, electric bulb and electric switches etc.

9.4 DISTANCE FORMULA

Rene Descartes was a French mathematician who was the principal individual to apply Algebra to Geometry. In the seventeenth century, Descartes and Pierre de Fermat, a French legal advisor with an enthusiasm for mathematics, demonstrated how a point could be spoken to by a couple of numbers, giving its good ways from two tomahawks.

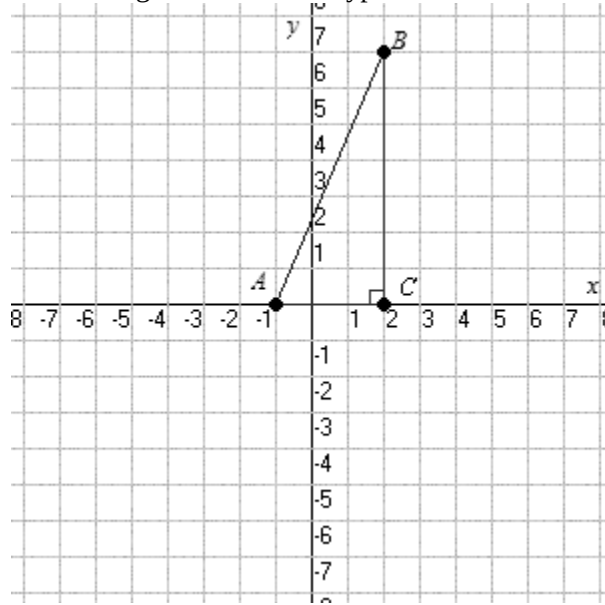
Distance between two points is the length of the line segment that connects these points.

The formula for calculating the distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on plane:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The distance formula is really just the **Pythagorean Theorem** in disguise.

To calculate the distance AB between point $A(x_1, y_1)$ and $B(x_2, y_2)$, first draw a right triangle which has the segment AB as its hypotenuse.



then by the Pythagorean Theorem,

$$(AB)^2 = (AC)^2 + (BC)^2$$

Solving for the distance AB , we have:

$$AB = \sqrt{(AC)^2 + (BC)^2}$$

Since AC is a horizontal distance, it is just the difference between the x -coordinates: $|x_2 - x_1|$. Similarly, BC is the vertical distance $|y_2 - y_1|$.

Since we're squaring these distances anyway (and squares are always non-negative), we don't need to worry about those absolute value signs.

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Find the distance between points A and B in the figure above.

In the above example, we have:

$$A(x_1, y_1) = (-1, 0), B(x_2, y_2) = (2, 7)$$

so

$$AB = \sqrt{(2 - (-1))^2 + (7 - 0)^2} = \sqrt{9 + 49} = \sqrt{58}$$

9.5 APPLICATIONS OF DISTANCE FORMULA

We use the distance formula to show that the given three non collinear points form:

- a right angle triangle
- an isosceles triangle
- an equilateral triangle
- a scalene triangle

Mobile phone network use GPS or Global positioning Systems for specifying location.

Maps are used to represent the world and specifically ordnance survey maps include a two dimensional co-ordinate system for specifying location .

Example

Show that the points A(-1,2) , B(7,5) and C(2,-6) are vertices of a right triangle.

Solution: Given A(-1,2) , B(7,5), C(2,-6).

Let a,b,c denote the lengths of the sides BC , CA , and AB respectively of triangle ABC , using formula

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We have

$$a = |BC| = \sqrt{(2 - 7)^2 + (-6 - 5)^2} = \sqrt{(5)^2 + (11)^2} = \sqrt{146}$$

$$b = |CA| = \sqrt{(2 - (-1))^2 + (-6 - 2)^2} = \sqrt{(3)^2 + (8)^2} = \sqrt{73}$$

$$c = |AB| = \sqrt{(7 - (-1))^2 + (5 - 2)^2} = \sqrt{(8)^2 + (3)^2} = \sqrt{64 + 9} = \sqrt{73}$$

$$c^2 + b^2 = (\sqrt{73})^2 + (\sqrt{73})^2$$

$$= 73 + 73$$

$$= 146$$

$$= a^2$$

9.5.1 Self-Assessment

1. What Kind of Triangle has vertices A(6,-2), B(1,-2) and C (-2,2) ?
2. Find the distance between A (2,1) and B (-4,3)
3. Find the distance between A (-1,3) and B (-2,-1)
4. Find the distance between A (a,-b) and B (b,-a)
5. Show that the points A(0,2) , B (3, -2) and C(0,-2) are vertices of a right triangle.
6. Show that the points A(6,1) , B(2,7) and C(-6,-7) are vertices of a right triangle.
7. Show that the points A (4,-2) , B(-2 , 4) and C(5,5) are vertices of an isosceles triangle.
8. Find the distance from the point (-4 , 8) to the point (-1,12)
9. Show that the points A(2,2) , B (6,6) and C (11,1) are vertices of a right angled triangle
10. For what value of x will the distance between the points (x,-1) and (3,2) be equal to 5 ?
11. Prove that (2,3) , (0,-1) and (-2,1) can be used as the vertices of an isosceles triangle

9.5.2 Teaching suggestion

The teacher should make realize the students that in the human history there had been always need to represent the position of places and to calculate the distance between them . He should , for showing the importance Of the distance formula , tell the student about the maps preserved on ancient clay tablets from Babylonian times . The teacher should ask the question from different students about the distance of school from their houses. The teacher should ask the students how do we usually measure distance between two points. The teacher should divide the class in groups and give each group the measurement activity of different lengths for example length of a pen , length of a side of a desk , the length of a classroom and length of school pitch etc. After the completion of this activity the teacher should tell the students that today we are going to discover a formula to calculate distances without having to physically measure the distance or use a scale or physically travel the route .

9.6 MIDPOINT OF TWO GIVEN POINTS

The midpoint is **halfway** between the two end points:

- Its **x value** is halfway between the two x values
- Its **y value** is halfway between the two y values

To calculate it:

- Add both "x" coordinates, divide by 2
- Add both "y" coordinates, divide by 2

Mathematically it is expressed as

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Example:

Find the midpoint of the two points A(1, -3) and B(4, 5).

Solution:

As a formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Midpoint} = (2.5, 1)$$

Find an endpoint when given a midpoint and another endpoint

We can use the midpoint formula to find an endpoint when given a midpoint and another endpoint.

Example:

M(3, 8) is the midpoint of the line AB. A has the coordinates (-2, 3), Find the coordinates of B.

Solution:

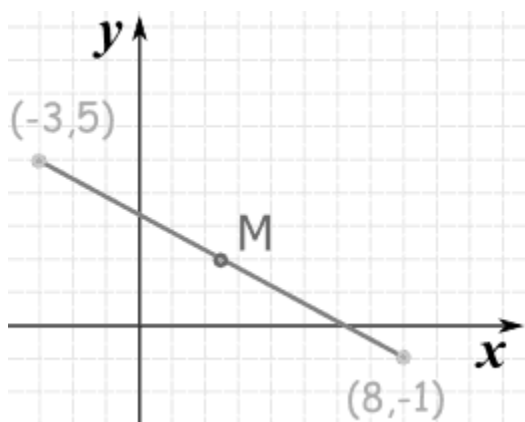
Let the coordinates of B be (x, y)

$$\frac{-2+x}{2} = 3 \Rightarrow x = 8$$

$$\frac{3+y}{2} = 8 \Rightarrow y = 13$$

Coordinates of B = (8, 13)

Example: What is the midpoint here?



Using the formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$M = \left(\frac{-3+8}{2}, \frac{5-1}{2} \right)$$

$$M = (5/2, 4/2)$$

$$M = (2.5, 2)$$

9.6.1 Self Assessment

1. A line in the standard (x, y) coordinate plane contains the points P(3, -5) and Q (-7, 9). Find the midpoint of PQ
2. Find the Midpoint between (4, -3) and (-3, 8).
3. What is the midpoint of the line segment whose endpoints are (6, 0) and (3, 7)?
4. Find the midpoint of the segment defined by points C (4, -5) and D (-8, 9).
5. What is the midpoint of the two points: (-4, 3) and (2, 5)?

9.6.2 Teaching Suggestions

The teacher should divide his /her class in to groups .Give task of measuring physically different objects in the class room and in the school. For example length of desk , length of wall ,length of white board ,length of school pitch and length of lawn etc. Then teacher ask each group to locate the Midpoint of each object to whom they have worked during

task . On the basis of this activity the students will be enabled to grasp the idea of midpoint.

9.7 EQUATION OF A STRAIGHT LINE

A straight line can be defined as a curve with constant slope.

The equation of a line is an equation in x and y which is satisfied by every point of the line and by no other point.

9.7.1 Equation of a straight line when parallel to x-axis

Let l be a line parallel to the X -axis at a distance of b units. we can regard it as the locus of the point $P(x, y)$ which moves such that it remains at a constant distance b units from the X -axis that is y -coordinate of P is always equal to b . Therefore $y=b$ is required equation

9.7.2 Equation of a straight line when parallel to Y-axis

Let l be a line parallel to the Y -axis at a distance of a units .we can regard it as the locus of the point $P(x, y)$ which moves such that it remains at a constant distance a units from the Y -axis that is x -coordinate of P is always equal to a . Therefore $x=a$ is the required equation.

Example:

Find the equation of the line

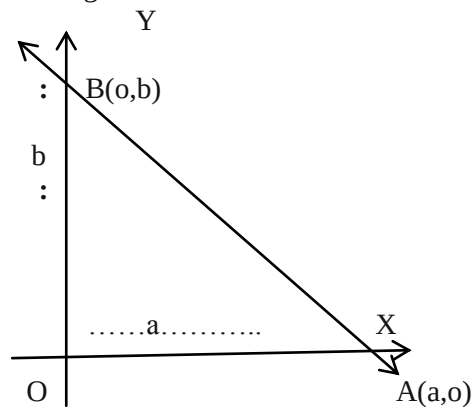
- Parallel to x -axis and 7 units below it.
- Parallel to Y -axis and 3 units to the right

Solution:

- Using the formula : $y=b$, we have , $y=7$
- Using the formula: $x=a$, we have $x=3$

9.7.3 Intercepts

Equation of a straight line has following five forms



If a line l intersects the X-axis in a point $(a,0)$, then it is said to make an intercept a on the X-axis. Similarly, if l intersects the Y-axis in a point $(0,b)$, then b is called the intercept of l on the Y-axis.

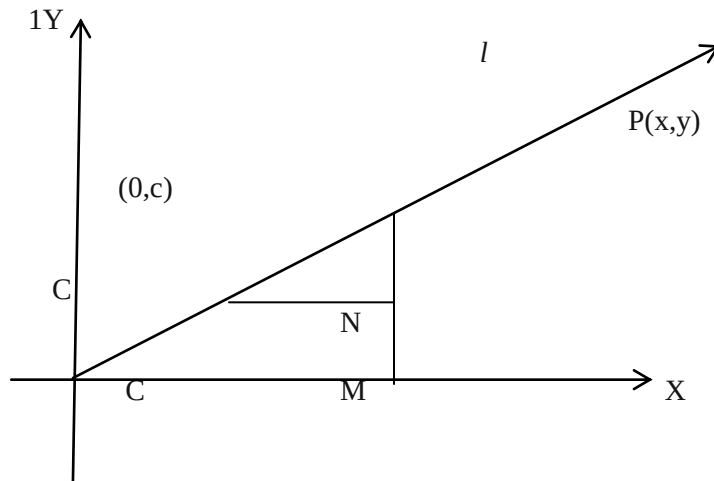
NOTE 1. The intercept on X-axis is positive if it is measured to the right of the origin and negative if measured to the left of the origin.

2. The intercept on Y-axis is positive if it is measured above the origin and negative if measured below the origin.

9.7.4 Equation of a Straight line in Various forms

i. Slope-intercept form of the equation of a straight line

Let the given line be l and $P(x,y)$ be any point on it. Then



$$\begin{aligned} m = \tan \theta &= \frac{NP}{CN} = \frac{MP - MN}{OM} \\ &= \frac{PM - OC}{x} = \frac{y - c}{x} \\ &= \frac{y - c}{x} \end{aligned}$$

$$y = mx + c$$

Such is the required equation. Since the above equation involves the slope m

Example 1

Find the equation of the straight line having y-intercept 5 and slope 3

Solution

Given that

$$C = 5$$

$$m = 3$$

using the formula $y = mx + c$

we have $y = 3x + 5$ as solution

Example 2

Find the equation of the straight line making of an angle of 135° with the X-axis and cutting the Y-axis at distance 2 units below the origin.

Solution

Here

$$m = \tan 135^\circ = -1, c = -2$$

Using the equation in slope-intercept form:

$$y = mx + c$$

$$\text{or } y = (-1)x + (-2)$$

$$\text{or } 3y = 2x - 2$$

$$\text{or } 2x + y + 2 = 0$$

is the required straight line.

ii. One-Point Form of the Equation of a Straight Line.

Since the equation of the line is

$$y = mx + c$$

If it passes through the point (x_1, y_1) , then (x_1, y_1) satisfies (1), giving

$$y_1 = mx_1 + c$$

Subtracting equation (2) from equation (1), we have

$$y - y_1 = m(x - x_1)$$

is the equation of a line passing through the point (x_1, y_1) and having slope as m .

Example

Find the equation of the line passing through the point $(2, 3)$ and whose slope is $-\frac{1}{3}$.

Solution.

Given that $(x_1, y_1) = (2, 3)$, $m = -\frac{1}{3}$

Using the formula

$$y - y_1 = m(x - x_1)$$

we have $y - 3 = -\frac{1}{3}(x - 2)$

$$\text{or } 3y - 9 = -x + 2 \text{ or } x + 3y - 11 = 0$$

iii. Two-Points Form of the Equation of a Straight Line .

Let the line pass through the points (x_1, y_1) and (x_2, y_2) , then its slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} \dots (i)$$

Since a line passing through the point (x_1, y_1) is

$$y - y_1 = m(x - x_1)$$

□ Substituting the values of m from (i), we have

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\text{or } \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$

i.e.

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

is the required equation.

Example

Find the equation of the line passing through (5 , 7) and (1 , 1).

Solution.

Equation of the line in two-point form is

$$\begin{aligned} \frac{y - y_1}{y_2 - y_1} &= \frac{x - x_1}{x_2 - x_1} \\ \text{So } \frac{y - 7}{1 - 7} &= \frac{x - 5}{1 - 5} \\ \text{or } \frac{y - 7}{-6} &= \frac{x - 5}{-4} \end{aligned}$$

$$\text{or } -4(y - 7) = -6(x - 5)$$

$$\text{or } -4y + 28 = -6x + 30$$

$$\text{or } 6x - 4y - 2 = 0$$

$$\text{or } 3x - 2y - 1 = 0$$

is the required equation.

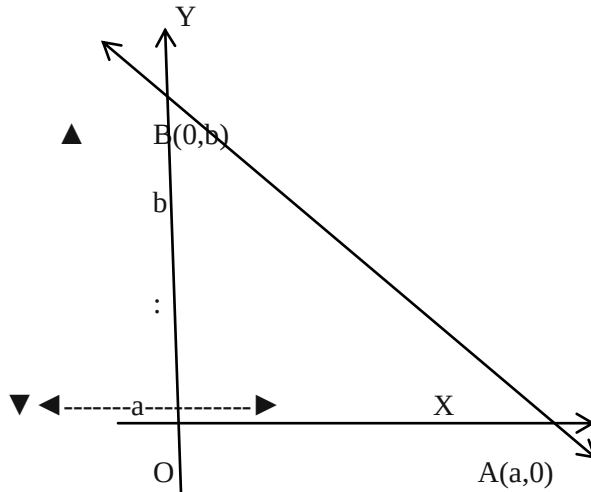
iv. Intercepts Form of the Equation of a Straight Line.

Let the line cut intercepts $a = OA$ and $b = OB$ on the axes then the coordinates of A and B are $(a, 0)$, $(0, b)$ respectively.

Therefore

Slope of line AB = m

$$= \frac{b - 0}{0 - a} = -\frac{b}{a}$$



Also, A $(a, 0)$ is a point of the line, therefore, by using equation of one-point form

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{b}{a}(x - a)$$

$$\text{or } ay = -bx + ab$$

$$\text{or } bx + ay = ab$$

Dividing both sides by ab , we get

$$\frac{x}{a} + \frac{y}{b} = 1$$

is the required equation in intercepts form .

v. Perpendicular (or Normal) Form Equation of a Straight Line

In the right-angled triangle

OAC , we have

$$\frac{p}{OA} = \cos a$$

$$\blacktriangleright OA = \frac{p}{\cos a}$$

Again , in the right-angled

Triangle OBC ,we have

$$\frac{p}{OB} = \sin a \quad \blacktriangleright OB = \frac{p}{\sin a}$$

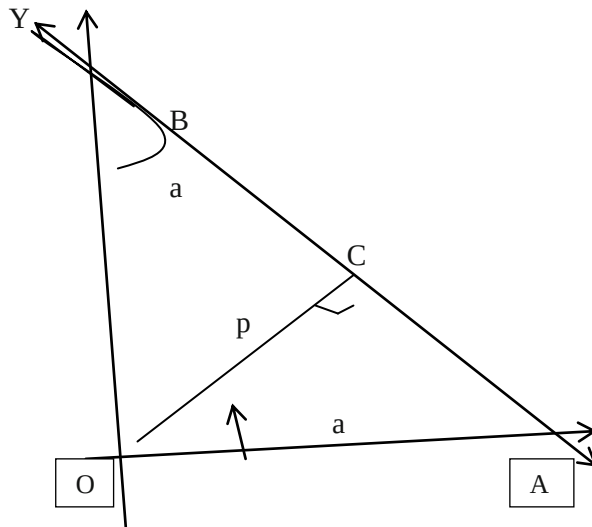
Using intercepts form, we have

$$\frac{x}{OA} + \frac{y}{OB} = 1$$

$$\text{or } \frac{x}{p/\cos a} + \frac{y}{p/\sin a} = 1$$

$$\text{or } x \cos a + y \sin a = p$$

is the required equation in perpendicular or normal form.



9.7.5 Self-Assessment

1. Reduce the equation $8x+6y-1=0$ to the
 - i. slope intercept form

- ii. intercepts form
 - iii. normal form.
2. Reduce the equation $3x+y-2=0$ to
- i. slope intercept form
 - ii. intercepts form
 - iii. normal form.

9.7.6 Teaching Suggestions

The teacher should explain the students that today we are going to use a system of co-ordinates to find various properties of the straight line between two points. We find the distance between the two points of the line joining the two points. Let's start by revising some facts about the coordinates of points.

Teacher should draw a point marked as O on a plane, together with a pair of perpendicular lines which pass through it, each with a uniform scale. Then label the lines with the letters x and y, and told the students that these are x and y axes. The fixed point O is called the origin, and it is the intersection point of the two axes. The teacher should mention any point, A, with ordered pair of numbers (x,y), he should call any student from class to measure the distance of A first from the y axis, and then from the x axis. Teacher should explain that the ordered pair (x,y) are called the coordinates of the point A. Teacher should draw cartesian system on board and should explain that any points to the left of the y, or vertical axis will have a negative x coordinate. Any points below the x or horizontal axis, will have a negative y coordinate. A is the point (3,2), as it is 3 units along in the x direction and 2 units up in the y direction. Similarly, B is the point (2,3), with the same numbers but in a different order: 2 units along in the x direction and 3 up in the y direction. Teacher should categorically show that the x co-ordinate comes first, followed by the y co-ordinate, just like the order in the alphabet. The teacher should give some examples to the class to solve.

9.8 TEACHING OF THEOREMS

Theorems are carefully worded articulations about scientific realities that have been demonstrated to be valid. Significant (and some not all that significant) thoughts in geometry and all of science are abridged as theorems.

Theorems have the type of IF at least one things are valid (called the theory), THEN some other thing is valid (called the end). Theorems are also known as conditional statements. For example : your friend says to you, IF it rains after school, THEN I will give you a ride home.

In a specific example, when you are certain every one of the states of the theory are valid, at that point you might be sure beyond a shadow of a doubt the end is likewise valid. When attempting to decide whether something is valid about some capacity, verify whether the states of the speculation are for the most part obvious.

Understandings utilizing a hypothesis need to realize the speculations just as the end.

When educating about a specific hypothesis, one thing that is regularly useful to understandings is to "play" with the speculation and perceive how it influences the end. In the event that there are a few sections to the speculation, see what occurs in the event that either is changed. An adjustment in some portion of the speculation, will have some effect – great hypotheses don't have extra, unneeded, or pointless conditions.

To make them read better, a few hypotheses are not expressed in on the off chance that ... , at that point... structure. On the off chance that there is any perplexity, repeat the hypothesis in on the off chance that... , at that point... structure.

- The geometry theorem, "The diagonals of a rhombus are perpendicular," really means: **IF** a quadrilateral is a rhombus, **THEN** its diagonals are perpendicular.

9.8.1 Types of theorems

i. Converse

The converse of a statement is formed by interchanging hypothesis and conclusion. A statement and its converse say different things . In fact some true statements have false converses

Statement: If Zahid lives in Islamabad, Then he lives in Pakistan

False converse : If Zahid lives in Pakistan , Then he lives in Islamabad

An if-then statement is false if an example can be found for which the hypothesis is true and the conclusion is false . Such an example is called **counter example**.

The converse in geometry applies to a conditional statement . In a conditional statement , the words if and "then" are used to show assumption and conclusion that are to be arrived at using logical reasoning . This is often used in theorem and problems involving proofs in geometry.

Example:

IF two parallel lines are intersected by a third line in two points , then the pairs of alternate interior angles are congruent . This is conditional statement and uses the word "if " followed by the word 'then' in the same sentence. When this relationship is reversed ,the result is a converse statement . If a pair of alternate interior angles is congruent , then the two lines are parallel. The assumption and the required conclusion are interchanged.

- IF the diagonals of a quadrilateral are opposite, Then the quadrilateral is a rhombus, assuming false. The quadrilateral may have opposite diagonals, yet except if they converge at their midpoints the figure isn't a rhombus (it is a kite shape).

ii. The inverse

The inverse is the contrapositive of the opposite or IF not p, THEN not q (or). The opposite will be valid if the opposite is valid, and false if the opposite is false.

- **IF** a quadrilateral isn't a rhombus, **THEN** the diagonals are not opposite (false – the kite once more).

iii. The Contrapositive

A proposition or theorem formed by contradicting both the subject and predicate or both the hypothesis and conclusion of a given proposition or theorem and interchanging them "if not-B then not-A " is the **contrapositive** of "if A then B "

There are a few theorems in the geometry where the contrapositive is by all accounts utilized more regularly than the theorems itself.

- IF the diagonals of a quadrilateral are not opposite, THEN the quadrilateral isn't a rhombus.

Example 1:

Statement	If two angles are congruent, then they have the same measure.
Converse	If two angles have the same measure, then they are congruent.
Inverse	If two angles are not congruent, then they do not have the same measure.
Contrapositive	If two angles do not have the same measure, then they are not congruent.

Example 2:

Statement	If a quadrilateral is a rectangle, then it has two pairs of parallel sides.
Converse	If a quadrilateral has two pairs of parallel sides, then it is a rectangle. (FALSE!)
Inverse	If a quadrilateral is not a rectangle, then it does not have two pairs of parallel sides. (FALSE!)
Contrapositive	If a quadrilateral does not have two pairs of parallel sides, then it is not a rectangle.

Example 3:

Statement	If you live in Islamabad you live in Pakistan.
Converse	If you live in Pakistan you live in Islamabad.
Inverse	If you do not live in Islamabad you do not live in Pakistan.
Contrapositive	If you do not live in Pakistan you do not live in Islamabad.

Example 4:

Statement	If I am hungry , then I will eat pizza.
Converse	If I eat pizza , then I am hungry.
Inverse	If I am not hungry , then I will not eat pizza.
Contrapositive	If I don't eat pizza , then I am not hungry.

Biconditional statements

A *biconditional theorem* is a theorem whose converse is also true (and therefore its inverse and contrapositive are true).

- An example from Geometry: **IF** two sides of a triangle are congruent, **THEN** the angles opposite them are congruent. The converse of this theorem is, **if** two angles of a triangle are congruent, **THEN** the sides opposite them are congruent. Since both the theorem and its converse (and its inverse, and its contrapositive) are true, you may write, “Two sides of a triangle are **congruent if, and only if**, the angles opposite them are congruent.”
- A rectangle is defined as a quadrilateral with four right angles or A quadrilateral is a rectangle **if, and only if**, it has four right angles.
- Definitions are always biconditional statements. They are always true and do not need to be proved; in fact they cannot be proved.

9.8.2 Proof

A proof is a logical argument that consists of a series of steps using propositions in such a way that the truth of the theorem is established.

9.8.3 Learning About Proof

Proof in geometry is commonly presented as a means of verifying the truth of conjectures, but it is usually something that is pursued after evidence has established that a conjecture is at least plausible, if not so convincing that the possibility of it being wrong seems very improbably. Mathematical proof involves a certainty, subject to agreement on initial premises, that does not apply to proofs in a scientific or legal context , where the argument hinges upon the evidence that is available and is always subject to revision if new evidence emerges.

9.8.4 Planning a proof

The proof of a theorem consists of five parts

- A statement of the theorem
- A diagram
- A list of what is given, in terms of diagram
- A statement of what is to be proved, in terms of the diagram
- A proof, which is a series of statements , supported by reasons, that lead to the desired conclusion

9.8.5 Types of proofs

There are four types of proofs namely

- Proof by construction drawing actual drawing.**

If we have to prove that if two perpendicular lines intersect each other, then corresponding angles are congruent. For proof we will have to draw two perpendicular lines and then measure corresponding angles. The equality of these angles give the proof.

ii. Logical proof

Some theorems are solved with the help of logical proofs in which with the help of logics students move from unknown to known facts. To provide logical proof one has to fix the practical work. There are some branches of geometry in which proof is sought only by logics without using any practical work.

iii. Indirect proof

It may be considered as a type of logical proof. For those theorems which have no established facts may be proved by indirect method.

If two perpendicular lines are intersected by a third perpendicular line such that pair of alternate angles are congruent then the perpendicular lines will be parallel.

To prove this theorem we will suppose that the given perpendicular lines are not parallel. At the end our assumption will be proved as wrong which shows that lines are perpendicular.

iv. Proof by contradiction

This is a type of indirect proof method. When we have no clear logics to prove the theorem in the light of established facts then a contradictory assumption against “To prove” is supposed to be right. This contradictory assumption creates different types of logical possibilities and these all logical possibilities are rejected after careful and detailed analysis as in each case an illogical result appeared. In this way the given original “TO PROVE” itself proved.

If in a triangle two angles are not congruent then the side in front of greater angle will be greater than the side in front of smaller angle. To prove this we will suppose that The length of side in front of greater angle is not greater than the side opposite to smaller angle, then there will be two options

- a. This side is equal to other side
- b. Or this side is smaller than other side.

In both the above cases, the result will be illogical. Hence our assumption will be proved as wrong.

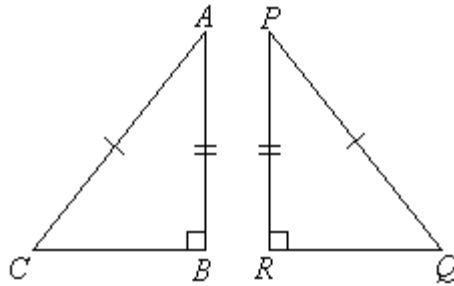
Example:

If the hypotenuse and one leg of a right triangle are equal to the hypotenuse and one leg of another right triangle, then the two right triangles are congruent.

In the following right triangles $\triangle ABC$ and $\triangle PQR$, if $AB = PR$, $AC = QR$ then $\triangle ABC \cong \triangle RPQ$.

Example:

State whether the following pair of triangles are congruent. If so, state the triangle congruence and the postulate that is used.



Solution:

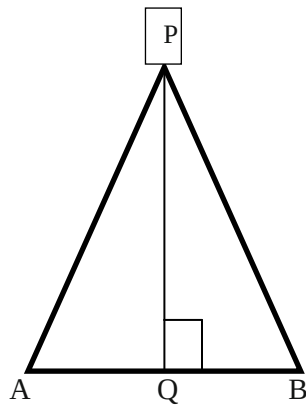
From the diagram, we can see that

- $\triangle ABC$ and $\triangle PQR$ are right triangles
- $AC = PQ$ (hypotenuse)
- $AB = PR$

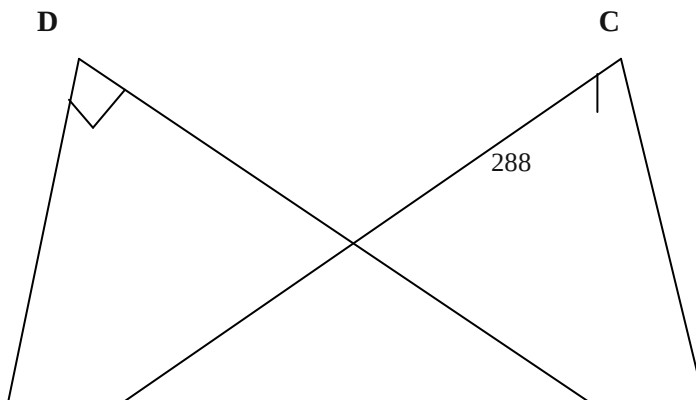
So, triangle ABC and triangle PQR are congruent.

9.8.6 Self-Assessment

1. Prove if two angles and the included side of one triangle are equal to two angles and the included side of other triangle, then triangles are congruent
2. Prove that if perpendicular from two vertices of a triangle to the opposite sides are congruent, then the triangle is isosceles.
3. In $\triangle PAB$ of figure, $\overline{PQ} \perp \overline{AB}$ and $\overline{PA} \cong \overline{PB}$, Prove that $\overline{AQ} \cong \overline{BQ}$ and $\angle APQ \cong \angle BPQ$



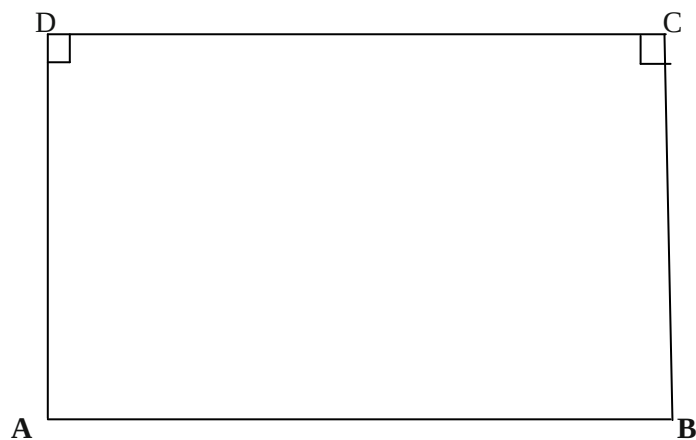
4. In the figure, $m\angle C = m\angle D = 90^\circ$ and $BC \cong AD$. Prove that $AC \cong BD$, and $\angle BAC \cong \angle ABD$.



—

A—————B

6. In the figure, $m\angle B = m\angle D = 90^\circ$ and $AD \cong BC$. Prove that ABCD is a rectangle.



9.8.7 Teaching Suggestions

When a teacher does a demonstration proof, assure his/her students that the order of statements and the approach to the proof may differ from one student to the other. Often students think there is only one way to prove something or that a proof must contain a specific number of steps. The teacher should remind his/her students that when beginning a proof, they should ask themselves, “What is given and what do I have to prove? How can I relate the given information to what I am asked to prove?” It is also helpful if you point out that the use of the Reflexive Property when two triangles share a common side is sometimes the final link in a congruence proof. Students don’t think of using it.

SUMMARY

Geometry is a branch of mathematics that focuses on study of shape, size ,relative configuration and position of geometric figures. Geometry has two branches pure geometry and coordinate geometry. Application of geometry are spread over from daily life to modern technologies.

Some of the fundamental geometry terms are

Point--- shown as dot and denoted by number or letter

Line--- a geometric figure that consists of an infinite number of points lined up straight that extends in both directions

Line segment--- a part of a line with two end points

Ray--- part of a line with one end point and extends in one direction forever

End points--- mark the beginning or end of a line segment or ray

Mid point--- the point at which the segment is divided into two equal segments

Intersection--- lines. Line segments and rays that meet or cross at a certain point is called point of intersection

Parallel lines--- Two lines that will never intersect

Transversal--- intersects at least two other lines

Plane--- two dimensional flat surface having length and width

Angle--- consists of two rays with a common end point

Complementary angle--- the sum of measures of two angles is 90

Supplementary angle--- the sum of measure of two angles is 180

Right angle---- a right angle a 90 degree angle.

Acute angle--- an acute angle is between 0 and 90 degree.

Obtuse angle--- an obtuse angle is between 90 and 180 degrees.

Straight angle--- a straight angle is a 180 degree angle.

Space--- set of all possible points on an infinite number of planes

Proof--- A proof is a logical argument that consists of a series of steps using propositions in such a way that the truth of the theorem is established.

Types of proof--- There are four types of proofs

Proof by construction drawing actual drawing, Logical proof ,Indirect proof and Proof by contradiction

Congruency of triangles--Two triangles are congruent if and only if their vertices can be matched up so that the corresponding parts (angles and sides) of the triangles are congruent .

Similarity: Mathematical principle of similar shape but not exact in size.

Angle-angle similarity--- two corresponding angles of two triangles are congruent

Side-side-side similarity--- the corresponding sides of two triangles are proportional

Side-angle-side similarity---two sides of two triangles are proportional and they have one corresponding angle congruent

Collinear points---a set of points that all lie on the same line

Non- Collinear points---the points that do not lie on the same line

Distance Formula--- the distance between two points P(x1,y1) and Q(x2,y2) is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance formula is used to show that the given three non collinear points form which type of triangle

Mid-point--- mid-point of two points A(x1,y1) ,B(x2,y2) is calculated as

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Straight line---a curve with a constant slope

Slope-intercept form--- $y = mx + c$

One point form-- $y - y_1 = m(x - x_1)$

Two points form--The line pass through the points (x_1, y_1) and (x_2, y_2) ,

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Intercepts form

$$\frac{x}{a} + \frac{y}{b} = 1$$

Perpendicular (normal) form--- $x \cos a + y \sin a = p$

Theorem---carefully worded statements about mathematical facts that have been proved to be true

Types of theorems

Converse-- The converse of a statement is formed by interchanging hypothesis and conclusion. A statement and its converse say different things . In fact some true statements have false converses

Inverse--The inverse is the contrapositive of the converse or **IF** not p , **THEN** not q (or $\sim p \rightarrow \sim q$). The inverse will be true if the converse is true, and false if the converse is false.

Contrapositive-- A proposition or theorem formed by contradicting both the subject and predicate or both the hypothesis and conclusion of a given proposition or theorem and interchanging them "if not-B then not-A " is the **contrapositive** of "if A then B "

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